# DEPENDENCE OF THE SIVERS TMD SHIFT ON QUARK LONGITUDINAL MOMENTUM FRACTION $x$ FROM LATTICE QCD

BY

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# DEDICATION

Dedicated to my wife, Nikitha, and my parents, Ravikumar and Menaka, for their unconditional love, sacrifices, and unwavering support throughout this journey.

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ABSTRACT

DEPENDENCE OF THE SIVERS TMD SHIFT ON QUARK LONGITUDINAL MOMENTUM FRACTION $x$ FROM LATTICE QCD

BY

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We compute the dependence of the isovector $(u-d)$ Sivers shift on the quark longitudinal momentum fraction $x$ in lattice QCD with 2+1 flavors of clover-improved Wilson fermions at a lattice spacing of $a \approx 0.114$ fm. The transverse momentum dependent parton distributions (TMDs) are defined through quark bilinears with staple-shaped Wilson lines. The generalized Sivers shift $\langle k_y \rangle_{TU} = m_N \, \tilde{f}_{1T}^{\perp(1)} / \tilde{f}_1^{(0)}$, the ratio of the Sivers TMD to the unpolarized TMD, is obtained from a ratio of Lorentz-invariant amplitudes. To extract the $x$-dependence, we scan the full accessible range of $b \cdot P$, the variable Fourier conjugate to $x$. This requires staples along off-axis lattice directions, which we obtain by interpolation. Using the PySR symbolic regression and statistical modeling, we systematically explored several ansätze that satisfy physical constraints, and we describe the amplitudes at four nucleon momenta with a global fit in $b \cdot P$, $b^2$, and the staple length $\eta$. After extrapolation to the light-cone limit $\hat{\zeta} \to \infty$, the magnitude of the $x$-weighted isovector Sivers shift $x\langle k_y \rangle_{TU}$ at $|b_T| = 0.34$ fm, for SIDIS/DY, is about 0.006–0.011 GeV for $0.3 \leq x \leq 0.4$. For $0.34 \leq |b_T| \leq 0.94$ fm, the isovector Sivers shift is close to, but not exactly, a product of a function of $x$ and a function of $|b_T|$, with a relative distance of 6–8% from the best factorized form.

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# 1 INTRODUCTION AND MOTIVATION

Protons and neutrons make up almost all of the visible mass in the universe, but their internal structure is only partly understood. The governing theory is known: Quantum Chromodynamics (QCD) describes quarks and gluons interacting through the exchange of color charge, and its Lagrangian is compact. In general, however, it is not solvable. At the nucleon scale, about one femtometer, the QCD coupling is large, perturbation theory does not converge, and quarks and gluons are permanently confined inside hadrons. Therefore, the nucleon's properties must be extracted from experiment via well-defined observables or computed numerically from first principles.

This thesis is about one such observable, the Sivers function $f_{1T}^{\perp}$. It describes how the transverse momentum of an unpolarized quark is correlated with the transverse spin of the nucleon that contains it. We compute the Sivers function with lattice QCD and, unlike earlier lattice calculations restricted to $b{\cdot}P = 0$, we resolve its dependence on the quark's longitudinal momentum fraction $x$. This chapter explains the origin of the Sivers function, why it matters, what experiments have measured, and what a lattice calculation can add. It closes with a summary of this work's contributions and an outline of the remaining chapters.

## 1.1 From form factors to partons

The first direct evidence that the proton has a finite size came from elastic electron scattering. In the 1950s, Hofstadter and collaborators measured the angular distribution of electrons scattered off hydrogen and found that it fell below the prediction for a point charge. The deviation is described by form factors, which are Fourier transforms of the charge and magnetization distributions, and it corresponds to a proton radius of roughly 0.8 fm [4].

A decade later, the SLAC-MIT experiments pushed the electron energy high enough to break the proton apart. In this deep inelastic scattering (DIS) regime, the cross section showed a surprising feature: at large momentum transfer $Q^2$ it depended, to a good approximation, only on the dimensionless ratio $x_B = Q^2/(2P \cdot q)$ and not on $Q^2$ separately [5, 6]. Bjorken had predicted this scaling behavior from

current algebra [7]. Feynman [8] and Bjorken and Paschos [9] gave it a simple physical meaning: in a frame where the proton moves very fast, the virtual photon scatters incoherently off pointlike constituents, the partons, and $x_B$ is the fraction of the proton momentum carried by the struck parton. Callan and Gross showed that charged spin-1/2 partons give the relation $F_2 = 2xF_1$ between the two DIS structure functions [10], and the measured structure functions follow it.

The parton picture raised a puzzle. If quarks are bound so tightly that no free quark has ever been observed, why do they behave as nearly free particles when struck hard? The answer came with non-Abelian gauge theory. Quarks carry a threefold color charge, and the color interaction is mediated by eight gluons that are themselves charged [11, 12]. Gross and Wilczek [13] and Politzer [14] showed that in such a theory the effective coupling decreases logarithmically at short distances. This property, asymptotic freedom, explains Bjorken scaling and its slow logarithmic violation. The same running makes the coupling grow at long distances, which is consistent with confinement. QCD therefore describes both regimes, and the strong coupling at long distances is the reason nucleon structure has to be studied with non-perturbative tools.

### 1.2 Beyond one dimension

The central objects of the parton model are the parton distribution functions (PDFs). The unpolarized quark PDF $f_1^q(x)$ gives the number density of quarks of flavor $q$ carrying momentum fraction $x$ in a fast-moving nucleon. PDFs are universal: the same functions describe DIS, the Drell-Yan production of lepton pairs in hadron collisions [15], jet production at the LHC, and many other processes. This universality rests on collinear factorization theorems, which separate a hard partonic cross section, calculable in perturbation theory, from the non-perturbative distributions. PDFs have been determined from global fits to experimental data with high precision over a wide range of $x$.

A PDF, however, is a one-dimensional picture. It integrates over all the transverse degrees of freedom of the parton, both its position and its momentum in the plane perpendicular to the nucleon's direction of motion. Two extensions restore this information. Generalized parton distributions (GPDs),

which appear in exclusive processes such as deeply virtual Compton scattering, encode the correlation between $x$ and the transverse position $\boldsymbol{b}_\mathrm{T}$ of the parton [16, 17, 18, 19]. This impact parameter differs from the transverse separation of the quark fields in a TMD correlator, which this thesis also denotes as $\boldsymbol{b}_\mathrm{T}$ (see Fig. 4). At zero skewness, their Fourier transform in the momentum transfer gives impact-parameter distributions, which have a density interpretation [20]. Transverse momentum dependent parton distributions (TMDs) instead describe the joint distribution in $x$ and the intrinsic transverse momentum $\boldsymbol{k}_\mathrm{T}$ [21, 22, 23]. They appear in processes where a transverse momentum is measured that is small compared to the hard scale, such as semi-inclusive DIS (SIDIS) and Drell-Yan production at low transverse momentum of the lepton pair.

The idea that partons move transversely inside the nucleon is almost as old as the parton model. Cahn showed that intrinsic transverse momentum produces a $\cos\phi_h$ modulation of the SIDIS cross section [24]. Collins and Soper formulated the factorization of processes sensitive to transverse momentum [25, 21], and Collins, Soper and Sterman used it to resum large logarithms in the transverse momentum distribution of Drell-Yan pairs [26]. The T-even TMDs at leading power were classified for Drell-Yan by Ralston and Soper [27] and by Tangerman and Mulders [28], and for SIDIS by Kotzinian [29] and by Mulders and Tangerman [22]. Boer and Mulders added the two naive T-odd distributions, $f_{1T}^\perp$ and $h_1^\perp$ [30]. Their operator definitions, including the treatment of gauge links and rapidity divergences, reached their current form much later [31, 23, 32].

Both GPDs and TMDs can be obtained from a common parent, the generalized TMDs (GTMDs), whose Fourier transforms are quantum phase-space (Wigner) distributions in $x$, $\boldsymbol{b}_\mathrm{T}$ and $\boldsymbol{k}_\mathrm{T}$ [33, 34, 35]. Figure 1 sketches these relations. Together, GPDs and TMDs make up the program of nucleon "tomography", a central goal of the Electron-Ion Collider (EIC) [36, 1, 37].

### 1.3 Spin and single-spin asymmetries

Spin measurements showed that this picture of the nucleon is incomplete. In 1988, the European Muon Collaboration measured the polarized structure function $g_1$ of the proton and concluded that the quark

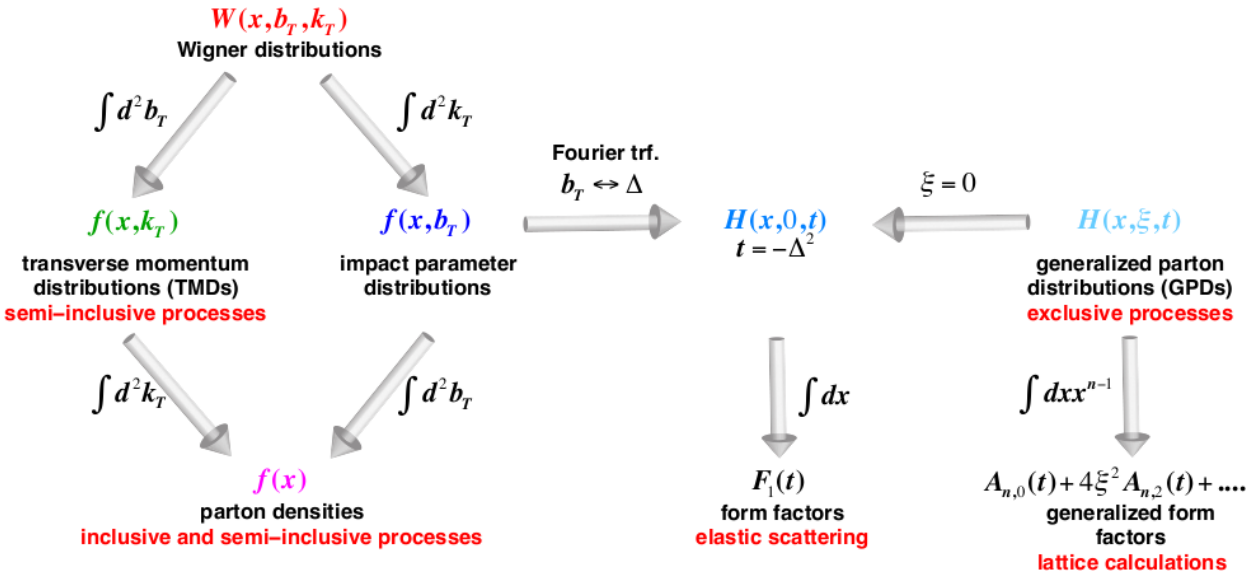



Figure 1: Relations between the distributions of unpolarized partons in an unpolarized nucleon, with the processes that probe each one. The Wigner distribution yields TMDs and impact-parameter distributions (Fourier transforms of GPDs at $\xi = 0$), and both reduce to the collinear PDF. Moments of GPDs give the elastic and generalized form factors. Reproduced from Ref. [1] (Eur. Phys. J. A **52**, 268 (2016); Fig. 2.2 of arXiv:1212.1701) under the CC BY 4.0 license.

spins account for only a small fraction of the proton spin [38]. The result, the "proton spin crisis", started a long program to understand how the spin of the nucleon is shared among quark spin, gluon spin, and orbital angular momentum [39, 18, 40]. Quark orbital motion involves transverse momentum, so TMDs are sensitive to it.

Another puzzle came from hadron collisions with a transversely polarized beam or target. Large transverse polarizations of $\Lambda$ hyperons produced in unpolarized proton collisions were observed in the 1970s [41], and large left-right asymmetries in the production of pions off transversely polarized protons were measured by the E704 experiment at Fermilab [42]. Such transverse single-spin asymmetries (SSAs) were not expected. In collinear perturbative QCD, Kane, Pumplin, and Repko showed that they are suppressed by the quark mass relative to the hard scale and by a power of $\alpha_s$ [43], making them far too small to explain the data.

In 1990, Sivers proposed a mechanism beyond the collinear framework [44]. If the transverse momentum distribution of quarks in a transversely polarized proton is not symmetric, but shifted in a direction perpendicular to the spin, then the produced hadrons inherit a preferred side, and a left-right asymmetry follows. The size of this effect is set by a new distribution function, today called the Sivers

function $f_{1T}^{\perp}(x, \boldsymbol{k}_{\mathrm{T}}^2)$. For a nucleon moving along the $z$ direction with transverse spin $\boldsymbol{S}_{\mathrm{T}}$, the number density of unpolarized quarks reads

$$f_{q/N^\uparrow}(x, \boldsymbol{k}_{\mathrm{T}}) = f_1^q(x, \boldsymbol{k}_{\mathrm{T}}^2) - \frac{\epsilon_{ij}\, k_{T,i}\, S_{T,j}}{m_N}\, f_{1T}^{\perp q}(x, \boldsymbol{k}_{\mathrm{T}}^2)\ , \tag{1.1}$$

with $\epsilon_{12} = +1$, so that $\epsilon_{ij} k_{T,i} S_{T,j} = (\hat{\boldsymbol{z}} \times \boldsymbol{k}_{\mathrm{T}}) \cdot \boldsymbol{S}_{\mathrm{T}}$. This is the "Trento" convention [45], and it is the form used throughout this thesis (compare Eq. (4.7)). Figure 2 illustrates the resulting distortion of the quark distribution in transverse momentum space.

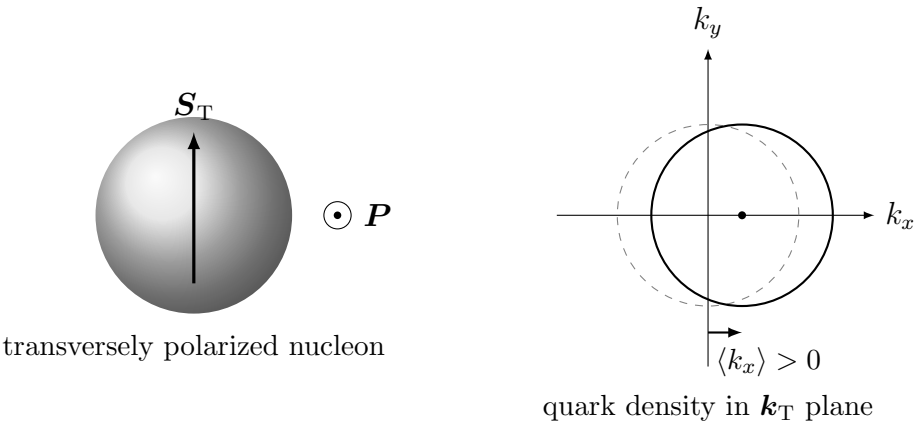



Figure 2: Schematic picture of the Sivers effect. For a nucleon moving out of the page ($\boldsymbol{P}$ along $\hat{\boldsymbol{z}}$) with transverse spin $\boldsymbol{S}_{\mathrm{T}}$ along $\hat{\boldsymbol{y}}$, the second term of Eq. (1.1) becomes $-\,k_x f_{1T}^{\perp}/m_N$. A nonzero Sivers function therefore shifts the density of unpolarized quarks along $\hat{\boldsymbol{x}}$, perpendicular to both $\boldsymbol{P}$ and $\boldsymbol{S}_{\mathrm{T}}$ (solid circle), relative to the symmetric unpolarized distribution (dashed). The shift shown, toward $+\hat{\boldsymbol{x}}$, corresponds to $f_{1T}^{\perp} < 0$.

The Sivers function soon ran into a theoretical objection. Collins argued in 1993 that time-reversal invariance of QCD forces it to vanish [46]. Under naive time reversal (time reversal applied to the operator definition without regard to the gauge link) the term $\epsilon_{ij} k_{T,i} S_{T,j}$ changes sign. In contrast, the rest of the correlator does not, so the Sivers term had to be zero. Functions with this property are still called "naively T-odd". For almost ten years, Collins's argument was taken to rule out the Sivers mechanism.

The situation changed in 2002. Brodsky, Hwang and Schmidt computed the SIDIS single-spin asymmetry in a simple spectator model and found a nonzero, leading-power result [47]. The asymmetry came from the final-state interaction between the struck quark and the target remnant: a single gluon exchange provides the phase needed for a spin asymmetry. Collins then showed where the 1993 argument

had gone wrong [48]. A gauge-invariant definition of a TMD requires a Wilson line (gauge link) that connects the two quark fields in the correlator. The link resums the gluon exchanges between the struck quark and the remnant, and its path depends on the process. In SIDIS, the struck quark leaves the nucleon, the interactions are in the final state, and the link points to the future. In Drell-Yan production, the quark annihilates with an antiquark from the other hadron; the interactions are in the initial state, and the link points to the past [49, 50, 51]. Time reversal does not force the Sivers function to vanish. It exchanges the two link paths, and the result is a definite prediction,

$$f_{1T}^{\perp}(x, \boldsymbol{k}_{\mathrm{T}}^2)\Big|_{\mathrm{SIDIS}} = -f_{1T}^{\perp}(x, \boldsymbol{k}_{\mathrm{T}}^2)\Big|_{\mathrm{DY}} \,. \tag{1.2}$$

The Sivers function is therefore universal only up to a sign fixed by the color flow of the process [48, 52]. Measuring this sign change tests the gauge-link structure of TMDs and TMD factorization. The same reasoning applies to the Boer-Mulders function $h_1^{\perp}$, which correlates the transverse spin of a quark with its transverse momentum in an unpolarized nucleon [30].

The Sivers function is connected to several other parts of hadron physics. Burkardt pointed out that the Sivers shift can be understood as the combination of a spatial distortion of the quark distribution in a transversely polarized nucleon, which is related to the GPD $E$ and thus to the anomalous magnetic moments, and an attractive final-state interaction. This "chromodynamic lensing" picture, which is model-dependent, predicts opposite signs for up and down quarks [53, 54]. Because the transverse momentum of the nucleon as a whole must vanish, the Sivers shifts of all quarks and gluons have to sum to zero (the Burkardt sum rule) [55]. The Sivers function also links to the collinear twist-3 quark-gluon correlation of Qiu and Sterman [56], which describes SSAs at large transverse momentum. The two descriptions agree in their overlap region [57]. Finally, in the light-front wave function picture, the Sivers effect requires interference between amplitudes in which the quark carries different orbital angular momentum [47], which ties it to the spin question raised by the EMC.

### 1.4 Experimental and phenomenological status

In SIDIS, $\ell N^\uparrow \to \ell' h X$, the Sivers function produces a characteristic azimuthal modulation $\sin(\phi_h - \phi_S)$ in the angle between the transverse momentum of the detected hadron and the transverse spin of the target [58]. The HERMES experiment at DESY reported the first clear observation of a nonzero Sivers asymmetry for positive pions produced off a transversely polarized proton target [59], with a final analysis of the full data set published later [60]. The COMPASS experiment at CERN found asymmetries compatible with zero on a deuteron target [61], as expected if the up and down quark Sivers functions have opposite signs and nearly cancel in an isoscalar target, and clear nonzero asymmetries on a proton target [62]. Measurements on a polarized $^3$He target at Jefferson Lab, which acts as an effective polarized neutron, added information on the flavor structure [63].

The sign change of Eq. (1.2) requires a Drell-Yan-like measurement. The STAR collaboration at RHIC measured the transverse single-spin asymmetry in $W^\pm$ and $Z^0$ production in polarized proton collisions [64], and COMPASS performed the first polarized Drell-Yan measurement with a pion beam on a transversely polarized proton target [65], together with a SIDIS analysis at matching hard scales [66]. Both results favor the predicted sign change, but the statistical significance is still limited.

Phenomenological analyses combine these data to extract the Sivers function. Early fits used the parton model with Gaussian transverse momentum dependence [67, 68]. Later analyses included TMD evolution at increasing perturbative accuracy [69, 70, 71, 72], and a global analysis showed that the SSAs in SIDIS, Drell-Yan, $e^+e^-$ annihilation and proton-proton collisions can be described with a common set of non-perturbative functions [73]. The extractions agree that the up-quark Sivers function is negative and the down-quark function is positive in SIDIS, consistent with the lensing picture. They are much less certain about the shape. The data cover a limited range of $x$, and the $x$ dependence and the $\boldsymbol{k}_\mathrm{T}$ dependence enter the fits through assumed functional forms, so the extracted shapes carry a model bias [74]. Sea quark and gluon Sivers functions are poorly constrained.

New data are coming. The 12 GeV upgrade of Jefferson Lab extends SIDIS measurements to high $x$

with high luminosity [75], and the Electron-Ion Collider will measure the Sivers asymmetry over a wide kinematic range with polarized protons and light nuclei [1, 37]. A first-principles determination of the $x$-dependence of the Sivers function would provide a benchmark for these measurements, independent of the functional forms assumed in phenomenological fits. The lattice result has its own dependence on how the data are parametrized, which we quantify in Chapter 6.

## 1.5 Lattice QCD and the $x$-dependence problem

Lattice QCD computes hadron matrix elements from the QCD Lagrangian with systematically improvable uncertainties [76, 77, 3]. Space-time is discretized on a four-dimensional Euclidean grid, the path integral is evaluated by Monte Carlo sampling, and the continuum theory is recovered as the lattice spacing goes to zero. Lattice calculations of nucleon form factors and of low Mellin moments of PDFs are well established [78, 79, 80].

Parton distributions are harder. They are defined through operators whose fields are separated along a light-like direction. Such separations cannot be represented at a single Euclidean time, so a lattice calculation cannot evaluate the defining matrix elements directly. The traditional way around this problem uses the operator product expansion: Mellin moments $\int dx\, x^{n-1} f(x)$ are matrix elements of local operators and can be computed on the lattice, but operator mixing on the hypercubic lattice limits this to the lowest few moments [78], which are not enough to reconstruct a function of $x$ (Section 2.5).

For TMDs, the situation is different. The quark fields in a TMD correlator are also separated in the transverse direction, and when the gauge link is taken slightly off the light cone, the whole operator can be boosted to a frame where it lives at a single time. Hägler, Musch, Negele and Schäfer turned this observation into a lattice method with straight links [81, 82], and Musch et al. extended it to staple-shaped links [2]. The TMD correlator is decomposed into Lorentz-invariant amplitudes $\widetilde{A}_i$, which are computed in a convenient lattice frame and then used in the frame that defines the TMDs. Ratios of amplitudes then give observables such as the generalized Sivers shift (Section 2.12). With staple-shaped gauge links this approach gave the first lattice results for the naively T-odd Sivers and

Boer-Mulders shifts, including the sign change between SIDIS and Drell-Yan [2]. Later work extended these calculations to larger values of $\hat{\zeta}$, to the pion, and to different fermion discretizations [83, 84, 85]. A complementary line of research, large-momentum effective theory (LaMET), computes equal-time "quasi" distributions in highly boosted hadrons and matches them to light-cone distributions [86, 87, 88]. Among other results, it has led to a lattice method for the Collins-Soper evolution kernel [89] and to determinations of that kernel [90, 91, 92].

The published Sivers shift calculations in the Lorentz-invariant approach are restricted to $b{\cdot}P = 0$. The variable $b{\cdot}P$ is Fourier conjugate to $x$, so setting it to zero amounts to integrating over $x$. These calculations therefore give $x$-integrated Sivers shifts, ratios in which quarks and antiquarks enter with different signs (Section 2.10), and not the dependence on the momentum fraction. The TMD Handbook, the community review written by the TMD Topical Collaboration, lists the extension to scans in $b{\cdot}P$ as a necessary next step for lattice TMD studies and shows only a preliminary exploration of it [74]. Taking that step is the subject of this thesis.

### 1.6 This thesis

We compute the isovector generalized Sivers shift $\langle k_y \rangle_{TU}$ in lattice QCD as a function of the momentum fraction $x$. The calculation uses staple-shaped gauge links and the Lorentz-invariant amplitude method, and it scans nonzero values of $b{\cdot}P$ by giving the quark separation $b$ a component along the nucleon momentum. We work with the isovector $(u - d)$ flavor combination: in the isospin limit $m_u = m_d$ the disconnected quark-loop contributions cancel in $u - d$ (Section 3.6), so the calculation involves only connected diagrams. All results in this thesis are for this combination. The main elements of the work are the following.

1. We derive the kinematic constraints the staple direction must satisfy when $b{\cdot}P \neq 0$. These constraints require off-axis staples. We construct them on the lattice by combining lattice paths and interpolating between the accessible directions. We separate the $b$-even (real) and $b$-odd (imaginary) parts of the amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ (Chapter 5).

2. We map the lattice data, taken at discrete Cartesian separations, onto the Lorentz-invariant variables $b{\cdot}P$ and $b^2$, restricting the data to $|\boldsymbol{b}| \geq 3a$ to reduce discretization artifacts (Chapters 5 and 6).

3. We show that a two-dimensional Gaussian fitted to $\widetilde{A}^{\mathrm{Re}}_{2B}$, as used in earlier lattice TMD studies [82], falls off faster in the transverse than in the longitudinal direction. It is then not integrable in $b{\cdot}P$, and the Fourier transform to $x$ does not exist. Using physics-informed symbolic regression [93], we find a power-law form $(1 + |\boldsymbol{b}|^2)^{-j}$, with $|\boldsymbol{b}|^2 = b_L^2 + b_T^2$ the squared spatial separation in the lattice frame, that decays fast enough to be absolutely integrable in $b{\cdot}P$ and describes the data well. We then fit all amplitudes simultaneously with their full covariance (Chapter 6).

4. We Fourier transform the fitted amplitudes to obtain continuous functions of $x$, extrapolate them in the Collins-Soper parameter toward the light-cone limit $\hat{\zeta} \to \infty$, and test whether the result factorizes into separate functions of $x$ and $|\boldsymbol{b}_{\mathrm{T}}|$ (Chapter 7).

At $|\boldsymbol{b}_{\mathrm{T}}| = 0.34$ fm the isovector Sivers shift has a magnitude of about 0.21 GeV at small $x$, falls to near zero by $x \approx 0.5$, and is close to a product of a function of $x$ and a function of $|\boldsymbol{b}_{\mathrm{T}}|$. Because the Sivers shift is T-odd, the opposite staple orientation gives the same magnitude with opposite sign (Eq. (1.2)). The magnitude and the $x$-shape are therefore the result; we make no claim about the overall sign.

### 1.7 Outline

Chapter 2 collects the theoretical background: our conventions, the path integral formulation of quantum field theory, QCD and its gauge symmetry, Wilson lines, asymptotic freedom and confinement, deep inelastic scattering and collinear PDFs, form factors and GPDs, SIDIS and Drell-Yan kinematics, the definition of TMDs with gauge links, the sign change of the Sivers function, rapidity divergences and the soft factor, the Sivers shift in $\boldsymbol{b}_{\mathrm{T}}$ space, the relation between $b{\cdot}P$ and $x$, and the approaches used to compute TMDs on the lattice.

Chapter 3 describes lattice QCD as used in this work: the discretized gauge and fermion actions, two- and three-point correlation functions, non-local operators with staple-shaped gauge links, and the

ratio method used to extract nucleon matrix elements.

Chapter 4 presents the theoretical framework of the calculation in detail. It defines the TMD correlator and its soft factor, the staple link geometry and the Collins-Soper parameter $\hat{\zeta}$, the symmetry constraints on the correlator, and the decomposition into the Lorentz-invariant amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ from which $f_1$ and $f_{1T}^{\perp}$ are reconstructed.

Chapter 5 explains how the $x$ dependence is extracted. It covers the kinematic constraints for $b{\cdot}P \neq 0$, the off-axis staple directions and their interpolation, the solution of the linear system for the amplitudes, and the transformation of the lattice data to Lorentz-invariant variables.

Chapter 6 develops the fit models. It explains why a Gaussian ansatz fails, introduces the short-distance cut, describes the symbolic regression search, tests factorization in $b{\cdot}P$ and $b^2$, and presents the simultaneous correlated fits.

Chapter 7 extrapolates the results to the light-cone limit $\hat{\zeta} \to \infty$ and tests the factorization of the $x$-dependent Sivers shift in $x$ and $|\boldsymbol{b}_{\mathrm{T}}|$.

Chapter 8 summarizes the results and discusses future directions.

# 2 QCD AND HADRON STRUCTURE

This chapter collects the theoretical background for the rest of the thesis, from conventions and the path integral to QCD and hadron structure, to the definition of TMDs and how they can be accessed in lattice QCD (see the outline in Section 1.7). The detailed parametrization used in our calculation is given in Chapter 4, and the lattice methods in Chapter 3.

## 2.1 Conventions

We use natural units, $\hbar = c = 1$, and the Minkowski metric $g^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$. Light-cone components of a four-vector $w$ are defined by

$$w^{\pm} = \frac{1}{\sqrt{2}} \left( w^0 \pm w^3 \right) , \tag{2.1}$$

and we write $w = (w^+, w^-, \boldsymbol{w}_{\mathrm{T}})$, where the bold transverse vector $\boldsymbol{w}_{\mathrm{T}} = (w^1, w^2)$ is a Euclidean two-vector. Scalar products read

$$a{\cdot}b = a^+ b^- + a^- b^+ - \boldsymbol{a}_{\mathrm{T}}{\cdot}\boldsymbol{b}_{\mathrm{T}} \ . \tag{2.2}$$

It is convenient to introduce the light-like vectors $\bar{n} = (1, 0, 0, 1)/\sqrt{2}$ and $n = (1, 0, 0, -1)/\sqrt{2}$, which satisfy $\bar{n}^2 = n^2 = 0$ and $\bar{n}{\cdot}n = 1$, so that $w^+ = n{\cdot}w$ and $w^- = \bar{n}{\cdot}w$. A nucleon of mass $m_N$ moving along the positive $z$ axis with no transverse momentum has

$$P = P^+ \, \bar{n} + \frac{m_N^2}{2P^+} \, n \ , \tag{2.3}$$

and its spin vector is

$$S = \Lambda \, \frac{P^+}{m_N} \, \bar{n} - \Lambda \, \frac{m_N}{2P^+} \, n + \boldsymbol{S}_{\mathrm{T}} \ , \tag{2.4}$$

where $\Lambda$ is the helicity and $\boldsymbol{S}_{\mathrm{T}}$ the transverse spin, with $\Lambda^2 + \boldsymbol{S}_{\mathrm{T}}^2 = 1$ for a pure state. The Levi-Civita tensor is normalized as $\epsilon^{0123} = +1$, and the transverse tensor $\epsilon_{ij} \equiv \epsilon^{-+ij}$ has $\epsilon_{12} = +1$. We use $\gamma^{\pm} = (\gamma^0 \pm \gamma^3)/\sqrt{2}$, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. Nucleon states are normalized covariantly, $\langle P', S' | P, S \rangle = 2E_P \, (2\pi)^3 \delta^3(\boldsymbol{P} - \boldsymbol{P'}) \, \delta_{SS'}$. These conventions follow Ref. [2]. The relations between Minkowski and Euclidean gamma matrices used in the lattice analysis are listed in Section 5.6.

In this convention, a parton with light-cone momentum fraction $x$ has $k^+ = xP^+$. Some references, including the TMD Handbook [74], place the antiquark field at $b$ and the quark field at the origin, which reverses the sign of $b$ relative to our definitions. We always write the operator as $\bar{q}(0)\,\Gamma\,\mathcal{U}\,q(b)$, as in Refs. [82, 2].

### 2.2 Quantum field theory and the path integral

A quantum field theory is fully specified by its correlation functions, the vacuum expectation values of time-ordered products of field operators. For a scalar field $\phi$ with action $S[\phi] = \int d^4x\,\mathcal{L}(\phi, \partial\phi)$, the path integral expresses them as [94]

$$\langle\Omega|\,T\{\phi(x_1)\cdots\phi(x_n)\}\,|\Omega\rangle = \frac{\int \mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\;e^{iS[\phi]}}{\int \mathcal{D}\phi\;e^{iS[\phi]}}\;. \tag{2.5}$$

The oscillating weight $e^{iS}$ makes this expression hard to define and unsuitable for numerical evaluation. Both problems are removed by a rotation to imaginary time. Setting $x^0 = -ix_4$ with real $x_4$, the action turns into $iS \to -S_E$, where $S_E$ is the Euclidean action, and correlation functions become

$$\langle\mathcal{O}_1\cdots\mathcal{O}_n\rangle_E = \frac{1}{Z}\int \mathcal{D}\phi\;\mathcal{O}_1\cdots\mathcal{O}_n\;e^{-S_E[\phi]}\;, \qquad Z = \int \mathcal{D}\phi\;e^{-S_E[\phi]}\;. \tag{2.6}$$

For a bosonic theory with a real, bounded-below action, $e^{-S_E}/Z$ is a probability distribution over field configurations. This is what allows Monte Carlo methods to be used once space-time is discretized, as described in Chapter 3.

The Euclidean formulation has a simple connection to the spectrum of the Hamiltonian $H$. For an operator $\mathcal{O}$ with Euclidean time dependence $\mathcal{O}(\tau) = e^{H\tau}\mathcal{O}(0)e^{-H\tau}$, inserting a complete set of energy eigenstates gives

$$\langle\Omega|\,\mathcal{O}(\tau)\,\mathcal{O}^\dagger(0)\,|\Omega\rangle = \sum_n \left|\langle\Omega|\mathcal{O}|n\rangle\right|^2 e^{-(E_n - E_\Omega)\tau}\;. \tag{2.7}$$

At large Euclidean time, the lowest state with nonzero overlap dominates. Hadron masses and matrix elements are extracted from lattice correlation functions in this way (Chapter 3).

Fermions are described by anticommuting Grassmann fields $\psi$ and $\bar{\psi}$. For an action bilinear in the

fermion fields, $S_F = \int d^4x\, \bar\psi\, M\, \psi$, the Grassmann integral can be done exactly,

$$\int \mathcal{D}\bar\psi\, \mathcal{D}\psi\; e^{-\bar\psi M \psi} = \det M \;, \qquad \langle \psi(x)\, \bar\psi(y) \rangle = M^{-1}(x,y) \;. \tag{2.8}$$

In QCD, $M$ is the Dirac operator in a background gluon field, so integrating out the quarks leaves a path integral over gluon fields only, weighted by $\det M$ times the exponential of the gauge action. Correlation functions of quark operators become products of quark propagators $M^{-1}$ averaged over gluon configurations.

### 2.3 Gauge invariance and QCD

QCD is a non-Abelian gauge theory with gauge group $SU(3)$ [11, 12, 13, 14]. Each quark flavor $f$ is a Dirac spinor carrying a color index in the fundamental representation. The gluon field $A_\mu = A_\mu^a t^a$ takes values in the Lie algebra, where the generators $t^a = \lambda^a/2$, $a = 1, \ldots, 8$, are built from the Gell-Mann matrices $\lambda^a$ and satisfy

$$[t^a, t^b] = i f^{abc}\, t^c \;, \qquad \mathrm{tr}\left(t^a t^b\right) = \frac{1}{2}\delta^{ab} \;, \tag{2.9}$$

with the $SU(3)$ structure constants $f^{abc}$. The QCD Lagrangian density is

$$\mathcal{L}_{\mathrm{QCD}} = \sum_f \bar q_f(x)\, (i\gamma^\mu D_\mu - m_f)\, q_f(x) - \frac{1}{4} F^a_{\mu\nu}(x) F^{a\,\mu\nu}(x) \;, \tag{2.10}$$

with the covariant derivative and field strength tensor

$$D_\mu = \partial_\mu - i g\, A^a_\mu\, t^a \;, \tag{2.11}$$

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu \;. \tag{2.12}$$

Here $g$ is the gauge coupling and $m_f$ the quark mass of flavor $f$. The last term in $F^a_{\mu\nu}$ makes the gluons interact with each other. The photon of quantum electrodynamics carries no charge and has no such self-interaction.

The Lagrangian is invariant under local $SU(3)$ transformations $V(x) = \exp\left(i\alpha^a(x) t^a\right)$,

$$\begin{aligned} q(x) &\to V(x)\, q(x) \;, \qquad \bar q(x) \to \bar q(x)\, V^\dagger(x) \;, \\ A_\mu(x) &\to V(x) A_\mu(x) V^\dagger(x) - \frac{i}{g}\big(\partial_\mu V(x)\big) V^\dagger(x) \;, \end{aligned} \tag{2.13}$$

under which $D_\mu q \to V D_\mu q$ and $F_{\mu\nu} \to V F_{\mu\nu} V^\dagger$. Physical observables must be gauge invariant.

Gauge invariance has an immediate consequence for the operators that define parton distributions. A local bilinear such as $\bar{q}(x)\Gamma q(x)$ is gauge invariant, but a non-local bilinear $\bar{q}(x)\Gamma q(y)$ with $x \neq y$ is not, because the two fields transform with $V$ at different points. The fields have to be connected by a Wilson line, the parallel transporter of color along a path. For a path $\mathcal{C}$ that starts at $x$ and ends at $y$ we define

$$\mathcal{U}[\mathcal{C}] = \mathcal{P} \exp\left( -ig \int_{\mathcal{C}} dz^\mu \, A_\mu(z) \right), \tag{2.14}$$

where the path ordering $\mathcal{P}$ places fields at points closer to the start of the path to the left. This Wilson line transforms as

$$\mathcal{U}[\mathcal{C}] \to V(x)\, \mathcal{U}[\mathcal{C}]\, V^\dagger(y) \ , \tag{2.15}$$

so the operator

$$\bar{q}(x)\, \Gamma\, \mathcal{U}[\mathcal{C}]\, q(y) \tag{2.16}$$

is gauge invariant for any path from $x$ to $y$ and any Dirac matrix $\Gamma$. For a path made of straight segments through the points $x, z_1, \ldots, z_n, y$ we write $\mathcal{U}[x, z_1, \ldots, z_n, y] = \mathcal{U}[x, z_1]\, \mathcal{U}[z_1, z_2] \cdots \mathcal{U}[z_n, y]$, the notation used for the staple-shaped links in Eq. (4.11). The operator depends on the path, not only on the endpoints. Which path is appropriate cannot be decided by gauge invariance alone; for parton distributions it is fixed by the scattering process, as we explain in Section 2.8.

A Wilson line along a closed path is a Wilson loop, $W(\mathcal{C}) = \frac{1}{3}\mathrm{tr}\,\mathcal{U}[\mathcal{C}]$, which is gauge invariant by itself. Wilson loops are the basic objects of Wilson's lattice formulation [76]. On the lattice, the gluon field is represented by link variables $U_\mu(x)$, which are the Wilson lines of single lattice links. The non-local operators of Eq. (2.16) are then products of link variables along a path on the lattice.

The smallest Wilson loop on a lattice with spacing $a$ is the plaquette $U_{\mu\nu}(x)$, the product of the four links around an elementary square in the $\mu\nu$ plane. For small $a$, it can be written as $U_{\mu\nu}(x) = \exp\big( \pm iga^2 F_{\mu\nu}(x) + \mathcal{O}(a^3) \big)$, with the sign fixed by the orientation of the loop, and expanding the

exponential gives

$$1 - \frac{1}{3}\,\mathrm{Re}\,\mathrm{tr}\,U_{\mu\nu}(x) = \frac{g^2 a^4}{6}\,\mathrm{tr}\big(F_{\mu\nu}(x)F_{\mu\nu}(x)\big) + \mathcal{O}(a^5) \qquad \text{(no sum over } \mu,\nu) \ . \tag{2.17}$$

Summing this expression over all plaquettes reproduces the Euclidean Yang-Mills action in the continuum limit. This is the origin of the Wilson gauge action used in Chapter 3.

### 2.4 Asymptotic freedom, confinement and chiral symmetry

Quantum corrections make the QCD coupling depend on the momentum scale $\mu$ at which it is defined. With $\alpha_s = g^2/(4\pi)$, the renormalization group equation reads

$$\mu^2 \frac{d\alpha_s}{d\mu^2} = \beta(\alpha_s) = -\left(b_0\,\alpha_s^2 + b_1\,\alpha_s^3 + \cdots\right), \qquad b_0 = \frac{33 - 2n_f}{12\pi} \ , \tag{2.18}$$

where $n_f$ is the number of quark flavors lighter than $\mu$. For $n_f \leq 16$ the coefficient $b_0$ is positive, and at leading order

$$\alpha_s(\mu^2) = \frac{1}{b_0\,\ln\left(\mu^2/\Lambda_{\mathrm{QCD}}^2\right)} \ . \tag{2.19}$$

The coupling decreases logarithmically at large $\mu$. This is asymptotic freedom [13, 14], and it is why processes with a large momentum transfer can be computed in perturbation theory. At the $Z$ boson mass the coupling is $\alpha_s(M_Z) = 0.1180(9)$ [95]. The scale $\Lambda_{\mathrm{QCD}}$, a few hundred MeV, is generated dynamically; the classical Lagrangian with massless quarks contains no dimensionful parameter. As $\mu$ approaches $\Lambda_{\mathrm{QCD}}$, the coupling becomes order one, and the perturbative expansion breaks down.

At long distances, QCD confines color. No isolated quark or gluon has been observed, and the hadron spectrum consists of color-singlet states. On the lattice, confinement shows up as an area law for large Wilson loops. For a rectangular loop of spatial extent $R$ and temporal extent $T$,

$$\langle W(R,T)\rangle \xrightarrow{T\to\infty} e^{-V(R)\,T} \ , \qquad V(R) \approx \sigma R \quad (R \text{ large}) \ , \tag{2.20}$$

where $V(R)$ is the potential between a static quark and antiquark and $\sigma$ is the string tension [76, 96]. This linear rise holds in pure Yang-Mills theory; with dynamical quarks, the string breaks at large $R$. The same kind of Wilson line self-interaction appears in the staple-shaped links of TMD operators,

where it produces a multiplicative dependence on the staple length. We remove it by forming ratios of amplitudes at fixed staple geometry (Section 2.9).

The third non-perturbative feature is the spontaneous breaking of chiral symmetry. For massless $u$ and $d$ quarks, the Lagrangian is invariant under separate $SU(2)$ rotations of left- and right-handed quark fields. The QCD vacuum has a nonzero quark condensate $\langle\bar{q}q\rangle$, which breaks this symmetry down to the isospin subgroup. The pions are the corresponding pseudo-Goldstone bosons, light because the quark masses are small, and the Gell-Mann-Oakes-Renner relation $f_\pi^2 m_\pi^2 = -(m_u + m_d)\langle\bar{q}q\rangle$, in the normalization where $f_\pi \approx 92$ MeV and $\langle\bar{q}q\rangle$ is the condensate of a single light flavor, links their mass to the condensate. The nucleon, with a mass of about 0.94 GeV, is much heavier than the sum of its valence quark masses. Its mass, like its internal structure, comes from the same non-perturbative dynamics that breaks chiral symmetry.

Asymptotic freedom lets experiments be interpreted in terms of parton distributions, while confinement and chiral symmetry breaking make the distributions themselves non-perturbative, which is why this thesis computes them with lattice QCD.

### 2.5 Deep inelastic scattering and the parton model

In deep inelastic lepton-nucleon scattering, $\ell(l) + N(P) \to \ell(l') + X$, the lepton exchanges a virtual photon of momentum $q = l - l'$ with the nucleon, and the final hadronic state $X$ is not observed. The standard kinematic variables are

$$Q^2 = -q^2 \ , \qquad x_B = \frac{Q^2}{2P\cdot q} \ , \qquad y = \frac{P\cdot q}{P\cdot l} \ . \tag{2.21}$$

The hadronic part of the cross section is contained in the hadronic tensor,

$$W^{\mu\nu} = \frac{1}{4\pi}\int d^4\xi \ e^{iq\cdot\xi} \, \langle P| \, [J^\mu(\xi), J^\nu(0)] \, |P\rangle \ , \tag{2.22}$$

where $J^\mu$ is the electromagnetic current. Current conservation, parity and Lorentz invariance allow two structure functions for an unpolarized target,

$$W^{\mu\nu} = \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2}\right) F_1(x_B, Q^2) + \left(P^\mu - \frac{P\cdot q}{q^2}\, q^\mu\right)\left(P^\nu - \frac{P\cdot q}{q^2}\, q^\nu\right)\frac{F_2(x_B, Q^2)}{P\cdot q} \ , \tag{2.23}$$

and, neglecting the nucleon mass, the cross section is

$$\frac{d^2\sigma}{dx_B\, dQ^2} = \frac{4\pi\alpha^2}{x_B\, Q^4}\left[(1-y)\, F_2(x_B, Q^2) + x_B\, y^2\, F_1(x_B, Q^2)\right]. \tag{2.24}$$

In the parton model, the photon scatters elastically off a quark that carries a fraction $x$ of the nucleon's light-cone momentum. The on-shell condition for the struck quark, $(k+q)^2 \approx 0$ with the quark virtuality and transverse momentum neglected, then sets $x = x_B$ up to corrections suppressed by $1/Q^2$, and the structure functions become

$$F_2(x_B) = 2x_B F_1(x_B) = x_B \sum_q e_q^2 \left[f_1^q(x_B) + f_1^{\bar{q}}(x_B)\right], \tag{2.25}$$

where $e_q$ is the quark charge in units of the proton charge and $f_1^q(x)$ is the unpolarized PDF of quark flavor $q$ [8, 7, 10]. The first equality, the Callan-Gross relation, reflects the spin 1/2 of the quarks. The corresponding "handbag" diagram is shown in Figure 3.

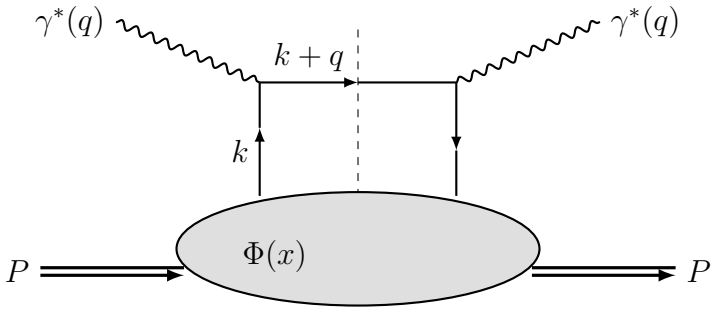



Figure 3: Handbag diagram for the leading contribution to the DIS hadronic tensor. The virtual photon scatters off a quark of momentum $k$ with $k^+ = xP^+$, and the lower blob is the quark-quark correlator whose $\gamma^+$ projection gives the PDF $f_1(x)$. The dashed line indicates the sum over final states.

QCD turns the parton model into a controlled approximation. The collinear factorization theorem states that, up to power corrections in $1/Q$, the structure functions are convolutions of perturbatively calculable coefficient functions with PDFs defined as matrix elements of quark and gluon operators [21, 23]. For the unpolarized quark distribution, this definition is

$$f_1^q(x) = \frac{1}{2}\int \frac{db^-}{2\pi}\, e^{ixP^+b^-} \langle P|\, \bar{q}(0)\, \gamma^+\, \mathcal{U}[0, nb^-]\, q(nb^-)\, |P\rangle\ . \tag{2.26}$$

The quark fields are separated along the light-cone direction $n$, and the straight Wilson line along $n$ between them makes the operator gauge invariant. In light-cone gauge $A^+ = 0$, this Wilson line is

equal to one, and $f_1^q(x)$ has the interpretation of a number density of quarks with plus-momentum $xP^+$ [23, 97]. Eq. (2.26) has support on $-1 \le x \le 1$. Positive $x$ corresponds to quarks, and negative $x$ to antiquarks, with $f_1^{\bar{q}}(x) = -f_1^q(-x)$ for $x > 0$.

Replacing $\gamma^+$ in Eq. (2.26) by $\gamma^+\gamma_5$ or by $i\sigma^{i+}\gamma_5$ defines the helicity distribution $g_1(x)$ of quarks in a longitudinally polarized nucleon and the transversity distribution $h_1(x)$ of transversely polarized quarks in a transversely polarized nucleon. Their integrals over $x$ are local matrix elements. For the isovector combination $u-d$ they give the nucleon axial charge, $\int dx\, g_1^{u-d}(x) = g_A$, and the tensor charge, $\int dx\, h_1^{u-d}(x) = g_T$. These three collinear distributions reappear, formally, as the $\boldsymbol{k}_\mathrm{T}$-integrals of TMDs in Section 2.8.

The operator in Eq. (2.26) contains ultraviolet divergences, and its renormalization introduces a scale $\mu$. The resulting dependence of $f_1(x,\mu)$ on $\mu$ is governed by the DGLAP equations [98, 99, 100],

$$\frac{df_1^q(x,\mu)}{d\ln\mu^2} = \frac{\alpha_s(\mu^2)}{2\pi}\int_x^1 \frac{dz}{z}\left[P_{qq}(z)\, f_1^q\Big(\frac{x}{z},\mu\Big) + P_{qg}(z)\, f_1^g\Big(\frac{x}{z},\mu\Big)\right], \tag{2.27}$$

with perturbatively known splitting functions $P_{ij}(z)$. Evolution in $\mu$ is perturbative, but the distributions at any one scale are not.

For lattice QCD, the relevant property of Eq. (2.26) is its relation to local operators. Expanding the non-local operator around $b^- = 0$ produces the Mellin moments,

$$\int_{-1}^1 dx\; x^{n-1} f_1^q(x) = \langle x^{n-1}\rangle_q\;, \qquad \langle P|\,\mathcal{O}_q^{\{\mu_1\cdots\mu_n\}}\,|P\rangle = 2\,\langle x^{n-1}\rangle_q\; P^{\{\mu_1}\cdots P^{\mu_n\}}\;, \tag{2.28}$$

where $\mathcal{O}_q^{\{\mu_1\cdots\mu_n\}} = \bar{q}\,\gamma^{\{\mu_1} iD^{\mu_2}\cdots iD^{\mu_n\}}\,q$ is a local twist-two operator, symmetrized in its indices and with traces removed. Local operators have no extent in time and can be computed on a Euclidean lattice, which is how lattice QCD has traditionally accessed PDFs [78, 79]. In practice, only the first few moments are available. On a hypercubic lattice, the operators for higher moments mix with operators of lower dimension, with coefficients that diverge as powers of the inverse lattice spacing. A handful of moments does not determine the $x$ dependence.

### 2.6 Form factors, GPDs and the transverse structure of the nucleon

PDFs describe the longitudinal momentum of partons. Information about their transverse position comes from elastic form factors and their generalization to GPDs. Because the lensing picture of the Sivers effect (Section 1.3) is built on this spatial information, we summarize the relevant definitions here.

The matrix element of the electromagnetic current between nucleon states of momenta $P$ and $P' = P + \Delta$ is parametrized by the Dirac and Pauli form factors,

$$\langle P'|\, J^\mu(0)\,|P\rangle = \bar{u}(P')\left[\gamma^\mu F_1(t) + \frac{i\sigma^{\mu\nu}\Delta_\nu}{2m_N}\,F_2(t)\right]u(P)\ , \qquad t=\Delta^2\ . \tag{2.29}$$

At $t=0$, $F_1$ gives the electric charge and $F_2$ the anomalous magnetic moment $\kappa$. The Sachs combinations $G_E = F_1 + \frac{t}{4m_N^2}F_2$ and $G_M = F_1 + F_2$ are what elastic scattering experiments measure, and the slope of $G_E$ at $t=0$ defines the charge radius, $\langle r^2\rangle = 6\,dG_E/dt|_{t=0}$.

GPDs are off-forward generalizations of PDFs. For unpolarized quarks they are defined by [16, 18, 17, 19]

$$\begin{aligned}&\frac{1}{2}\int\frac{dz^-}{2\pi}\,e^{ixP^+z^-}\,\langle P'|\,\bar{q}\big(-\tfrac{z}{2}\big)\,\gamma^+\,\mathcal{U}\,q\big(\tfrac{z}{2}\big)\,|P\rangle\Big|_{z^+=0,\,\boldsymbol{z}_{\rm T}=0}\\ &\qquad=\frac{1}{2\bar{P}^+}\left[H^q\,\bar{u}(P')\gamma^+u(P)+E^q\,\bar{u}(P')\frac{i\sigma^{+\nu}\Delta_\nu}{2m_N}u(P)\right],\end{aligned} \tag{2.30}$$

where $\bar{P} = (P+P')/2$, $\mathcal{U}$ is the straight light-like Wilson line between the quark fields, and $H^q$ and $E^q$ depend on $x$, the skewness $\xi = -\Delta^+/(2\bar{P}^+)$, and $t$. GPDs connect several quantities discussed so far. In the forward limit $H^q(x,0,0) = f_1^q(x)$. Their first moments are the form factors of Eq. (2.29), $\int dx\,H^q = F_1^q(t)$ and $\int dx\,E^q = F_2^q(t)$. Their second moments give the total angular momentum carried by quarks through Ji's sum rule [18],

$$J^q = \frac{1}{2}\int_{-1}^{1} dx\ x\,[H^q(x,\xi,0)+E^q(x,\xi,0)]\,, \tag{2.31}$$

which has been one of the main routes to the spin decomposition of the nucleon [40].

At $\xi = 0$, the momentum transfer is purely transverse, and the Fourier transform of $H^q$ with respect to $\boldsymbol{\Delta}_{\rm T}$ gives the density of quarks with momentum fraction $x$ at transverse distance $\boldsymbol{b}_{\rm T}$ from the center

of momentum of the nucleon [20],

$$\rho_q(x, \boldsymbol{b}_\mathrm{T}) = \int \frac{d^2\boldsymbol{\Delta}_\mathrm{T}}{(2\pi)^2}\, e^{-i\boldsymbol{b}_\mathrm{T}\cdot\boldsymbol{\Delta}_\mathrm{T}}\, H^q(x, 0, -\boldsymbol{\Delta}_\mathrm{T}^2)\ . \tag{2.32}$$

Figure 4 shows this picture: a parton carrying momentum fraction $x$ sits at transverse position $\boldsymbol{b}_\mathrm{T}$ inside the nucleon.

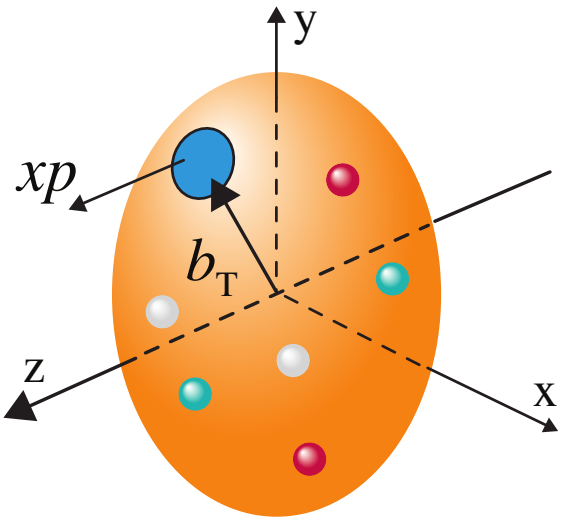



Figure 4: A parton with longitudinal momentum fraction $x$ and transverse position $\boldsymbol{b}_\mathrm{T}$ in the nucleon. Here $\boldsymbol{b}_\mathrm{T}$ is the impact parameter, the Fourier conjugate of the momentum transfer $\boldsymbol{\Delta}_\mathrm{T}$. It is not the transverse separation of the quark fields used for TMDs in Section 2.10 and in the lattice calculation, and $\rho_q(x, \boldsymbol{b}_\mathrm{T})$ and $f(x, \boldsymbol{k}_\mathrm{T})$ are not related by a Fourier transform. Reproduced from Ref. [1] (Eur. Phys. J. A **52**, 268 (2016); Fig. 2.1 of arXiv:1212.1701) under the CC BY 4.0 license.

In a transversely polarized nucleon, the density acquires a dipole term proportional to the Fourier transform of $E^q$. The quark distribution is shifted perpendicular to the spin, and the average shift of flavor $q$ is set by its contribution to the anomalous magnetic moment, $\kappa_q/(2m_N)$ [53]. From the measured magnetic moments of the proton and neutron, one finds $\kappa_u \approx 1.67$ and $\kappa_d \approx -2.03$, so up and down quarks are shifted in opposite directions. Burkardt argued that the final-state interaction in SIDIS is, on average, attractive, so it converts this spatial shift into a transverse-momentum shift in the opposite direction. This "chromodynamic lensing" picture relates the sign of the Sivers function of each flavor to the sign of $\kappa_q$ [53, 54]. It is a model-dependent argument, but it predicts opposite signs for the up and down quark Sivers functions, in agreement with the experimental extractions (Section 1.4).

### 2.7 SIDIS, Drell-Yan and TMD factorization

Transverse momentum dependent distributions enter processes in which a transverse momentum much smaller than the hard scale is measured. The two standard examples are semi-inclusive DIS and the Drell-Yan process.

In semi-inclusive DIS, $\ell(l) + N(P,S) \to \ell(l') + h(P_h) + X$, one hadron $h$ in the final state is detected in addition to the scattered lepton. Besides $x_B$, $y$ and $Q^2$, the kinematics are described by the fraction $z = P{\cdot}P_h/P{\cdot}q$ of the virtual photon energy carried by the hadron and by the transverse momentum $\boldsymbol{P}_{hT}$ of the hadron in the frame where the nucleon and the virtual photon move along the $z$ axis. The azimuthal angles $\phi_h$ of $\boldsymbol{P}_{hT}$ and $\phi_S$ of the target spin $\boldsymbol{S}_{\rm T}$ are measured around the photon direction, following the Trento conventions [45]. For $|\boldsymbol{P}_{hT}| \ll Q$, the cross section on a transversely polarized target has the structure [58]

$$d\sigma \propto F_{UU,T} + \ldots + |\boldsymbol{S}_{\rm T}| \, \sin(\phi_h - \phi_S) \, F_{UT,T}^{\sin(\phi_h - \phi_S)} + \ldots \ , \tag{2.33}$$

where the dots denote other azimuthal modulations and polarization states. At leading power, the structure functions are convolutions of a TMD with a transverse momentum-dependent fragmentation function $D_1$,

$$F_{UU,T} = \mathcal{C}\left[f_1 D_1\right] \ , \qquad\qquad F_{UT,T}^{\sin(\phi_h - \phi_S)} = \mathcal{C}\left[-\frac{\hat{\boldsymbol{h}}{\cdot}\boldsymbol{k}_{\rm T}}{m_N} \, f_{1T}^{\perp} D_1\right], \tag{2.34}$$

with $\hat{\boldsymbol{h}} = \boldsymbol{P}_{hT}/|\boldsymbol{P}_{hT}|$ and

$$\mathcal{C}[w\, f\, D] = x_B \sum_q e_q^2 \int d^2\boldsymbol{k}_{\rm T} \, d^2\boldsymbol{p}_{\rm T} \, \delta^{(2)}\left(\boldsymbol{k}_{\rm T} - \boldsymbol{p}_{\rm T} - \frac{\boldsymbol{P}_{hT}}{z}\right) w(\boldsymbol{k}_{\rm T}, \boldsymbol{p}_{\rm T}) \ f^q(x_B, \boldsymbol{k}_{\rm T}^2) \ D^q(z, \boldsymbol{p}_{\rm T}^2) \ . \tag{2.35}$$

Here $\boldsymbol{k}_{\rm T}$ is the transverse momentum of the quark in the nucleon and $\boldsymbol{p}_{\rm T}$ the transverse momentum of the fragmenting quark relative to the observed hadron. The ratio $A_{UT}^{\sin(\phi_h - \phi_S)} = F_{UT,T}^{\sin(\phi_h - \phi_S)}/F_{UU,T}$ is the Sivers asymmetry measured by HERMES, COMPASS and Jefferson Lab (Section 1.4). The transverse momentum of the observed hadron receives contributions from the quark's intrinsic motion in the nucleon and from the fragmentation process, and the convolution cannot be undone without

additional input. This is one reason why independent information on TMDs, for example from lattice QCD, is useful.

In the Drell-Yan process, $h_a(P_a) + h_b(P_b) \to \ell^+\ell^-(q) + X$, a quark from one hadron annihilates with an antiquark from the other into a virtual photon or $Z$ boson, which decays into a lepton pair of invariant mass $Q$ and transverse momentum $\boldsymbol{q}_{\rm T}$ [15]. No fragmentation is involved, and for $|\boldsymbol{q}_{\rm T}| \ll Q$ the cross section is determined by two TMDs. The TMD factorization theorem for this process takes the form [26, 23, 74]

$$\frac{d\sigma}{dQ^2\,dy\,d^2\boldsymbol{q}_{\rm T}} = \sigma_0 \sum_q e_q^2\, H(Q,\mu) \int \frac{d^2\boldsymbol{b}_{\rm T}}{(2\pi)^2}\, e^{-i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{q}_{\rm T}}\, \tilde{f}_1^q(x_a, \boldsymbol{b}_{\rm T}; \mu, \zeta_a)\, \tilde{f}_1^{\bar{q}}(x_b, \boldsymbol{b}_{\rm T}; \mu, \zeta_b) + (q \leftrightarrow \bar{q}) + \mathcal{O}\!\left(\frac{|\boldsymbol{q}_{\rm T}|}{Q}\right), \tag{2.36}$$

where $\sigma_0$ is the Born cross section, $H$ is a perturbative hard factor, $y$ is the rapidity of the lepton pair, and $\tilde{f}_1(x, \boldsymbol{b}_{\rm T})$ is the Fourier transform of the TMD in the transverse momentum, Eq. (4.8). The scales $\zeta_a$ and $\zeta_b$, with $\zeta_a\zeta_b = Q^4$, arise from the rapidity divergences discussed in Section 2.9. These rapidity scales have mass dimension two and are not the same as the Lorentz-invariant $\zeta = 2v\cdot P/\sqrt{|v^2|}$ of Chapter 4; only the dimensionless $\hat{\zeta}$ enters the lattice calculation. Transverse position space is the natural setting for TMD factorization: the convolution in $\boldsymbol{k}_{\rm T}$ becomes a product, and evolution is multiplicative. Similar factorization theorems hold for SIDIS and for $e^+e^-$ annihilation into two hadrons [31, 23]. A single-spin asymmetry in Drell-Yan with a transversely polarized hadron measures the Sivers function of that hadron in the Drell-Yan process, which makes the sign change of Eq. (1.2) testable.

### 2.8 The TMD correlator and gauge links

TMDs are defined through the quark-quark correlator

$$\Phi^{[\Gamma]}(x, \boldsymbol{k}_{\rm T}; P, S; \mathcal{C}) = \int \frac{db^-\, d^2\boldsymbol{b}_{\rm T}}{(2\pi)^3}\, e^{ixP^+b^- - i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\, \frac{1}{2}\langle P,S|\,\bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\,q(b)\,|P,S\rangle\Big|_{b^+=0}, \tag{2.37}$$

which generalizes Eq. (2.26) by allowing a transverse separation $\boldsymbol{b}_{\rm T}$ between the quark fields. Integrating over $\boldsymbol{k}_{\rm T}$ sets $\boldsymbol{b}_{\rm T} = 0$ and formally returns the collinear correlator; in QCD the $\boldsymbol{k}_{\rm T}$ integral is ultraviolet divergent (Section 2.10). Eq. (2.37) is schematic: it omits the soft factor and the regularization of

rapidity divergences, which are included in the full definition, Eq. (4.1). It is enough to discuss the gauge link and the classification of TMDs.

### 2.8.1 Process-dependent gauge links

For collinear PDFs, the fields sit on a light-like line, and the straight Wilson line along that line is the natural gauge-invariant choice. When the fields are separated transversely, no path is singled out by gauge invariance. The path is determined instead by the physics of the process [48, 50, 51]. In SIDIS, the quark struck by the virtual photon moves out of the nucleon with a large momentum along the light-cone direction $n$. Before it hadronizes, it can exchange gluons with the colored remnant of the target. In the eikonal approximation, each such exchange with a gluon of momentum $l$ contributes a factor proportional to $n{\cdot}A/(n{\cdot}l + i\epsilon)$, and summing all exchanges builds a Wilson line from the position of the quark field to infinity along $n$. Connecting the two quark fields, together with a transverse link at infinity that is needed for gauge invariance in light-cone gauge [50], gives the staple-shaped gauge link

$$\mathcal{U}[\mathcal{C}_b]_{\mathrm{SIDIS}} = \mathcal{U}[0,\ +\infty n,\ +\infty n + b,\ b]\ . \tag{2.38}$$

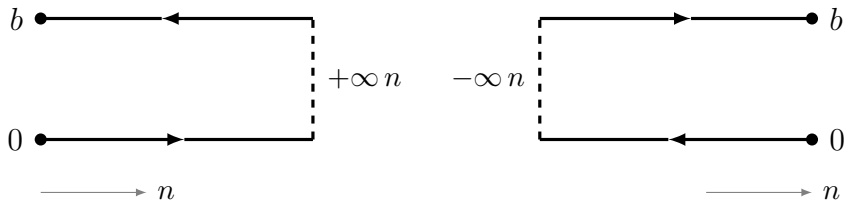
In Drell-Yan, the active quark interacts with the other hadron before it annihilates. The gluon exchanges are in the initial state; the eikonal propagators become $1/(n{\cdot}l - i\epsilon)$, and the link runs to past infinity,

$$\mathcal{U}[\mathcal{C}_b]_{\mathrm{DY}} = \mathcal{U}[0,\ -\infty n,\ -\infty n + b,\ b]\ . \tag{2.39}$$

Both paths are illustrated in Figure 5. The two gauge links differ only in the sign of the $i\epsilon$ in the eikonal propagators. Writing $1/(n{\cdot}l \pm i\epsilon) = \mathrm{PV}\, 1/(n{\cdot}l) \mp i\pi\, \delta(n{\cdot}l)$ shows that the imaginary part, the phase that a single-spin asymmetry requires, is opposite in the two processes. This is the diagrammatic origin of the sign change found by Brodsky, Hwang and Schmidt in their model calculation [47, 49].

### 2.8.2 Leading-twist TMDs

At leading twist, the correlator is projected with the Dirac structures $\Gamma = \gamma^+$, $\gamma^+\gamma_5$ and $i\sigma^{i+}\gamma_5$, which select the "good" light-cone components of the quark field. Parity, hermiticity and Lorentz covariance




(a) SIDIS: future-pointing (b) Drell-Yan: past-pointing

Figure 5: Staple-shaped gauge links connecting the quark fields at 0 and $b$ in the TMD correlator, Eq. (2.37). (a) In SIDIS, the link runs to $+\infty$ along the light-cone direction $n$ of the outgoing quark, Eq. (2.38). (b) In Drell-Yan it runs to $-\infty$, Eq. (2.39). The dashed segment is the transverse link at infinity. In the lattice calculation the legs have finite length $\eta|v|$ along a space-like direction $v$, and the two cases correspond to $\eta v{\cdot}P \to +\infty$ and $\eta v{\cdot}P \to -\infty$ [2].

then allow eight functions of $x$ and $\boldsymbol{k}_{\mathrm{T}}^2$ [27, 28, 22, 30, 101]:

$$\Phi^{[\gamma^+]} = f_1 - \left[\frac{\epsilon_{ij}\, k_{T,i}\, S_{T,j}}{m_N}\, f_{1T}^{\perp}\right]_{\mathrm{odd}} , \tag{2.40}$$

$$\Phi^{[\gamma^+\gamma_5]} = \Lambda\, g_1 + \frac{\boldsymbol{k}_{\mathrm{T}}{\cdot}\boldsymbol{S}_{\mathrm{T}}}{m_N}\, g_{1T} , \tag{2.41}$$

$$\Phi^{[i\sigma^{i+}\gamma_5]} = S_{T,i}\, h_1 + \frac{\Lambda\, k_{T,i}}{m_N}\, h_{1L}^{\perp} - \frac{\left(k_{T,i}k_{T,j} - \frac{1}{2}\boldsymbol{k}_{\mathrm{T}}^2\delta_{ij}\right) S_{T,j}}{m_N^2}\, h_{1T}^{\perp} + \left[\frac{\epsilon_{ij}\, k_{T,j}}{m_N}\, h_1^{\perp}\right]_{\mathrm{odd}} . \tag{2.42}$$

The eight functions are summarized in Table 1. Three survive integration over $\boldsymbol{k}_{\mathrm{T}}$ and have collinear counterparts: the unpolarized distribution $f_1$, the helicity distribution $g_1$, and the transversity distribution $h_1$. The other five exist only because of the transverse momentum. The worm-gear functions $g_{1T}$ and $h_{1L}^{\perp}$ and the pretzelosity $h_{1T}^{\perp}$ correlate the quark transverse momentum with the quark and nucleon polarizations. The Sivers function $f_{1T}^{\perp}$ and the Boer-Mulders function $h_1^{\perp}$, marked by the brackets $[\ ]_{\mathrm{odd}}$, are the naively T-odd functions.

| nucleon \ quark | U | L | T |
|---|---|---|---|
| U | $f_1$ | | $\left[h_1^{\perp}\right]_{\mathrm{odd}}$ |
| L | | $g_1$ | $h_{1L}^{\perp}$ |
| T | $\left[f_{1T}^{\perp}\right]_{\mathrm{odd}}$ | $g_{1T}$ | $h_1,\ h_{1T}^{\perp}$ |

Table 1: Leading-twist quark TMDs, arranged by the polarization of the nucleon (rows) and of the quark (columns): unpolarized (U), longitudinally polarized (L) and transversely polarized (T). The Sivers function $f_{1T}^{\perp}$ and the Boer-Mulders function $h_1^{\perp}$ are naively T-odd.

The $\gamma^+$ projection has a density interpretation. In light-cone gauge, $\Phi^{[\gamma^+]}(x, \boldsymbol{k}_{\mathrm{T}}; P, S)$ counts quarks with momentum fraction $x$ and transverse momentum $\boldsymbol{k}_{\mathrm{T}}$ in a nucleon with spin $S$, summed over quark

helicities [23, 102]. Eq. (2.40) is therefore the density of Eq. (1.1). Positivity of this density implies the bound $(|\boldsymbol{k}_{\mathrm{T}}|/m_N)\,|f_{1T}^{\perp}(x,\boldsymbol{k}_{\mathrm{T}}^2)| \leq f_1(x,\boldsymbol{k}_{\mathrm{T}}^2)$ [103].

### 2.8.3 Time reversal and the sign change

Time reversal is a symmetry of QCD, but it also acts on the gauge link. Combined with parity, it maps the correlator with a future-pointing staple onto the correlator with a past-pointing staple, $n \to -n$ [48, 51]. Under this map the T-even functions are unchanged, whereas the terms in brackets in Eqs. (2.40) and (2.42) change sign. Combined with the link directions of Eqs. (2.38) and (2.39), this gives

$$f_{1T}^{\perp}\big|_{\mathrm{SIDIS}} = -f_{1T}^{\perp}\big|_{\mathrm{DY}}\,, \qquad h_1^{\perp}\big|_{\mathrm{SIDIS}} = -h_1^{\perp}\big|_{\mathrm{DY}}\,, \qquad f^{\text{T-even}}\big|_{\mathrm{SIDIS}} = f^{\text{T-even}}\big|_{\mathrm{DY}}\,. \tag{2.43}$$

This resolves the objection described in Section 1.3: Collins's 1993 argument [46] ignored the gauge link in the operator definition. For a straight link, which time reversal maps onto itself, the T-odd functions do vanish. The Sivers function is nonzero only because the physical gauge link is staple-shaped, along a direction set by the process, and so is not invariant under time reversal.

The same structure appears in the lattice calculation. The staple of Eq. (4.11) has finite legs of length $\eta|v|$ along a direction $v$, and the SIDIS and DY cases correspond to $\eta v{\cdot}P \to +\infty$ and $\eta v{\cdot}P \to -\infty$ [2]. T-even quantities are even functions of $\eta$, T-odd quantities are odd, and T-odd quantities vanish for a straight link, $\eta = 0$. We derive the symmetry constraints on the correlator that lead to these statements in Chapter 4.

## 2.9 Rapidity divergences, the soft factor and evolution

The staple-shaped links of Eqs. (2.38) and (2.39) cause a problem that does not arise for collinear PDFs. Wilson lines along an exactly light-like direction produce divergences from gluons with large rapidity, moving parallel to the Wilson line, which are not regulated by dimensional regularization [102, 104, 23]. These rapidity divergences cancel in the cross section between the TMDs and a soft factor, but they must be regulated in each TMD definition.

Collins and Soper introduced the regulator used in this work [25, 21]: the Wilson lines are tilted

away from the light cone into a space-like direction $v$. The approach to the light cone is measured by the Lorentz-invariant parameter

$$\hat{\zeta} = \frac{v\cdot P}{\sqrt{|v^2|}\,\sqrt{P^2}} = \sinh(y_P - y_v) \ , \tag{2.44}$$

Eq. (4.12), where $y_P$ and $y_v$ are the rapidities of $P$ and $v$, and the sign of $v$ is fixed by $v^- > 0$. The light-cone limit is $\hat{\zeta} \to \infty$. A space-like $v$ matters for the lattice because one can always boost to a frame where $v$ has no time component, so the entire operator lies at a single time [2].

The soft factor $\tilde{\mathcal{S}}$ is a vacuum expectation value of Wilson lines, essentially a Wilson loop made from two staples. Dividing the unsubtracted correlator by an appropriate combination of soft factors, Eq. (4.1), removes the rapidity divergences and the dependence on the regulator in the limit where the Wilson lines approach the light cone [23, 32]. The Wilson lines also have a self-energy that diverges linearly with the ultraviolet cutoff, proportional to the path length. On the lattice, this divergence appears as a factor $e^{-\delta m\, L}$ for a path of length $L$, with $\delta m \propto 1/a$, together with additional divergences from the corners of the path [105, 82]. Continuum analyses suggest, and we assume as in Ref. [2], that for a fixed path the soft factor and these renormalization factors multiply the correlator independently of the Dirac structure $\Gamma$. Under this assumption, they cancel in ratios of amplitudes taken at the same $b$, $P$ and staple geometry. Whether this multiplicative structure carries over exactly to the lattice is not guaranteed. The observables computed in this thesis are built from ratios of this type.

After the rapidity divergences are removed, the TMDs depend on two scales: the renormalization scale $\mu$ and the rapidity scale $\zeta$. In the convention of Collins, the dependence on them is described by the Collins-Soper equations [25, 26, 23],

$$\frac{\partial \ln \tilde{f}(x, \boldsymbol{b}_\mathrm{T}; \mu, \zeta)}{\partial \ln \sqrt{\zeta}} = \tilde{K}(|\boldsymbol{b}_\mathrm{T}|; \mu) \ , \qquad \frac{d\tilde{K}(|\boldsymbol{b}_\mathrm{T}|; \mu)}{d \ln \mu} = -\gamma_K\big(\alpha_s(\mu)\big) \ , \tag{2.45}$$

together with a renormalization group equation in $\mu$. The Collins-Soper kernel $\tilde{K}$ is independent of $x$ and of the hadron, and $\gamma_K$ is, in this normalization, twice the cusp anomalous dimension. For $|\boldsymbol{b}_\mathrm{T}| \ll 1/\Lambda_\mathrm{QCD}$ the kernel can be computed in perturbation theory. At larger $|\boldsymbol{b}_\mathrm{T}|$, it is non-perturbative. A lattice method for it was proposed in Ref. [89], and recent lattice calculations have determined it [90, 91, 92].

The evolution of the Sivers function at these scales has been studied in Refs. [106, 69]. In the lattice calculation, the approach to the light cone is instead parametrized by $\hat{\zeta}$ of the tilted Wilson lines, which is finite at accessible nucleon momenta, and the light-cone limit $\hat{\zeta} \to \infty$ has to be reached by extrapolation (Chapter 7).

### 2.10 Transverse position space and the Sivers shift

Lattice calculations produce matrix elements at fixed separations $b$, so it is natural to work with TMDs in transverse position space. The Fourier transform $\tilde{f}(x, \boldsymbol{b}_\mathrm{T}^2)$ and its derivatives $\tilde{f}^{(n)}(x, \boldsymbol{b}_\mathrm{T}^2)$ with respect to $\boldsymbol{b}_\mathrm{T}^2$ are defined in Eqs. (4.8) and (4.9). As $|\boldsymbol{b}_\mathrm{T}| \to 0$ they reduce to the transverse momentum moments

$$f^{(n)}(x) = \int d^2\boldsymbol{k}_\mathrm{T} \left( \frac{\boldsymbol{k}_\mathrm{T}^2}{2m_N^2} \right)^n f(x, \boldsymbol{k}_\mathrm{T}^2) \;, \tag{2.46}$$

Eq. (4.10). For the Sivers function, the first moment $f_{1T}^{\perp(1)}(x)$ is the quantity that enters the transverse momentum weighted Sivers asymmetry. It is also ultraviolet divergent in QCD, because the perturbative large-$\boldsymbol{k}_\mathrm{T}$ tail of the Sivers function falls off too slowly [107]. A finite $|\boldsymbol{b}_\mathrm{T}|$ acts as a regulator, and we keep $|\boldsymbol{b}_\mathrm{T}|$ finite throughout [2, 108]. We also use the notation $f^{[m]} \equiv \int_{-1}^{1} dx\, x^{m-1} f(x)$ for $x$-moments, so that $f^{[1]}$ is the $x$-integral.

In $\boldsymbol{b}_\mathrm{T}$ space, the $\gamma^+$ projection of the correlator contains two Lorentz-invariant amplitudes [2],

$$\frac{1}{2P^+}\, \tilde{\Phi}^{[\gamma^+]}_{\text{unsubtr.}} = \widetilde{A}_{2B} + i m_N\, \epsilon_{ij}\, b_i\, S_j\; \widetilde{A}_{12B} \;, \tag{2.47}$$

where $\widetilde{A}_{2B}$ gives the unpolarized TMD and $\widetilde{A}_{12B}$ the Sivers function after the Fourier transform in $b{\cdot}P$ and division by the soft factor. Their definitions in terms of the full set of invariant amplitudes are given in Eq. (4.34). At $b{\cdot}P = 0$ the Fourier transform in $b{\cdot}P$ reduces to an integral over $x$, and one finds [2]

$$\tilde{f}_1^{[1](0)}(\boldsymbol{b}_\mathrm{T}^2) = \frac{2\widetilde{A}_{2B}}{\tilde{\mathcal{S}}} \;, \qquad \tilde{f}_{1T}^{\perp[1](1)}(\boldsymbol{b}_\mathrm{T}^2) = -\frac{2\widetilde{A}_{12B}}{\tilde{\mathcal{S}}} \;, \tag{2.48}$$

with both amplitudes evaluated at $b^2 = -\boldsymbol{b}_\mathrm{T}^2$ and $b{\cdot}P = 0$.

Since $b$ is space-like, $b^2 < 0$ in the Minkowski metric. The lattice frame, used from Chapter 5 on, is one in which $b$ has no time component, so $-b^2 = |\boldsymbol{b}|^2$. In that frame we write $b_L$ and $b_T$ for

the components of $\boldsymbol{b}$ along and perpendicular to the nucleon momentum $\boldsymbol{P}$, and $b^2$ for the positive quantity $|\boldsymbol{b}|^2 = b_L^2 + b_T^2$. Because the TMD frame has $b^+ = 0$, the TMD transverse separation satisfies $\boldsymbol{b}_{\mathrm{T}}^2 = -b^2 = b_L^2 + b_T^2$. It coincides with the lattice component $b_T$ only when $b_L = 0$, that is, at $b{\cdot}P = 0$.

The Sivers function describes the average transverse momentum of unpolarized quarks in the direction perpendicular to the nucleon spin. For a nucleon polarized along $\hat{\boldsymbol{x}}$, $\boldsymbol{S}_{\mathrm{T}} = (1, 0)$, Eq. (2.40) gives $\Phi^{[\gamma^+]} = f_1 + (k_y/m_N)\, f_{1T}^{\perp}$, and the average $k_y$ is $m_N f_{1T}^{\perp(1)}/f_1^{(0)}$. The generalized Sivers shift extends this quantity to finite $|\boldsymbol{b}_{\mathrm{T}}|$ [2, 108],

$$\langle k_y \rangle_{TU}(\boldsymbol{b}_{\mathrm{T}}^2; \hat{\zeta}, \eta v{\cdot}P) \equiv m_N \, \frac{\tilde{f}_{1T}^{\perp[1](1)}(\boldsymbol{b}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{[1](0)}(\boldsymbol{b}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v{\cdot}P)} = -m_N \, \frac{\widetilde{A}_{12B}(-\boldsymbol{b}_{\mathrm{T}}^2, 0, 0, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)}{\widetilde{A}_{2B}(-\boldsymbol{b}_{\mathrm{T}}^2, 0, 0, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)} \ . \quad (2.49)$$

Under the assumption stated in Section 2.9, the soft factor and the multiplicative renormalization factors cancel in this ratio, so $\langle k_y \rangle_{TU}$ can be computed on the lattice without constructing either. In the limit $|\boldsymbol{b}_{\mathrm{T}}| \to 0$, it would become the average transverse momentum of the quarks, with the numerator summing quark and antiquark contributions and the denominator counting quarks minus antiquarks [74]. Because $b{\cdot}P = 0$, Eq. (2.49) is integrated over $x$.

The $x$ dependence enters through the second argument of the amplitudes. Since $x$ and $b{\cdot}P$ are Fourier conjugate variables (Chapter 4), keeping $b{\cdot}P$ as a variable and Fourier transforming the numerator and denominator separately gives an $x$-dependent generalized Sivers shift,

$$\langle k_y \rangle_{TU}(x, \boldsymbol{b}_{\mathrm{T}}^2; \hat{\zeta}, \eta v{\cdot}P) = -m_N \, \frac{\int d(b{\cdot}P)\ e^{ix(b\cdot P)}\ \widetilde{A}_{12B}(b^2, b{\cdot}P, \ldots)}{\int d(b{\cdot}P)\ e^{ix(b\cdot P)}\ \widetilde{A}_{2B}(b^2, b{\cdot}P, \ldots)} \ . \quad (2.50)$$

We form this ratio from amplitudes computed at the same $b$, $P$ and staple geometry, and we obtain its $x$ dependence by scanning the full accessible range of $b{\cdot}P$. This is the quantity computed in this thesis. The full expression, including the kinematic constraint that relates $v{\cdot}b$ to $b{\cdot}P$, is Eq. (5.6) in Chapter 5.

## 2.11 From $b{\cdot}P$ to $x$

Because the $x$ dependence is obtained through a Fourier transform, we first state what the $b{\cdot}P$ dependence of an amplitude tells us about the corresponding distribution. We take the unpolarized case as the example. At fixed $b^2 = -\boldsymbol{b}_{\mathrm{T}}^2$, the relation $f_1 = 2\int_{\mathcal{F}} \widetilde{A}_{2B}$ of Eq. (4.36) can be inverted in the longitudinal

variable. We assume, as holds for light-cone correlators, that $\tilde{f}_1(x, \boldsymbol{b}_{\mathrm{T}}^2)$ vanishes outside $-1 \le x \le 1$; at finite $\hat{\zeta}$ and $\eta$ this support property has not been proven. Then

$$\widetilde{A}_{2B}(-\boldsymbol{b}_{\mathrm{T}}^2, b{\cdot}P) = \frac{\tilde{\mathcal{S}}}{2} \int_{-1}^{1} dx \; e^{-ix(b\cdot P)} \; \tilde{f}_1(x, \boldsymbol{b}_{\mathrm{T}}^2) \; , \tag{2.51}$$

where we have suppressed the remaining arguments of the amplitude and of the soft factor. Three properties follow directly from this representation.

First, the Taylor coefficients of the amplitude in $b{\cdot}P$ are the Mellin moments of the distribution at fixed $\boldsymbol{b}_{\mathrm{T}}$,

$$\frac{\widetilde{A}_{2B}(-\boldsymbol{b}_{\mathrm{T}}^2, b{\cdot}P)}{\widetilde{A}_{2B}(-\boldsymbol{b}_{\mathrm{T}}^2, 0)} = \sum_{n=0}^{\infty} \frac{(-i\, b{\cdot}P)^n}{n!} \; \overline{\langle x^n \rangle}_{\boldsymbol{b}_{\mathrm{T}}} \; , \qquad \overline{\langle x^n \rangle}_{\boldsymbol{b}_{\mathrm{T}}} = \frac{\int dx \, x^n \, \tilde{f}_1(x, \boldsymbol{b}_{\mathrm{T}}^2)}{\int dx \, \tilde{f}_1(x, \boldsymbol{b}_{\mathrm{T}}^2)} \; . \tag{2.52}$$

The value at $b{\cdot}P = 0$ is the $x$-integrated quantity studied in earlier lattice work [2, 85]. The first few derivatives give $x$-moments at fixed $|\boldsymbol{b}_{\mathrm{T}}|$, which are analogous to, but not the same as, the moments that local operators provide. Musch used this relation with straight links to estimate the lowest moments of the unpolarized distribution [105].

Second, the real and imaginary parts of the amplitude separate quark and antiquark contributions. Using $f_1^{\bar{q}}(x) = -f_1^{q}(-x)$ for $x > 0$, and taking the soft factor to be real,

$$\begin{aligned} \mathrm{Re}\, \widetilde{A}_{2B} &= \frac{\tilde{\mathcal{S}}}{2} \int_0^1 dx \; \cos\big(x\, b{\cdot}P\big) \Big[ \tilde{f}_1^{q}(x) - \tilde{f}_1^{\bar{q}}(x) \Big] \, , \\ \mathrm{Im}\, \widetilde{A}_{2B} &= -\frac{\tilde{\mathcal{S}}}{2} \int_0^1 dx \; \sin\big(x\, b{\cdot}P\big) \Big[ \tilde{f}_1^{q}(x) + \tilde{f}_1^{\bar{q}}(x) \Big] \, . \end{aligned} \tag{2.53}$$

The real part, which is even in $b{\cdot}P$, is determined by the quark-minus-antiquark (valence-like) combination, and the imaginary part, which is odd, by the quark-plus-antiquark combination. This is the parity structure used in Chapter 5. For the naively T-odd Sivers function, the antiquark enters with the opposite relative sign [74], so the roles of the two combinations are interchanged for $\widetilde{A}_{12B}$.

Third, the support property constrains how the amplitude can behave at large $|b{\cdot}P|$. If $\tilde{f}_1$ is integrable, the Riemann-Lebesgue lemma implies $\widetilde{A}_{2B} \to 0$ as $|b{\cdot}P| \to \infty$, and since the support is bounded, $\widetilde{A}_{2B}$ extends to an entire function of $b{\cdot}P$ of exponential type no larger than one. A parametrization of lattice data that grows with $|b{\cdot}P|$ at fixed $b^2$ is therefore inconsistent with a physical distribution and

has no Fourier transform in the ordinary sense. This criterion is used in Section 6.1 to reject the Gaussian ansatz. A parametrization that decays like a power of $b{\cdot}P$ fast enough to be absolutely integrable does have a Fourier transform. Such a power law is not an entire function, so by the same theorem the resulting distribution extends beyond $|x| \leq 1$, and the size of this tail is one measure of the model dependence.

Lattice data cover only a finite range of $b{\cdot}P$, because $|b{\cdot}P| \leq |\boldsymbol{P}|\,|\boldsymbol{b}|$ in a frame where $b$ has no time component (Section 2.12). The behavior of $\tilde{f}$ at small $|x|$ is governed by the amplitude at large $|b{\cdot}P|$, outside the range of the data. The $x$ dependence reconstructed from a finite range of $b{\cdot}P$ therefore depends on the functional form used to describe the data, and it depends on it most strongly at small $x$. This is why the choice and validation of fit models take up a large part of this thesis (Chapter 6).

### 2.12 TMDs from lattice QCD

Lattice QCD works in Euclidean space-time. A non-local operator can be evaluated directly only if all of its points lie at a single Euclidean time, which corresponds to purely space-like separations in Minkowski space. Operators that extend along a light-like direction, like those in Eq. (2.26) and Eq. (2.37), cannot be placed on the lattice as they stand. Three strategies have been developed to deal with this.

The first is the local-operator approach in Section 2.5. Only the lowest Mellin moments are accessible. For TMDs, this approach does not apply, because the staple-shaped link cannot be expanded into local operators.

The second is the Lorentz-invariant amplitude approach, introduced with straight links by Hägler, Musch, Negele and Schäfer [81, 82] and extended to staple links by Musch et al. [2], which is used in this thesis. The correlator with a finite, space-like staple is decomposed into amplitudes $\widetilde{A}_i$ and $\widetilde{B}_i$ that depend only on Lorentz invariants: $b^2$, $b{\cdot}P$, $\hat{\zeta}$, the staple length through $\eta v{\cdot}P$ [101, 2], and $v{\cdot}b/v{\cdot}P$, which the TMD kinematics fix in terms of $b{\cdot}P$ and $\hat{\zeta}$ (Eq. (4.32)). When $b$ and $v$ are space-like and also span a space-like plane, $(b{\cdot}v)^2 < b^2 v^2$, there is a frame in which both have vanishing time components, and in that frame the operator can be evaluated at a single Euclidean time. The amplitudes computed

there are, by Lorentz invariance, the same amplitudes that define TMDs in the light-cone frame. The accessible kinematics are limited by the lattice. In a frame with $b^0 = v^0 = 0$, the invariants become $b{\cdot}P = -\boldsymbol{b}{\cdot}\boldsymbol{P}$ and $\hat{\zeta} = -\boldsymbol{v}{\cdot}\boldsymbol{P}/(|\boldsymbol{v}|\, m_N)$, so that

$$|b{\cdot}P| \leq |\boldsymbol{P}|\,|\boldsymbol{b}| \;, \qquad |\hat{\zeta}| \leq \frac{|\boldsymbol{P}|}{m_N} \;. \tag{2.54}$$

Nonzero $b{\cdot}P$ requires a component of $\boldsymbol{b}$ along the nucleon momentum, and large $\hat{\zeta}$ requires large nucleon momenta, where the statistical signal deteriorates. The light-cone limit $\hat{\zeta} \to \infty$ and the SIDIS and DY limits $\eta v{\cdot}P \to \pm\infty$ are reached by extrapolation. The observables are ratios of amplitudes at fixed geometry (Sections 2.9 and 2.10). With this approach, the Sivers and Boer-Mulders shifts, the worm-gear shift, and the generalized tensor charge have been computed for the nucleon and the pion [2, 83, 84, 85].

The third strategy is large-momentum effective theory (LaMET) [86, 88]. Instead of Lorentz invariance, it uses the fact that an equal-time correlator in a hadron with large momentum $P^z$ approaches the light-cone correlator, up to a perturbative matching and corrections suppressed by powers of $1/P^z$. For TMDs, the quasi-TMD is defined with a staple along the $z$ direction and a Fourier transform in the longitudinal separation, and it is related to the light-cone TMD by a matching coefficient, a factor involving the Collins-Soper kernel, and a reduced soft function that can itself be computed on the lattice [87, 89, 109, 110]. Ratios of quasi-TMDs at different momenta give the Collins-Soper kernel [89, 90, 92]. A related approach uses pseudo-distributions, which are functions of the Lorentz invariants $b{\cdot}P$ and $b^2$ [111]. Reviews of these methods can be found in Refs. [79, 80, 74].

The Lorentz-invariant approach is well suited to ratio observables such as the Sivers shift, because it works directly with the amplitudes from which the ratios are formed, and it makes the dependence on $b{\cdot}P$ explicit. That dependence is what this thesis uses to reach the $x$ dependence. The next chapter describes how the correlation functions are computed on the lattice.

# 3 LATTICE QCD METHODOLOGY

This chapter describes how the QCD correlation functions used in this thesis are defined and computed on a Euclidean spacetime lattice. We follow Lin and Meyer [3] and Gattringer and Lang [77]. Sections 3.1–3.3 set up the lattice, the Wilson gauge action with its continuum limit, and the Wilson-clover fermion action. Section 3.4 integrates out the quarks and introduces propagators and the Monte Carlo method. The nucleon two-point function (Section 3.5) and three-point functions of local operators (Section 3.6) then lead to the ratio that isolates a matrix element, and Section 3.7 extends this construction to the staple-link operators that define TMDs.

## 3.1 The Euclidean lattice

The Euclidean path integral of Section 2.2 is regulated by restricting spacetime to a four-dimensional hypercubic lattice [76],

$$\Lambda_4 = \{n = (n_1, n_2, n_3, n_4) \,|\, n_i \in \{0, 1, \ldots, L_i - 1\}\} \ , \tag{3.1}$$

where $L_i$ is the number of sites in direction $i$ and a site $n$ sits at the spacetime point $x = an$, with $a$ the lattice spacing. We write $n + \hat{\mu}$ for the neighboring site in direction $\mu$, which is the point $x + a\hat{\mu}$. Spatial boundary conditions are periodic for all fields; in time they are periodic for the gauge field and anti-periodic for the quark fields [77]. The lattice spacing acts as an ultraviolet regulator, with momenta restricted to the Brillouin zone $p_\mu \in (-\pi/a, \pi/a]$, and the finite volume acts as an infrared regulator. Physical results are recovered in the continuum limit $a \to 0$ and the infinite-volume limit [77].

### 3.1.1 Euclidean Dirac matrices

The Euclidean Dirac matrices used in this thesis are related to the Minkowski ones by

$$\gamma_E^1 = +i\gamma_M^1 \ , \quad \gamma_E^2 = -i\gamma_M^2 \ , \quad \gamma_E^3 = +i\gamma_M^3 \ , \quad \gamma_E^4 = \gamma_M^0 \ . \tag{3.2}$$

This is the convention of the lattice code that produced the matrix elements analyzed in Section 5.6. These matrices are hermitian and satisfy $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$, and $\gamma_5 \equiv \gamma_1\gamma_2\gamma_3\gamma_4 = -\gamma_M^5$ is hermitian and

anticommutes with every $\gamma_\mu$. Products of Dirac matrices are labeled by a four-bit integer, $\Gamma(n) = \gamma_1^{n_1}\gamma_2^{n_2}\gamma_3^{n_3}\gamma_4^{n_4}$ with $n = n_1 + 2n_2 + 4n_3 + 8n_4$, so that $\Gamma(1) = \gamma_1$, $\Gamma(8) = \gamma_4$ and $\Gamma(15) = \gamma_5$. For $\gamma_1$ and $\gamma_3$, Eq. (3.2) agrees with the convention $\gamma_i^M = -i\gamma_i^E$ of Ref. [3]; for $\gamma_2$ the sign is opposite. For a Dirac spinor $u(p,s)$ of mass $m$ the spin sum reads

$$\sum_s u(p,s)\,\bar{u}(p,s) = -i\,\not{p} + m\ , \qquad \not{p} \equiv \gamma_\mu p_\mu\ , \qquad p_\mu = (-p^1,\, p^2,\, -p^3,\, iE)\ , \tag{3.3}$$

where $(E, p^1, p^2, p^3)$ are the Minkowski components. Equation (3.3) follows from the Minkowski relation $\sum_s u\bar{u} = \gamma_M^\mu p_\mu + m$ and Eq. (3.2), with $\bar{u}u = 2m$. It has the form of Eq. (3.78) of Ref. [3], with the components of $p_\mu$ fixed by Eq. (3.2). For momenta along the 1-axis, $p_\mu = (-p^1, 0, 0, iE)$.

### 3.2 Gauge fields on the lattice

On the lattice the gauge field is represented by link variables, $SU(3)$ matrices that live on the links between neighboring sites. In this chapter we use the lattice convention of Ref. [77], in which the coupling is absorbed into the gauge field,

$$U_\mu(n) = \exp\big(iaA_\mu(n)\big)\ , \tag{3.4}$$

with $A_\mu(n)$ evaluated at the starting point of the link, which is accurate to $\mathcal{O}(a)$ [77]. This convention is related to that of Eqs. (2.10)–(2.14) by $A_\mu \to -gA_\mu$ and does not affect gauge-invariant quantities. The link $U_\mu(x)$ is the lattice version of the Wilson line between $x$ and $x + a\hat{\mu}$ [3]. Under a gauge transformation $\Lambda(x) \in SU(3)$, the quark fields and links transform as [3]

$$\psi(x) \to \Lambda(x)\psi(x)\ , \qquad \bar{\psi}(x) \to \bar{\psi}(x)\Lambda(x)^{-1}\ , \qquad U_\mu(x) \to \Lambda(x)\,U_\mu(x)\,\Lambda(x+a\hat{\mu})^{-1}\ , \tag{3.5}$$

so that $U_\mu(x)\psi(x+a\hat{\mu})$ transforms like $\psi(x)$. A product of links along a path transforms with $\Lambda$ at its two end points only. The trace of the product around a closed loop is therefore gauge invariant [77], and so is a product of links along an open path with a quark field attached at each end, which is how the non-local operators of Section 3.7 are built.

The shortest non-trivial closed loop is the plaquette (Fig. 6),

$$P_{\mu\nu}(x) = U_\mu(x)\,U_\nu(x+a\hat{\mu})\,U_\mu(x+a\hat{\nu})^\dagger\,U_\nu(x)^\dagger\ . \tag{3.6}$$

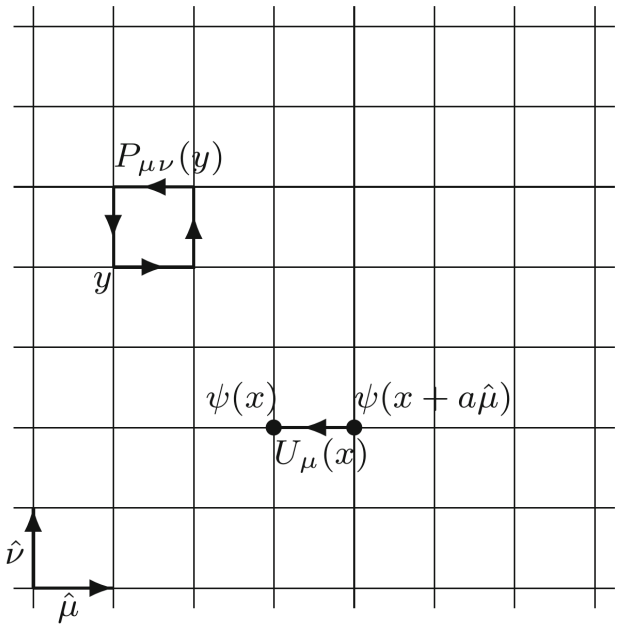



Figure 6: Quark fields $\psi(x)$ and $\psi(x+a\hat{\mu})$ on the sites of a cubic spacetime lattice, the link variable $U_\mu(x)$ between them, and the plaquette $P_{\mu\nu}(y)$ of Eq. (3.6) at the site $y$. The product $U_\mu(x)\psi(x+a\hat{\mu})$ transforms in the same way as $\psi(x)$. Adapted from Ref. [3], Fig. 1.1.

Wilson's gauge action is a sum over all plaquettes, each counted with one orientation [76, 3],

$$S_G[U] = \frac{2}{g^2} \sum_{n\in\Lambda_4} \sum_{\mu<\nu} \mathrm{Re}\,\mathrm{tr}\big[\mathbb{1} - P_{\mu\nu}(n)\big] \ . \tag{3.7}$$

Written as $\beta\sum[1 - \frac{1}{3}\mathrm{Re}\,\mathrm{tr}P_{\mu\nu}]$, the prefactor is the inverse coupling $\beta = 6/g^2$ [77].

### 3.2.1 Continuum limit of the Wilson action

Equation (2.17) quoted the small-$a$ behavior of the plaquette; here we derive it. To show that Eq. (3.7) approaches the Yang-Mills action, we insert Eq. (3.4) into the plaquette and combine the four exponentials with the Baker-Campbell-Hausdorff formula [77],

$$e^A e^B = \exp\Big(A + B + \tfrac{1}{2}[A,B] + \cdots\Big) \ . \tag{3.8}$$

Applying Eq. (3.8) three times gives

$$\begin{aligned} P_{\mu\nu}(n) = \exp\Big( & iaA_\mu(n) + iaA_\nu(n+\hat{\mu}) - iaA_\mu(n+\hat{\nu}) - iaA_\nu(n) \\ & - \tfrac{a^2}{2}[A_\mu(n), A_\nu(n+\hat{\mu})] - \tfrac{a^2}{2}[A_\mu(n+\hat{\nu}), A_\nu(n)] + \tfrac{a^2}{2}[A_\nu(n+\hat{\mu}), A_\mu(n+\hat{\nu})] \\ & + \tfrac{a^2}{2}[A_\mu(n), A_\nu(n)] + \tfrac{a^2}{2}[A_\mu(n), A_\mu(n+\hat{\nu})] + \tfrac{a^2}{2}[A_\nu(n+\hat{\mu}), A_\nu(n)] + \mathcal{O}(a^3)\Big) \ . \end{aligned} \tag{3.9}$$

The fields at shifted points are Taylor expanded, $A_\nu(n+\hat{\mu}) = A_\nu(n) + a\,\partial_\mu A_\nu(n) + \mathcal{O}(a^2)$. The linear terms combine into $ia^2(\partial_\mu A_\nu - \partial_\nu A_\mu)$, the last two commutators vanish at this order, and the remaining four combine into $-a^2[A_\mu, A_\nu]$, so that [77]

$$P_{\mu\nu}(n) = \exp\Big(ia^2\big(\partial_\mu A_\nu(n) - \partial_\nu A_\mu(n) + i[A_\mu(n), A_\nu(n)]\big) + \mathcal{O}(a^3)\Big) = \exp\big(ia^2 F_{\mu\nu}(n) + \mathcal{O}(a^3)\big)\ . \quad (3.10)$$

Expanding the exponential, $\mathbb{1} - P_{\mu\nu} = -ia^2 F_{\mu\nu} + \frac{a^4}{2}F_{\mu\nu}^2 + \ldots$. The $\mathcal{O}(a^2)$ term $-ia^2 F_{\mu\nu}$ drops out of $\mathrm{Re}\,\mathrm{tr}$, because $F_{\mu\nu}$ is traceless (equivalently, $\mathrm{tr}[P_{\mu\nu}]^* = \mathrm{tr}[P_{\nu\mu}]$), and the $\mathcal{O}(a^3)$ terms cancel in a similar way [77]. Using $\sum_{\mu<\nu} = \frac{1}{2}\sum_{\mu,\nu}$ for the symmetric summand,

$$S_G[U] = \frac{a^4}{2g^2}\sum_{n\in\Lambda_4}\sum_{\mu,\nu}\mathrm{tr}\big[F_{\mu\nu}(n)^2\big] + \mathcal{O}(a^2) \ \xrightarrow{a\to 0}\ \frac{1}{2g^2}\int d^4x \sum_{\mu,\nu}\mathrm{tr}\big[F_{\mu\nu}(x)^2\big]\ , \quad (3.11)$$

since $a^4\sum_n$ becomes $\int d^4x$. This is the Euclidean Yang-Mills action [3], and the corrections of Eq. (3.7) start at $\mathcal{O}(a^2)$ [77].

The integration over gauge fields uses the product of Haar measures of all links, $\mathcal{D}[U] = \prod_{x,\mu} dU_\mu(x)$, which is invariant under the gauge transformation of Eq. (3.5) [3].

### 3.3 Fermions on the lattice

#### 3.3.1 Naive discretization and doublers

Replacing the derivative in the Euclidean Dirac action by a symmetric, gauge-covariant difference gives the naive lattice action [77]

$$S_F[\psi,\bar{\psi},U] = a^4\sum_{n\in\Lambda_4}\bar{\psi}(n)\left(\sum_{\mu=1}^{4}\gamma_\mu\frac{U_\mu(n)\psi(n+\hat{\mu}) - U_{-\mu}(n)\psi(n-\hat{\mu})}{2a} + m\,\psi(n)\right), \quad (3.12)$$

with $U_{-\mu}(n) \equiv U_\mu(n-\hat{\mu})^\dagger$. The action is bilinear, $S_F = a^4\sum_{n,m}\bar{\psi}(n)D(n|m)\psi(m)$, with the naive Dirac operator

$$D(n|m)^{\alpha\beta}_{ab} = \sum_{\mu=1}^{4}(\gamma_\mu)_{\alpha\beta}\,\frac{U_\mu(n)_{ab}\,\delta_{n+\hat{\mu},m} - U_{-\mu}(n)_{ab}\,\delta_{n-\hat{\mu},m}}{2a} + m\,\delta_{\alpha\beta}\,\delta_{ab}\,\delta_{n,m}\ . \quad (3.13)$$

For the free field, $U_\mu(n) = \mathbb{1}$, the Fourier transform of Eq. (3.13) is diagonal in momentum, with [77]

$$\widetilde{D}(p) = m\,\mathbb{1} + \frac{i}{a}\sum_{\mu=1}^{4}\gamma_\mu\sin(p_\mu a)\ , \qquad \widetilde{D}(p)^{-1} = \frac{m\,\mathbb{1} - ia^{-1}\sum_\mu\gamma_\mu\sin(p_\mu a)}{m^2 + a^{-2}\sum_\mu\sin^2(p_\mu a)}\ . \quad (3.14)$$

For $m = 0$ the propagator has the correct continuum limit at fixed $p$, $\widetilde{D}(p)^{-1} \to -i \sum_\mu \gamma_\mu p_\mu / p^2$, with a pole at $p = (0,0,0,0)$. On the lattice, however, $\sin(p_\mu a)$ also vanishes at $p_\mu = \pi/a$. Every momentum with components equal to 0 or $\pi/a$ is a pole [77],

$$p = (\pi/a, 0, 0, 0)\,,\ (0, \pi/a, 0, 0)\,,\ \ldots\,,\ (\pi/a, \pi/a, \pi/a, \pi/a)\ , \tag{3.15}$$

which gives 15 unphysical poles, the doublers, in addition to the physical one. By the Nielsen-Ninomiya theorem [112, 3], a local, translation-invariant lattice Dirac operator with the correct continuum limit, no doublers and exact chiral symmetry in the form $\gamma_5 D + D\gamma_5 = 0$ cannot be constructed.

### 3.3.2 Wilson fermions

Wilson removed the doublers by adding a term that vanishes for $p_\mu = 0$ [76, 77],

$$\widetilde{D}(p) = m\,\mathbb{1} + \frac{i}{a}\sum_{\mu=1}^{4} \gamma_\mu \sin(p_\mu a) + \mathbb{1}\,\frac{1}{a}\sum_{\mu=1}^{4}\big(1 - \cos(p_\mu a)\big)\ . \tag{3.16}$$

Each component $p_\mu = \pi/a$ contributes $2/a$, so a doubler with $\ell$ such components has mass $m + 2\ell/a$. In the limit $a \to 0$ the doublers become infinitely heavy and decouple. In position space the Wilson term is a discretization of $-\frac{a}{2}\partial_\mu\partial_\mu$, and combining it with Eq. (3.13) gives the Wilson-Dirac operator [77]

$$D(n|m)^{\alpha\beta}_{ab} = \Big(m + \frac{4}{a}\Big)\delta_{\alpha\beta}\,\delta_{ab}\,\delta_{n,m} - \frac{1}{2a}\sum_{\mu=\pm1}^{\pm4}(\mathbb{1} - \gamma_\mu)_{\alpha\beta}\,U_\mu(n)_{ab}\,\delta_{n+\hat{\mu},m}\ , \qquad \gamma_{-\mu} = -\gamma_\mu\ . \tag{3.17}$$

Rescaling the quark fields by $\sqrt{m + 4/a}$ brings it to the hopping form [77]

$$D = \mathbb{1} - \kappa H\ , \qquad \kappa = \frac{1}{2(am+4)}\ , \qquad H(n|m)^{\alpha\beta}_{ab} = \sum_{\mu=\pm1}^{\pm4}(\mathbb{1} - \gamma_\mu)_{\alpha\beta}\,U_\mu(n)_{ab}\,\delta_{n+\hat{\mu},m}\ , \tag{3.18}$$

where $H$ collects the nearest-neighbor terms and $\kappa$ is the hopping parameter. The Wilson-Dirac operator is $\gamma_5$-hermitian [3, 77],

$$D^\dagger = \gamma_5\, D\, \gamma_5\ , \tag{3.19}$$

which follows from $\gamma_5^\dagger = \gamma_5$, $\gamma_5\gamma_\mu\gamma_5 = -\gamma_\mu$ and $U_{-\mu}(n) = U_\mu(n-\hat{\mu})^\dagger$, and which holds in the basis of Eq. (3.2). The Wilson term breaks chiral symmetry explicitly, even for massless quarks, and introduces discretization errors of $\mathcal{O}(a)$ [77].

### 3.3.3 Clover improvement

The $\mathcal{O}(a)$ errors of the Wilson action are removed by adding the Sheikholeslami-Wohlert or clover term [113, 77],

$$S_{\text{clover}} = S_{\text{Wilson}} + c_{\text{sw}}\, a^5 \sum_{n\in\Lambda_4} \sum_{\mu<\nu} \bar{\psi}(n)\, \frac{1}{2}\, \sigma_{\mu\nu}\, \widehat{F}_{\mu\nu}(n)\, \psi(n) \;, \qquad \sigma_{\mu\nu} = \frac{1}{2i}[\gamma_\mu, \gamma_\nu] \;, \tag{3.20}$$

with the Euclidean $\sigma_{\mu\nu}$ of Ref. [77], which is not the Minkowski $\sigma^{\mu\nu}$ of Section 2.1. The lattice field strength is built from the four plaquettes around the site $n$,

$$\widehat{F}_{\mu\nu}(n) = \frac{-i}{8a^2}\big(Q_{\mu\nu}(n) - Q_{\nu\mu}(n)\big) \;, \qquad Q_{\mu\nu}(n) = P_{\mu\nu}(n) + P_{\nu,-\mu}(n) + P_{-\mu,-\nu}(n) + P_{-\nu,\mu}(n) \;, \tag{3.21}$$

with $P_{\mu\nu}(n)$ the plaquette of Eq. (3.6) and $P_{\nu,-\mu}$, $P_{-\mu,-\nu}$, $P_{-\nu,\mu}$ the plaquettes with the indicated orientations. The four plaquettes form the clover leaf that gives the term its name, and the coefficient $c_{\text{sw}}$ is tuned to cancel the $\mathcal{O}(a)$ terms [3, 77].

### 3.3.4 Other formulations

Twisted-mass fermions remove the doublers with a Wilson term, while staggered (Kogut-Susskind) fermions reduce the sixteen naive species to four tastes and keep a remnant chiral symmetry. Overlap fermions satisfy the Ginsparg-Wilson relation, and domain-wall fermions, which introduce a fifth dimension, satisfy it approximately at finite fifth-dimension extent [77]. The gauge ensemble used in this work was generated with $n_f = 2 + 1$ flavors of clover-improved Wilson fermions [85].

## 3.4 Path integral, Wick contractions and quark propagators

With gauge action $S_G$ and fermion action $S_F = a^4 \sum_f \sum_{n,m} \bar{\psi}^{(f)}(n) D^{(f)}(n|m) \psi^{(f)}(m)$, the lattice expectation value of an observable is [77, 3]

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}[U]\, \mathcal{D}[\psi, \bar{\psi}]\; \mathcal{O}[\psi, \bar{\psi}, U]\, e^{-S_F[\psi,\bar{\psi},U] - S_G[U]} \;, \qquad Z = \int \mathcal{D}[U]\, \mathcal{D}[\psi, \bar{\psi}]\; e^{-S_F - S_G} \;. \tag{3.22}$$

The quark fields are Grassmann variables (Eq. (2.8)). For a quadratic form, the Grassmann integral with sources is [3]

$$\int \prod_i d\eta_i\, d\bar{\eta}_i\, \exp\Big( -\sum_{ij} \bar{\eta}_i A_{ij} \eta_j + \sum_i (\bar{\eta}_i \zeta_i + \bar{\zeta}_i \eta_i) \Big) = c\, \det(A)\, \exp\Big( \sum_{ij} \bar{\zeta}_i (A^{-1})_{ij} \zeta_j \Big) \;, \tag{3.23}$$

so the quark integration produces the fermion determinant, and derivatives with respect to the sources turn every pair $\psi\bar{\psi}$ into an element of $A^{-1}$. With $A = a^4 D$ this is Wick's theorem: the quark fields in $\mathcal{O}$ are contracted in all possible ways and replaced by quark propagators [77],

$$\left\langle \psi(n)_{\alpha a}\, \bar{\psi}(m)_{\beta b} \right\rangle_F = a^{-4} D^{-1}(n|m)^{\alpha\beta}_{ab} \equiv S(n,m)^{\alpha\beta}_{ab} \ . \tag{3.24}$$

After integrating out the quarks,

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}[U]\ \widetilde{\mathcal{O}}[U] \prod_f \det\left[D^{(f)}[U]\right] e^{-S_G[U]} \ , \tag{3.25}$$

where $\widetilde{\mathcal{O}}[U]$ is the contracted observable, a function of the propagators in the gauge field $U$ [77].

### 3.4.1 Monte Carlo evaluation

For real and positive determinants, $\mathcal{P}[U] = Z^{-1} \prod_f \det[D^{(f)}[U]]\, e^{-S_G[U]}$ is a probability distribution. A Markov chain whose transition probabilities satisfy detailed balance produces gauge configurations $U_i$ distributed according to $\mathcal{P}[U]$ [77], and the expectation value becomes an ensemble average [3],

$$\langle \mathcal{O} \rangle = \frac{1}{N} \sum_{i=1}^{N} \widetilde{\mathcal{O}}[U_i] + \mathcal{O}\big(N^{-1/2}\big) \ . \tag{3.26}$$

With dynamical quarks the standard algorithm is hybrid Monte Carlo [114]. Statistical uncertainties in this thesis are estimated with the Jackknife method (Section 6.8).

### 3.4.2 Computing propagators

Rather than computing the full matrix $D^{-1}$, one solves the linear system [77]

$$\sum_m D(n|m)\, G(m) = S_0(n) \ , \qquad S_0(m)^{\beta}_{b} = \delta_{m,m_0}\, \delta_{\beta\beta_0}\, \delta_{bb_0} \ , \tag{3.27}$$

for a point source $S_0$ at $m_0$ with iterative Krylov-space solvers. The twelve solutions for $\beta_0 = 1, \ldots, 4$ and $b_0 = 1, 2, 3$ give the point-to-all propagator $D^{-1}(n|m_0)$ from $m_0$ to every site $n$. Propagators in the opposite direction follow from Eq. (3.19) without further inversions [77],

$$S(m_0, n) = \gamma_5\, S(n, m_0)^\dagger\, \gamma_5 \ . \tag{3.28}$$

### 3.4.3 Example: the pion two-point function

As an example of Wick contraction [77, 3], consider $\chi_\pi = \bar{d}\,\gamma_5\,u$. Its conjugate is $\bar{\chi}_\pi = (\bar{d}\gamma_5 u)^\dagger = -\bar{u}\gamma_5 d$, using $\bar{\psi} = \psi^\dagger\gamma_4$ and $\gamma_4\gamma_5\gamma_4 = -\gamma_5$. Wick's theorem contracts $u(x)\bar{u}(0)$ and $d(0)\bar{d}(x)$; moving $d(0)$ to the front past three Grassmann fields gives a sign, and the minus sign of $\bar{\chi}_\pi$ cancels it:

$$\big\langle \chi_\pi(x)\,\bar{\chi}_\pi(0)\big\rangle_F = \mathrm{tr}\big[S_d(0,x)\,\gamma_5\,S_u(x,0)\,\gamma_5\big] = \mathrm{tr}\big[S_d(x,0)^\dagger\,S_u(x,0)\big]\ , \tag{3.29}$$

where the second step uses Eq. (3.28) and the trace runs over Dirac and color indices. For degenerate $u$ and $d$ quarks this is $\sum |S_u(x,0)|^2$, so a single point-to-all propagator per configuration determines the pion correlator. The nucleon correlators below are built in the same way.

## 3.5 The nucleon two-point function

### 3.5.1 Interpolating operator and contraction

A standard interpolating operator with the quantum numbers of the proton is [3]

$$\chi_\alpha(\boldsymbol{x},t) = \epsilon^{ijk}\,u^i_\alpha(\boldsymbol{x},t)\,u^j_\gamma(\boldsymbol{x},t)\,[C^{-1}\gamma_5]_{\gamma\delta}\,d^k_\delta(\boldsymbol{x},t)\ , \tag{3.30}$$

with $C$ the charge-conjugation matrix (not to be confused with the correlator $C_{\alpha\beta}$ below). The momentum-projected two-point function is

$$C_{\alpha\beta}(t,\boldsymbol{p}) = a^3\sum_{\boldsymbol{x}} e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\,\big\langle \chi_\alpha(\boldsymbol{x},t)\,\bar{\chi}_\beta(\boldsymbol{0},0)\big\rangle\ , \tag{3.31}$$

and Wick's theorem expresses it through $u$ and $d$ propagators from the source [77],

$$C_{\alpha\beta}(t,\boldsymbol{p}) = -a^3\sum_{\boldsymbol{x}} e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\,\epsilon^{ijk}\epsilon^{i'j'k'}\,[C^{-1}\gamma_5]_{\alpha'\alpha''}\,[\gamma_5 C]_{\beta'\beta''}\,\Big\langle \big[S_d\big]^{ki'}_{\alpha''\beta'}\Big\{\big[S_u\big]^{jj'}_{\alpha'\beta''}\big[S_u\big]^{ik'}_{\alpha\beta} - \big[S_u\big]^{ij'}_{\alpha\beta''}\big[S_u\big]^{jk'}_{\alpha'\beta}\Big\}\Big\rangle\ , \tag{3.32}$$

with $S_f \equiv S_f(\boldsymbol{x},t;\boldsymbol{0},0)$. The two terms correspond to the two ways of contracting the two $u$ quarks.

### 3.5.2 Spectral decomposition

On a lattice with time extent $T$, the two-point function is a trace over states [3],

$$\big\langle \chi(\boldsymbol{x},t)\,\bar{\chi}(\boldsymbol{0},0)\big\rangle = \frac{\mathrm{Tr}\big[e^{-H(T-t)}\chi(\boldsymbol{x})\,e^{-Ht}\,\bar{\chi}(\boldsymbol{0})\big]}{\mathrm{Tr}\big[e^{-HT}\big]}\ . \tag{3.33}$$

For $T \to \infty$ only the vacuum survives in the outer trace, and inserting a complete set of states between the operators gives the spectral sum of Eq. (2.7). We use relativistically normalized states, $\langle \boldsymbol{p}'|\boldsymbol{p}\rangle = (2\pi)^3 2E_{\boldsymbol{p}}\,\delta^{(3)}(\boldsymbol{p}-\boldsymbol{p}')$, for which the momentum projection in Eq. (3.31) selects states of momentum $\boldsymbol{p}$ weighted by $1/(2E_n(\boldsymbol{p}))$ [3]. For a nucleon state the overlap with the interpolating operator is proportional to a Dirac spinor,

$$\langle \Omega|\chi_\alpha(0)|N_n(\boldsymbol{p},s)\rangle = \sqrt{Z_n(\boldsymbol{p})}\;u_\alpha(n,\boldsymbol{p},s)\;, \qquad \langle N_n(\boldsymbol{p},s)|\bar\chi_\beta(0)|\Omega\rangle = \sqrt{Z_n(\boldsymbol{p})}^{\,*}\;\bar u_\beta(n,\boldsymbol{p},s)\;, \tag{3.34}$$

where $n$ labels the nucleon and its excitations. If source and sink operators differ, for example through smearing, $Z_n$ stands for $\sqrt{Z_n^{\rm snk}\bar Z_n^{\rm src}}$ [3]. Then

$$C_{\alpha\beta}(t,\boldsymbol{p}) = \sum_n \frac{|Z_n(\boldsymbol{p})|}{2E_n(\boldsymbol{p})}\,e^{-E_n(\boldsymbol{p})t}\sum_s u_\alpha(n,\boldsymbol{p},s)\,\bar u_\beta(n,\boldsymbol{p},s) = \sum_n \frac{|Z_n(\boldsymbol{p})|}{2E_n(\boldsymbol{p})}\,e^{-E_n(\boldsymbol{p})t}\left(-i\not{p}_n + m_n\right)_{\alpha\beta}\,, \tag{3.35}$$

where we used Eq. (3.3), with $p_n$ the Euclidean four-momentum of Eq. (3.3) for energy $E_n(\boldsymbol{p})$. The correlator is projected onto positive parity with $\Gamma^+ = \frac{1}{2}(1+\gamma_4)$ [77]. Using $\mathrm{tr}[\gamma_\mu]=0$, $\mathrm{tr}[\gamma_4\gamma_4]=4$ and $p_4 = iE$,

$$\mathrm{tr}\big[\Gamma^+\big(-i\not{p}+m\big)\big] = \frac{1}{2}\Big(-i\,\mathrm{tr}\big[\gamma_4\gamma_4\big]\,iE + 4m\Big) = 2(E+m)\;, \tag{3.36}$$

so that, for $0 \ll t \ll T/2$,

$$C_{\Gamma^+}(t,\boldsymbol{p}) \equiv \Gamma^+_{\beta\alpha}C_{\alpha\beta}(t,\boldsymbol{p}) = |Z(\boldsymbol{p})|\,\frac{E(\boldsymbol{p})+m}{E(\boldsymbol{p})}\,e^{-E(\boldsymbol{p})t}\left[1+\mathcal{O}\big(e^{-\Delta E\,t}\big)\right]\;, \tag{3.37}$$

where $E$ and $m$ refer to the nucleon ground state and $\Delta E$ is the gap to the first excitation. This agrees with Eq. (3.79) of Ref. [3] and Eq. (3.46) of Ref. [105]. On a finite lattice, the backward-propagating contribution $\propto e^{-E(T-t)}$ and the negative-parity states are not negligible near $t \approx T/2$ [3]. Fits of Eq. (3.37) at large $t$ determine the nucleon mass and dispersion relation.

### 3.6 Three-point functions of local operators

#### 3.6.1 Definition and spectral decomposition

To probe nucleon structure, an operator is inserted at time $\tau$ between the source at time 0 and the sink at time $t$. For a local quark bilinear,

$$\mathcal{O}_\Gamma(x) = \bar q(x)\,\Gamma\,q(x)\;, \tag{3.38}$$

where $\Gamma$ is any product of Dirac matrices, the three-point function is [3]

$$G(t,\tau;\boldsymbol{p},\boldsymbol{p}') = a^6 \sum_{\boldsymbol{x}_2,\boldsymbol{x}_1} e^{-i\boldsymbol{p}'\cdot(\boldsymbol{x}_2-\boldsymbol{x}_1)}\, e^{-i\boldsymbol{p}\cdot\boldsymbol{x}_1}\, \Gamma^{\mathrm{3pt}}_{\beta\alpha} \left\langle \Omega \middle| T\big\{\chi_\alpha(\boldsymbol{x}_2,t)\,\mathcal{O}(\boldsymbol{x}_1,\tau)\,\bar{\chi}_\beta(\boldsymbol{0},0)\big\} \middle| \Omega \right\rangle , \tag{3.39}$$

with spin projector $\Gamma^{\mathrm{3pt}}$ and momentum transfer $\boldsymbol{q} = \boldsymbol{p}' - \boldsymbol{p}$. One factor $a^3$, from the sink projection, cancels against the two-point function in the ratio below; the other defines the momentum-projected operator $\mathcal{O}(\boldsymbol{q}) = a^3 \sum_{\boldsymbol{x}_1} e^{i\boldsymbol{q}\cdot\boldsymbol{x}_1}\mathcal{O}(\boldsymbol{x}_1)$. Inserting complete sets of states before and after $\mathcal{O}$ and using the translation $\chi(\boldsymbol{x},t) = e^{Ht}e^{-i\boldsymbol{P}\cdot\boldsymbol{x}}\chi(0)e^{i\boldsymbol{P}\cdot\boldsymbol{x}}e^{-Ht}$ [3],

$$\begin{aligned} G(t,\tau;\boldsymbol{p},\boldsymbol{p}') = \sum_{n,n'}\sum_{s,s'} & \frac{e^{-E_{n'}(\boldsymbol{p}')(t-\tau)}\, e^{-E_n(\boldsymbol{p})\tau}}{2E_{n'}(\boldsymbol{p}')\,2E_n(\boldsymbol{p})}\, \Gamma^{\mathrm{3pt}}_{\beta\alpha} \\ & \times \langle\Omega|\chi_\alpha|N_{n'}(\boldsymbol{p}',s')\rangle \, \langle N_{n'}(\boldsymbol{p}',s')|\mathcal{O}(\boldsymbol{q})|N_n(\boldsymbol{p},s)\rangle \, \langle N_n(\boldsymbol{p},s)|\bar{\chi}_\beta|\Omega\rangle \ . \end{aligned} \tag{3.40}$$

For $0 \ll \tau \ll t$ the ground state dominates both sums [3]. The matrix element is written as a spinor sandwich,

$$\langle N(\boldsymbol{p}',s')|\mathcal{O}(\boldsymbol{q})|N(\boldsymbol{p},s)\rangle = \bar{u}(\boldsymbol{p}',s')\,\mathcal{J}\,u(\boldsymbol{p},s) \ , \tag{3.41}$$

where $\mathcal{J}$ is the combination of Dirac matrices and form factors allowed by the symmetries of $\mathcal{O}$ [3]. Inserting Eqs. (3.34) and (3.41), the spin sums close into Eq. (3.3),

$$\begin{aligned} G(t,\tau;\boldsymbol{p},\boldsymbol{p}') &= \frac{\sqrt{Z(\boldsymbol{p}')}\sqrt{Z(\boldsymbol{p})}^{\,*}}{4E(\boldsymbol{p}')E(\boldsymbol{p})}\, e^{-E(\boldsymbol{p}')(t-\tau)}\, e^{-E(\boldsymbol{p})\tau}\, \Gamma^{\mathrm{3pt}}_{\beta\alpha} \sum_{s'} u_\alpha(\boldsymbol{p}',s')\bar{u}(\boldsymbol{p}',s')\,\mathcal{J}\sum_s u(\boldsymbol{p},s)\bar{u}_\beta(\boldsymbol{p},s) \\ &= \frac{\sqrt{Z(\boldsymbol{p}')}\sqrt{Z(\boldsymbol{p})}^{\,*}}{4E(\boldsymbol{p}')E(\boldsymbol{p})}\, e^{-E(\boldsymbol{p}')(t-\tau)}\, e^{-E(\boldsymbol{p})\tau}\, \mathrm{tr}\Big[\Gamma^{\mathrm{3pt}}\big(-i\not{p}'+m\big)\,\mathcal{J}\,\big(-i\not{p}+m\big)\Big] \ , \end{aligned} \tag{3.42}$$

(compare Eqs. (3.80)–(3.81) of Ref. [3] and Eq. (3.57) of Ref. [105]).

### 3.6.2 Quark-level contraction and sequential propagators

After Wick contraction, the three-point function contains a connected part, in which the operator acts on a valence quark line, and a disconnected part, in which it closes a separate quark loop [3]. In the isospin limit $m_u = m_d$, the disconnected part cancels in the isovector combination $u - d$ [74, 85]. In the connected part, the propagator from the operator at $(\boldsymbol{x}_1,\tau)$ to the sink at $(\boldsymbol{x}_2,t)$ is summed over $\boldsymbol{x}_2$ and cannot be computed from point-to-all propagators. The sequential-source method solves this problem [3]. For a fixed sink time $t$, sink momentum $\boldsymbol{p}'$ and projector $\Gamma^{\mathrm{3pt}}$, the forward propagators from the source are combined into a sequential source at the sink, and one further inversion gives the

sequential propagator $\Sigma(\mathbf{0}, 0; \boldsymbol{x}_1, \tau; \boldsymbol{p}', t)$ from the sink to every point $(\boldsymbol{x}_1, \tau)$. The three-point function then becomes, up to the factors of $a^3$ and with the phase convention of Eq. (3.39) [3],

$$G_\Gamma(t, \tau; \boldsymbol{p}, \boldsymbol{p}') = \sum_{\boldsymbol{x}_1} e^{+i\boldsymbol{q}\cdot\boldsymbol{x}_1} \left\langle \mathrm{tr}\big[\Sigma(\mathbf{0}, 0; \boldsymbol{x}_1, \tau; \boldsymbol{p}', t)\, \Gamma\, S(\boldsymbol{x}_1, \tau; \mathbf{0}, 0)\big] \right\rangle , \tag{3.43}$$

a contraction of the sequential propagator, the operator and an ordinary propagator. Because the sum over $\boldsymbol{x}_1$ is done last, any operator and any momentum transfer can be used with the same sequential propagator. A new inversion is needed for each sink momentum, sink time, projector and quark flavor [3, 105].

### 3.6.3 Ratio and zero momentum transfer

The overlaps $Z(\boldsymbol{p})$ and the exponential time dependence are removed with the ratio [3]

$$R(t, \tau; \boldsymbol{p}', \boldsymbol{p}) = \frac{G_\Gamma(t, \tau; \boldsymbol{p}, \boldsymbol{p}')}{C_{\Gamma^+}(t, \boldsymbol{p}')} \left[ \frac{C_{\Gamma^+}(\tau, \boldsymbol{p}')\, C_{\Gamma^+}(t, \boldsymbol{p}')\, C_{\Gamma^+}(t-\tau, \boldsymbol{p})}{C_{\Gamma^+}(\tau, \boldsymbol{p})\, C_{\Gamma^+}(t, \boldsymbol{p})\, C_{\Gamma^+}(t-\tau, \boldsymbol{p}')} \right]^{1/2} . \tag{3.44}$$

For the forward matrix elements that define TMDs, the source and sink momenta are equal, $\boldsymbol{p}' = \boldsymbol{p}$, so $\boldsymbol{q} = 0$. Every two-point function in the numerator of the bracket then appears in the denominator with the same arguments,

$$\left[ \frac{C_{\Gamma^+}(\tau, \boldsymbol{p})\, C_{\Gamma^+}(t, \boldsymbol{p})\, C_{\Gamma^+}(t-\tau, \boldsymbol{p})}{C_{\Gamma^+}(\tau, \boldsymbol{p})\, C_{\Gamma^+}(t, \boldsymbol{p})\, C_{\Gamma^+}(t-\tau, \boldsymbol{p})} \right]^{1/2} = 1 , \tag{3.45}$$

and the ratio reduces to $R = G_\Gamma(t, \tau; \boldsymbol{p}, \boldsymbol{p})/C_{\Gamma^+}(t, \boldsymbol{p})$. Inserting Eqs. (3.37) and (3.42) with $\boldsymbol{p}' = \boldsymbol{p}$, the overlaps and the factors $e^{-E(t-\tau)}e^{-E\tau} = e^{-Et}$ cancel,

$$R(t, \tau; \boldsymbol{p}, \boldsymbol{p}) \xrightarrow{0 \ll \tau \ll t} \frac{\mathrm{tr}\big[\Gamma^{3\mathrm{pt}}\big(-i\not{p} + m\big)\, \mathcal{J}\, \big(-i\not{p} + m\big)\big]}{4E(\boldsymbol{p})\big(E(\boldsymbol{p}) + m\big)} = \frac{\mathrm{tr}\big[\Gamma^{3\mathrm{pt}}\big(-i\not{p} + m\big)\, \mathcal{J}\, \big(-i\not{p} + m\big)\big]}{2E(\boldsymbol{p})\, \mathrm{tr}\big[\Gamma^{+}\big(-i\not{p} + m\big)\big]} , \tag{3.46}$$

where the second form uses Eq. (3.36) and agrees with Eq. (3.61) of Ref. [105]. Equivalently, the kinematic prefactor $\sqrt{E_{p'}E_p/[(E_{p'}+m)(E_p+m)]}$ of Eq. (3.93) in Ref. [3] becomes $E/(E+m)$ at $\boldsymbol{p}' = \boldsymbol{p}$. The ratio is independent of $\tau$ and $t$ in the region where the ground state dominates. Excited states are suppressed by increasing the source-sink separation, but the statistical noise grows with it, so the separation is a compromise between the two [105].

### 3.7 Non-local staple-link operators

A local operator such as Eq. (3.38) places the quark and antiquark fields at the same point. The TMD correlator of Eq. (4.1) instead contains the non-local operator $\bar{q}(0)\,\Gamma\,\mathcal{U}[0, \eta v, \eta v + b, b]\,q(b)$, in which the quark fields are separated by $b$ and connected by the staple-shaped gauge link of Eq. (4.11) to make the operator gauge invariant. As discussed in Sections 2.8 and 4.3, the staple legs of the continuum definition extend to infinity; on the lattice $v$ is space-like, $\eta$ is finite, and the results are extrapolated to $\eta \to \infty$ [74, 85].

#### 3.7.1 Lattice operator and three-point function

The operator is evaluated in a Lorentz frame in which $b$ and $v$ are purely spatial, so that the whole operator lies in one time slice and can be computed with the Euclidean path integral [2, 85]. On the lattice the gauge link is a product of link variables along a step-wise path $\mathcal{C}^{\text{lat}}$ that approximates the staple [105],

$$O^{\text{lat}}_{\Gamma,q}(\mathcal{C}^{\text{lat}}; z) \equiv \bar{q}(z)\,\Gamma\,\mathcal{U}^{\text{lat}}\big[\mathcal{C}^{\text{lat}} + z\big]\,q\big(z + \ell[\mathcal{C}^{\text{lat}}]\big)\ , \tag{3.47}$$

where $\ell[\mathcal{C}^{\text{lat}}]$ is the displacement between the two ends of the path. We place $\bar{q}$ at the start of the path and $q$ at its end, as in Eq. (4.1); Ref. [105] uses the opposite labeling. The paths used in this work for $\boldsymbol{b}$ and $\eta\boldsymbol{v}$ are given in Sections 5.4 and 5.5. The three-point function is [105]

$$C^{\text{3pt}}_{\Gamma,q}(\tau, \boldsymbol{P}; \mathcal{C}^{\text{lat}}) = \frac{1}{\hat{L}^3} \sum_{z} \sum_{\beta\alpha} \Gamma^{\text{3pt}}_{\beta\alpha} \left\langle \chi_\alpha(t_{\text{snk}}, \boldsymbol{P})\, O^{\text{lat}}_{\Gamma,q}(\mathcal{C}^{\text{lat}}; z, \tau)\, \bar{\chi}_\beta(t_{\text{src}}, \boldsymbol{P}) \right\rangle\ , \tag{3.48}$$

with $\hat{L}$ the spatial extent in lattice units. Here $P$ denotes the nucleon momentum and $u(P, S)$ the corresponding spinor. Source and sink carry the same momentum $\boldsymbol{P}$: TMDs are defined by diagonal matrix elements, $P' = P$, whereas generalized TMDs also depend on the momentum transfer $\Delta = P' - P$ [74]. The spectral decomposition of Section 3.6 therefore applies with $\boldsymbol{p}' = \boldsymbol{p} = \boldsymbol{P}$, and Eq. (3.45) holds for every link path.

### 3.7.2 Contraction

The connected contraction of Eq. (3.48) uses the forward propagator from the source to the quark field at one end of the link path and the sequential propagator from the sink to the antiquark field at the other end, joined by the link product [105]. The sequential propagator enters as a backward propagator, but it is stored in the format of a forward propagator, so Eq. (3.28) is applied before the contraction [105]. Schematically, for every site $z$ in the operator time slice,

$$C_{\Gamma}^{\mathrm{3pt}}(z) = \mathrm{tr}\Big[\Gamma\, \mathcal{U}^{\mathrm{lat}}\big[\mathcal{C}^{\mathrm{lat}} + z\big]\; S_{\mathrm{fw}}\big(z+\ell\big)\; \gamma_5\, S_{\mathrm{bw}}(z)^{\dagger}\, \gamma_5\Big]\;, \tag{3.49}$$

where $S_{\mathrm{fw}}$ is the forward propagator from the source, $S_{\mathrm{bw}}$ the sequential propagator, and the trace runs over Dirac and color indices. In the lattice code used for this work, this contraction is implemented in Qlua as follows, with `gamma{n=15}` $= \Gamma(15) = \gamma_5$:

Listing 1: Contraction of the staple-link three-point function in Qlua.

```
local stap_fw_shifts_bw = (staple * fw_p:shift(b_dir, "from_forward")
                           * gamma{n=15} * bw_p:adjoin() * gamma{n=15})
for i_gn, gn in pairs(gamma_list) do
  -- bw_p is conjugated to do dot(bw, gamma{n=gn} * stap_fw_shifts)
  local bb = (gamma{n=gn} * stap_fw_shift_bw):trace()
  ...
end
```

The forward propagator `fw_p` is shifted along the separation direction `b_dir`, multiplied by the staple link product, and contracted with the $\gamma_5$-conjugated sequential propagator `bw_p`; the trace is then taken for every Dirac structure $\Gamma(n)$ in `gamma_list`. Ref. [105] gives an equivalent implementation in Chroma. In the isospin limit the disconnected contribution again cancels in the isovector combination $u-d$ [74, 85].

### 3.7.3 Ratio and matrix element

Since source and sink momenta are equal, the matrix element is extracted from the plateau of the forward ratio of Eq. (3.46) [105, 2],

$$\begin{aligned} R_{\Gamma,q}(\boldsymbol{P};\mathcal{C}^{\mathrm{lat}}) &\equiv \mathrm{plateau}\left[\frac{C_{\Gamma,q}^{\mathrm{3pt}}(\tau,\boldsymbol{P};\mathcal{C}^{\mathrm{lat}})}{C^{\mathrm{2pt}}(t_{\mathrm{snk}}-t_{\mathrm{src}},\boldsymbol{P})}\right] \\ &= \sum_{S,S'} \frac{\bar{u}(P,S)\,\Gamma^{\mathrm{3pt}}\,u(P,S')}{2E_P\,\mathrm{tr}\big[\Gamma^{+}\big(-i\not{P}+m_N\big)\big]}\,\big\langle N(P,S')\big|O^{\mathrm{lat}}\big|N(P,S)\big\rangle\;. \end{aligned} \tag{3.50}$$

The ratios $R_{\Gamma(n)}$ for $\Gamma(1), \Gamma(2), \Gamma(4), \Gamma(8) = \gamma_1, \gamma_2, \gamma_3, \gamma_4$ are the input from which the invariant amplitudes are solved in Section 5.6. Under the assumption stated in Section 2.9, the soft factor and the renormalization of the gauge link cancel in ratios of amplitudes taken at the same $b$, $P$ and staple geometry [74, 85]; Ref. [85] examined whether this multiplicative structure carries over to the lattice. The following chapters work with such ratios. Chapter 4 gives the Lorentz-invariant parametrization of $\langle N|O^{\mathrm{lat}}|N\rangle$ that turns the ratios $R_{\Gamma(n)}$ into invariant amplitudes.

# 4 TMD CORRELATOR FOR LATTICE CALCULATIONS

This chapter defines the terms used in the rest of the thesis. Section 4.1 defines the TMD correlator as the Fourier transform of a bare non-local matrix element divided by the soft factor, and identifies the unpolarized distribution $f_1$ and the T-odd Sivers function $f_{1T}^{\perp}$ in its $\gamma^+$ projection. Section 4.2 introduces $\boldsymbol{b}_{\mathrm{T}}$-dependent TMDs and their derivatives, which the lattice evaluates directly, keeping $|\boldsymbol{b}_{\mathrm{T}}|$ finite because $f_1^{(0)}(x)$ and $f_{1T}^{\perp(1)}(x)$ diverge without additional regularization. Section 4.3 introduces the staple-shaped gauge link and the Collins-Soper parameter $\hat{\zeta}$. Section 4.4 lists the Lorentz, parity, time-reversal, and Hermitian conjugation transformations, parametrizes the $b$-dependent correlator in terms of the Lorentz invariant amplitudes $\widetilde{A}_i$ and $\widetilde{B}_i$, and uses hermiticity to show that their real and imaginary parts are even and odd in $b$. Section 4.6 combines these amplitudes into $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$, from which $f_1$ and $f_{1T}^{\perp}$ follow by Fourier transformation.

## 4.1 Definition of TMDs

Consider a frame of reference where the nucleon has large momentum in the $z$-direction, i.e., $P^+ \gg m_N$, $\boldsymbol{P}_{\mathrm{T}} = 0$. In light cone coordinates, the components $k^+$, $\boldsymbol{k}_{\mathrm{T}}$, $k^-$ of the quark momentum $k$ scale as $P^+/m_N$, 1, $m_N/P^+$, respectively, under boosts along the $z$-axis. Thus, the quark longitudinal momentum fraction $x = k^+/P^+$ and the transverse momentum $\boldsymbol{k}_{\mathrm{T}}$ are invariant under boosts along the $z$-axis, while the $k^-$ component is suppressed. This leads to transverse momentum-dependent parton distribution functions (TMDs), which depend on the longitudinal momentum fraction $x \equiv k^+/P^+$ and the quark transverse momentum $\boldsymbol{k}_{\mathrm{T}}$. The transverse momentum components $\boldsymbol{k}_{\mathrm{T}}$ describe an intrinsic motion of the quarks inside the proton that occurs independently of the momentum of the proton itself. TMDs are an important ingredient for understanding the origin of the large angular and spin asymmetries observed in experiments such as semi-inclusive deep inelastic scattering (SIDIS) and the Drell-Yan process (DY).

We construct a correlator from the gauge-invariant bilocal operator $\bar{q}(0)\Gamma\mathcal{U}[\mathcal{C}_b]q(b)$ acting on the nucleon state $|P,S\rangle$, where $\bar{q}$ and $q$ are local quark field operators, and the Wilson line $\mathcal{U}[\mathcal{C}_b]$ makes

the bi-local operator gauge invariant. The Fourier transform of such a position space correlator $\langle P,S|\ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle$ with respect to $b$ gives the momentum space correlator $\Phi^{[\Gamma]}(k,P,S;\ldots)$, or equivalently the TMDs. The Wilson line (to be specified in more detail below) introduces divergences that conventional dimensional regularization cannot treat [2], so we divide by the soft factor, $\widetilde{\mathcal{S}}(b^2;\ldots)$,

$$\Phi^{[\Gamma]}(k,P,S;\ldots)\ \equiv\ \int \frac{d^4b}{(2\pi)^4}\,e^{ik\cdot b}\,\frac{\overbrace{\frac{1}{2}\ \langle P,S|\ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle}^{\equiv\ \widetilde{\Phi}^{[\Gamma]}_{\text{unsubtr.}}(b,P,S;\ldots)}}{\widetilde{\mathcal{S}}(b^2;\ldots)}\ . \tag{4.1}$$

The numerator, $\widetilde{\Phi}^{[\Gamma]}_{\text{unsubtr.}}(b,P,S;\ldots)$, is the bare non-local matrix element in which a gauge link $\mathcal{U}[\mathcal{C}_b]$ connects the quark fields. The denominator is the soft factor, $\widetilde{\mathcal{S}}(b^2;\ldots)$, which regulates rapidity divergences. The ellipses ("...") denote additional dependence on parameters governing the Wilson line geometry and the soft factor, as well as an implicit dependence on the ultraviolet renormalization scale $\mu$.

The soft factor, $\tilde{\mathcal{S}}$, stands symbolically for an expression that can, depending on the formalism, involve several vacuum expectation values. In the framework of Collins and Aybat [32, 23, 106], it takes the form

$$\widetilde{\mathcal{S}}(b^2;\ldots)=\sqrt{\frac{\tilde{S}_{(0)}(\boldsymbol{b}_{\rm T},+\infty,-\infty)\ \tilde{S}_{(0)}(\boldsymbol{b}_{\rm T},y_s,-\infty)}{\tilde{S}_{(0)}(\boldsymbol{b}_{\rm T},+\infty,y_s)}}\ , \tag{4.2}$$

here each of the objects $\tilde{S}_{(0)}(\boldsymbol{b}_{\rm T},\ldots)$ is a vacuum expectation value of Wilson line structures. In this framework, we start with space-like Wilson lines. Some Wilson lines remain tilted away from the light cone, leading to the dependence on the rapidity parameter $y_s$ in the above expression. We need not specify $\tilde{\mathcal{S}}$ further, since it cancels in the observables we consider.

To project the correlator onto the relevant kinematics, we integrate over the suppressed momentum

component $k^-$, which sets the position-space separation to $b^+ = 0$:

$$\begin{aligned}
\Phi^{[\Gamma]}(x, \boldsymbol{k}_{\rm T}; P, S; \ldots) &\equiv \int dk^- \Phi^{[\Gamma]}(k, P, S; \ldots) \\
&= \int dk^- \int \frac{d^4b}{(2\pi)^4}\, e^{ik\cdot b} \frac{\frac{1}{2}\ \langle P,S|\ \ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2;\ldots)} \qquad (4.3) \\
&= \int dk^- \int \frac{db^+}{2\pi} e^{ik^-b^+} \int \frac{d^2\mathbf{b}_T}{(2\pi)^2} \int \frac{db^-}{2\pi} e^{ik^+b^- - i\mathbf{k}_T\cdot\mathbf{b}_T} \frac{\frac{1}{2}\ \langle P,S|\ \ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2;\ldots)} \qquad (4.4) \\
&= \int db^+ \delta(b^+) \int \frac{d^2\mathbf{b}_T}{(2\pi)^2} \int \frac{db^-}{2\pi} e^{ik^+b^- - i\mathbf{k}_T\cdot\mathbf{b}_T} \frac{\frac{1}{2}\ \langle P,S|\ \ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2;\ldots)} \qquad (4.5) \\
&= \int \frac{d^2\boldsymbol{b}_{\rm T}}{(2\pi)^2} \int \frac{d(b\cdot P)}{(2\pi)P^+}\ e^{ix(b\cdot P) - i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\ \left.\frac{\frac{1}{2}\ \langle P,S|\ \ \bar{q}(0)\,\Gamma\,\mathcal{U}[\mathcal{C}_b]\ q(b)\ \ |P,S\rangle}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2;\ldots)}\right|_{b^+=0} . \qquad (4.6)
\end{aligned}$$

As a consequence, $x \leftrightarrow (b\cdot P)$ and $\boldsymbol{k}_{\rm T} \leftrightarrow \boldsymbol{b}_{\rm T}$ are independent pairs of Fourier conjugate variables. With the Dirac matrix $\Gamma = \gamma^+$, which projects onto the leading-twist unpolarized quark distribution, the correlator becomes [27, 28, 22, 101]:

$$\Phi^{[\gamma^+]}(x, \boldsymbol{k}_{\rm T}; P, S, \ldots) = f_1 - \left[\frac{\epsilon_{ij}\,\boldsymbol{k}_i\,\boldsymbol{S}_j}{m_N}\ f_{1T}^{\perp}\right]_{\rm odd} . \qquad (4.7)$$

The term enclosed in $[\,]_{\rm odd}$ isolates the naively time-reversal odd (T-odd) Sivers function $f_{1T}^{\perp}$ [44], which describes the distribution of unpolarized quarks inside a transversely polarized nucleon. The origin of this parametrization and the special role of T-odd TMDs become clear once we discuss the geometry of the gauge link path $\mathcal{C}_b$ and symmetry transformation properties.

### 4.2 TMDs in Fourier space

Our lattice method evaluates the $b$-dependent matrix elements $\widetilde{\Phi}^{[\Gamma]}_{\rm unsubtr.}(b, P, S; \ldots)$ introduced in Eq. (4.1). It is therefore more direct and natural to state our results in terms of Fourier-transformed, $\boldsymbol{b}_{\rm T}$-dependent TMDs and their $\boldsymbol{b}_{\rm T}$-derivatives. For a generic TMD $f$, we define the Fourier transform and its derivatives:

$$\tilde{f}(x, \boldsymbol{b}_{\rm T}^2; \ldots) \equiv \int d^2\boldsymbol{k}_{\rm T}\, e^{i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\ f(x, \boldsymbol{k}_{\rm T}^2; \ldots) = 2\pi \int d|\boldsymbol{k}_{\rm T}||\boldsymbol{k}_{\rm T}|\ J_0(|\boldsymbol{b}_{\rm T}||\boldsymbol{k}_{\rm T}|)\ f(x, \boldsymbol{k}_{\rm T}^2; \ldots)\ , \qquad (4.8)$$

$$\begin{aligned}
\tilde{f}^{(n)}(x, \boldsymbol{b}_{\rm T}^2 \ldots) &\equiv n! \left(-\frac{2}{m_N^2}\partial_{\boldsymbol{b}_{\rm T}^2}\right)^n\ \tilde{f}(x, \boldsymbol{b}_{\rm T}^2; \ldots) \\
&= \frac{2\pi\ n!}{(m_N^2)^n} \int d|\boldsymbol{k}_{\rm T}||\boldsymbol{k}_{\rm T}| \left(\frac{|\boldsymbol{k}_{\rm T}|}{|\boldsymbol{b}_{\rm T}|}\right)^n J_n(|\boldsymbol{b}_{\rm T}||\boldsymbol{k}_{\rm T}|)\ f(x, \boldsymbol{k}_{\rm T}^2; \ldots)\ , \qquad (4.9)
\end{aligned}$$

where $J_n$ are the Bessel functions of the first kind, as shown in Ref. [108]; these Bessel-weighted quantities can be used to structure evolution equations.

In the formal limit $|\boldsymbol{b}_{\rm T}| \to 0$, they reduce to the conventional $\boldsymbol{k}_{\rm T}$-moments:

$$\tilde{f}^{(n)}(x,0;\ldots) = \int d^2\boldsymbol{k}_{\rm T} \left(\frac{\boldsymbol{k}_{\rm T}^2}{2m_N^2}\right)^n f(x,\boldsymbol{k}_{\rm T}^2;\ldots) \equiv f^{(n)}(x) \ . \tag{4.10}$$

However, the first transverse moment of the Sivers function, $f_{1T}^{\perp(1)}(x)$, diverges without additional regularization, because the perturbatively calculable high-$\boldsymbol{k}_{\rm T}$ tails do not decay fast enough [107]. We therefore do not extrapolate to $\boldsymbol{b}_{\rm T} = 0$. At finite $|\boldsymbol{b}_{\rm T}|$, the integrals converge, and we access the $x$-dependence of the Sivers TMD through the conjugate variable $b{\cdot}P$ [81, 82].

### 4.3 Link geometry

The geometry of the gauge link path $\mathcal{C}_b$ affects both the number of allowed TMD structures and the numerical result.

The simplest link geometry is a straight line connecting the quark fields at 0 and $b$. Refs. [81, 82] have studied TMDs with straight gauge links on the lattice. These "process-independent" TMDs are theoretically interesting in their own right, but it is not yet known how to relate them quantitatively to the TMDs that appear in scattering experiments. The operator with straight gauge links has the greatest possible symmetry; as a result, T-odd TMDs vanish for straight gauge links.

To accommodate the symmetries required to isolate the Sivers effect and its $x$ dependence, we use a staple-shaped gauge link; see Fig. 7:

$$\mathcal{U}[\mathcal{C}_b^{(\eta v)}] = \mathcal{U}[0,\eta v,\eta v+b,b] \ . \tag{4.11}$$

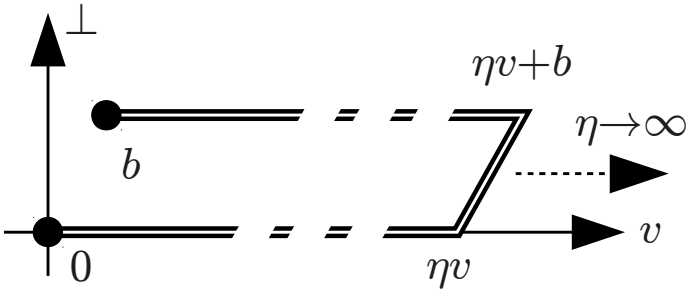



Figure 7: Staple-shaped gauge link as in SIDIS and DY, following Ref. [2].

In Fig. 7, the two parallel Wilson lines are connected at the far end by another straight Wilson line. The complete gauge link thus has a staple-like shape. The vector $v$ sets the staple direction, and $\eta$ sets its extent. Lattice calculations are typically performed without any gauge fixing. We therefore prefer the notation with an explicitly gauge-invariant operator. Moreover, in our study we explicitly take the limit of an infinite "staple extent" $\eta$. Even at finite $\eta$, this gauge-link geometry satisfies the desired symmetry transformation rules mentioned in Section 4.4 below. In this phenomenology frame, quark separation $b$ and staple extent $v$ are spacelike; in the lattice frame, however, they become purely spatial, as we discuss in the upcoming chapters.

The Lorentz-invariant Collins-Soper parameter, $\zeta \equiv 2v\cdot P/\sqrt{|v^2|}$, parameterizes the approach toward the physical light-like limit ($v = n$). As $v$ becomes light-like, $\zeta \to \infty$; the parameter acts as an artificial scale that regulates the rapidity divergences [25, 115, 106]. Expressing the vectors $P$ and $v$ via rapidities $y_P$ and $y_v$ (we use $P^\pm = m_N e^{\pm y_P}/\sqrt{2}$ and $v^+/v^- = -e^{2y_v}$, with $v^- > 0$), we define the dimensionless variable:

$$\hat{\zeta} \equiv \zeta/2m_N = \frac{v\cdot P}{\sqrt{|v^2|}\sqrt{P^2}} = \frac{v^+P^- + v^-P^+}{\sqrt{|v^2|}m_N} = \sinh(y_P - y_v)\ , \tag{4.12}$$

So $\hat{\zeta}$ measures the rapidity difference between the nucleon momentum and the Wilson line direction. This allows us to boost to a frame where $v$ is purely spatial ($v^0 = 0, y_v = 0$), which will be suited for lattice calculations.

### 4.4 Symmetry Constraints

Discrete symmetries constrain the correlation function. Consider the unsubtracted correlator [108]:

$$\Phi^{[\Gamma]}_{\text{unsub}}(k,P,S;v,\mu) = \int \frac{d^4b}{(2\pi)^4}\, e^{ik\cdot b} \underbrace{\frac{1}{2}\,\langle P,S|\ \bar{q}(0)\ \overbrace{\mathcal{U}[0,\infty v]\,\mathcal{U}[\infty v,b]}^{\mathcal{U}[\mathcal{C}_b]}\,\Gamma\ q(b)\ \ |P,S\rangle}_{\widetilde{\Phi}^{[\Gamma]}_{\text{unsub}}(b,P,S;v,\mu)}\ . \tag{4.13}$$

For unpolarized quarks ($\Gamma = \gamma^\mu$), its decomposition into real-valued Lorentz-invariant amplitudes reads

$$\begin{aligned}\frac{1}{2}\Phi^{[\gamma^\mu]}_{\text{unsub}} &= P^\mu\, A_2^{(+)} + k^\mu\, A_3^{(+)} + \frac{1}{m_N}\epsilon^{\mu\nu\alpha\beta} P_\nu k_\alpha S_\beta\, A_{12}^{(+)} + \frac{m_N^2}{(v\cdot P)} v^\mu\, B_1^{(+)} \\ &+ \frac{m_N}{v\cdot P}\epsilon^{\mu\nu\alpha\beta} P_\nu v_\alpha S_\beta\, B_7^{(+)} + \frac{m_N}{v\cdot P}\epsilon^{\mu\nu\alpha\beta} k_\nu v_\alpha S_\beta\, B_8^{(+)} \\ &+ \frac{1}{m_N(v\cdot P)}(k\cdot S)\epsilon^{\mu\nu\alpha\beta} P_\nu k_\alpha v_\beta\, B_9^{(+)} + \frac{m_N}{(v\cdot P)^2}(v\cdot S)\epsilon^{\mu\nu\alpha\beta} P_\nu k_\alpha v_\beta B_{10}^{(+)}\ . \end{aligned} \tag{4.14}$$

On applying Lorentz transformations ($L$), parity transformation ($P$), time-reversal ($T$), and Hermitian conjugation ($\dagger$) to the matrix elements, we find that the correlator fulfills [2, 108]:

$$(L): \qquad \Phi^{[\Gamma]}_{\text{unsub}}(k,P,S;v,\mu) = \Phi^{[\Lambda_{1/2}^{-1}\Gamma\Lambda_{1/2}]}_{\text{unsub}}(\Lambda k,\Lambda P,\Lambda S;\Lambda v,\mu)\ , \tag{4.15}$$

$$(P): \qquad \Phi^{[\Gamma]}_{\text{unsub}}(k,P,S;v,\mu) = \Phi^{[\gamma^0\Gamma\gamma^0]}_{\text{unsub}}(\overline{k},\overline{P},-\overline{S};\overline{v},\mu)\ , \tag{4.16}$$

$$(T): \qquad \Big[\Phi^{[\Gamma]}_{\text{unsub}}(k,P,S;v,\mu)\Big]^* = \Phi^{[\gamma^1\gamma^3\Gamma^*\gamma^3\gamma^1]}_{\text{unsub}}(\overline{k},\overline{P},\overline{S};-\overline{v},\mu)\ , \tag{4.17}$$

$$(\dagger): \qquad \Big[\Phi^{[\Gamma]}_{\text{unsub}}(k,P,S;v,\mu)\Big]^* = \Phi^{[\gamma^0\Gamma^\dagger\gamma^0]}_{\text{unsub}}(k,P,S;v,\mu)\ . \tag{4.18}$$

where, for a vector $c$, $\overline{c} \equiv (c^0,-c^1,-c^2,-c^3)$ denotes the vector with its spatial components reversed in sign. Hermiticity ($\dagger$) implies that the $A_i^{(+)}$ and $B_i^{(+)}$ in Eq. (4.14) are real valued. Time reversal ($T$) does not constrain the number of allowed structures, because it changes the sign of $(v\cdot P)$. Instead, time reversal ($T$) relates the SIDIS amplitudes $A_i^{(+)}$, $B_i^{(+)}$ to the Drell-Yan amplitudes $A_i^{(-)}$, $B_i^{(-)}$.

For any of the transformations $\mathcal{T} \in \{L,P,T,\dagger\}$, the Eqs. (4.15)-(4.18) are of the general form

$$\mathcal{T}_\Phi\left(\Phi(k,w)\right) = \Phi\left(\mathcal{T}_k(k),\mathcal{T}_w(w)\right) \tag{4.19}$$

where we omit the subscript "unsub" and the renormalization scale $\mu$, and the symbol $w$ collects all dependence on $\Gamma$, $P$, $S$ and $v$. Here $\mathcal{T}_\Phi$ is either the identity function or complex conjugation. The transformation rule $\mathcal{T}_k(k)$ maps onto $\Lambda k$, $k$ or $\overline{k}$ and thus fulfills $a\cdot b = \mathcal{T}_k(a)\cdot\mathcal{T}_k(b)$ for any two vectors $a$ and $b$. The Fourier-transformed correlator

$$\widetilde{\Phi}(b,w) = \int d^4k\ e^{-ik\cdot b}\Phi(k,w) \tag{4.20}$$

transforms according to

$$\begin{aligned}
\mathcal{T}_\Phi\left(\widetilde{\Phi}(b,w)\right) &= \int d^4k\; e^{\mathcal{T}_\Phi(-i)\,k\cdot b}\;\mathcal{T}_\Phi\left(\Phi(k,w)\right) \\
&= \int d^4q\; e^{\mathcal{T}_\Phi(-i)\,\mathcal{T}_k^{-1}(q)\cdot b}\;\Phi\left(q,\mathcal{T}_w(w)\right) \\
&\qquad\qquad \text{(change of variables: } k\to q=\mathcal{T}_k(k)\text{, i.e. } k=\mathcal{T}_k^{-1}(q)\text{)} \\
&= \int d^4q\; e^{\mathcal{T}_\Phi(-i)\,q\cdot\mathcal{T}_k(b)}\;\Phi\left(q,\mathcal{T}_w(w)\right) \qquad (\because q\cdot b=\mathcal{T}_k(q)\cdot\mathcal{T}_k(b)) \\
&= \int d^4q\; e^{-iq\left(\frac{\mathcal{T}_\Phi(i)}{i}\cdot\mathcal{T}_k(b)\right)}\;\Phi\left(q,\mathcal{T}_w(w)\right) \\
&= \widetilde{\Phi}\left(\frac{\mathcal{T}_\Phi(i)}{i}\mathcal{T}_k(b),\mathcal{T}_w(w)\right)\,. \qquad (4.21)
\end{aligned}$$

In particular, under Hermitian conjugation the position-space correlator transforms as

$$(\dagger): \qquad \left[\widetilde{\Phi}^{[\Gamma]}_{\rm unsub}(b,P,S;v)\right]^* = \widetilde{\Phi}^{[\gamma^0\Gamma^\dagger\gamma^0]}_{\rm unsub}(-b,P,S;v)\;. \qquad (4.22)$$

Let $f(k,w)$ be any of the structures preceding the invariant amplitudes in the parameterization of $\Phi$. The structure $f(k,w)$ is a homogeneous function of some degree $n$ in $k$, i.e., $f(\alpha k,w)=\alpha^n f(k,w)$ for any number $\alpha$. For example, the structure $f(k,w)=\frac{1}{m_N(v\cdot P)}(k\cdot S)\epsilon^{\mu\nu\alpha\beta}P_\nu k_\alpha v_\beta$ preceding $B_9^{(+)}$ in Eq. (4.14) has degree $n=2$. If we define $\tilde{f}(b,w)\equiv f(-im_N^2 b,w)$, then

$$\mathcal{T}_\Phi\left(\tilde{f}(b,w)\right)=\mathcal{T}_\Phi(-im_N^2)^n\,\mathcal{T}_\Phi\left(f(b,w)\right)=f\left(\mathcal{T}_\Phi(-im_N^2)\mathcal{T}_k(b),\mathcal{T}_w(w)\right)=\tilde{f}\left(\frac{\mathcal{T}_\Phi(i)}{i}b,w\right)\,. \qquad (4.23)$$

This shows that $\tilde{f}$ transforms like $\widetilde{\Phi}$ in Eq. (4.21). The parameterization of $\widetilde{\Phi}$ therefore follows from the substitution $k\to -im_N^2 b$ in the structures parameterizing $\Phi$, which gives

$$\begin{aligned}
\frac{1}{2}\widetilde{\Phi}^{[\gamma^\mu]}_{\rm unsub} =& P^\mu\,\widetilde{A}_2 - im_N^2 b^\mu\,\widetilde{A}_3 - im_N\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha S_\beta\,\widetilde{A}_{12} + \frac{m_N^2}{(v\cdot P)}v^\mu\,\widetilde{B}_1 \\
&+\frac{m_N}{v\cdot P}\epsilon^{\mu\nu\alpha\beta}P_\nu v_\alpha S_\beta\,\widetilde{B}_7 - \frac{im_N^3}{v\cdot P}\epsilon^{\mu\nu\alpha\beta}b_\nu v_\alpha S_\beta\,\widetilde{B}_8 \\
&-\frac{m_N^3}{v\cdot P}(b\cdot S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha v_\beta\,\widetilde{B}_9 - \frac{im_N^3}{(v\cdot P)^2}(v\cdot S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha v_\beta\widetilde{B}_{10}\;. \qquad (4.24)
\end{aligned}$$

The amplitudes $\widetilde{A}_i$ and $\widetilde{B}_i$ introduced this way are no longer constrained to be real-valued functions. Instead, Hermitian conjugation, Eq. (4.22), gives the relation

$$\left[\widetilde{A}_i(b^2,b\cdot P,v\cdot b/(v\cdot P),\zeta^{-2},\mu^2)\right]^* = \widetilde{A}_i(b^2,-b\cdot P,-v\cdot b/(v\cdot P),\zeta^{-2},\mu^2)\,. \qquad (4.25)$$

In general, the amplitudes are complex,

$$\widetilde{A}_i = \widetilde{A}_i^{\mathrm{Re}} + i\widetilde{A}_i^{\mathrm{Im}} \tag{4.26}$$

$$\widetilde{B}_i = \widetilde{B}_i^{\mathrm{Re}} + i\widetilde{B}_i^{\mathrm{Im}} \tag{4.27}$$

Given the amplitudes at both $\pm b$, Eq. (4.25) implies

$$\widetilde{A}_i^{\mathrm{Re}} = \widetilde{A}_i^{(\text{b-even})} \tag{4.28a}$$

$$\widetilde{A}_i^{\mathrm{Im}} = \widetilde{A}_i^{(\text{b-odd})} \tag{4.28b}$$

$$\widetilde{B}_i^{\mathrm{Re}} = \widetilde{B}_i^{(\text{b-even})} \tag{4.28c}$$

$$\widetilde{B}_i^{\mathrm{Im}} = \widetilde{B}_i^{(\text{b-odd})} \tag{4.28d}$$

### 4.5 Correlator Parametrization

To translate Euclidean space lattice calculations into TMDs, which are defined and interpreted in light cone coordinates, we need the parametrization of the correlator $\tilde{\Phi}_{\text{unsubtr.}}$ in terms of manifestly Lorentz-invariant amplitudes. A parametrization of the correlator $\Phi_{\text{unsubtr.}}(k, P, S; \infty v, \mu)$ for link paths that extend to infinity in a direction $v$ has been worked out in Ref. [101] and involves 32 independent amplitudes $A_i$ and $B_i$ that depend on the Lorentz-invariant quantities $k^2$, $k \cdot P$, $k{\cdot}v/v{\cdot}P$ and $\hat{\zeta}$. Section 4.4 shows that a parametrization of the corresponding $b$-dependent correlator $\tilde{\Phi}_{\text{unsubtr.}}$ is of the same form as the parametrization of $\Phi_{\text{unsubtr.}}$ if we substitute $k \to -im_N^2 b$, which is Eq. (4.24) (for $\Gamma = \gamma^\mu$):

$$\begin{aligned}\frac{1}{2}\widetilde{\Phi}_{\text{unsub}}^{[\gamma^\mu]} =& P^\mu\, \widetilde{A}_2 - im_N^2 b^\mu\, \widetilde{A}_3 - im_N \epsilon^{\mu\nu\alpha\beta} P_\nu b_\alpha S_\beta\, \widetilde{A}_{12} + \frac{m_N^2}{(v{\cdot}P)} v^\mu\, \widetilde{B}_1 \\ &+ \frac{m_N}{v{\cdot}P}\epsilon^{\mu\nu\alpha\beta} P_\nu v_\alpha S_\beta\, \widetilde{B}_7 - \frac{im_N^3}{v{\cdot}P}\epsilon^{\mu\nu\alpha\beta} b_\nu v_\alpha S_\beta\, \widetilde{B}_8 \\ &- \frac{m_N^3}{v{\cdot}P}(b{\cdot}S)\epsilon^{\mu\nu\alpha\beta} P_\nu b_\alpha v_\beta\, \widetilde{B}_9 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\epsilon^{\mu\nu\alpha\beta} P_\nu b_\alpha v_\beta \widetilde{B}_{10}\ .\end{aligned} \tag{4.29}$$

For the lattice calculations, it is advantageous to start with an equivalent parametrization in which the structures explicitly depend on the staple extent $\eta$ and which is still well-defined for $v{\cdot}P = 0$. Such a parametrization can be obtained from the parametrization above by replacing $v \to \eta v$ and by leaving

out the factors $(v{\cdot}P)^{-n}$.

$$\begin{aligned}\frac{1}{2}\widetilde{\Phi}^{[\gamma^\mu]}_{\text{unsub}} =& P^\mu\,\tilde{a}_2 - im_N^2 b^\mu\,\tilde{a}_3 - im_N\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha S_\beta\,\tilde{a}_{12}\\ &+ m_N^2\eta v^\mu\,\tilde{b}_1 + m_N\epsilon^{\mu\nu\alpha\beta}P_\nu\eta v_\alpha S_\beta\,\tilde{b}_7 - im_N^3\epsilon^{\mu\nu\alpha\beta}b_\nu\eta v_\alpha S_\beta\,\tilde{b}_8\\ &- m_N^3(b{\cdot}S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha\eta v_\beta\,\tilde{b}_9 - im_N^3(v{\cdot}S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha\eta^2 v_\beta\tilde{b}_{10}\end{aligned} \tag{4.30}$$

Here we have used lowercase amplitudes to distinguish the two parametrizations. The relation to the uppercase amplitudes is given by

$$\begin{aligned}\widetilde{A}_i\left(b^2, b{\cdot}P, \frac{v{\cdot}b}{v{\cdot}P}, \frac{v^2}{(v{\cdot}P)^2}, \eta v{\cdot}P\right) &= \tilde{a}_i(b^2, b{\cdot}P, \eta v{\cdot}b, (\eta v)^2, \eta v{\cdot}P)\,,\\ \widetilde{B}_i\left(b^2, b{\cdot}P, \frac{v{\cdot}b}{v{\cdot}P}, \frac{v^2}{(v{\cdot}P)^2}, \eta v{\cdot}P\right) &= (\eta v{\cdot}P)^n\,\tilde{b}_i(b^2, b{\cdot}P, \eta v{\cdot}b, (\eta v)^2, \eta v{\cdot}P)\,.\end{aligned} \tag{4.31}$$

where $n$ is the power with which $v{\cdot}P$ appears in the denominator in front of the corresponding amplitude $\widetilde{B}_i$ in the parametrization. Notice that the $\tilde{a}_i$ and $\tilde{b}_i$ are functions of all the Lorentz-invariant products of $b$, $P$ and $\eta v$. For the uppercase amplitudes, however, we choose to represent the dependence on these invariants in the third and fourth arguments by $\eta$-independent expressions, to facilitate taking the limit $\eta \to \pm\infty$. The dependence on the Collins-Soper parameter $\hat{\zeta}$ is given by the fourth argument, $v^2/(v{\cdot}P)^2 = -1/(m_N\hat{\zeta})^2$, while the fifth argument, $\eta v{\cdot}P$, characterizes the length of the gauge link and distinguishes between future- and past-pointing Wilson lines. In our frame ($b^+ = 0$, $\boldsymbol{v}_\text{T} = \boldsymbol{P}_\text{T} = 0$), the kinematics fixes the relation between the invariants [105, 116]:

$$\frac{v{\cdot}b}{v{\cdot}P} = b{\cdot}P\frac{R(\hat{\zeta}^2)}{m_N^2}\,, \quad \text{where} \quad R(\hat{\zeta}^2) \equiv 1 - \sqrt{1+\hat{\zeta}^{-2}} = \frac{m_N^2}{v{\cdot}P}\frac{v^+}{P^+}\,. \tag{4.32}$$

The relation in Eq. (4.32) shows that the third argument of the $\widetilde{A}_i$ and $\widetilde{B}_i$ is not independent of the others in the context of TMDs. Moreover, in our lattice calculations, we have to choose the link directions $b$ and $v$ such that Eq. (4.32) is fulfilled. As a side remark, the parameter corresponding to Eq. (4.32) in momentum space is $v{\cdot}k/v{\cdot}P \approx x$, i.e., the amplitudes $A_i$ and $B_i$ acquire an explicit $x$-dependence, which has already been pointed out in Refs. [105, 116].

For the $\Gamma = \gamma^+$ structure at leading twist, the correlator can be written in the form

$$\frac{1}{2P^+}\widetilde{\Phi}^{[\gamma^+]}_{\text{unsubtr.}} = \widetilde{A}_{2B} + im_N\epsilon_{ij}\boldsymbol{b}_i\boldsymbol{S}_j\,\widetilde{A}_{12B}\,, \tag{4.33}$$

where the indices $i, j$ correspond to transverse directions, $i, j \in \{1, 2\}$ (we use the convention $\epsilon^{0123} = 1$ for the totally antisymmetric Levi-Civita symbol, and introduce $\epsilon_{ij} \equiv \epsilon^{-+ij}$ such that $\epsilon_{12} = 1$), and where we have introduced the following abbreviations for combinations of amplitudes:

$$\begin{aligned} \widetilde{A}_{2B} &\equiv \widetilde{A}_2 + R(\hat{\zeta}^2)\widetilde{B}_1 \\ \widetilde{A}_{12B} &\equiv \widetilde{A}_{12} - R(\hat{\zeta}^2)\widetilde{B}_8 \,. \end{aligned} \tag{4.34}$$

The unpolarized distribution $f_1$ and the Sivers distribution $f_{1T}^{\perp}$ follow from the Fourier integration operator $\int_{\mathcal{F}}$:

$$\begin{aligned} \int_{\mathcal{F}} \widetilde{A}_i &\equiv \int \frac{d^2\boldsymbol{b}_{\mathrm{T}}}{(2\pi)^2}\, e^{-i\boldsymbol{b}_{\mathrm{T}}\cdot\boldsymbol{k}_{\mathrm{T}}}\, \frac{1}{\widetilde{\mathcal{S}}(b^2;\ldots)} \int \frac{d(b\cdot P)}{(2\pi)}\, e^{ix(b\cdot P)}\, \widetilde{A}_i(-\boldsymbol{b}_{\mathrm{T}}^2, b\cdot P, (b\cdot P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v\cdot P) \\ &= \int_0^\infty \frac{d(-b^2)}{2(2\pi)}\, \frac{J_0(\sqrt{-b^2}\,|\boldsymbol{k}_{\mathrm{T}}|)}{\widetilde{\mathcal{S}}(b^2;\ldots)} \int \frac{d(b\cdot P)}{(2\pi)}\, e^{ix(b\cdot P)}\, \widetilde{A}_i(b^2, b\cdot P, (b\cdot P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v\cdot P) \end{aligned} \tag{4.35}$$

Applied to the combined amplitudes, it gives the leading-twist distributions:

$$\begin{aligned} f_1(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v\cdot P) &= 2\int_{\mathcal{F}} \widetilde{A}_{2B}\,, \\ f_{1T}^{\perp}(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v\cdot P) &= 4m_N^2 \partial_{\boldsymbol{k}_{\mathrm{T}}^2} \int_{\mathcal{F}} \widetilde{A}_{12B}\,. \end{aligned} \tag{4.36}$$

Here, the dots "..." indicate further parameters that specify the geometry of the soft factor. The $f_1$ is a T-even distribution which fulfills

$$f^{\text{T-even}}(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v\cdot P) = f^{\text{T-even}}(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, -\eta v\cdot P) \tag{4.37}$$

while the Sivers function $f_{1T}^{\perp}$ is a T-odd distribution which fulfills

$$f^{\text{T-odd}}(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, \eta v\cdot P) = -f^{\text{T-odd}}(x, \boldsymbol{k}_{\mathrm{T}}^2; \hat{\zeta}, \ldots, -\eta v\cdot P) \tag{4.38}$$

As a result, T-odd distributions must vanish for $\eta = 0$, which corresponds to straight gauge links. TMDs for SIDIS and DY are obtained for $\eta v\cdot P \to \infty$ and $\eta v\cdot P \to -\infty$, respectively. In the following, we choose $v\cdot P \geq 0$, such that the SIDIS and DY limits for space-like $v$ can also be written as $\eta|v| \to \infty$ and $\eta|v| \to -\infty$, respectively.

Equation (4.36) shows that the $x$-dependent TMDs in Fourier space, $\tilde{f}(x, \boldsymbol{b}_{\rm T}^2)$ as defined in Eqs. (4.8) and (4.9), are given directly by the Fourier transform of the amplitudes $\widetilde{A}_{iB}$ with respect to $b{\cdot}P$. For brevity, define the $x-(b{\cdot}P)$-Fourier-transformed, soft-factor-divided amplitudes

$$\widetilde{A}_i^{[x]}(x, \boldsymbol{b}_{\rm T}^2) \equiv \frac{1}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2; \dots)} \int_{-\infty}^{\infty} \frac{d(b{\cdot}P)}{2\pi}\, e^{ix(b\cdot P)}\, \widetilde{A}_i\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, \dots\big)\,, \tag{4.39}$$

so that, by Eq. (4.35), $\int_{\mathcal{F}} \widetilde{A}_i = \int \frac{d^2\boldsymbol{b}'_{\rm T}}{(2\pi)^2}\, e^{-i\boldsymbol{b}'_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\, \widetilde{A}_i^{[x]}(x, \boldsymbol{b}'^2_{\rm T})$. Inserting $f_1 = 2\int_{\mathcal{F}} \widetilde{A}_{2B}$ into Eq. (4.8), we have for the unpolarized TMD,

$$\begin{aligned}\tilde{f}_1(x, \boldsymbol{b}_{\rm T}^2) &= 2\int d^2\boldsymbol{k}_{\rm T}\, e^{i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}} \int \frac{d^2\boldsymbol{b}'_{\rm T}}{(2\pi)^2}\, e^{-i\boldsymbol{b}'_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\, \widetilde{A}_{2B}^{[x]}(x, \boldsymbol{b}'^2_{\rm T}) = 2\int d^2\boldsymbol{b}'_{\rm T}\, \delta^{(2)}(\boldsymbol{b}_{\rm T} - \boldsymbol{b}'_{\rm T})\, \widetilde{A}_{2B}^{[x]}(x, \boldsymbol{b}'^2_{\rm T}) \\ &= 2\, \widetilde{A}_{2B}^{[x]}(x, \boldsymbol{b}_{\rm T}^2)\,,\end{aligned} \tag{4.40}$$

For Sivers TMD, on starting from $f_{1T}^{\perp} = 4m_N^2\, \partial_{\boldsymbol{k}_{\rm T}^2} \int_{\mathcal{F}} \widetilde{A}_{12B}$ and using $\boldsymbol{k}_j\, \partial_{\boldsymbol{k}_{\rm T}^2} = \frac{1}{2}\, \partial_{\boldsymbol{k}_j}$, differentiating Eq. (4.8) with respect to $\boldsymbol{b}_j$, we have

$$\begin{aligned}\partial_{\boldsymbol{b}_j} \tilde{f}_{1T}^{\perp} &= \int d^2\boldsymbol{k}_{\rm T}\, (i\boldsymbol{k}_j)\, e^{i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\, 4m_N^2\, \partial_{\boldsymbol{k}_{\rm T}^2} \int_{\mathcal{F}} \widetilde{A}_{12B} = 2im_N^2 \int d^2\boldsymbol{k}_{\rm T}\, e^{i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}}\, \partial_{\boldsymbol{k}_j} \int_{\mathcal{F}} \widetilde{A}_{12B} \\ &= -2im_N^2\, (i\boldsymbol{b}_j) \int d^2\boldsymbol{k}_{\rm T}\, e^{i\boldsymbol{b}_{\rm T}\cdot\boldsymbol{k}_{\rm T}} \int_{\mathcal{F}} \widetilde{A}_{12B} = 2m_N^2\, \boldsymbol{b}_j\, \widetilde{A}_{12B}^{[x]}(x, \boldsymbol{b}_{\rm T}^2)\,,\end{aligned} \tag{4.41}$$

where we integrated by parts in $\boldsymbol{k}_j$ (the surface term vanishes for $|\boldsymbol{b}_{\rm T}| > 0$) and used the same $\delta$-function step as for $f_1$. Since $\partial_{\boldsymbol{b}_j} \tilde{f} = 2\boldsymbol{b}_j\, \partial_{\boldsymbol{b}_{\rm T}^2} \tilde{f}$, this gives $\partial_{\boldsymbol{b}_{\rm T}^2} \tilde{f}_{1T}^{\perp} = m_N^2\, \widetilde{A}_{12B}^{[x]}$. Eq. (4.9) with $n = 1$ then yields

$$\tilde{f}_{1T}^{\perp(1)}(x, \boldsymbol{b}_{\rm T}^2) = -\frac{2}{m_N^2}\, \partial_{\boldsymbol{b}_{\rm T}^2} \tilde{f}_{1T}^{\perp}(x, \boldsymbol{b}_{\rm T}^2) = -2\, \widetilde{A}_{12B}^{[x]}(x, \boldsymbol{b}_{\rm T}^2)\,. \tag{4.42}$$

In summary, we have,

$$\begin{aligned}\tilde{f}_1^{(0)}(x, \boldsymbol{b}_{\rm T}^2; \hat{\zeta}, \dots, \eta v{\cdot}P) &= 2\int_{-\infty}^{\infty} \frac{d(b{\cdot}P)}{2\pi}\, e^{ix(b\cdot P)}\, \frac{\widetilde{A}_{2B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2; \dots)}\,, \\ &= 2\, \widetilde{A}_{2B}^{[x]}(x, \boldsymbol{b}_{\rm T}^2)\,.\end{aligned} \tag{4.43}$$

$$\begin{aligned}\tilde{f}_{1T}^{\perp(1)}(x, \boldsymbol{b}_{\rm T}^2; \hat{\zeta}, \dots, \eta v{\cdot}P) &= -2\int_{-\infty}^{\infty} \frac{d(b{\cdot}P)}{2\pi}\, e^{ix(b\cdot P)}\, \frac{\widetilde{A}_{12B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)}{\widetilde{\mathcal{S}}(-\boldsymbol{b}_{\rm T}^2; \dots)}\,, \\ &= -2\, \widetilde{A}_{12B}^{[x]}(x, \boldsymbol{b}_{\rm T}^2)\,.\end{aligned} \tag{4.44}$$

This framework lays the groundwork for extracting its $x$-dependence. In subsequent chapters, for brevity, we may drop the soft-factor term from the definitions above, since it ultimately cancels in the ratios we consider.

### 4.6 Generalized Sivers shift from amplitudes

Eqs. (4.43)-(4.44) identify $x$-dependent derivatives of Fourier-transformed TMDs with simple linear combinations of amplitudes $\widetilde{A}_i$ and $\widetilde{B}_i$ evaluated at the same values of $\boldsymbol{b}_{\rm T}^2$, $b\cdot P$, $\hat{\zeta}$ and $\eta v{\cdot}P$. Forming ratios of these objects thus amounts to taking ratios of linear combinations of the fundamental correlators $\tilde{\Phi}^{[\Gamma]}_{\rm unsubtr.}$ evaluated at the same point, i.e., with the same values of $b$, $P$, and $\eta v$. For a discussion of the renormalization properties of ratios of the objects in (4.43)-(4.44), it is thus sufficient to understand the renormalization properties of (ratios formed from) the correlators $\widetilde{\Phi}^{[\Gamma]}_{\rm unsubtr.} = \frac{1}{2}\,\langle P,S|\;\bar{q}(0)\,\Gamma\,\mathcal{U}\;q(b)\;\;|P,S\rangle$.

As discussed in Refs. [2, 85], analytical studies of the operator $\bar{q}(0)\,\Gamma\,\mathcal{U}q(b)$ in the continuum suggest that for $b^2\neq 0$ the renormalization factors are multiplicative and $\Gamma$-independent. The basic reason is that the quark field operators are at different locations and undergo wave function renormalization separately. We will assume here that our lattice representation of $\bar{q}(0)\,\Gamma\,\mathcal{U}\,q(b)$ is renormalized multiplicatively independent of $\Gamma$ as long as we keep $\boldsymbol{b}_{\rm T}^2$ larger than a few lattice spacings. A more detailed discussion and numerical studies of this operator's renormalization properties appear in Ref. [82]. Under the assumption of multiplicative renormalization, the soft factor and the renormalization factors cancel in ratios of amplitudes evaluated at the same $b$, $P$ and $\eta v$, and such ratios depend only on $\boldsymbol{b}_{\rm T}^2$, $P\cdot b$, $\hat{\zeta}$, and the staple extent $\eta v{\cdot}P$.

From phenomenology, the Sivers shift is the ratio defined as

$$\langle \boldsymbol{k}_y\rangle_{TU} \equiv m_N\frac{f_{1T}^{\perp(1)}(x)}{f_1^{(0)}(x)} = \left.\frac{\int d^2\boldsymbol{k}_{\rm T}\,\boldsymbol{k}_y\;\Phi^{[\gamma^+]}(x,\boldsymbol{k}_{\rm T},P,S;\ldots)}{\int d^2\boldsymbol{k}_{\rm T}\quad\;\Phi^{[\gamma^+]}(x,\boldsymbol{k}_{\rm T},P,S;\ldots)}\right|_{\boldsymbol{S}_{\rm T}\,=\,(1,0)}\,, \tag{4.45}$$

where $\gamma^+$ projects onto the leading twist. The ratio above yields the average transverse momentum in the $y$-direction, for quarks with a given longitudinal momentum fraction $x$ inside a proton polarized in the $x$-direction. As mentioned before, understanding $f_{1T}^{\perp(1)}$ simply as a $\boldsymbol{k}_{\rm T}$-weighted TMD is problematic, since the $\boldsymbol{k}_{\rm T}$-integral is expected to be UV divergent. A natural way of circumventing this divergence

is to generalize the Sivers shift to an expression in terms of the Fourier-transformed TMDs:

$$\langle \boldsymbol{k}_y \rangle_{TU}(\boldsymbol{b}_{\mathrm{T}}^2, x, \hat{\zeta}, \eta v{\cdot}P) \;\equiv\; m_N \frac{\tilde{f}_{1T}^{\perp(1)}(\boldsymbol{b}_{\mathrm{T}}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{(0)}(\boldsymbol{b}_{\mathrm{T}}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)} \tag{4.46}$$

$$= -m_N \frac{\int d(b\cdot P) e^{ix(b\cdot P)} \widetilde{A}_{12B}(b^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)}{\int d(b\cdot P) e^{ix(b\cdot P)} \widetilde{A}_{2B}(b^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)} \tag{4.47}$$

$$\xrightarrow{\boldsymbol{b}_{\mathrm{T}}^2=0} \left. \frac{\int d^2\boldsymbol{k}_{\mathrm{T}}\, \boldsymbol{k}_y\; \Phi^{[\gamma^+]}(x, \boldsymbol{k}_{\mathrm{T}}, P, S; \ldots)}{\int d^2\boldsymbol{k}_{\mathrm{T}} \qquad \Phi^{[\gamma^+]}(x, \boldsymbol{k}_{\mathrm{T}}, P, S; \ldots)} \right|_{\boldsymbol{S}_{\mathrm{T}} = (1,0)} \tag{4.48}$$

In the limit $\boldsymbol{b}_{\mathrm{T}}^2 = 0$ we recover the Sivers shift (4.45), because the Fourier transformed TMDs $\tilde{f}_{1T}^{\perp(1)}$ and $\tilde{f}_1^{(0)}$ coincide with the moments $f_{1T}^{\perp(1)}$ and $f_1^{(0)}$, respectively. However, we are interested in the generalized Sivers shift for non-zero $\boldsymbol{b}_{\mathrm{T}}^2$, where this UV divergence disappears. The variable $\boldsymbol{b}_{\mathrm{T}}^2$ effectively acts as a regulator. Moreover, the $\boldsymbol{b}_{\mathrm{T}}$-dependence allows us to study differences in the widths of distributions on a qualitative level.

Furthermore, the generalized Sivers shift is T-odd, i.e., we expect to obtain results of opposite sign in the SIDIS and DY limits $\eta v{\cdot}P \to \infty$ and $\eta v{\cdot}P \to -\infty$, respectively, as demonstrated in Refs. [2, 85].

In summary, the generalized Sivers shift describes a feature of the transverse momentum distribution of (unpolarized) quarks in a transversely polarized proton. In the formal limit $\boldsymbol{b}_{\mathrm{T}}^2 = 0$, it measures the dipole moment of that distribution orthogonal to the polarization of the proton.

In the next chapter, we explore the lattice QCD calculations of the generalized Sivers shift and extract its dependence on the longitudinal momentum fraction $x$.

# 5 LATTICE DATA & EXTRACTING THE $x-$DEPENDENCE OF SIVERS SHIFT

This Chapter connects the lattice correlators to the $x$-dependent Sivers shift. All lattice data and results in this chapter are for the isovector $u-d$ combination. Section 5.1 writes the Sivers shift as a ratio of Fourier transforms over $b \cdot P$ of the amplitudes $\widetilde{A}_{12B}$ and $\widetilde{A}_{2B}$. Section 5.2 explains how extrapolating the Collins-Soper parameter $\hat{\zeta} \to \infty$ is performed. Sections 5.3 and 5.4 fix the staple direction $v$ for each nucleon momentum so that we scan the whole $b \cdot P$ range, and describe how the correlator at the off-axis target $\eta \boldsymbol{v}/a$ is obtained by path averaging and quadratic interpolation. Section 5.5 describes the quark separation vectors $\boldsymbol{b}$, and the gauge link paths that correspond to them.

Section 5.6 projects the correlator parametrization onto the lattice kinematics and solves for the real and imaginary amplitudes from the symmetric and antisymmetric combinations of the longitudinal $b$ data. Section 5.7 combines them into $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$, and Section 5.8 maps these from $(b_L, b_T)$ to $(P \cdot b, b^2)$. Section 5.9 checks the parity of the extracted amplitudes, shows them at the four simulated momenta, and discards $\widetilde{A}^{\mathrm{Im}}_{12B}$, which is statistically compatible with zero.

## 5.1 The Sivers Shift as a Function of Momentum Fraction $x$

To connect our lattice observables to the physical, $x$-dependent Transverse Momentum Dependent Parton Distribution Functions (TMDs), we must isolate the relevant amplitudes from the correlation function. Focusing on the leading-twist unpolarized quark distribution, the correlator is parameterized by the Lorentz-invariant amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$, which we construct from the primitive correlator amplitudes in Section 5.7. Our primary goal is to extract the generalized Sivers shift, $\langle \boldsymbol{k}_y \rangle_{TU}$, as a continuous function of the longitudinal momentum fraction $x = k^+/P^+$. The Sivers shift is defined phenomenologically as the ratio of the first transverse momentum moment of the Sivers function to the unpolarized distribution as discussed in Section 4.6.

Because $x$ and the longitudinal spatial separation $b \cdot P$ are Fourier conjugate variables, mapping the $x$-dependence requires a Fourier transform of these amplitudes with respect to $b \cdot P$. Analytically, the

$x$-dependent Sivers shift takes the form:

$$\langle \boldsymbol{k}_y \rangle_{TU}(\boldsymbol{b}_{\mathrm{T}}^2, x, \hat{\zeta}, \eta v{\cdot}P) \;\equiv\; m_N \frac{\tilde{f}_{1T}^{\perp(1)}(\boldsymbol{b}_{\mathrm{T}}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{(0)}(\boldsymbol{b}_{\mathrm{T}}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)} \tag{5.1}$$

$$= -m_N \frac{\int d(b\cdot P) e^{ix(b\cdot P)} \widetilde{A}_{12B}(b^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)}{\int d(b\cdot P) e^{ix(b\cdot P)} \widetilde{A}_{2B}(b^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P)} \tag{5.2}$$

$$= -m_N \frac{\int d(b\cdot P) e^{ix(b\cdot P)} \left[\widetilde{A}_{12B}^{\mathrm{Re}}(\cdots) + i\widetilde{A}_{12B}^{\mathrm{Im}}(\cdots)\right]}{\int d(b\cdot P) e^{ix(b\cdot P)} \left[\widetilde{A}_{2B}^{\mathrm{Re}}(\cdots) + i\widetilde{A}_{2B}^{\mathrm{Im}}(\cdots)\right]} \tag{5.3}$$

$$= -m_N \frac{\int d(b\cdot P) \left[\cos\left(x(b\cdot P)\right) + i\sin\left(x(b\cdot P)\right)\right] \left[\widetilde{A}_{12B}^{\mathrm{Re}}(\cdots) + i\widetilde{A}_{12B}^{\mathrm{Im}}(\cdots)\right]}{\int d(b\cdot P) \left[\cos\left(x(b\cdot P)\right) + i\sin\left(x(b\cdot P)\right)\right] \left[\widetilde{A}_{2B}^{\mathrm{Re}}(\cdots) + i\widetilde{A}_{2B}^{\mathrm{Im}}(\cdots)\right]} \tag{5.4}$$

$$= -m_N \frac{\int d(b\cdot P) \left[\cos\left(x(b\cdot P)\right)\widetilde{A}_{12B}^{\mathrm{Re}}(\cdots) - \sin\left(x(b\cdot P)\right)\widetilde{A}_{12B}^{\mathrm{Im}}(\cdots)\right]}{\int d(b\cdot P) \left[\cos\left(x(b\cdot P)\right)\widetilde{A}_{2B}^{\mathrm{Re}}(\cdots) - \sin\left(x(b\cdot P)\right)\widetilde{A}_{2B}^{\mathrm{Im}}(\cdots)\right]} \tag{5.5}$$

$$= -m_N \frac{\left[\widetilde{A}_{12B}^{[x]\mathrm{Re}}(b^2, x, \hat{\zeta}, \eta v{\cdot}P) - \widetilde{A}_{12B}^{[x]\mathrm{Im}}(b^2, x, \hat{\zeta}, \eta v{\cdot}P)\right]}{\left[\widetilde{A}_{2B}^{[x]\mathrm{Re}}(b^2, x, \hat{\zeta}, \eta v{\cdot}P) - \widetilde{A}_{2B}^{[x]\mathrm{Im}}(b^2, x, \hat{\zeta}, \eta v{\cdot}P)\right]} \tag{5.6}$$

Expanding the Fourier phase factor into its trigonometric components isolates the real and imaginary contributions of the amplitudes to the Sivers shift. Because the integration over $d(b\cdot P)$ is performed over a symmetric domain, the non-vanishing terms of the integral are dictated by the parity of the integrands with respect to $b\cdot P$. The trigonometric weights $\cos\left(x(b\cdot P)\right)$ and $\sin\left(x(b\cdot P)\right)$ are even and odd, respectively.

This harmonic symmetry filters the invariant amplitudes based on their behavior under Hermitian conjugation. By Eqs. (4.25) and (4.28), the real components $\widetilde{A}_{2B}^{\mathrm{Re}}$ and $\widetilde{A}_{12B}^{\mathrm{Re}}$ are even functions of $b\cdot P$, while the imaginary components $\widetilde{A}_{2B}^{\mathrm{Im}}$ and $\widetilde{A}_{12B}^{\mathrm{Im}}$ are odd. (Parity in $b$ is parity in $b\cdot P$ here, because $b^2$ is unchanged under $b\to -b$, $v\cdot b$ is tied to $b\cdot P$ by Eq. (4.32), and $R(\hat{\zeta}^2)$ is real.) Only the symmetric products therefore survive the integration, i.e., the even real amplitudes coupled with the cosine term and the odd imaginary amplitudes coupled with the sine term. Here, $\widetilde{A}_{iB}^{[x]\mathrm{Re}}$ and $\widetilde{A}_{iB}^{[x]\mathrm{Im}}$ are the Fourier-transformed amplitudes. Note that, since $\widetilde{A}_{iB}^{\mathrm{Im}}$ is odd in $(P\cdot b)$, its Fourier transform gives an imaginary

solution, i. e. $\int \frac{d(b\cdot P)}{2\pi} e^{ix(b\cdot P)} \widetilde{A}_{iB}^{\mathrm{Im}} = i\widetilde{A}_{iB}^{[x]\mathrm{Im}}$, this results in

$$\int \frac{d(b\cdot P)}{2\pi} e^{ix(b\cdot P)} \left[\widetilde{A}_{iB}^{\mathrm{Re}} + i\widetilde{A}_{iB}^{\mathrm{Im}}\right] = \left[\widetilde{A}_{iB}^{[x]\mathrm{Re}} + i^2 \widetilde{A}_{iB}^{[x]\mathrm{Im}}\right] \tag{5.7}$$

$$= \left[\widetilde{A}_{iB}^{[x]\mathrm{Re}} - \widetilde{A}_{iB}^{[x]\mathrm{Im}}\right] \tag{5.8}$$

for both unpolarized and Sivers TMD amplitudes.

Next, we extract these parity components from the discrete lattice data; Sections 5.5 and 5.6 describe how we build correlation functions to separate them.

### 5.2 The Collins-Soper Parameter ($\hat{\zeta}$)

We defined the Collins-Soper parameter $\hat{\zeta}$ in the previous Chapter. 4 (Eq. (4.12)):

$$\hat{\zeta} \equiv \zeta/2m_N = \frac{v\cdot P}{\sqrt{|v^2|}\sqrt{P^2}}. \tag{5.9}$$

On the lattice, the staple direction $v$ must be space-like ($v^2 < 0$), so the physical TMD limit is recovered only by extrapolating $\hat{\zeta} \to \infty$. In practice, since $v$ is a purely spatial vector on the lattice ($v_0 = 0$), increasing $\hat{\zeta}$ requires simulating the nucleon at progressively higher spatial momenta $\boldsymbol{P}$. Tracking the evolution of the Sivers shift across multiple $\hat{\zeta}$ values is therefore necessary for phenomenological predictions.

### 5.3 Kinematic Constraints to scan full $b\cdot P$

We now move to the frame in which the lattice calculation is performed. To perform the Fourier transform over $b\cdot P$, we require a lattice geometry with both non-zero $b\cdot P$ and zero $b\cdot P$. However, the Lorentz-invariant arguments of the amplitudes are not independent. The phenomenological frame ($b^+ = 0, \boldsymbol{P}_{\mathrm{T}} = 0$) enforces the constraint of Eq. (4.32). Expanding the dot products in the case where the momentum is directed along the 1-axis ($P = (E, P_1, 0, 0)$), we find:

$$b_1|v|\left[\frac{P_1}{m_N}\left(\hat{\zeta} - \sqrt{\hat{\zeta}^2+1}\right) + \hat{\zeta}\frac{m_N}{P_1}\right] = \vec{v}_T \cdot \vec{b}_T \ . \tag{5.10}$$

To satisfy this equation over a continuous range of $b\cdot P$ without coupling the transverse separation $\vec{b}_T$ directly to the longitudinal staple geometry, we orient the transverse separation vectors orthogonally.

By choosing $\vec{v}_T = (0, v_3)$ and $\vec{b}_T = (b_2, 0)$, the right-hand side identically vanishes ($\vec{v}_T \cdot \vec{b}_T = 0$). This is not the only choice, but it is convenient for a lattice calculation because both $b$ and $v$ can remain in 2-dimensional lattice planes. In general, $b$ and $v$ may have to point in arbitrary directions in three-dimensional space.

### 5.3.1 Explicit Calculation for Hadron Momentum $P_1 = -1$

Setting the right-hand side of Eq. (5.10) to zero implies that the bracketed term

$$\left[\frac{P_1}{m_N}\left(\hat{\zeta} - \sqrt{\hat{\zeta}^2 + 1}\right) + \hat{\zeta}\frac{m_N}{P_1}\right]$$

must vanish. This fixes the required Collins-Soper parameter, up to a sign, for a given lattice momentum. Using the lattice ensemble parameters from our lattice size of $\{\hat{L}^3 \times \hat{T}\} = \{32^3 \times 96\}$ and lattice unit of $a = 0.11403$ fm, the nucleon mass and the lowest momentum mode are:

$$m_N = \frac{0.6228(28)}{0.11403 \text{ fm}} = \frac{0.6228(28) \times 197.32698 \text{ MeV}}{0.11403} = 1077.8(48) \text{ MeV} , \tag{5.11}$$

$$P_1 = -\frac{2\pi}{L} = -\frac{2\pi}{32 \times 0.11403 \text{ fm}} = -339.8 \text{ MeV} . \tag{5.12}$$

Substituting these values into the constraint equation yields:

$$\begin{aligned} \Longrightarrow \left[\frac{P_1}{m_N}\left(\hat{\zeta} - \sqrt{\hat{\zeta}^2 + 1}\right) + \hat{\zeta}\frac{m_N}{P_1}\right] &= 0 \\ \Longrightarrow \hat{\zeta} &= \sqrt{\frac{P_1^2}{m_N^2\left(\frac{m_N^2}{P_1^2} + 2\right)}} \\ \hat{\zeta} &= \pm 0.09077(75) \\ \boxed{\hat{\zeta} = \pm 0.090772} &. \end{aligned} \tag{5.13}$$

With $\hat{\zeta}$ determined, we can derive the specific orientation of the space-like staple vector $v$. From the definition of $\hat{\zeta}$:

$$\hat{\zeta} = \frac{-v_1 P_1}{|v| m_N} = \frac{-v_1 P_1}{\left(\sqrt{v_1^2 + 0^2 + v_3^2}\right) m_N} . \tag{5.14}$$

Solving for the longitudinal staple component $v_1$ in terms of the transverse component $v_3$:

$$\Longrightarrow \boxed{v_1 = \pm 0.301 \; v_3} . \tag{5.15}$$

### 5.4 Lattice Parameter Space and Off-Axis Links

This geometric calculation is repeated over a range of nucleon momenta to increase $\hat{\zeta}$ systematically.

The resulting constraint parameters are summarized in Table 2.

| $\boldsymbol{P} \cdot a\hat{L}/(2\pi)$ | $\hat{\zeta}$ | $v_1$ |
|---|---|---|
| $(-1, 0, 0)$ | $\pm 0.090772$ | $v_1 = \pm 0.301\ v_3$ |
| $(-2, 0, 0)$ | $\pm 0.294367$ | $v_1 = \pm 0.531\ v_3$ |
| $(-3, 0, 0)$ | $\pm 0.539684$ | $v_1 = \pm 0.689\ v_3$ |
| $(-4, 0, 0)$ | $\pm 0.785756$ | $v_1 = \pm 0.786\ v_3$ |

Table 2: Resulting Collins-Soper parameters ($\hat{\zeta}$) and the necessary geometric constraint on the staple vector components to ensure the full $b \cdot P$ range is scanned.

| $\boldsymbol{b}/a$ | $\eta \boldsymbol{v}/a$ | $\boldsymbol{P} \cdot a\hat{L}/(2\pi)$ |
|---|---|---|
| $(b_1, b_2, 0)$ | $\eta \cdot (v_1, 0, v_3)$, $\eta = 0, \cdots, 10$ | $(n, 0, 0)$, $n = -1, -2, -3, -4$ |

Table 3: Configuration parameters for the Lattice QCD non-local matrix element extraction. Our lattice size is $\{\hat{L}^3 \times \hat{T}\} = \{32^3 \times 96\}$ and the lattice spacing is $a = 0.11403$ fm.

A consequence of Table 2 is that the staple vector $v$ cannot align cleanly with the Cartesian axes of the Euclidean grid. Instead, it must point in an off-axis direction within the $v_1 - v_3$ plane. For example, to achieve $v_1 = 0.301\ v_3$ at $|P| = 1$, the Wilson line must trace an oblique, step-wise path across the discrete lattice sites (e.g., traversing roughly three lattice spacings in the 1-direction for every ten in the 3-direction). Generating these off-axis Wilson lines without injecting excessive discretization noise guides our strategies for link smearing and gauge path selection. Figure 8 plots the resulting staple directions in the $v_1 - v_3$ plane for all four Collins-Soper values in Table 2.

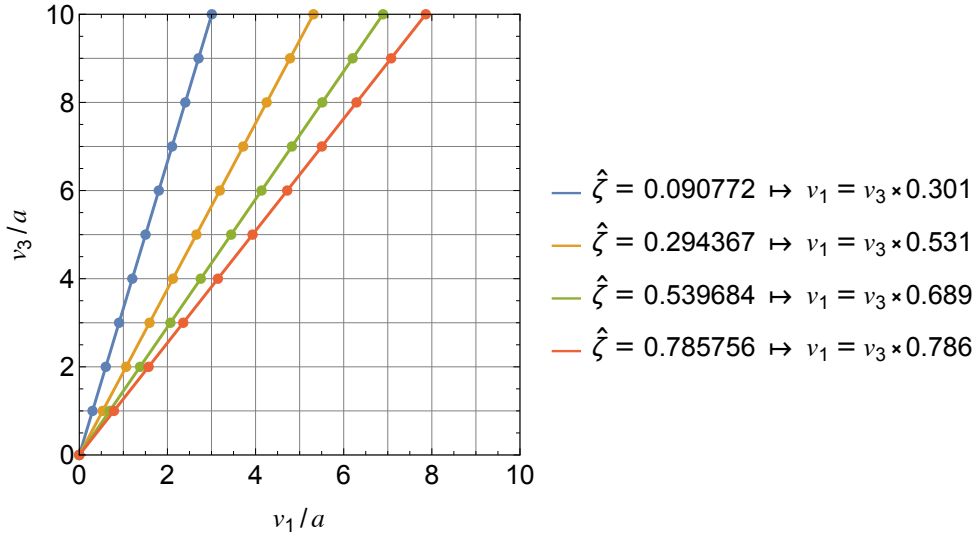



Figure 8: Directions of the spatial staple vectors $\boldsymbol{v}$ mapped in the $v_1 - v_3$ plane. Each vector corresponds to the requisite geometry to satisfy the $b \cdot P \neq 0$ constraint for the four sequential Collins-Soper parameters $\hat{\zeta}$.

Section 5.4.1 identifies the integer lattice sites that bound each off-axis target, and Section 5.4.2 averages over degenerate paths and interpolates to the target.

### 5.4.1 Off-Axis $v$ and Spatial Interpolation

To evaluate the correlation function as a function of the staple extent, we scale the direction vector $\boldsymbol{v}$ by an increment $\eta$, and $\boldsymbol{v}$ is a unit vector. The physical length of the staple is thus governed by the coordinate $\eta\boldsymbol{v}/a$. Because $\boldsymbol{v}$ points in a continuous, off-axis direction, scaling it by integer values of $\eta$ generally yields target coordinates that fall somewhere between the discrete Cartesian lattice sites.

To access the continuous off-axis coordinate $\eta\boldsymbol{v}/a$, we employ a spatial bounding and interpolation strategy. Rather than constructing a single, potentially biased step-wise path (a "staircase") that roughly approximates the vector, we compute the observable at multiple distinct integer lattice sites surrounding the exact ideal target coordinate. We reach these bounding coordinates through specific permutations of unit links. A path sequence denoted $\{1, 3, 1\}$ is a single link in the $\hat{x_1}$ direction, followed by one in the $\hat{x_3}$ direction and a final link in the $\hat{x_1}$ direction, which lands at the integer coordinate $(2, 0, 1)$. By varying these step sequences, we gather data at three to four adjacent lattice sites (e.g., on either side of the target along the $x^3$ slice).

Once these bounding matrix elements are computed, we perform a spatial interpolation inside the Jackknife resampling loop to extract the value at the continuous target coordinate $\eta\boldsymbol{v}/a$. The interpolation reduces discretization artifacts and the staircase errors of off-axis Wilson lines; the next section describes the procedure. Figures 9–12 and Tables 4–7 show the ideal continuous trajectories for each nucleon momentum, together with the corresponding discrete bounding paths and link sequences.

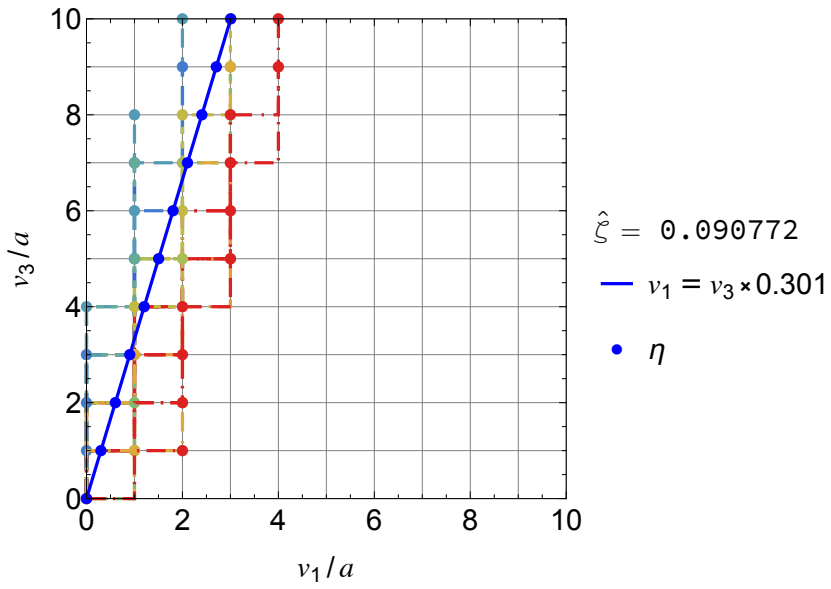



Figure 9: Ideal continuous trajectory of the off-axis staple vector (solid blue) for $P_1 = -1$. Circular markers indicate target continuous coordinates for sequential values of $\eta$. Dashed lines illustrate the discrete, step-wise permutations of gauge links used to reach the surrounding bounding integer lattice sites.

| $\eta\,\boldsymbol{v}/a$ | Gauge-link paths for interpolation ($\hat{\zeta} = 0.090772$) |
|---|---|
| 0·(0.301,0,1) | {{}} |
| 1·(0.301,0,1) | {{3}, {1, 3}, {3, 1}, {1, 3, 1}} |
| 2·(0.301,0,1) | {{3, 3}, {3, 1, 3}, {1, 3, 1, 3}, {3, 1, 3, 1}} |
| 3·(0.301,0,1) | {{3, 3, 3}, {3, 1, 3, 3}, {3, 3, 1, 3}, {3, 1, 3, 1, 3}} |
| 4·(0.301,0,1) | {{3, 3, 3, 3}, {3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3}} |
| 5·(0.301,0,1) | {{3, 3, 1, 3, 3, 3}, {3, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 1, 3, 3}, {3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3}} |
| 6·(0.301,0,1) | {{3, 3, 3, 1, 3, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 3, 1, 3}} |
| 7·(0.301,0,1) | {{3, 3, 3, 1, 3, 3, 3, 3}, {3, 3, 3, 3, 1, 3, 3, 3}, {3, 3, 1, 3, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 1, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 1, 3, 3}} |
| 8·(0.301,0,1) | {{3, 3, 3, 3, 1, 3, 3, 3, 3}, {3, 3, 3, 1, 3, 3, 1, 3, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3}} |
| 9·(0.301,0,1) | {{3, 3, 3, 1, 3, 3, 3, 1, 3, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 3, 3, 1, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 3}} |
| 10·(0.301,0,1) | {{3, 3, 3, 1, 3, 3, 3, 3, 1, 3, 3, 3}, {3, 3, 1, 3, 3, 3, 1, 3, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3}} |

Table 4: Discrete gauge link configurations defining the interpolation bounds for $P_1 = -1$. Empty brackets {} denote a zero-length origin link, while numerical arrays (e.g., $\{1, 3, 1\}$) give the sequence of directional steps.

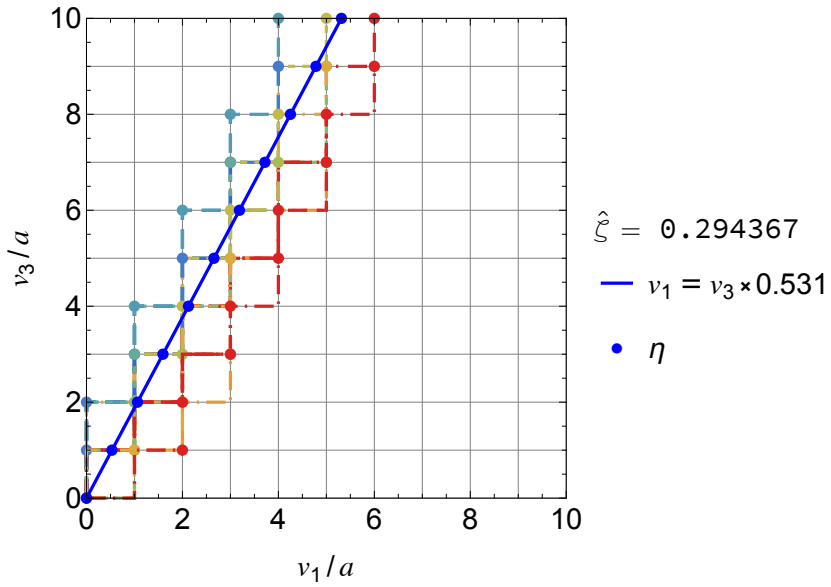



Figure 10: Ideal continuous trajectory and corresponding discrete bounding paths for the off-axis staple vector at $P_1 = -2$.

| $\eta \cdot \mathbf{v} / a$ | Gauge-link paths for interpolation ($\hat{\zeta} = 0.294367$) |
|---|---|
| 0·(0.531,0,1) | {{}} |
| 1·(0.531,0,1) | {{3}, {1, 3}, {3, 1}, {1, 3, 1}} |
| 2·(0.531,0,1) | {{3, 3}, {3, 1, 3}, {1, 3, 1, 3}, {3, 1, 3, 1}} |
| 3·(0.531,0,1) | {{3, 1, 3, 3}, {3, 3, 1, 3}, {3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1}} |
| 4·(0.531,0,1) | {{3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3}} |
| 5·(0.531,0,1) | {{3, 3, 1, 3, 1, 3, 3}, {3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 6·(0.531,0,1) | {{3, 3, 1, 3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3}} |
| 7·(0.531,0,1) | {{3, 3, 1, 3, 1, 3, 3, 1, 3, 3}, {3, 3, 1, 3, 3, 1, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3}} |
| 8·(0.531,0,1) | {{3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3}} |
| 9·(0.531,0,1) | {{3, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3}, {3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}} |
| 10·(0.531,0,1) | {{3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3}, {3, 1, 3, 3, 1, 3, 3, 1, 3, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}} |

Table 5: Discrete gauge link configurations defining the interpolation bounds for $P_1 = -2$.

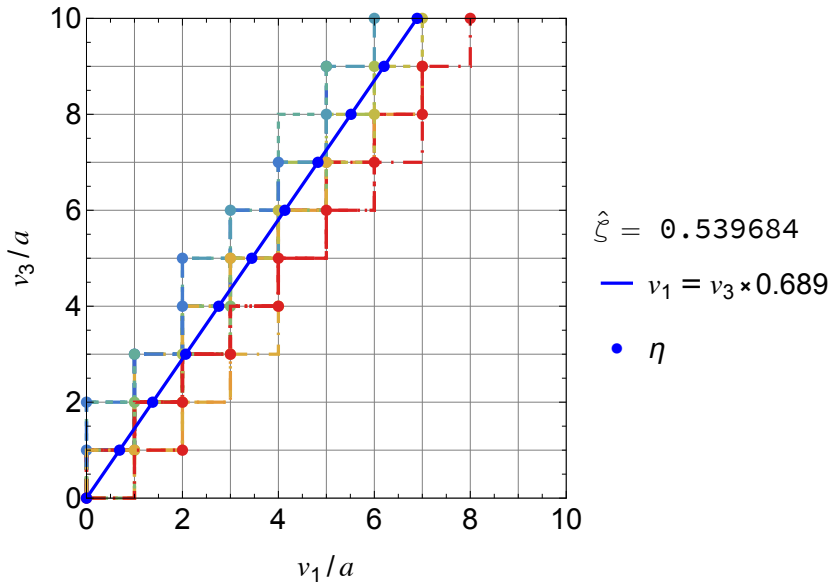



Figure 11: Ideal continuous trajectory and corresponding discrete bounding paths for the off-axis staple vector at $P_1 = -3$.

| $\eta\, \boldsymbol{\nu} / a$ | Gauge-link paths for interpolation ($\hat{\zeta} = 0.539684$) |
|---|---|
| 0·(0.689,0,1) | {{}} |
| 1·(0.689,0,1) | {{3}, {1, 3}, {3, 1}, {1, 3, 1}} |
| 2·(0.689,0,1) | {{3, 3}, {3, 1, 3}, {1, 3, 1, 3}, {3, 1, 3, 1}} |
| 3·(0.689,0,1) | {{3, 1, 3, 3}, {3, 3, 1, 3}, {3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1}} |
| 4·(0.689,0,1) | {{3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1}} |
| 5·(0.689,0,1) | {{3, 3, 1, 3, 1, 3, 3}, {3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 6·(0.689,0,1) | {{3, 1, 3, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 7·(0.689,0,1) | {{3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 8·(0.689,0,1) | {{3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 9·(0.689,0,1) | {{3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3}, {3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 10·(0.689,0,1) | {{3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |

Table 6: Discrete gauge link configurations defining the interpolation bounds for $P_1 = -3$.

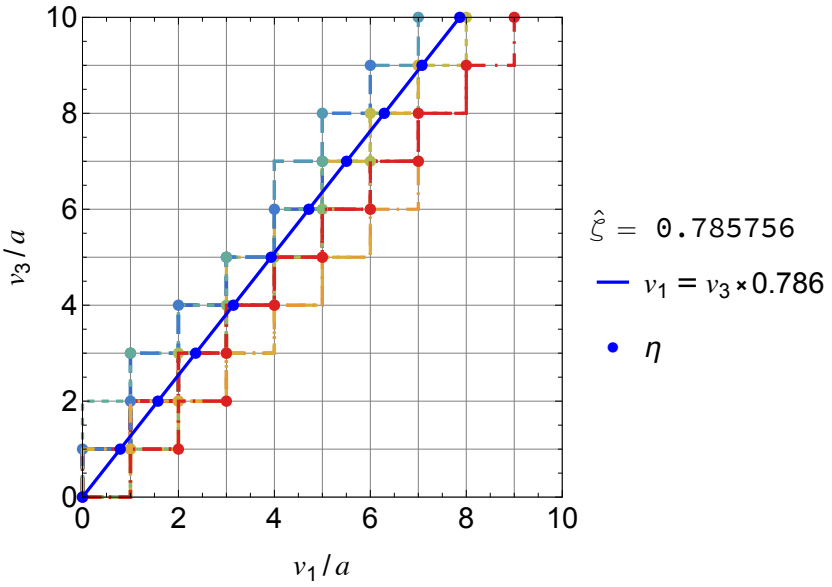



Figure 12: Ideal continuous trajectory and corresponding discrete bounding paths for the off-axis staple vector at $P_1 = -4$.

| $\eta \cdot \mathbf{v} / a$ | Gauge-link paths for interpolation ($\hat{\zeta} = 0.785756$) |
|---|---|
| 0·(0.786,0,1) | {{}} |
| 1·(0.786,0,1) | {{3}, {1, 3}, {3, 1}, {1, 3, 1}} |
| 2·(0.786,0,1) | {{3, 1, 3}, {1, 3, 1, 3}, {3, 1, 3, 1}, {1, 3, 1, 3, 1}} |
| 3·(0.786,0,1) | {{3, 1, 3, 3}, {3, 3, 1, 3}, {3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1}} |
| 4·(0.786,0,1) | {{3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1}} |
| 5·(0.786,0,1) | {{3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 3, 1, 3, 1, 3},<br>{3, 1, 3, 1, 3, 1, 3, 1, 3}, {1, 3, 1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1}} |
| 6·(0.786,0,1) | {{3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3},<br>{1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1}} |
| 7·(0.786,0,1) | {{3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3},<br>{1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1}} |
| 8·(0.786,0,1) | {{3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 9·(0.786,0,1) | {{3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3},<br>{3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |
| 10·(0.786,0,1) | {{3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3},<br>{3, 1, 3, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1, 3, 1, 3, 1, 3}, {3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3}} |

Table 7: Discrete gauge link configurations defining the interpolation bounds for $P_1 = -4$.

### 5.4.2 Spatial Interpolation of Off-Axis Coordinates

Before interpolating to the off-axis target $\eta \boldsymbol{v}/a$, we must account for path degeneracy. Because the lattice gauge links are restricted to the Cartesian axes, there are multiple step-wise permutations (different side-link paths) that terminate at the same physical lattice coordinate. While these paths are geometrically equivalent at their endpoints, they correspond to different local gauge field configurations and thus exhibit distinct discretization artifacts and gauge noise. As illustrated in Figure 13, to improve the symmetry, we average the correlation functions evaluated over all degenerate paths that share the same endpoint. Path averaging significantly improves the signal-to-noise ratio before any interpolation. It symmetry-improves the latticized TMD operators.

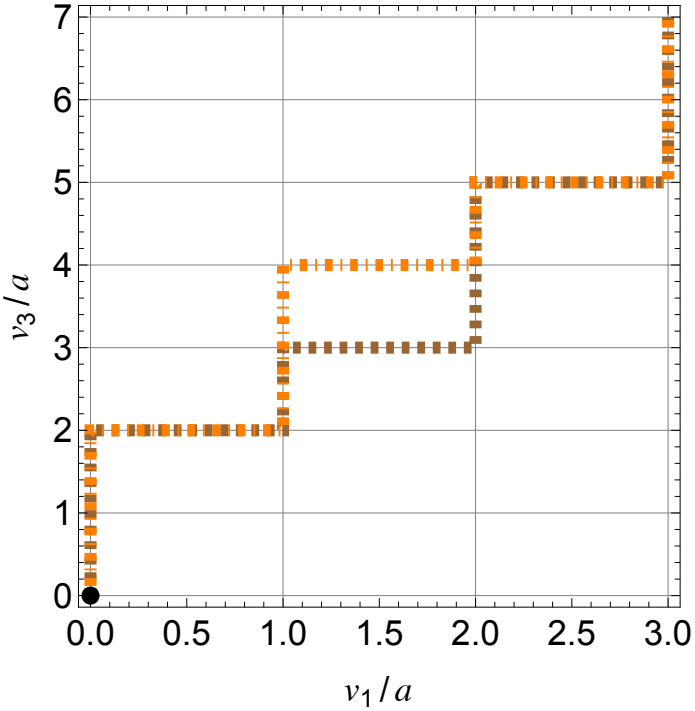



Figure 13: Illustration of path degeneracy. Two distinct step-wise gauge link permutations (e.g., solid and dashed lines) terminate at the identical discrete lattice coordinate. We average the correlators from these equivalent paths to suppress local gauge noise before interpolation.

Once we obtain the symmetry-improved (path-averaged) correlators at the discrete bounding sites, we perform a 1-dimensional interpolation. For a fixed transverse staple extent $v_3$, the kinematic constraint dictates an ideal continuous longitudinal component, $v_1^{\text{constraint}}$. The surrounding discrete lattice sites provide correlators at actual simulated coordinates, $v_1^{\text{lattice}}$. We define the fractional deviation from

the ideal path as:

$$\Delta v_1 = \frac{v_1^{\text{constraint}} - v_1^{\text{lattice}}}{a} \ . \tag{5.16}$$

By calculating the observable at three or more bounding lattice sites, we generate a set of data points as a function of this deviation $\Delta v_1$. Within each Jackknife resampling bin (to preserve the covariance and statistical error), we perform a quadratic interpolation over these points. We then extract the physical amplitude on the continuous constraint trajectory by evaluating the interpolating polynomial exactly at $\Delta v_1 = 0$.

The core Mathematica logic executed inside the data analysis pipeline to perform this quadratic interpolation is shown in the snippet below:

Listing 2: Core interpolation logic for evaluating the constrained correlator.

```
(* Pair fractional deviation (x) with path-averaged data (y) *)
plotforinterpolation = Transpose[{v1dataforInterpolation,
                                  dataforInterpolation}];

(* Construct quadratic interpolation over bounding points *)
fninterpol = Interpolation[plotforinterpolation,
                           InterpolationOrder -> 2];

(* Evaluate exactly on the constraint trajectory (deviation = 0) *)
interpol0 = {{0, fninterpol[0]}};
```

A representative example of this interpolation procedure is shown in Figure 14 for a specific staple extent ($\eta = 7$) and nucleon momentum ($P_1 = -1$). The red data points represent the path-averaged lattice evaluations at the discrete bounding sites, plotted against their deviation from the ideal trajectory. The solid gray curve denotes the quadratic interpolation, and the blue point highlights the final extracted physical value at $\Delta v_1 = 0$. We apply this procedure to all Dirac structures ($\Gamma$), separately for the real ($b \cdot P$-even) and imaginary ($b \cdot P$-odd) parts of the amplitudes.

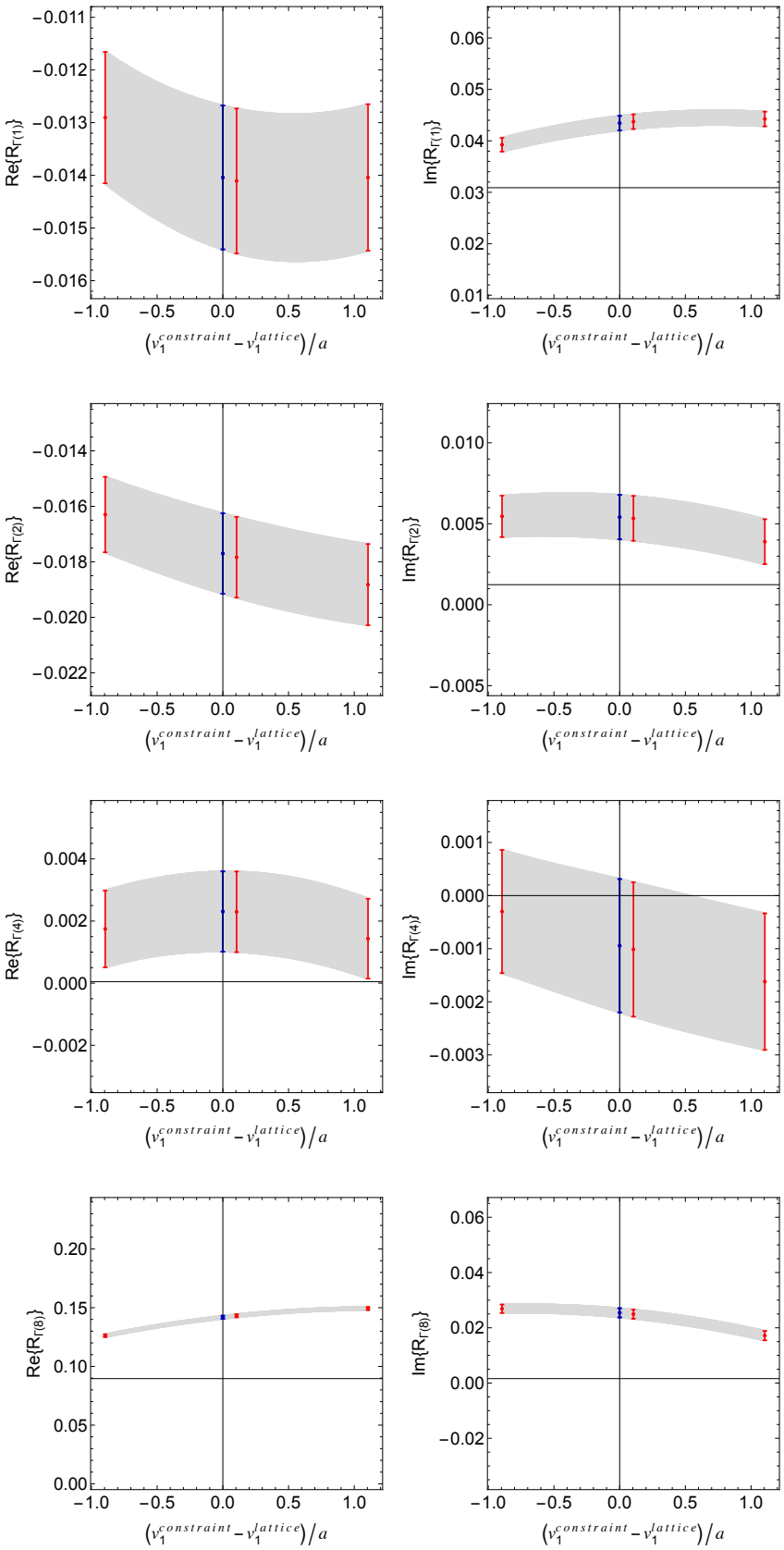



Figure 14: Quadratic interpolation of the real and imaginary parts of the correlation function for $\boldsymbol{b} = \{1, 2, 0\}$, $\eta = 7$, and $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = (-1, 0, 0)$. Red markers indicate the path-averaged lattice data points as a function of their deviation from the ideal constraint trajectory, $\Delta v_1 = (v_1^{\text{constraint}} - v_1^{\text{lattice}})/a$. The gray band indicates the quadratic fit ('InterpolationOrder $\rightarrow$ 2'), and the central blue marker represents the final extracted value at $\Delta v_1 = 0$, lying exactly on the required off-axis spatial trajectory.

### 5.5 Quark Separation Vectors ($b$) and Amplitude Parity

In addition to the staple vector $\boldsymbol{v}$, the correlation function depends on the spatial separation $\boldsymbol{b}$ between the initial and final quark fields. Recall that the real and imaginary parts of the invariant amplitudes ($\widetilde{A}_{iB}^{\mathrm{Re}}$ and $\widetilde{A}_{iB}^{\mathrm{Im}}$) are even and odd functions of the invariant dot product $b \cdot P$, respectively.

Because the nucleon momentum in our lattice setup is fixed along the $\hat{x}_1$-axis ($P = P_1\hat{x}_1$), the longitudinal separation is directly proportional to the $\hat{x}_1$-component of the separation vector: $b \cdot P = -P_1 b_1$. Therefore, to isolate the real and imaginary parts of the amplitudes, we must evaluate the correlator at both positive and negative values of $b_1$. By taking the symmetric (even) and antisymmetric (odd) linear combinations of the correlators evaluated at $+b_1$ and $-b_1$, we project out the real and imaginary amplitudes required for the $x$-dependent Fourier transform. The numerical Fourier transform over $b \cdot P$ requires a wide and dense grid of the separation vector $\boldsymbol{b}$. Figures 15-16 show the parameter space of our calculations. The left panel shows the discrete Cartesian grid in the $b_1 - b_2$ plane. The right panel shows how this grid maps onto the phenomenologically relevant invariant Variable Space spanned by $(b \cdot P)\hat{L}/(2\pi)$ and the absolute separation $|b|/a$; the mapping shifts with the target Collins-Soper parameter $\hat{\zeta}$. Where, $\hat{L}$ is the spatial lattice length $\hat{L} = 32$, and $a = 0.11403$ fm is the lattice spacing.

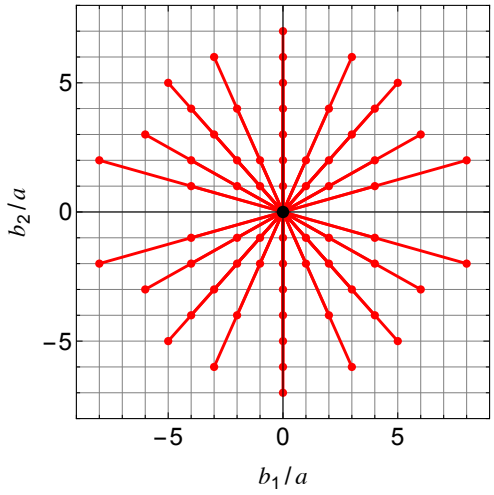



Figure 15: The sampled operator separation vectors mapped in the spatial $b_1 - b_2$ plane (where $b_1 \equiv b_L$ and $b_2 \equiv b_T$).

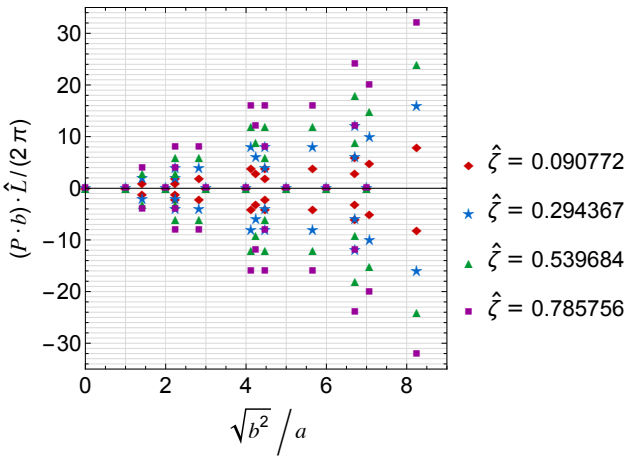



Figure 16: Mapping of the spatial lattice vectors into the physical invariant Variable Space $(b \cdot P/(2\pi), |b|/a)$. The trajectories diverge based on the selected target $\hat{\zeta}$.

Section 5.5.1 lists the gauge link paths used to reach each $\boldsymbol{b}$, and Section 5.5.2 averages over degen-

erate paths.

### 5.5.1 Gauge Link Construction for $b$

Constructing the direct Wilson line connecting the quark fields at 0 and $\boldsymbol{b}$ requires generating step-wise paths across the lattice. We categorize these paths into purely transverse separations ($b_1 = 0$) and off-axis separations with a longitudinal component ($b_1 \neq 0$). For purely transverse separations, where $b \cdot P = 0$, the gauge links lie entirely within the transverse plane. Table 8 lists the link sequences used to reach these target coordinates. An empty bracket $\{\}$ denotes the local origin, while arrays such as $\{2, 2\}$ indicate a sequence of two consecutive links in the $\hat{x_2}$-direction.

| $\{+b_1, b_3, b_3\}/a$ | Gauge-link paths | $\{-b_1, b_3, b_3\}/a$ | Gauge-link paths |
|---|---|---|---|
| {0, 0, 0} | {{}} | {0, 0, 0} | {{}} |
| {0, 1, 0} | {{2}} | {0, 1, 0} | {{2}} |
| {0, 2, 0} | {{2, 2}} | {0, 2, 0} | {{2, 2}} |
| {0, 3, 0} | {{2, 2, 2}} | {0, 3, 0} | {{2, 2, 2}} |
| {0, 4, 0} | {{2, 2, 2, 2}} | {0, 4, 0} | {{2, 2, 2, 2}} |
| {0, 5, 0} | {{2, 2, 2, 2, 2}} | {0, 5, 0} | {{2, 2, 2, 2, 2}} |
| {0, 6, 0} | {{2, 2, 2, 2, 2, 2}} | {0, 6, 0} | {{2, 2, 2, 2, 2, 2}} |
| {0, 7, 0} | {{2, 2, 2, 2, 2, 2, 2}} | {0, 7, 0} | {{2, 2, 2, 2, 2, 2, 2}} |
| {0, -1, 0} | {{-2}} | {0, -1, 0} | {{-2}} |
| {0, -2, 0} | {{-2, -2}} | {0, -2, 0} | {{-2, -2}} |
| {0, -3, 0} | {{-2, -2, -2}} | {0, -3, 0} | {{-2, -2, -2}} |
| {0, -4, 0} | {{-2, -2, -2, -2}} | {0, -4, 0} | {{-2, -2, -2, -2}} |
| {0, -5, 0} | {{-2, -2, -2, -2, -2}} | {0, -5, 0} | {{-2, -2, -2, -2, -2}} |
| {0, -6, 0} | {{-2, -2, -2, -2, -2, -2}} | {0, -6, 0} | {{-2, -2, -2, -2, -2, -2}} |
| {0, -7, 0} | {{-2, -2, -2, -2, -2, -2, -2}} | {0, -7, 0} | {{-2, -2, -2, -2, -2, -2, -2}} |

Table 8: Discrete gauge link configurations for purely transverse quark separations ($\boldsymbol{b} = \{0, b_2, 0\}$). The numerical arrays show the sequential lattice steps (e.g., '2' represents a shift in the $x_2$ direction). Our lattice size is $\{\hat{L}^3 \times \hat{T}\} = \{32^3 \times 96\}$ and the lattice spacing is $a = 0.11403$ fm.

To access finite $b \cdot P$, we construct off-axis paths with simultaneous steps in the longitudinal ($\hat{x_1}$) and transverse ($\hat{x_2}$) directions. To preserve the symmetry required for isolating parity states, every path that reaches a $+b_1$ coordinate has an exact mirror path reaching $-b_1$. Tables 9-10 show these mirrored sequences side by side. Sequences such as $\{2, 1\}$ indicate an alternating "staircase" step: one unit in the $\hat{x_2}$-direction followed by one unit in the $\hat{x_1}$-direction.

| $\{+b_1,\ b_3,\ b_3\}\ /\ a$ | Gauge-link paths | $\{-b_1,\ b_3,\ b_3\}\ /\ a$ | Gauge-link paths |
|---|---|---|---|
| {1, 1, 0} | {{2, 1}, {1, 2}} | {-1, 1, 0} | {{2, -1}, {-1, 2}} |
| {2, 2, 0} | {{2, 1, 2, 1}, {1, 2, 1, 2}} | {-2, 2, 0} | {{2, -1, 2, -1}, {-1, 2, -1, 2}} |
| {3, 3, 0} | {{2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2}} | {-3, 3, 0} | {{2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2}} |
| {4, 4, 0} | {{2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2}} | {-4, 4, 0} | {{2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2}} |
| {5, 5, 0} | {{2, 1, 2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2, 1, 2}} | {-5, 5, 0} | {{2, -1, 2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2, -1, 2}} |
| {1, 2, 0} | {{2, 1, 2}} | {-1, 2, 0} | {{2, -1, 2}} |
| {2, 4, 0} | {{2, 1, 2, 2, 1, 2}} | {-2, 4, 0} | {{2, -1, 2, 2, -1, 2}} |
| {3, 6, 0} | {{2, 1, 2, 2, 1, 2, 2, 1, 2}} | {-3, 6, 0} | {{2, -1, 2, 2, -1, 2, 2, -1, 2}} |
| {2, 1, 0} | {{1, 2, 1}} | {-2, 1, 0} | {{-1, 2, -1}} |
| {4, 2, 0} | {{1, 2, 1, 1, 2, 1}} | {-4, 2, 0} | {{-1, 2, -1, -1, 2, -1}} |
| {6, 3, 0} | {{1, 2, 1, 1, 2, 1, 1, 2, 1}} | {-6, 3, 0} | {{-1, 2, -1, -1, 2, -1, -1, 2, -1}} |
| {4, 1, 0} | {{1, 1, 2, 1, 1}} | {-4, 1, 0} | {{-1, -1, 2, -1, -1}} |
| {8, 2, 0} | {{1, 1, 2, 1, 1, 1, 1, 2, 1, 1}} | {-8, 2, 0} | {{-1, -1, 2, -1, -1, -1, -1, 2, -1, -1}} |
| {1, -1, 0} | {{-2, 1}, {1, -2}} | {-1, -1, 0} | {{-2, -1}, {-1, -2}} |
| {2, -2, 0} | {{-2, 1, -2, 1}, {1, -2, 1, -2}} | {-2, -2, 0} | {{-2, -1, -2, -1}, {-1, -2, -1, -2}} |
| {3, -3, 0} | {{-2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2}} | {-3, -3, 0} | {{-2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2}} |
| {4, -4, 0} | {{-2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2}} | {-4, -4, 0} | {{-2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2}} |
| {5, -5, 0} | {{-2, 1, -2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2, 1, -2}} | {-5, -5, 0} | {{-2, -1, -2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2, -1, -2}} |
| {1, -2, 0} | {{-2, 1, -2}} | {-1, -2, 0} | {{-2, -1, -2}} |
| {2, -4, 0} | {{-2, 1, -2, -2, 1, -2}} | {-2, -4, 0} | {{-2, -1, -2, -2, -1, -2}} |
| {3, -6, 0} | {{-2, 1, -2, -2, 1, -2, -2, 1, -2}} | {-3, -6, 0} | {{-2, -1, -2, -2, -1, -2, -2, -1, -2}} |
| {2, -1, 0} | {{1, -2, 1}} | {-2, -1, 0} | {{-1, -2, -1}} |
| {4, -2, 0} | {{1, -2, 1, 1, -2, 1}} | {-4, -2, 0} | {{-1, -2, -1, -1, -2, -1}} |
| {6, -3, 0} | {{1, -2, 1, 1, -2, 1, 1, -2, 1}} | {-6, -3, 0} | {{-1, -2, -1, -1, -2, -1, -1, -2, -1}} |
| {4, -1, 0} | {{1, 1, -2, 1, 1}} | {-4, -1, 0} | {{-1, -1, -2, -1, -1}} |
| {8, -2, 0} | {{1, 1, -2, 1, 1, 1, 1, -2, 1, 1}} | {-8, -2, 0} | {{-1, -1, -2, -1, -1, -1, -1, -2, -1, -1}} |

Table 9: Gauge link sequences for $+b_1$ (positive longitudinal separation). Symmetric sets of step-wise gauge link paths required to access finite $\pm b \cdot P$ and isolate the real and imaginary components of the invariant amplitudes. Our lattice size is $\{\hat{L}^3 \times \hat{T}\} = \{32^3 \times 96\}$ and the lattice spacing is $a = 0.11403$ fm.

| $\{+b_1, b_3, b_3\}/a$ | Gauge-link paths | $\{-b_1, b_3, b_3\}/a$ | Gauge-link paths |
|---|---|---|---|
| {-1, 1, 0} | {{2, -1}, {-1, 2}} | {1, 1, 0} | {{2, 1}, {1, 2}} |
| {-2, 2, 0} | {{2, -1, 2, -1}, {-1, 2, -1, 2}} | {2, 2, 0} | {{2, 1, 2, 1}, {1, 2, 1, 2}} |
| {-3, 3, 0} | {{2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2}} | {3, 3, 0} | {{2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2}} |
| {-4, 4, 0} | {{2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2}} | {4, 4, 0} | {{2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2}} |
| {-5, 5, 0} | {{2, -1, 2, -1, 2, -1, 2, -1, 2, -1}, {-1, 2, -1, 2, -1, 2, -1, 2, -1, 2}} | {5, 5, 0} | {{2, 1, 2, 1, 2, 1, 2, 1, 2, 1}, {1, 2, 1, 2, 1, 2, 1, 2, 1, 2}} |
| {-1, 2, 0} | {{2, -1, 2}} | {1, 2, 0} | {{2, 1, 2}} |
| {-2, 4, 0} | {{2, -1, 2, 2, -1, 2}} | {2, 4, 0} | {{2, 1, 2, 2, 1, 2}} |
| {-3, 6, 0} | {{2, -1, 2, 2, -1, 2, 2, -1, 2}} | {3, 6, 0} | {{2, 1, 2, 2, 1, 2, 2, 1, 2}} |
| {-2, 1, 0} | {{-1, 2, -1}} | {2, 1, 0} | {{1, 2, 1}} |
| {-4, 2, 0} | {{-1, 2, -1, -1, 2, -1}} | {4, 2, 0} | {{1, 2, 1, 1, 2, 1}} |
| {-6, 3, 0} | {{-1, 2, -1, -1, 2, -1, -1, 2, -1}} | {6, 3, 0} | {{1, 2, 1, 1, 2, 1, 1, 2, 1}} |
| {-4, 1, 0} | {{-1, -1, 2, -1, -1}} | {4, 1, 0} | {{1, 1, 2, 1, 1}} |
| {-8, 2, 0} | {{-1, -1, 2, -1, -1, -1, -1, 2, -1, -1}} | {8, 2, 0} | {{1, 1, 2, 1, 1, 1, 1, 2, 1, 1}} |
| {-1, -1, 0} | {{-2, -1}, {-1, -2}} | {1, -1, 0} | {{-2, 1}, {1, -2}} |
| {-2, -2, 0} | {{-2, -1, -2, -1}, {-1, -2, -1, -2}} | {2, -2, 0} | {{-2, 1, -2, 1}, {1, -2, 1, -2}} |
| {-3, -3, 0} | {{-2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2}} | {3, -3, 0} | {{-2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2}} |
| {-4, -4, 0} | {{-2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2}} | {4, -4, 0} | {{-2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2}} |
| {-5, -5, 0} | {{-2, -1, -2, -1, -2, -1, -2, -1, -2, -1}, {-1, -2, -1, -2, -1, -2, -1, -2, -1, -2}} | {5, -5, 0} | {{-2, 1, -2, 1, -2, 1, -2, 1, -2, 1}, {1, -2, 1, -2, 1, -2, 1, -2, 1, -2}} |
| {-1, -2, 0} | {{-2, -1, -2}} | {1, -2, 0} | {{-2, 1, -2}} |
| {-2, -4, 0} | {{-2, -1, -2, -2, -1, -2}} | {2, -4, 0} | {{-2, 1, -2, -2, 1, -2}} |
| {-3, -6, 0} | {{-2, -1, -2, -2, -1, -2, -2, -1, -2}} | {3, -6, 0} | {{-2, 1, -2, -2, 1, -2, -2, 1, -2}} |
| {-2, -1, 0} | {{-1, -2, -1}} | {2, -1, 0} | {{1, -2, 1}} |
| {-4, -2, 0} | {{-1, -2, -1, -1, -2, -1}} | {4, -2, 0} | {{1, -2, 1, 1, -2, 1}} |
| {-6, -3, 0} | {{-1, -2, -1, -1, -2, -1, -1, -2, -1}} | {6, -3, 0} | {{1, -2, 1, 1, -2, 1, 1, -2, 1}} |
| {-4, -1, 0} | {{-1, -1, -2, -1, -1}} | {4, -1, 0} | {{1, 1, -2, 1, 1}} |
| {-8, -2, 0} | {{-1, -1, -2, -1, -1, -1, -1, -2, -1, -1}} | {8, -2, 0} | {{1, 1, -2, 1, 1, 1, 1, -2, 1, 1}} |

Table 10: Mirrored gauge link sequences for $-b_1$ (negative longitudinal separation).Symmetric sets of step-wise gauge link paths required to access finite $\pm b \cdot P$ and isolate the real and imaginary components of the invariant amplitudes. Our lattice size is $\{\hat{L}^3 \times \hat{T}\} = \{32^3 \times 96\}$ and the lattice spacing is $a = 0.11403$ fm.

### 5.5.2 Symmetry Improving Operation (Path Averaging) for $\boldsymbol{b}$

Several quark separation vectors $\boldsymbol{b}$ have the same path degeneracy as the staple direction vector $\boldsymbol{v}$ (Section 5.4.2). We treat it the same way: we average the correlation function over all equivalent step-wise paths that share the same origin and endpoint, as illustrated for a single coordinate in Figure 17.

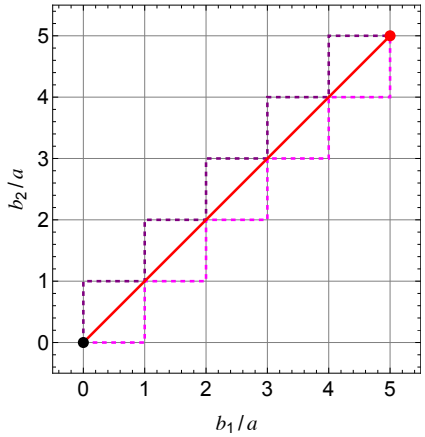



Figure 17: Illustration of path degeneracy for the quark separation vector $\boldsymbol{b}$. Distinct step-wise permutations (solid and dashed lines) terminating at the same Cartesian coordinate are averaged to suppress local gauge noise.

Applying this symmetry-improving operation (path-averaging) strategy across the entire target grid yields the full network of smoothed off-axis trajectories, visualized in Figure 18.

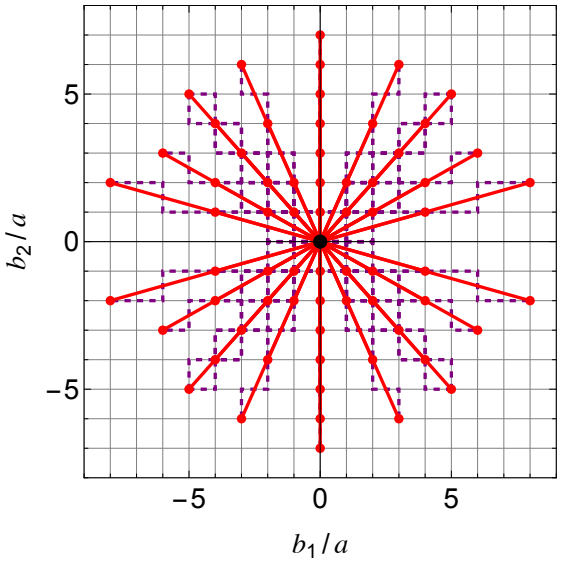



Figure 18: The full network of path-averaged $\boldsymbol{b}$ vectors in the $b_1/a - b_2/a$ plane. Solid red lines represent the ideal vectors, while dashed lines illustrate the underlying degenerate staircase paths calculated and averaged on the discrete lattice.

From this path-averaged dataset, we obtain the parity-separated amplitudes used in the final Fourier

transform.

### 5.6 Solving for Real and Imaginary Amplitudes $\widetilde{A}_{iB}^{\mathbf{Re}}$ and $\widetilde{A}_{iB}^{\mathbf{Im}}$ from correlator $\widetilde{\Phi}^{[\gamma^\mu]}_{\mathbf{unsubtr.}}$

To extract the physical Sivers shift from the lattice, we must isolate the specific invariant amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ from the raw correlator data. We start from the parametrization of the unsubtracted correlator in Eq. (4.24) [2]. While this parametrization describes a continuous gauge link extending to infinity, we adopt an equivalent representation for lattice calculations in which the structures depend explicitly on the finite staple extent $\eta$. This form remains well defined even in the $v \cdot P = 0$ limit. We obtain it by making the substitution $v \to \eta v$ and extracting the scaling factors $(v{\cdot}P)^{-n}$ as discussed in detail in Chapter 4:

$$\begin{aligned}\frac{1}{2}\widetilde{\Phi}^{[\gamma^\mu]}_{\text{unsubtr.}} = \frac{1}{2}R_{\gamma^\mu}E_N = P^\mu\,\tilde{a}_2 - im_N^2 b^\mu\,\tilde{a}_3 - im_N\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha S_\beta\,\tilde{a}_{12} \\ + m_N^2\eta v^\mu\,\tilde{b}_1 + m_N\epsilon^{\mu\nu\alpha\beta}P_\nu\eta v_\alpha S_\beta\,\tilde{b}_7 - im_N^3\epsilon^{\mu\nu\alpha\beta}b_\nu\eta v_\alpha S_\beta\,\tilde{b}_8 \\ - m_N^3(b{\cdot}S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha\eta v_\beta\,\tilde{b}_9 - im_N^3(v{\cdot}S)\epsilon^{\mu\nu\alpha\beta}P_\nu b_\alpha\eta^2 v_\beta\tilde{b}_{10}\end{aligned} \tag{5.17}$$

To evaluate Eq. (4.24), we project it onto the kinematics chosen for our lattice setup. We choose a frame where the nucleon moves in the negative $\hat{1}$-direction, the spin is polarized along the $\hat{3}$-axis, and the separation and staple vectors lie in the transverse/longitudinal planes. We set $P^\mu = (E_N, -|\vec{P}^1|, 0, 0)$, $b^\mu = (0, b_1, b_2, 0)$, $v^\mu = \pm n'(0, v_1, 0, v_3)$, and $S^\mu = (0, 0, 0, 1)$. Expanding the sum over Lorentz indices for the temporal component ($\mu = 0$) and canceling the terms that vanish (such as $b^0 = 0$, $v^0 = 0$, and

$b{\cdot}S = 0$), we find

$$\begin{aligned}
\frac{1}{2}\widetilde{\Phi}^{[\gamma^0]}_{\text{unsubtr.}} &= P^0\,\widetilde{A}_2 - \cancelto{0}{im_N^2 b^0\,\widetilde{A}_3} + im_N\underbrace{\epsilon^{0123}}_{1}|\vec{P}^1|b_2S_3\,\widetilde{A}_{12} \\
&+ \cancelto{0}{\frac{m_N^2}{(v{\cdot}P)}v^0\,\widetilde{B}_1} + \cancelto{0}{\frac{m_N}{v{\cdot}P}\epsilon^{0113}P_1v_1S_3\,\widetilde{B}_7} + \cancelto{0}{\frac{m_N}{v{\cdot}P}\epsilon^{0133}P_1v_3S_3\,\widetilde{B}_7} \\
&- \frac{im_N^3}{v{\cdot}P}\underbrace{\epsilon^{0213}}_{-1}b_2v_1S_3\,\widetilde{B}_8 - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{0113}b_1v_1S_3\,\widetilde{B}_8} - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{0133}b_1v_3S_3\,\widetilde{B}_8} - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{0233}b_2v_3S_3\,\widetilde{B}_8} \\
&+ \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)}\underbrace{\epsilon^{0123}}_{1}|\vec{P}_1|b_2v_3\,\widetilde{B}_9 - \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)\epsilon^{0121}P_1b_2v_1\,\widetilde{B}_9} - \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)\epsilon^{0113}P_1b_1v_3\,\widetilde{B}_9} \\
&- \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)\epsilon^{0111}P_1b_1v_1\,\widetilde{B}_9} + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\underbrace{\epsilon^{0123}}_{1}|\vec{P}_1|b_2v_3\widetilde{B}_{10} - \cancelto{0}{\frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\epsilon^{0121}P_1b_2v_1\widetilde{B}_{10}} \\
&- \cancelto{0}{\frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\epsilon^{0111}P_1b_1v_1\widetilde{B}_{10}} - \cancelto{0}{\frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\epsilon^{0113}P_1b_1v_3\widetilde{B}_{10}}
\end{aligned} \tag{5.18}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^0]}_{\text{unsubtr.}} = P^0\,\widetilde{A}_2 + im_N|\vec{P}^1|b_2S_3\,\widetilde{A}_{12} + \frac{im_N^3}{v{\cdot}P}b_2v_1S_3\,\widetilde{B}_8 + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)|P_1|b_2v_3\widetilde{B}_{10}} \tag{5.19}$$

Repeating this tensor contraction for the longitudinal spatial component ($\mu = 1$), taking care to track the signs of the Levi-Civita symbol, yields:

$$\begin{aligned}
\frac{1}{2}\widetilde{\Phi}^{[\gamma^1]}_{\text{unsubtr.}} &= -|\vec{P}^1|\,\widetilde{A}_2 - im_N^2b^1\,\widetilde{A}_3 - im_N\underbrace{\epsilon^{1023}}_{-1}P_0b_2S_3\,\widetilde{A}_{12} \\
&+ \frac{m_N^2}{(v{\cdot}P)}v^1\,\widetilde{B}_1 + \cancelto{0}{\frac{m_N}{v{\cdot}P}\epsilon^{1013}P_0v_1S_3\,\widetilde{B}_7} + \cancelto{0}{\frac{m_N}{v{\cdot}P}\epsilon^{1033}P_0v_3S_3\,\widetilde{B}_7} \\
&- \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{1213}b_2v_1S_3\,\widetilde{B}_8} - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{1233}b_2v_3S_3\,\widetilde{B}_8} - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{1133}b_1v_3S_3\,\widetilde{B}_8} - \cancelto{0}{\frac{im_N^3}{v{\cdot}P}\epsilon^{1113}b_1v_1S_3\,\widetilde{B}_8} \\
&- \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)}\underbrace{\epsilon^{1023}}_{1}P_0b_2v_3\,\widetilde{B}_9 - \cancelto{0}{\frac{m_N^3}{v{\cdot}P}(b{\cdot}S)\epsilon^{1021}P_0b_2v_1\,\widetilde{B}_9} - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\underbrace{\epsilon^{1023}}_{-1}P_0b_2v_3\widetilde{B}_{10} \\
&- \cancelto{0}{\frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\epsilon^{1021}P_0b_2v_1\widetilde{B}_{10}}
\end{aligned} \tag{5.20}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^1]}_{\text{unsubtr.}} = -|\vec{P}^1|\,\widetilde{A}_2 + \frac{m_N^2}{(v{\cdot}P)}v^1\,\widetilde{B}_1 + im_NP_0b_2S_3\,\widetilde{A}_{12} + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)P_0b_2v_3\widetilde{B}_{10} - im_N^2b^1\,\widetilde{A}_3} \tag{5.21}$$

Evaluating the transverse component ($\mu = 2$) isolates terms that depend on $P_0$ and the longitudinal separation $b_1$:

$$\begin{aligned}\frac{1}{2}\widetilde{\Phi}^{[\gamma^2]}_{\text{unsubtr.}} &= -im_N^2 b^2\,\widetilde{A}_3 - im_N \underbrace{\epsilon^{2013}}_{1} P_0 b_1 S_3\,\widetilde{A}_{12} \\ &\quad + \frac{m_N}{v{\cdot}P}\underbrace{\epsilon^{2013}}_{1} P_0 v_1 S_3\,\widetilde{B}_7 \\ &\quad - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\underbrace{\epsilon^{2013}}_{1} P_0 b_1 v_3 \widetilde{B}_{10}\end{aligned} \tag{5.22}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^2]}_{\text{unsubtr.}} = -im_N^2 b^2\,\widetilde{A}_3 - im_N P_0 b_1 S_3\,\widetilde{A}_{12} + \frac{m_N}{v{\cdot}P} P_0 v_1 S_3\,\widetilde{B}_7 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_1 v_3 \widetilde{B}_{10}} \tag{5.23}$$

Finally, the orthogonal transverse component ($\mu = 3$) simplifies because of the alignment of the spin vector:

$$\begin{aligned}\frac{1}{2}\widetilde{\Phi}^{[\gamma^3]}_{\text{unsubtr.}} &= \frac{m_N^2}{(v{\cdot}P)} v^3\,\widetilde{B}_1 \\ &\quad - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)\underbrace{\epsilon^{3021}}_{1} P_0 b_2 v_1 \widetilde{B}_{10}\end{aligned} \tag{5.24}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^3]}_{\text{unsubtr.}} = \frac{m_N^2}{(v{\cdot}P)} v^3\,\widetilde{B}_1 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_2 v_1 \widetilde{B}_{10}} \tag{5.25}$$

Collecting these boxed results gives the full system of equations mapping the infinite-link continuous amplitudes to the projected correlator components:

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^0]}_{\text{unsubtr.}} = P^0\,\widetilde{A}_2 + im_N|\vec{P}^1| b_2 S_3\,\widetilde{A}_{12} + \frac{im_N^3}{v{\cdot}P} b_2 v_1 S_3\,\widetilde{B}_8 + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)|P_1| b_2 v_3 \widetilde{B}_{10}} \tag{5.26}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^1]}_{\text{unsubtr.}} = -|\vec{P}^1|\,\widetilde{A}_2 - im_N^2 b^1\,\widetilde{A}_3 + \frac{m_N^2}{(v{\cdot}P)} v^1\,\widetilde{B}_1 + im_N P_0 b_2 S_3\,\widetilde{A}_{12} + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_2 v_3 \widetilde{B}_{10}} \tag{5.27}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^2]}_{\text{unsubtr.}} = -im_N^2 b^2\,\widetilde{A}_3 - im_N P_0 b_1 S_3\,\widetilde{A}_{12} + \frac{m_N}{v{\cdot}P} P_0 v_1 S_3\,\widetilde{B}_7 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_1 v_3 \widetilde{B}_{10}} \tag{5.28}$$

$$\boxed{\frac{1}{2}\widetilde{\Phi}^{[\gamma^3]}_{\text{unsubtr.}} = \frac{m_N^2}{(v{\cdot}P)} v^3\,\widetilde{B}_1 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_2 v_1 \widetilde{B}_{10}} \tag{5.29}$$

We structure this system as a column vector to streamline the upcoming algebraic inversions:

$$\begin{pmatrix} \widetilde{\Phi}^{[\gamma^0]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^1]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^2]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^3]}_{\text{unsubtr.}} \end{pmatrix} = 2 \begin{pmatrix} P^0\,\widetilde{A}_2 + im_N|\vec{P}^1| b_2 S_3\,\widetilde{A}_{12} + \frac{im_N^3}{v{\cdot}P} b_2 v_1 S_3\,\widetilde{B}_8 + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S)|P_1| b_2 v_3 \widetilde{B}_{10} \\ -|\vec{P}^1|\,\widetilde{A}_2 - im_N^2 b^1\,\widetilde{A}_3 + \frac{m_N^2}{(v{\cdot}P)} v^1\,\widetilde{B}_1 + im_N P_0 b_2 S_3\,\widetilde{A}_{12} + \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_2 v_3 \widetilde{B}_{10} \\ -im_N^2 b^2\,\widetilde{A}_3 - im_N P_0 b_1 S_3\,\widetilde{A}_{12} + \frac{m_N}{v{\cdot}P} P_0 v_1 S_3\,\widetilde{B}_7 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_1 v_3 \widetilde{B}_{10} \\ \frac{m_N^2}{(v{\cdot}P)} v^3\,\widetilde{B}_1 - \frac{im_N^3}{(v{\cdot}P)^2}(v{\cdot}S) P_0 b_2 v_1 \widetilde{B}_{10} \end{pmatrix} \tag{5.30}$$

To match standard contravariant Lorentz notation, we raise the spatial indices using the Minkowski metric $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$. This introduces a relative minus sign for the spatial components (e.g., $P_1 = -P^1$, $b_2 = -b^2$):

$$\begin{pmatrix} \widetilde{\Phi}^{[\gamma^0]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^1]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^2]}_{\text{unsubtr.}} \\ \widetilde{\Phi}^{[\gamma^3]}_{\text{unsubtr.}} \end{pmatrix} = 2 \begin{pmatrix} P^0\,\widetilde{A}_2 - im_N|\vec{P}^1|b^2S^3\,\widetilde{A}_{12} - \frac{im_N^3}{v\cdot P}b^2v^1S^3\,\widetilde{B}_8 - \frac{im_N^3}{(v\cdot P)^2}(v\cdot S)|\vec{P}^1|b^2v^3\widetilde{B}_{10} \\ -|\vec{P}^1|\,\widetilde{A}_2 - im_N^2b^1\,\widetilde{A}_3 + \frac{m_N^2}{(v\cdot P)}v^1\,\widetilde{B}_1 + im_NP^0b^2S^3\,\widetilde{A}_{12} + \frac{im_N^3}{(v\cdot P)^2}(v\cdot S)P^0b^2v^3\widetilde{B}_{10} \\ -im_N^2b^2\,\widetilde{A}_3 - im_NP^0b^1S^3\,\widetilde{A}_{12} + \frac{m_N}{v\cdot P}P^0v^1S^3\,\widetilde{B}_7 - \frac{im_N^3}{(v\cdot P)^2}(v\cdot S)P^0b^1v^3\widetilde{B}_{10} \\ \frac{m_N^2}{(v\cdot P)}v^3\,\widetilde{B}_1 - \frac{im_N^3}{(v\cdot P)^2}(v\cdot S)P^0b^2v^1\widetilde{B}_{10} \end{pmatrix} \tag{5.31}$$

To connect these continuous amplitudes to the discretized finite-link observables simulated on the lattice, we use the mapping of Eq. (4.31) between the infinite-link amplitudes ($\widetilde{A}_i$, $\widetilde{B}_i$) and the finite-link amplitudes ($\tilde{a}_i$, $\tilde{b}_i$). The finite-link amplitudes are complex-valued functions, which we decompose into their real and imaginary components:

$$\tilde{a}_i = \tilde{a}_i^{\text{Re}} + i\tilde{a}_i^{\text{Im}} \tag{5.32}$$

$$\tilde{b}_i = \tilde{b}_i^{\text{Re}} + i\tilde{b}_i^{\text{Im}} \tag{5.33}$$

The lattice evaluation also requires converting the continuous Minkowski gamma matrices ($\gamma_M$) into the Euclidean conventions ($\gamma_E$) used by the lattice simulation software:

- Correct Euclidean $\longleftrightarrow$ Minkowski $\gamma$-matrices notation:

$$\boxed{\gamma_E^1 = +i\gamma_M^1} \tag{5.34}$$

$$\boxed{\gamma_E^2 = -i\gamma_M^2} \tag{5.35}$$

$$\boxed{\gamma_E^3 = +i\gamma_M^3} \tag{5.36}$$

$$\boxed{\gamma_E^4 = \gamma_M^0} \tag{5.37}$$

$$\boxed{\gamma_E^5} = \gamma_E^1\gamma_E^2\gamma_E^3\gamma_E^4 = \boxed{-\gamma_M^5} = -i\gamma_M^0\gamma_M^1\gamma_M^2\gamma_M^3 \tag{5.38}$$

Substituting the Euclidean gamma structures and mapping $P^0 \to E_N$, we rewrite the column vector entirely in terms of the lattice ratio observables $R_{\Gamma(i)}$ and the finite-link amplitudes. The index $i \in \{1, 2, 4, 8\}$ labels each Dirac structure by its bit-encoded position (CHROMA & QLUA convention

discussed in Appendix A.) in our operator basis, matching the correspondence $\Gamma(1) = [+i\gamma^1]$, $\Gamma(2) = [-i\gamma^2]$, $\Gamma(4) = [+i\gamma^3]$, and $\Gamma(8) = [\gamma^0]$ used below:

$$\begin{pmatrix} R_{\Gamma(1)} \\ R_{\Gamma(2)} \\ R_{\Gamma(4)} \\ R_{\Gamma(8)} \end{pmatrix} = \begin{pmatrix} R_{[+i\gamma^1]} \\ R_{[-i\gamma^2]} \\ R_{[+i\gamma^3]} \\ R_{[\gamma^0]} \end{pmatrix} = 2 \begin{pmatrix} -i\frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2 + \frac{m_N^2}{E_N} b^1\,\tilde{a}_3 + i\frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1 - m_N b^2\,\tilde{a}_{12} + m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10} \\ -\frac{m_N^2}{E_N} b^2\,\tilde{a}_3 - m_N b^1 \tilde{a}_{12} - i m_N \eta v^1 \tilde{b}_7 + m_N^3 b^1 \eta^2 (v^3)^2 \tilde{b}_{10} \\ i\frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1 - m_N^3 b^2\eta^2 v^1 v^3 \tilde{b}_{10} \\ \tilde{a}_2 - i m_N \frac{|\vec{P}^1|}{E_N} b^2 \tilde{a}_{12} - \frac{i m_N^3}{E_N} b^2 \eta v^1 \tilde{b}_8 + \frac{i m_N^3}{E_N} |\vec{P}^1| b^2 \eta^2 (v^3)^2 \tilde{b}_{10} \end{pmatrix} \tag{5.39}$$

To separate the real and imaginary amplitude contributions analytically, we split this matrix equation into two sets by flipping the longitudinal separation component $b^1$. For a strictly positive longitudinal separation ($+b^1$), writing out the real and imaginary parts of the correlator ratios gives:

$$\begin{pmatrix} \mathrm{Re}\{R_{\Gamma(1)}\}^{[+b^1]} \\ \mathrm{Im}\{R_{\Gamma(1)}\}^{[+b^1]} \\ \mathrm{Re}\{R_{\Gamma(2)}\}^{[+b^1]} \\ \mathrm{Im}\{R_{\Gamma(2)}\}^{[+b^1]} \\ \mathrm{Re}\{R_{\Gamma(4)}\}^{[+b^1]} \\ \mathrm{Im}\{R_{\Gamma(4)}\}^{[+b^1]} \\ \mathrm{Re}\{R_{\Gamma(8)}\}^{[+b^1]} \\ \mathrm{Im}\{R_{\Gamma(8)}\}^{[+b^1]} \end{pmatrix} = 2 \begin{pmatrix} \frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Re}} - m_N b^2\,\tilde{a}_{12}^{\mathrm{Re}} + m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} + \frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Im}} - \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Im}} \\ -\frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Re}} + \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Re}} + \frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Im}} - m_N b^2\,\tilde{a}_{12}^{\mathrm{Im}} + m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -\frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Re}} - m_N b^1 \tilde{a}_{12}^{\mathrm{Re}} + m_N^3 b^1 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} + m_N \eta v^1 \tilde{b}_7^{\mathrm{Im}} \\ -m_N \eta v^1 \tilde{b}_7^{\mathrm{Re}} - \frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Im}} - m_N b^1 \tilde{a}_{12}^{\mathrm{Im}} + m_N^3 b^1 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -m_N^3 b^2\eta^2 v^1 v^3 \tilde{b}_{10}^{\mathrm{Re}} - \frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Im}} \\ \frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Re}} - m_N^3 b^2\eta^2 v^1 v^3 \tilde{b}_{10}^{\mathrm{Im}} \\ \tilde{a}_2^{\mathrm{Re}} + m_N \frac{|\vec{P}^1|}{E_N} b^2 \tilde{a}_{12}^{\mathrm{Im}} + \frac{m_N^3}{E_N} b^2 \eta v^1 \tilde{b}_8^{\mathrm{Im}} - \frac{m_N^3}{E_N} |\vec{P}^1| b^2 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -m_N \frac{|\vec{P}^1|}{E_N} b^2 \tilde{a}_{12}^{\mathrm{Re}} - \frac{m_N^3}{E_N} b^2 \eta v^1 \tilde{b}_8^{\mathrm{Re}} + \frac{m_N^3}{E_N} |\vec{P}^1| b^2 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} + \tilde{a}_2^{\mathrm{Im}} \end{pmatrix} \tag{5.40}$$

Evaluating the system for the negated separation ($-b^1$) flips the signs of selected terms according to the parity constraints of Eq. (4.28). By construction, the imaginary parts of the amplitudes are odd functions of $b^1$ ($\widetilde{A}_i^{\mathrm{Im}} = \widetilde{A}_i^{(\text{b-odd})} = \widetilde{A}_i^{(b^1\text{-odd})}$ and $\widetilde{B}_i^{\mathrm{Im}} = \widetilde{B}_i^{(\text{b-odd})} = \widetilde{B}_i^{(b^1\text{-odd})}$). Applying these parity reversals yields the conjugate matrix:

$$\begin{pmatrix} \mathrm{Re}\{R_{\Gamma(1)}\}^{[-b^1]} \\ \mathrm{Im}\{R_{\Gamma(1)}\}^{[-b^1]} \\ \mathrm{Re}\{R_{\Gamma(2)}\}^{[-b^1]} \\ \mathrm{Im}\{R_{\Gamma(2)}\}^{[-b^1]} \\ \mathrm{Re}\{R_{\Gamma(4)}\}^{[-b^1]} \\ \mathrm{Im}\{R_{\Gamma(4)}\}^{[-b^1]} \\ \mathrm{Re}\{R_{\Gamma(8)}\}^{[-b^1]} \\ \mathrm{Im}\{R_{\Gamma(8)}\}^{[-b^1]} \end{pmatrix} = 2 \begin{pmatrix} -\frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Re}} - m_N b^2\,\tilde{a}_{12}^{\mathrm{Re}} + m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} - \frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Im}} + \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Im}} \\ -\frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Re}} + \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Re}} + \frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Im}} + m_N b^2\,\tilde{a}_{12}^{\mathrm{Im}} - m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -\frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Re}} + m_N b^1 \tilde{a}_{12}^{\mathrm{Re}} - m_N^3 b^1 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} - m_N \eta v^1 \tilde{b}_7^{\mathrm{Im}} \\ -m_N \eta v^1 \tilde{b}_7^{\mathrm{Re}} + \frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Im}} - m_N b^1 \tilde{a}_{12}^{\mathrm{Im}} + m_N^3 b^1 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -m_N^3 b^2\eta^2 v^1 v^3 \tilde{b}_{10}^{\mathrm{Re}} + \frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Im}} \\ \frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Re}} + m_N^3 b^2\eta^2 v^1 v^3 \tilde{b}_{10}^{\mathrm{Im}} \\ \tilde{a}_2^{\mathrm{Re}} - m_N \frac{|\vec{P}^1|}{E_N} b^2 \tilde{a}_{12}^{\mathrm{Im}} - \frac{m_N^3}{E_N} b^2 \eta v^1 \tilde{b}_8^{\mathrm{Im}} + \frac{m_N^3}{E_N} |\vec{P}^1| b^2 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Im}} \\ -m_N \frac{|\vec{P}^1|}{E_N} b^2 \tilde{a}_{12}^{\mathrm{Re}} - \frac{m_N^3}{E_N} b^2 \eta v^1 \tilde{b}_8^{\mathrm{Re}} + \frac{m_N^3}{E_N} |\vec{P}^1| b^2 \eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}} - \tilde{a}_2^{\mathrm{Im}} \end{pmatrix} \tag{5.41}$$

By taking the symmetric average of the two matrices, $\frac{1}{2}\left(R^{[+b]}+R^{[-b]}\right)$, we project out the $b$-even combination, isolating the real parts of the requisite amplitudes:

$$\begin{pmatrix}\mathrm{Re}\{R_{\Gamma(1)}\}^{[b^1\text{-even}]}\\ \mathrm{Im}\{R_{\Gamma(1)}\}^{[b^1\text{-even}]}\\ \mathrm{Re}\{R_{\Gamma(2)}\}^{[b^1\text{-even}]}\\ \mathrm{Im}\{R_{\Gamma(2)}\}^{[b^1\text{-even}]}\\ \mathrm{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]}\\ \mathrm{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]}\\ \mathrm{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]}\\ \mathrm{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]}\end{pmatrix} = 2\begin{pmatrix} -m_N b^2\,\tilde{a}_{12}^{\mathrm{Re}} + m_N^3 b^2\eta^2 (v^3)^2 \tilde{b}_{10}^{\mathrm{Re}}\\ -\frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Re}} + \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Re}} + \frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Im}}\\ -\frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Re}}\\ -m_N\eta v^1\tilde{b}_7^{\mathrm{Re}} - m_N b^1\tilde{a}_{12}^{\mathrm{Im}} + m_N^3 b^1\eta^2(v^3)^2\tilde{b}_{10}^{\mathrm{Im}}\\ -m_N^3 b^2\eta^2 v^1 v^3\tilde{b}_{10}^{\mathrm{Re}}\\ \frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Re}}\\ \tilde{a}_2^{\mathrm{Re}}\\ -m_N\frac{|\vec{P}^1|}{E_N} b^2\tilde{a}_{12}^{\mathrm{Re}} - \frac{m_N^3}{E_N} b^2\eta v^1\tilde{b}_8^{\mathrm{Re}} + \frac{m_N^3}{E_N}|\vec{P}^1| b^2\eta^2(v^3)^2\tilde{b}_{10}^{\mathrm{Re}}\end{pmatrix} \tag{5.42}$$

Conversely, taking the antisymmetric half-difference of the two matrices, $\frac{1}{2}\left(R^{[+b]}-R^{[-b]}\right)$, projects out the $b$-odd combination, isolating the imaginary parts:

$$\begin{pmatrix}\mathrm{Re}\{R_{\Gamma(1)}\}^{[b^1\text{-odd}]}\\ \mathrm{Im}\{R_{\Gamma(1)}\}^{[b^1\text{-odd}]}\\ \mathrm{Re}\{R_{\Gamma(2)}\}^{[b^1\text{-odd}]}\\ \mathrm{Im}\{R_{\Gamma(2)}\}^{[b^1\text{-odd}]}\\ \mathrm{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]}\\ \mathrm{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]}\\ \mathrm{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]}\\ \mathrm{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]}\end{pmatrix} = 2\begin{pmatrix} \frac{m_N^2}{E_N} b^1\,\tilde{a}_3^{\mathrm{Re}} + \frac{|\vec{P}^1|}{E_N}\,\tilde{a}_2^{\mathrm{Im}} - \frac{m_N^2}{E_N}\eta v^1\,\tilde{b}_1^{\mathrm{Im}}\\ -m_N b^2\,\tilde{a}_{12}^{\mathrm{Im}} + m_N^3 b^2\eta^2(v^3)^2\tilde{b}_{10}^{\mathrm{Im}}\\ -m_N b^1\tilde{a}_{12}^{\mathrm{Re}} + m_N^3 b^1\eta^2(v^3)^2\tilde{b}_{10}^{\mathrm{Re}} + m_N\eta v^1\tilde{b}_7^{\mathrm{Im}}\\ -\frac{m_N^2}{E_N} b^2\,\tilde{a}_3^{\mathrm{Im}}\\ -\frac{m_N^2}{E_N}\eta v^3\,\tilde{b}_1^{\mathrm{Im}}\\ -m_N^3 b^2\eta^2 v^1 v^3\tilde{b}_{10}^{\mathrm{Im}}\\ m_N\frac{|\vec{P}^1|}{E_N} b^2\tilde{a}_{12}^{\mathrm{Im}} + \frac{m_N^3}{E_N} b^2\eta v^1\tilde{b}_8^{\mathrm{Im}} - \frac{m_N^3}{E_N}|\vec{P}^1| b^2\eta^2(v^3)^2\tilde{b}_{10}^{\mathrm{Im}}\\ \tilde{a}_2^{\mathrm{Im}}\end{pmatrix} \tag{5.43}$$

To analytically reconstruct the Sivers shift, we solve these projected systems for the amplitudes that drive the Sivers effect: $\tilde{a}_2$, $\tilde{b}_1$, $\tilde{a}_{12}$, and $\tilde{b}_8$. Reading directly from the $b$-even system, we isolate the real components:

$$\tilde{a}_2^{\mathrm{Re}} = \frac{1}{2}\mathrm{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]} \tag{5.44}$$

$$\tilde{b}_1^{\mathrm{Re}} = \frac{E_N}{2m_N^2\eta v^3}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]} \tag{5.45}$$

Substituting the appropriate matrix rows, we decouple $\tilde{a}_{12}$ and $\tilde{b}_8$:

$$\tilde{a}_{12}^{\mathrm{Re}} = \frac{-1}{2m_N b^2}\left[\mathrm{Re}\{R_{\Gamma(1)}\}^{[b^1\text{-even}]} + \frac{v^3}{v^1}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]}\right] \tag{5.46}$$

$$\tilde{b}_8^{\mathrm{Re}} = \frac{-E_N}{2m_N^3 b^2\eta v^1}\left[\mathrm{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]} + \frac{|\vec{P}^1| v^3}{E_N v^1}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]} + 2m_N\frac{|\vec{P}^1|}{E_N} b^2\tilde{a}_{12}^{\mathrm{Re}}\right] \tag{5.47}$$

Applying identical algebraic decoupling to the $b$-odd system, we isolate the corresponding imaginary components:

$$\tilde{a}_2^{\text{Im}} = \frac{1}{2}\text{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]} \tag{5.48}$$

$$\tilde{b}_1^{\text{Im}} = \frac{-E_N}{2m_N^2\eta v^3}\text{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]} \tag{5.49}$$

and the remaining transverse amplitudes:

$$\tilde{a}_{12}^{\text{Im}} = \frac{-1}{2m_N b^2}\left[\text{Im}\{R_{\Gamma(1)}\}^{[b^1\text{-odd}]} + \frac{v^3}{v^1}\text{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]}\right] \tag{5.50}$$

$$\tilde{b}_8^{\text{Im}} = \frac{E_N}{2m_N^3 b^2\eta v^1}\left[\text{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]} - \frac{|\vec{P}^1|v^3}{E_N v^1}\text{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]} - 2m_N\frac{|\vec{P}^1|}{E_N}b^2\tilde{a}_{12}^{\text{Im}}\right] \tag{5.51}$$

Finally, we convert these finite-link evaluations $(\tilde{a}_i, \tilde{b}_i)$ back into the standard continuum-invariant amplitudes $(\widetilde{A}_i, \widetilde{B}_i)$ using the proportionality dictionary:

$$\widetilde{A}_i = \tilde{a}_i \tag{5.52}$$

$$\widetilde{B}_i = (\eta v\cdot P)^n\ \tilde{b}_i \tag{5.53}$$

Recalling that $v\cdot P = v_1 P^1 = v^1|P^1|$ in our chosen frame, each amplitude structure has its own conversion factor. Applying this mapping to the real components yields the final accessible amplitudes on the lattice:

$$\boxed{\widetilde{A}_2^{\text{Re}} = \frac{1}{2}\text{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]}} \tag{5.54}$$

$$\boxed{\widetilde{B}_1^{\text{Re}} = \frac{E_N|P^1|v^1}{2m_N^2 v^3}\text{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]}} \tag{5.55}$$

$$\boxed{\widetilde{A}_{12}^{\text{Re}} = \frac{-1}{2m_N b^2}\left[\text{Re}\{R_{\Gamma(1)}\}^{[b^1\text{-even}]} + \frac{v^3}{v^1}\text{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]}\right]} \tag{5.56}$$

$$\boxed{\widetilde{B}_8^{\text{Re}} = \frac{-E_N|P^1|}{2m_N^3 b^2}\left[\text{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-even}]} + \frac{|\vec{P}^1|v^3}{E_N v^1}\text{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-even}]} + 2m_N\frac{|\vec{P}^1|}{E_N}b^2\widetilde{A}_{12}^{\text{Re}}\right]} \tag{5.57}$$

Applying the same scaling factors to the imaginary counterparts provides the complementary $b$-odd contributions:

$$\boxed{\widetilde{A}_2^{\mathrm{Im}} = \frac{1}{2}\mathrm{Im}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]}} \tag{5.58}$$

$$\boxed{\widetilde{B}_1^{\mathrm{Im}} = \frac{-E_N|P^1|v^1}{2m_N^2 v^3}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]}} \tag{5.59}$$

$$\boxed{\widetilde{A}_{12}^{\mathrm{Im}} = \frac{-1}{2m_N b^2}\left[\mathrm{Im}\{R_{\Gamma(1)}\}^{[b^1\text{-odd}]} + \frac{v^3}{v^1}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]}\right]} \tag{5.60}$$

$$\boxed{\widetilde{B}_8^{\mathrm{Im}} = \frac{E_N|P^1|}{2m_N^3 b^2}\left[\mathrm{Re}\{R_{\Gamma(8)}\}^{[b^1\text{-odd}]} - \frac{|\vec{P}^1|v^3}{E_N v^1}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b^1\text{-odd}]} - 2m_N\frac{|\vec{P}^1|}{E_N}b^2\widetilde{A}_{12}^{\mathrm{Im}}\right]} \tag{5.61}$$

### 5.7 Constructing the unpolarized $\widetilde{A}_{2B}$ and Sivers $\widetilde{A}_{12B}$ Amplitudes

With the invariant amplitudes isolated by parity projection, we now combine them into the amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ of Eq. (4.34), with $R(\hat{\zeta}^2)$ from Eq. (4.32). To apply these expressions in our simulated lattice geometry, we map the contravariant spatial indices to our physical grid parameters. We define the longitudinal spatial separation as $b_L \equiv b^1$ and the transverse spatial separation as $b_T \equiv b^2$. The squared spatial separation then expands to $b^2 = b_L^2 + b_T^2$, and we identify the longitudinal nucleon momentum as $P_L \equiv P^1$.

We begin with the defining linear combination for the unpolarized amplitude, $\widetilde{A}_{2B}$. Splitting this amplitude into its real and imaginary components yields:

$$\widetilde{A}_{2B}^{\mathrm{Re}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_2^{\mathrm{Re}} + R(\hat{\zeta}^2)\widetilde{B}_1^{\mathrm{Re}} \tag{5.62}$$

$$\widetilde{A}_{2B}^{\mathrm{Im}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_2^{\mathrm{Im}} + R(\hat{\zeta}^2)\widetilde{B}_1^{\mathrm{Im}} \tag{5.63}$$

Substituting the parity-projected lattice ratio observables derived in Section 5.6 (and applying our $b_L, P_L$ geometric mapping), we obtain the real part of $\widetilde{A}_{2B}$ as an expression computable directly from our grid coordinates:

$$\widetilde{A}_{2B}^{\mathrm{Re}}(b_L, b_T, \hat{\zeta}) = \frac{1}{2}\mathrm{Re}\{R_{\Gamma(8)}\}^{[b_L\text{-even}]} + R(\hat{\zeta}^2)\frac{E_N|P_L|v^1}{2m_N^2 v^3}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b_L\text{-even}]} \tag{5.64}$$

The same substitution for the imaginary component, now with the $b_L$-odd projected data sets so

that time-reversal parity is preserved, gives:

$$\widetilde{A}_{2B}^{\mathrm{Im}}(b_L, b_T, \hat{\zeta}) = \frac{1}{2}\mathrm{Im}\{R_{\Gamma(8)}\}^{[b_L\text{-odd}]} - R(\hat{\zeta}^2)\frac{E_N|P_L|v^1}{2m_N^2 v^3}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b_L\text{-odd}]} \tag{5.65}$$

The amplitude that drives the Sivers shift, $\widetilde{A}_{12B}$, combines the structures $\widetilde{A}_{12}$ and $\widetilde{B}_8$. Its real and imaginary decompositions are defined as:

$$\widetilde{A}_{12B}^{\mathrm{Re}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_{12}^{\mathrm{Re}} - R(\hat{\zeta}^2)\widetilde{B}_8^{\mathrm{Re}} \tag{5.66}$$

$$\widetilde{A}_{12B}^{\mathrm{Im}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_{12}^{\mathrm{Im}} - R(\hat{\zeta}^2)\widetilde{B}_8^{\mathrm{Im}} \tag{5.67}$$

Because the terms within $\widetilde{A}_{12B}$ are more complex, we first write out the inner constituent $\widetilde{A}_{12}^{\mathrm{Re}}$ in terms of the grid parameters $b_L$ and $b_T$, using the spatial kinematic constraint $b^2 = b_L^2 + b_T^2$ from above. The real part of the leading-twist term becomes:

$$\widetilde{A}_{12}^{\mathrm{Re}} = \frac{-1}{2m_N(b_L^2 + b_T^2)}\left[\mathrm{Re}\{R_{\Gamma(1)}\}^{[b_L\text{-even}]} + \frac{v^3}{v^1}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b_L\text{-even}]}\right] \tag{5.68}$$

Substituting this internal structure alongside the expansion for $\widetilde{B}_8^{\mathrm{Re}}$ gives the full equation for the real part entering the Sivers integration:

$$\begin{aligned}\widetilde{A}_{12B}^{\mathrm{Re}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_{12}^{\mathrm{Re}} - R(\hat{\zeta}^2)\Bigg(\frac{-E_N|P_L|}{2m_N^3(b_L^2 + b_T^2)}\Bigg[&\mathrm{Im}\{R_{\Gamma(8)}\}^{[b_L\text{-even}]} + \frac{|P_L|v^3}{E_N v^1}\mathrm{Re}\{R_{\Gamma(4)}\}^{[b_L\text{-even}]} \\ &+ 2m_N\frac{|P_L|}{E_N}(b_L^2 + b_T^2)\widetilde{A}_{12}^{\mathrm{Re}}\Bigg]\Bigg)\end{aligned} \tag{5.69}$$

Following the same procedure for the imaginary (T-odd) components, we first define the internal $b_L$-odd constituent $\widetilde{A}_{12}^{\mathrm{Im}}$:

$$\widetilde{A}_{12}^{\mathrm{Im}} = \frac{-1}{2m_N(b_L^2 + b_T^2)}\left[\mathrm{Im}\{R_{\Gamma(1)}\}^{[b_L\text{-odd}]} + \frac{v^3}{v^1}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b_L\text{-odd}]}\right] \tag{5.70}$$

We then insert it to obtain the full equation for the imaginary part of the Sivers amplitude:

$$\begin{aligned}\widetilde{A}_{12B}^{\mathrm{Im}}(b_L, b_T, \hat{\zeta}) = \widetilde{A}_{12}^{\mathrm{Im}} - R(\hat{\zeta}^2)\Bigg(\frac{E_N|P_L|}{2m_N^3(b_L^2 + b_T^2)}\Bigg[&\mathrm{Re}\{R_{\Gamma(8)}\}^{[b_L\text{-odd}]} - \frac{|P_L|v^3}{E_N v^1}\mathrm{Im}\{R_{\Gamma(4)}\}^{[b_L\text{-odd}]} \\ &- 2m_N\frac{|P_L|}{E_N}(b_L^2 + b_T^2)\widetilde{A}_{12}^{\mathrm{Im}}\Bigg]\Bigg)\end{aligned} \tag{5.71}$$

### 5.8 Transformation to the Lorentz-Invariant Variable Space

At this stage, the amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ are discrete functions of the Cartesian lattice grid parameters $b_L$ and $b_T$. Extracting the dependence on the longitudinal momentum fraction $x$, however, requires the Fourier transform over the Lorentz-invariant inner product $P \cdot b$. To perform this integration, we first show that our simulated Cartesian lattice coordinates map one-to-one to the invariant Variable Space.

In our selected kinematic frame, the 4-vectors for the nucleon momentum and spatial separation are defined as:

$$P^\mu = (E_N, P_L, 0, 0) \tag{5.72}$$

$$b^\mu = (0, b_L, b_T, 0) \tag{5.73}$$

Evaluating the inner product using the Minkowski metric signature $(+,-,-,-)$, the temporal and transverse orthogonal components vanish, leaving:

$$\begin{aligned} P \cdot b &= P_0 b^0 - P_1 b^1 - P_2 b^2 - P_3 b^3 \\ &= -P_L b_L \end{aligned} \tag{5.74}$$

Inverting this identity expresses the longitudinal Cartesian lattice coordinate $b_L$ as a continuous function of the invariant scalar $P \cdot b$ and the fixed lattice momentum $P_L$:

$$b_L = \frac{P \cdot b}{-P_L} \tag{5.75}$$

To complete the geometric mapping, we must also ensure the squared separation term $b^2$ present in the denominators of our amplitudes transforms invariantly. Substituting our Lorentz expression for $b_L$ into $b^2 = b_L^2 + b_T^2$ yields:

$$b^2 = \left(\frac{P \cdot b}{-P_L}\right)^2 + b_T^2 \tag{5.76}$$

By replacing $b_L$ and $b^2$ with these invariant expressions, we transform our discrete lattice amplitudes into continuous functions of the Lorentz-invariant variables:

$$\widetilde{A}_{iB}(b_L, b_T, \hat{\zeta}) \quad \xrightarrow{\text{Coordinate Transformation}} \quad \widetilde{A}_{iB}\left(P \cdot b, b^2, \hat{\zeta}\right) \tag{5.77}$$

With the amplitudes now expressed as functions of $P \cdot b$, the Fourier transform of Eq. (5.6) can be evaluated directly. This connects the Euclidean correlation ratios evaluated on the lattice to the physical, $x$-dependent phenomenological observables needed to study the Sivers effect.

### 5.9 Numerical Results and Parity Constraints of the Extracted Amplitudes

Before examining the lattice data across the different Collins-Soper scales ($\hat{\zeta}$), we check the parity constraints of Eq. (4.28). Since $P \cdot b = -P_L b_L$ (Eq. (5.74)), the real amplitudes $\widetilde{A}^{\mathrm{Re}}_{2B}$ and $\widetilde{A}^{\mathrm{Re}}_{12B}$ must be even in the Cartesian longitudinal separation $b_L$, and the imaginary amplitudes $\widetilde{A}^{\mathrm{Im}}_{2B}$ and $\widetilde{A}^{\mathrm{Im}}_{12B}$ must be odd. The lattice data in the figures below show these even and odd profiles along the $b_L$ axis, which validates the parity projection.

However, the statistical quality of the extracted signals differs between amplitudes. While the unpolarized amplitudes ($\widetilde{A}^{\mathrm{Re}}_{2B}$ and $\widetilde{A}^{\mathrm{Im}}_{2B}$) and the real Sivers amplitude ($\widetilde{A}^{\mathrm{Re}}_{12B}$) show clear, statistically resolvable signals across the simulated data, the imaginary Sivers amplitude ($\widetilde{A}^{\mathrm{Im}}_{12B}$) does not. Across all evaluated nucleon momenta and staple extents, the extracted lattice data for $\widetilde{A}^{\mathrm{Im}}_{12B}$ remains completely dominated by gauge noise and is statistically compatible with zero. With no distinguishable signal, we cannot meaningfully constrain or fit a phenomenological model to this component. We therefore discard $\widetilde{A}^{\mathrm{Im}}_{12B}$ from further analysis: we drop the $\sin(x(b \cdot P))\widetilde{A}^{\mathrm{Im}}_{12B}$ term in Eq. (5.6) and evaluate the $x$-dependent Sivers shift using only the $\widetilde{A}^{\mathrm{Re}}_{12B}$ data.

Sections 5.9.1–5.9.4 show the amplitudes at $P_L = -1, -2, -3, -4$.

#### 5.9.1 Amplitude Profiles at $\hat{\zeta} = 0.090772$ ($P_L = -1$)

Figures 19 through 22 show the extracted amplitudes at this momentum, first in the discrete $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ lattice grid and then mapped into the Lorentz-invariant $(P \cdot b, b^2)$ variable space of Section 5.8.

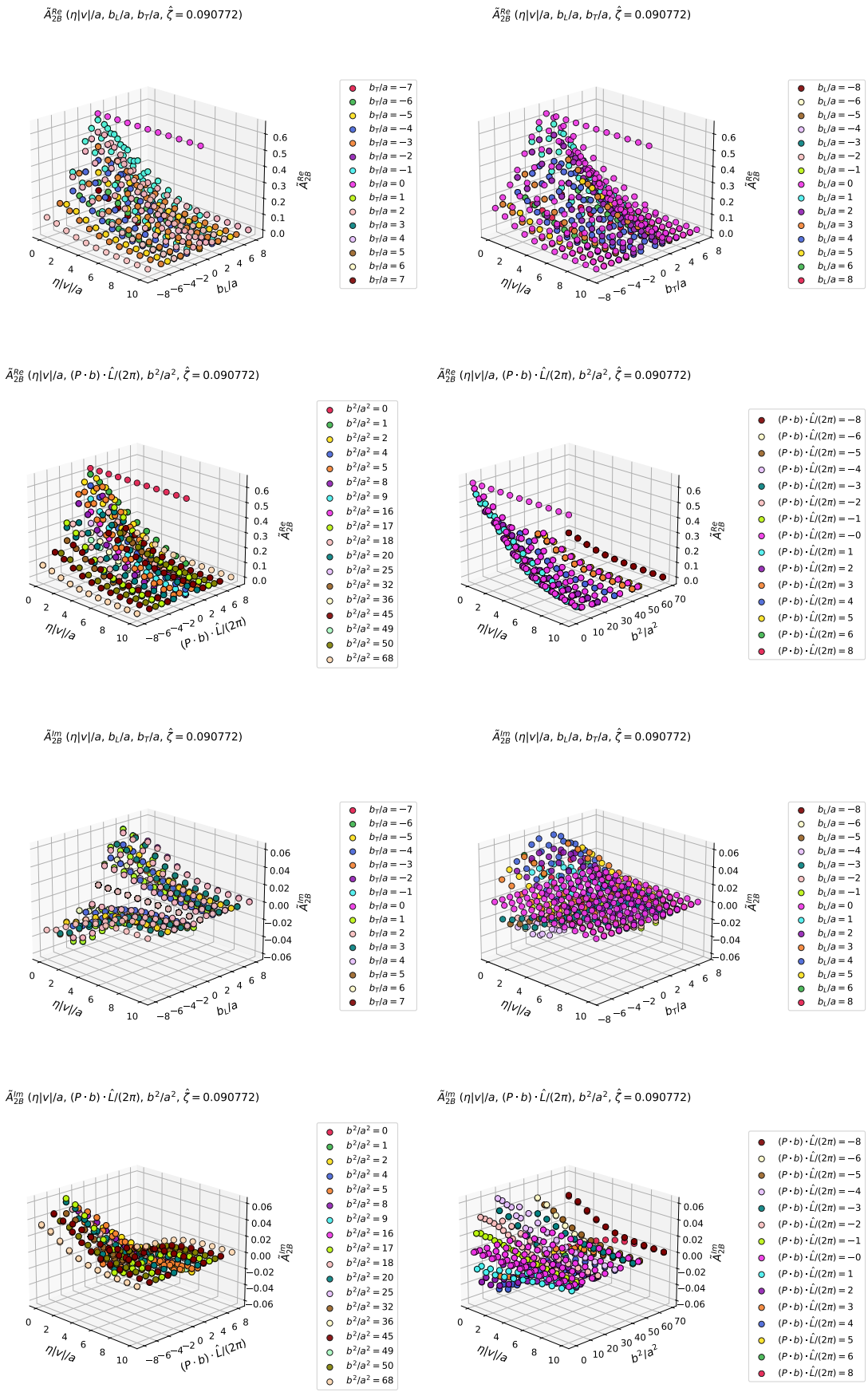



Figure 19: Real (top 2 rows) and imaginary (bottom 2 rows) parts of the $\widetilde{A}_{2B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$. The first row of each 2-row block displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the second row of each 2-row block displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space.

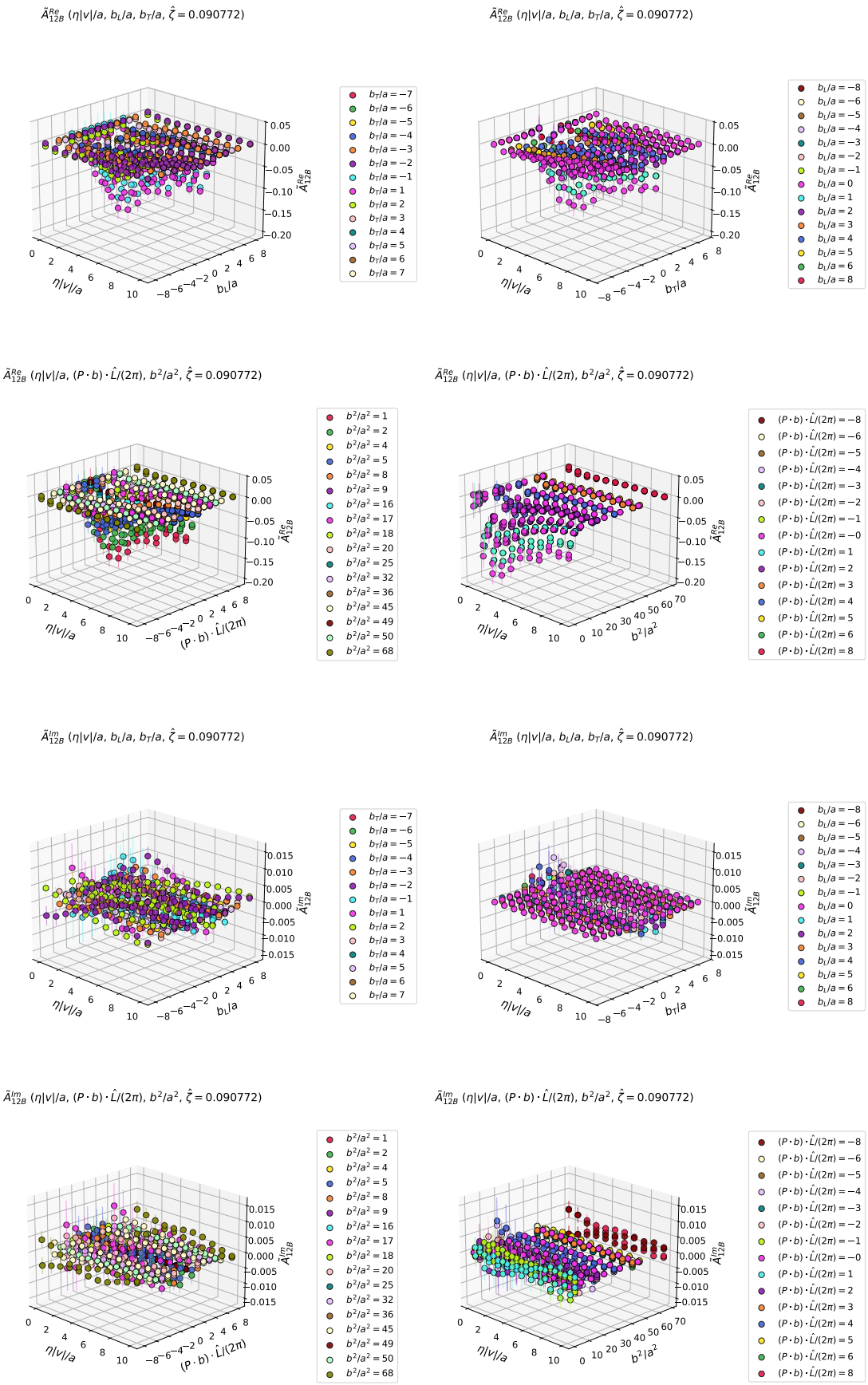



Figure 20: Real (top 2 rows) and imaginary (bottom 2 rows) parts of the $\widetilde{A}_{12B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$. The first row of each 2-row block displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the second row of each 2-row block displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space.

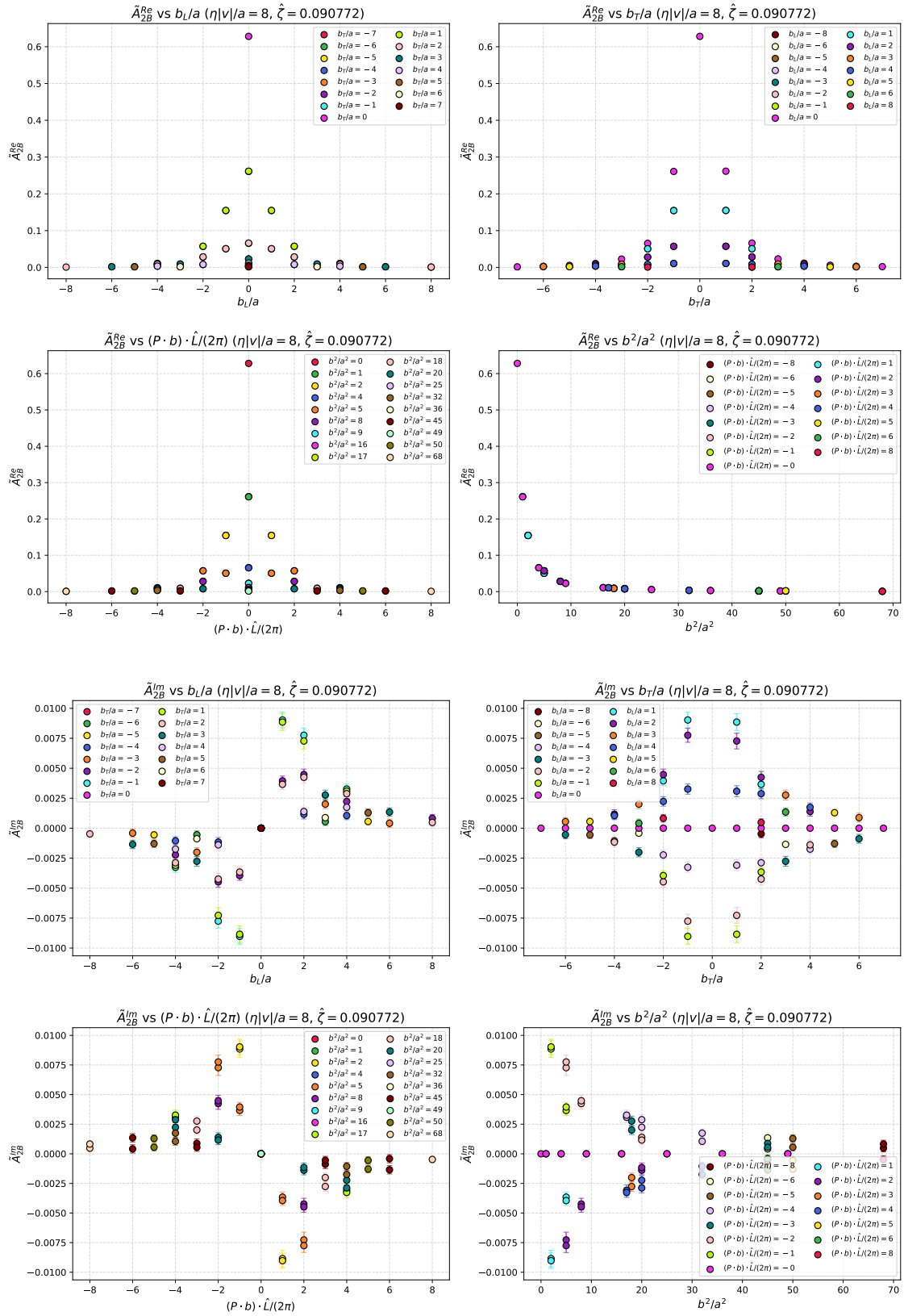



Figure 21: Real (top 2 rows) and imaginary (bottom 2 rows) parts of the $\widetilde{A}_{2B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$ at fixed $\eta|\boldsymbol{v}|/a = 8$. The first row of each 2-row block displays the $(b_L/a, b_T/a)$ space, while the second row of each 2-row block displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2)$ space.

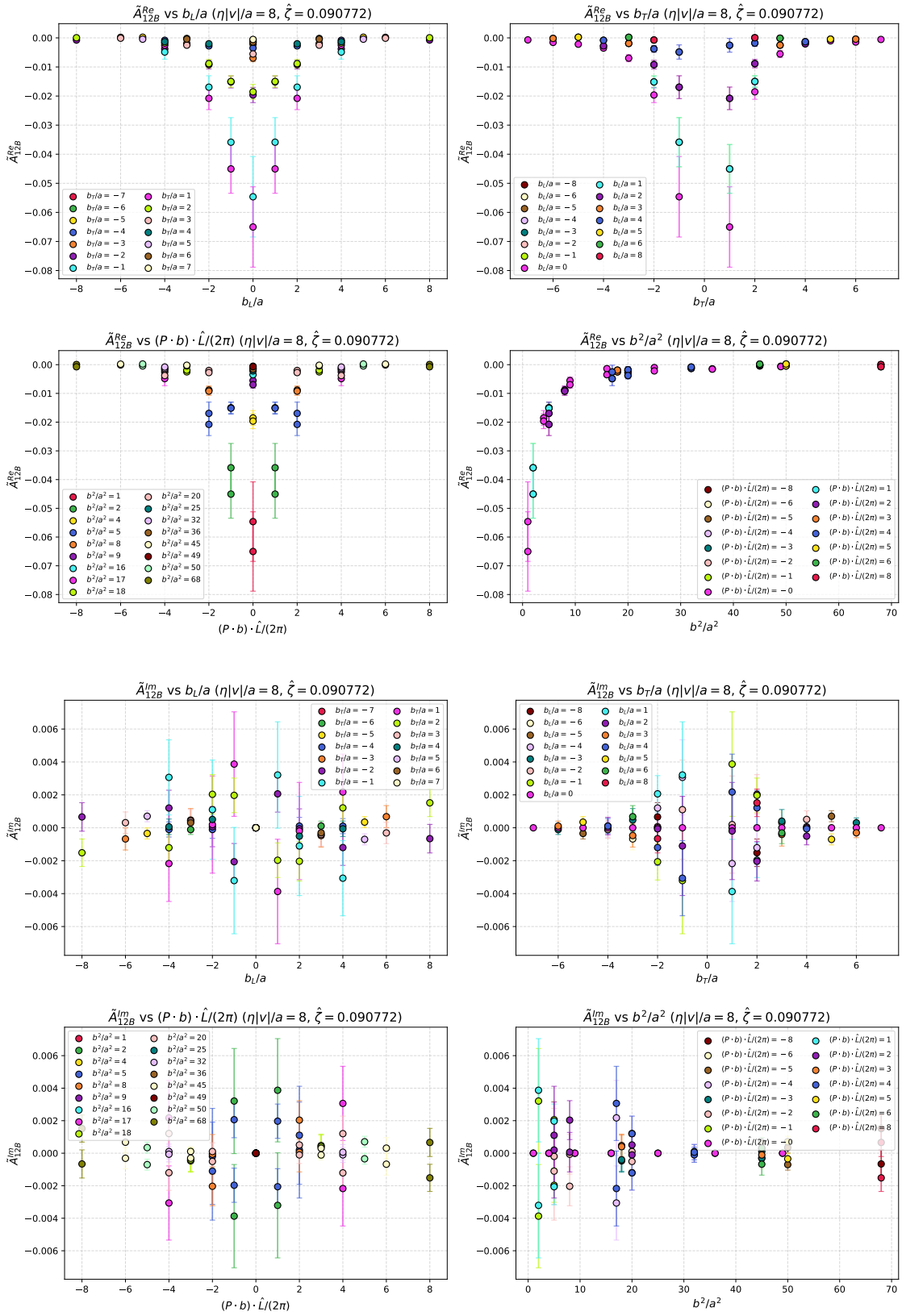



Figure 22: Real (top 2 rows) and imaginary (bottom 2 rows) parts of the $\widetilde{A}_{12B}$ amplitude for $\boldsymbol{P}\cdot a\hat{\boldsymbol{L}}/(2\pi) = \{-1, 0, 0\}$ at fixed $\eta|\boldsymbol{v}|/a = 8$. The first row of each 2-row block displays the $(b_L/a, b_T/a)$ space, while the second row of each 2-row block displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2)$ space. Notice the lack of discernible signal in $\widetilde{A}^{\rm Im}_{12B}$.

### 5.9.2 Amplitude Profiles at $\hat{\zeta} = 0.294367$ ($P_L = -2$)

Figures 23 through 26 show the corresponding amplitudes at this larger Collins-Soper value.

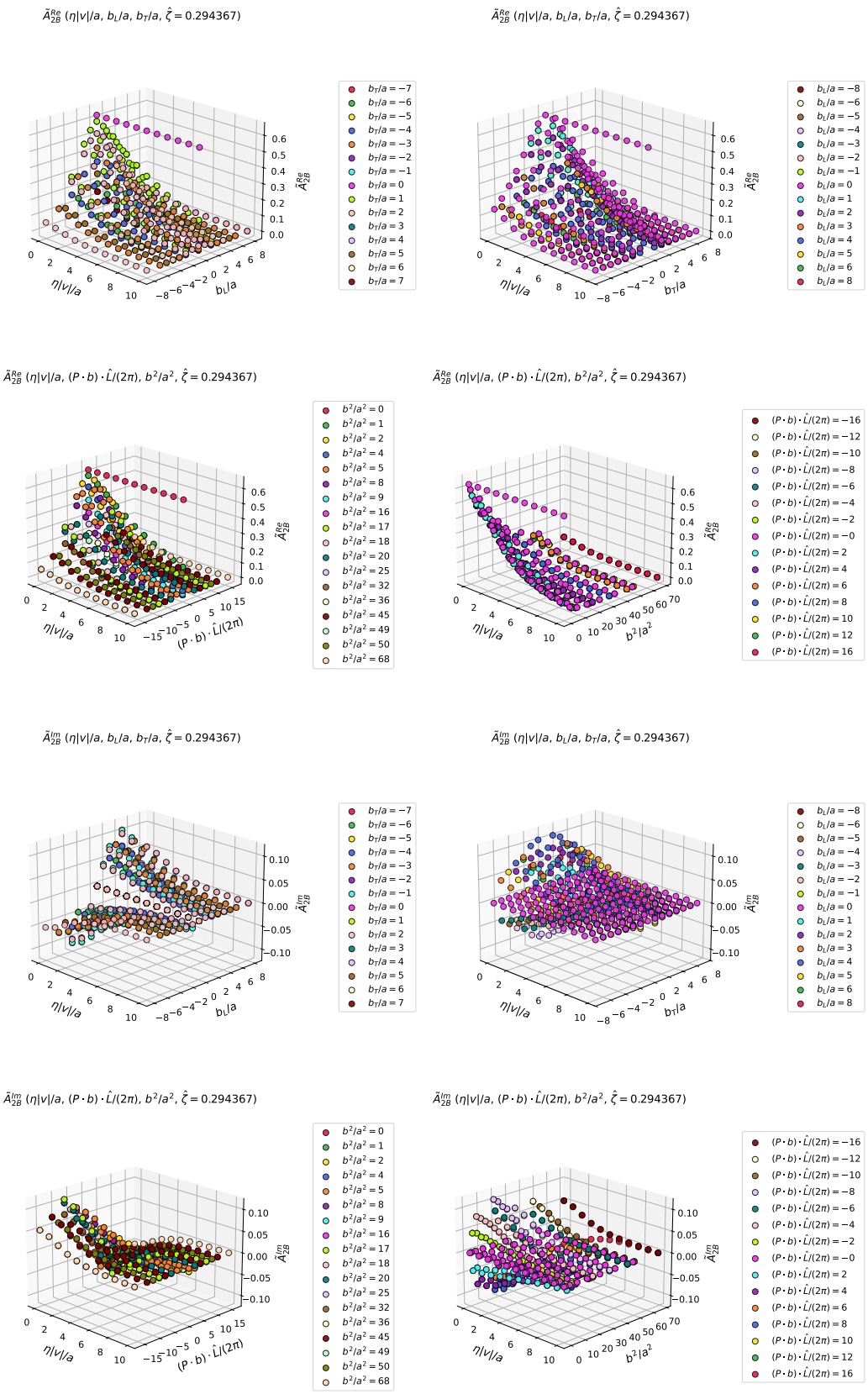



Figure 23: Same as Fig. 19, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ ($\hat{\zeta} = 0.294367$).

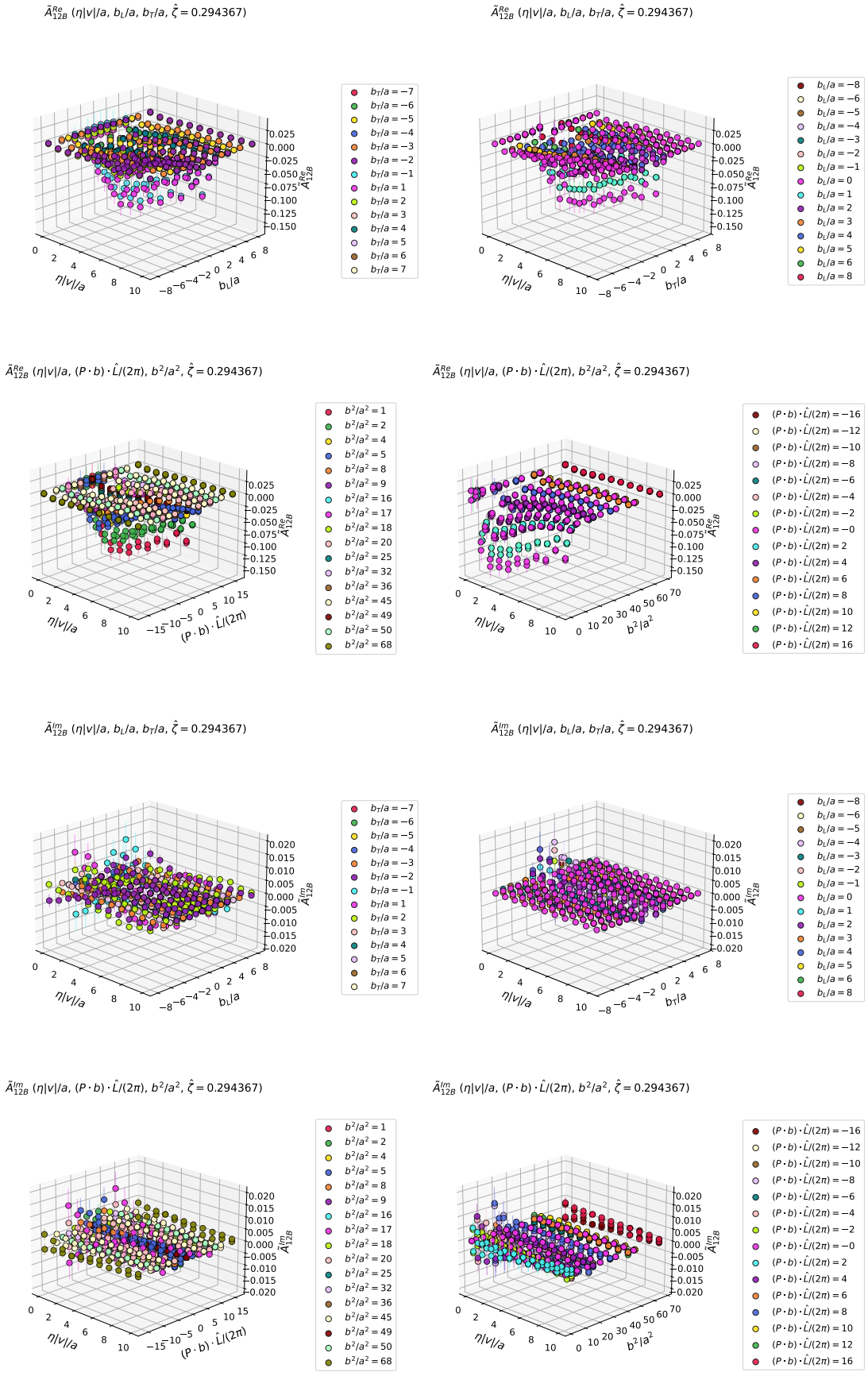



Figure 24: Same as Fig. 20, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi)=\{-2,0,0\}$ ($\hat{\zeta}=0.294367$).

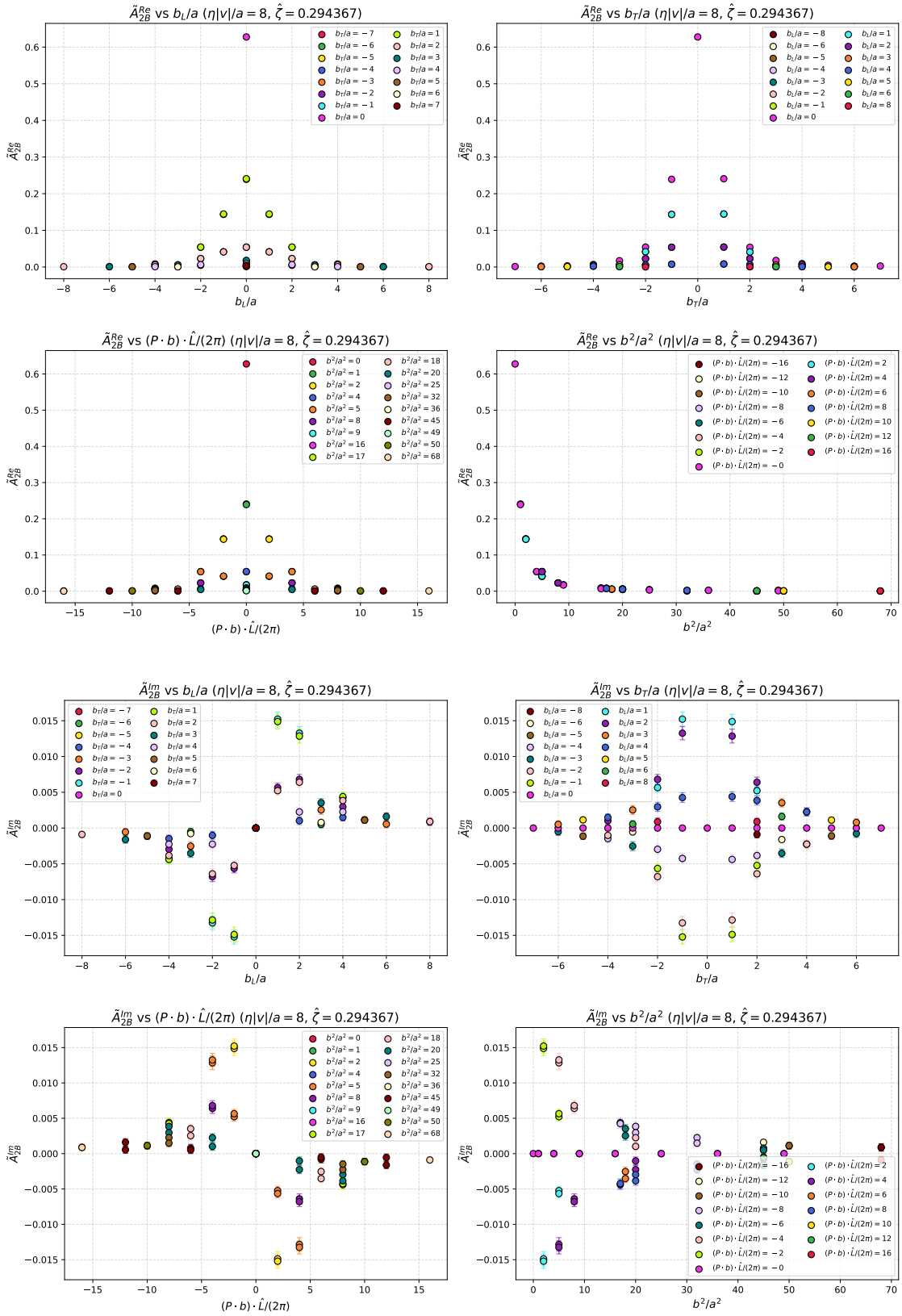



Figure 25: Same as Fig. 21, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ ($\hat{\zeta} = 0.294367$).

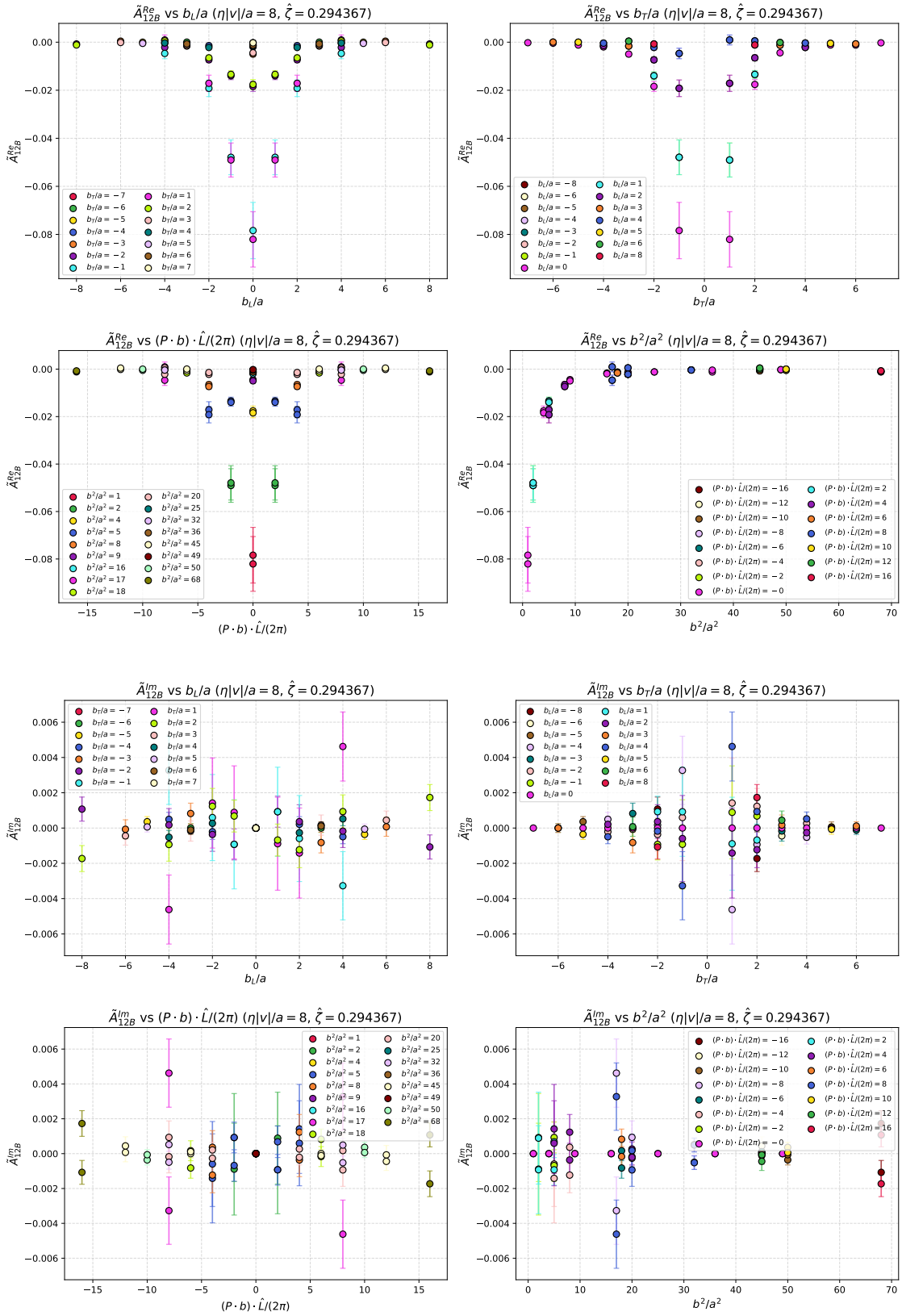



Figure 26: Same as Fig. 22, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ ($\hat{\zeta} = 0.294367$).

### 5.9.3 Amplitude Profiles at $\hat{\zeta} = 0.539684$ ($P_L = -3$)

Figures 27 through 30 give the same amplitude decomposition at $P_L = -3$.

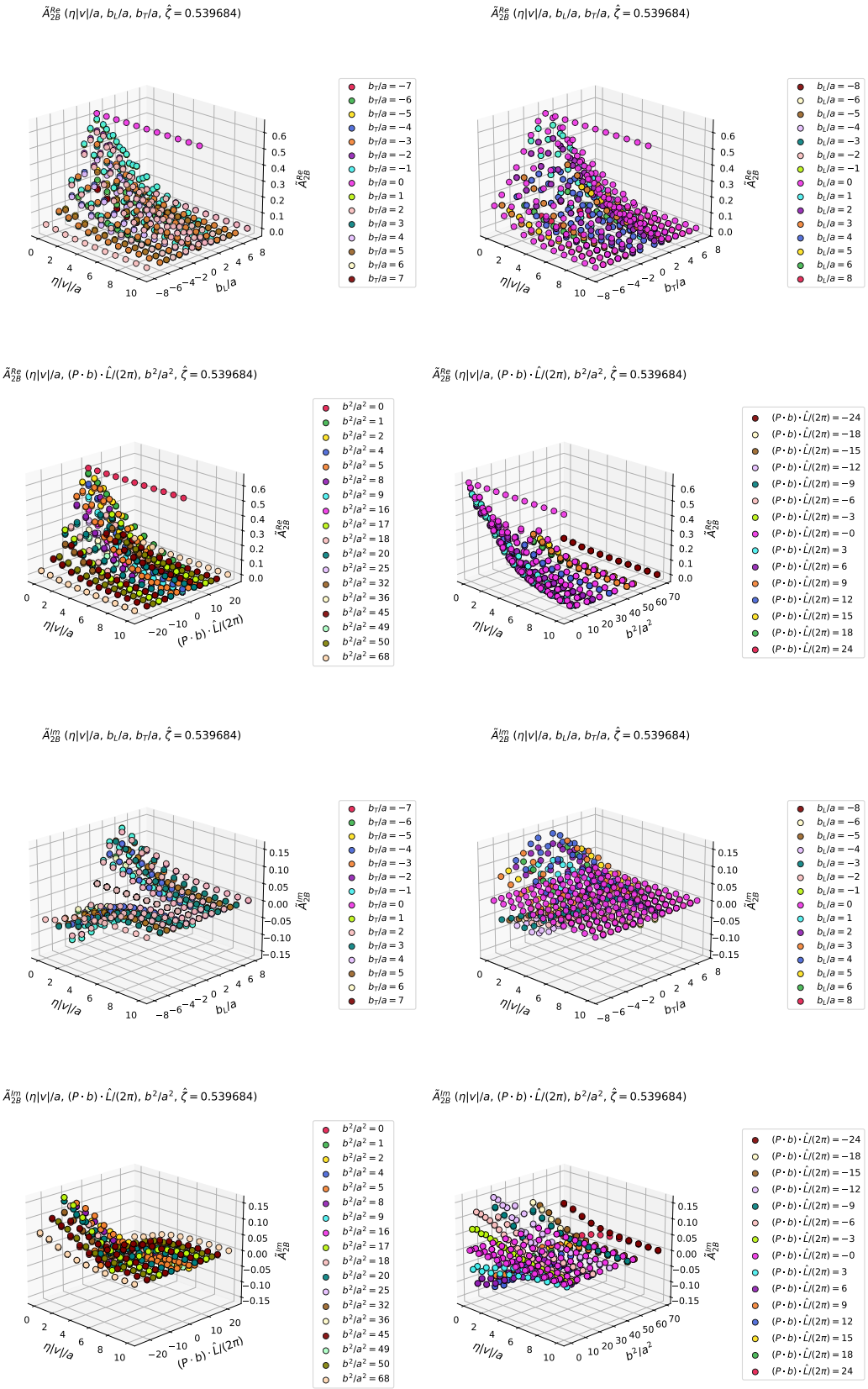



Figure 27: Same as Fig. 19, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$ ($\hat{\zeta} = 0.539684$).

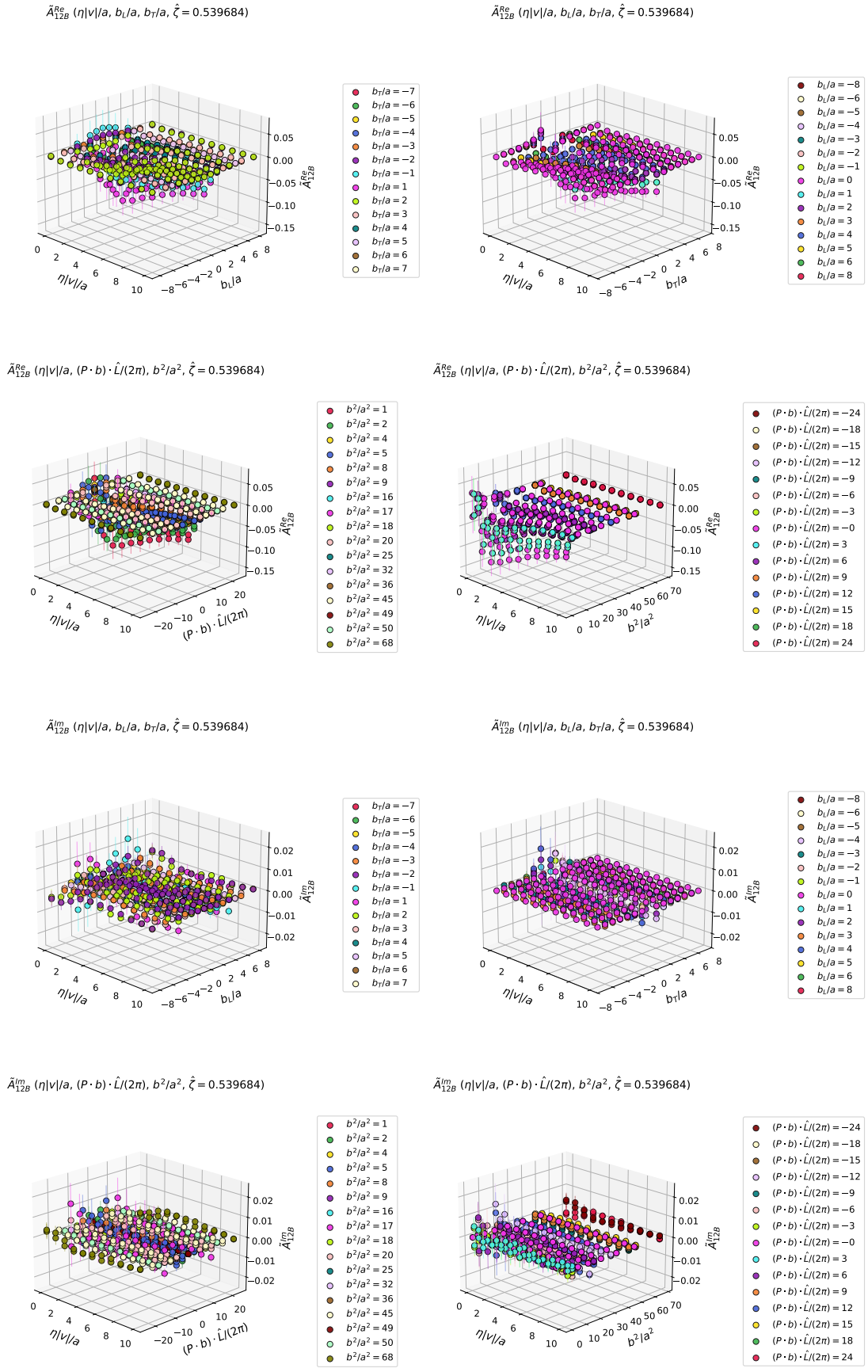



Figure 28: Same as Fig. 20, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$ ($\hat{\zeta} = 0.539684$).

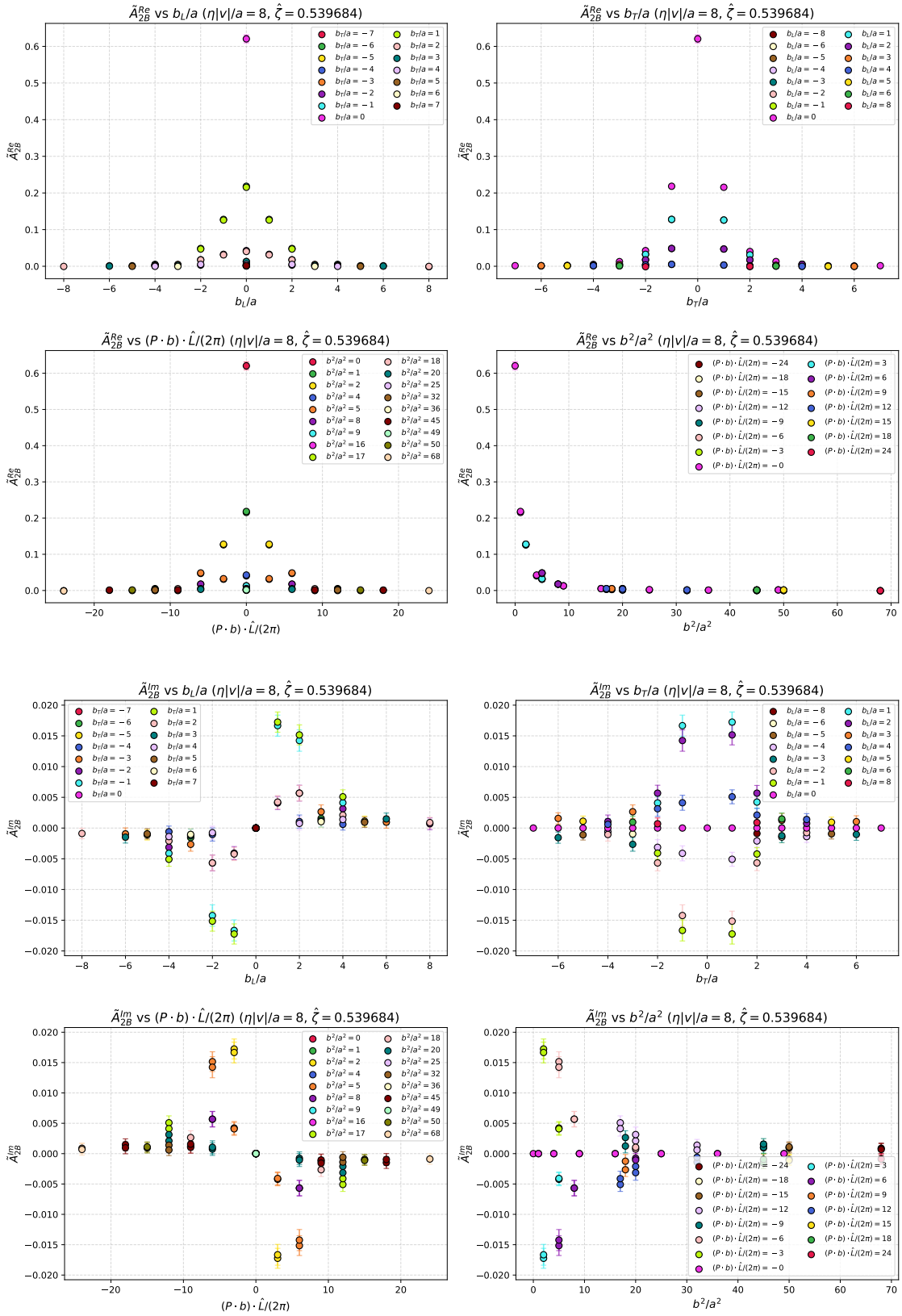



Figure 29: Same as Fig. 21, but for $\boldsymbol{P}\cdot a\hat{\boldsymbol{L}}/(2\pi) = \{-3, 0, 0\}$ ($\hat{\zeta} = 0.539684$).

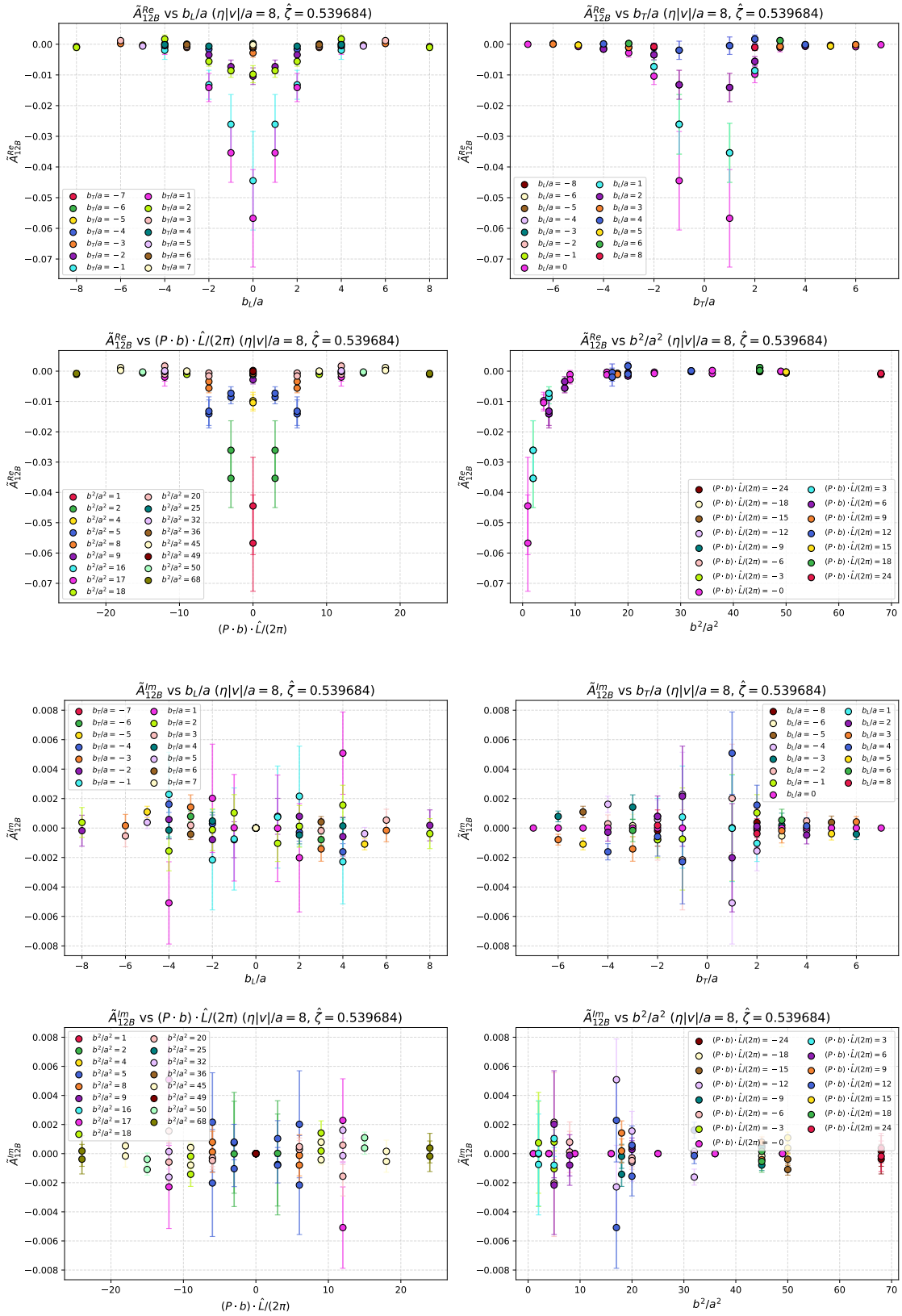



Figure 30: Same as Fig. 22, but for $\boldsymbol{P}\cdot a\hat{\boldsymbol{L}}/(2\pi) = \{-3, 0, 0\}$ ($\hat{\zeta} = 0.539684$).

### 5.9.4 Amplitude Profiles at $\hat{\zeta} = 0.785756$ ($P_L = -4$)

Figures 31 through 34 complete the set at our largest simulated momentum, $P_L = -4$.

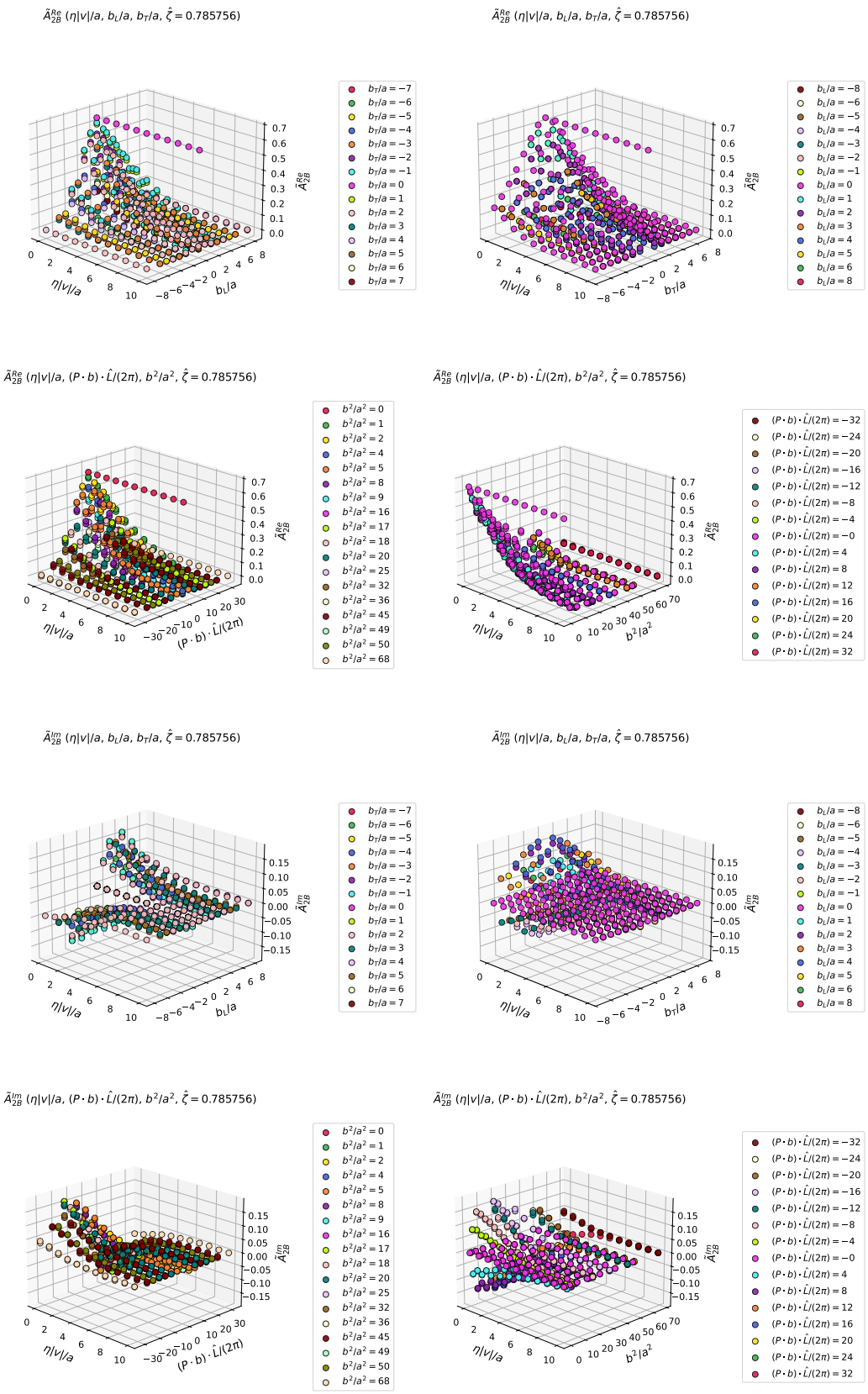



Figure 31: Same as Fig. 19, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ ($\hat{\zeta} = 0.785756$).

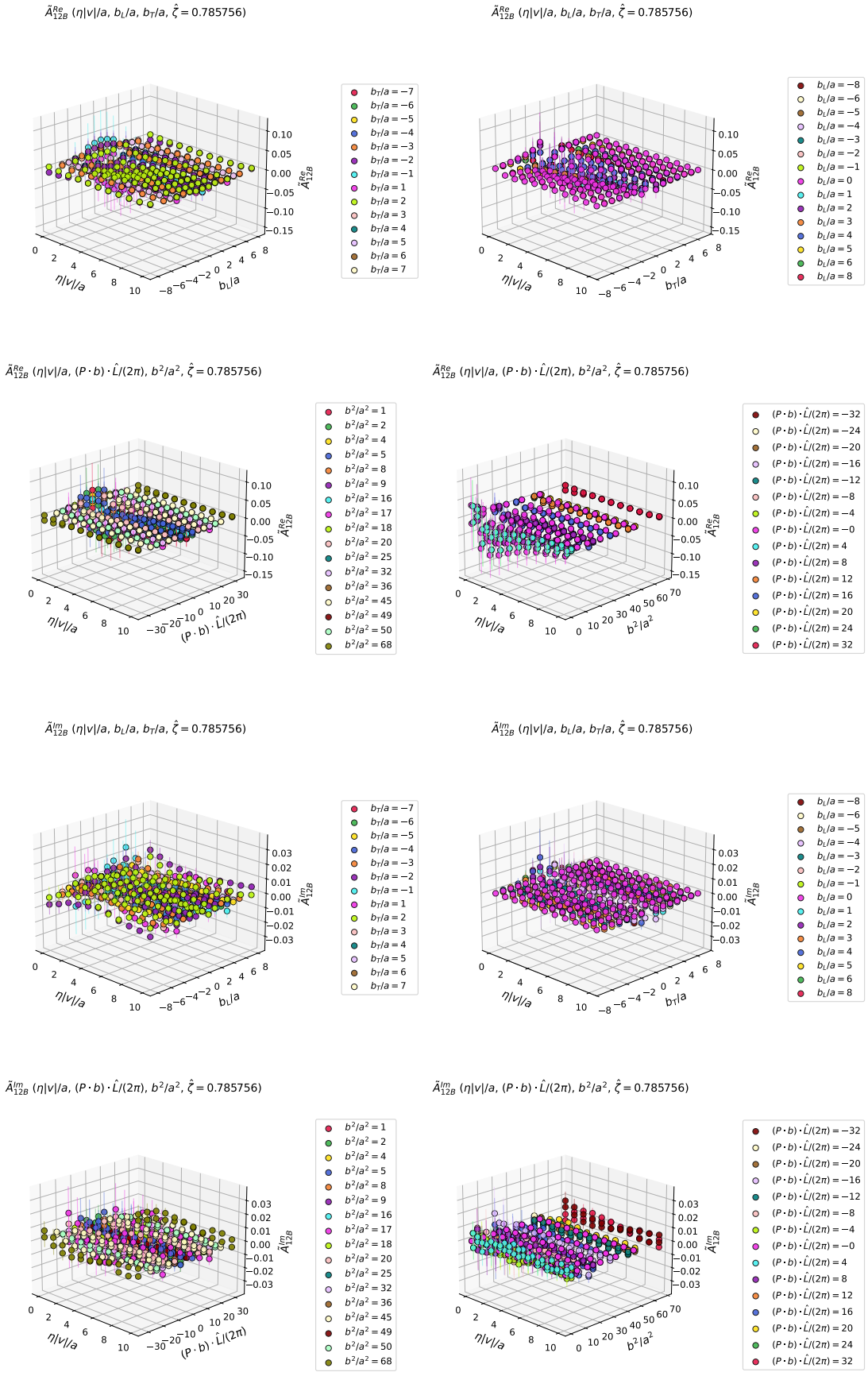



Figure 32: Same as Fig. 20, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ ($\hat{\zeta} = 0.785756$).

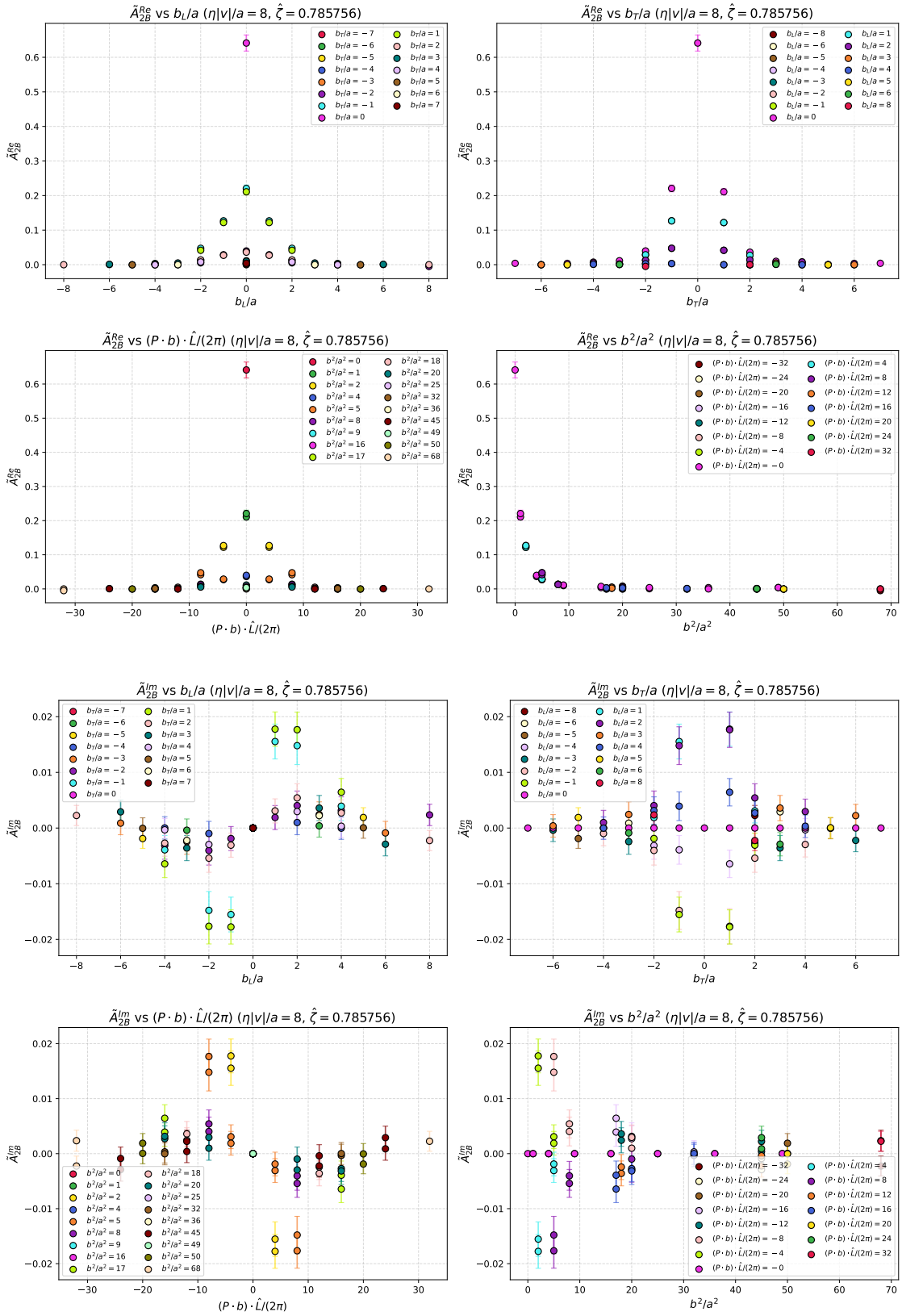



Figure 33: Same as Fig. 21, but for $\boldsymbol{P}\cdot a\hat{\boldsymbol{L}}/(2\pi) = \{-4, 0, 0\}$ $(\hat{\zeta} = 0.785756)$.

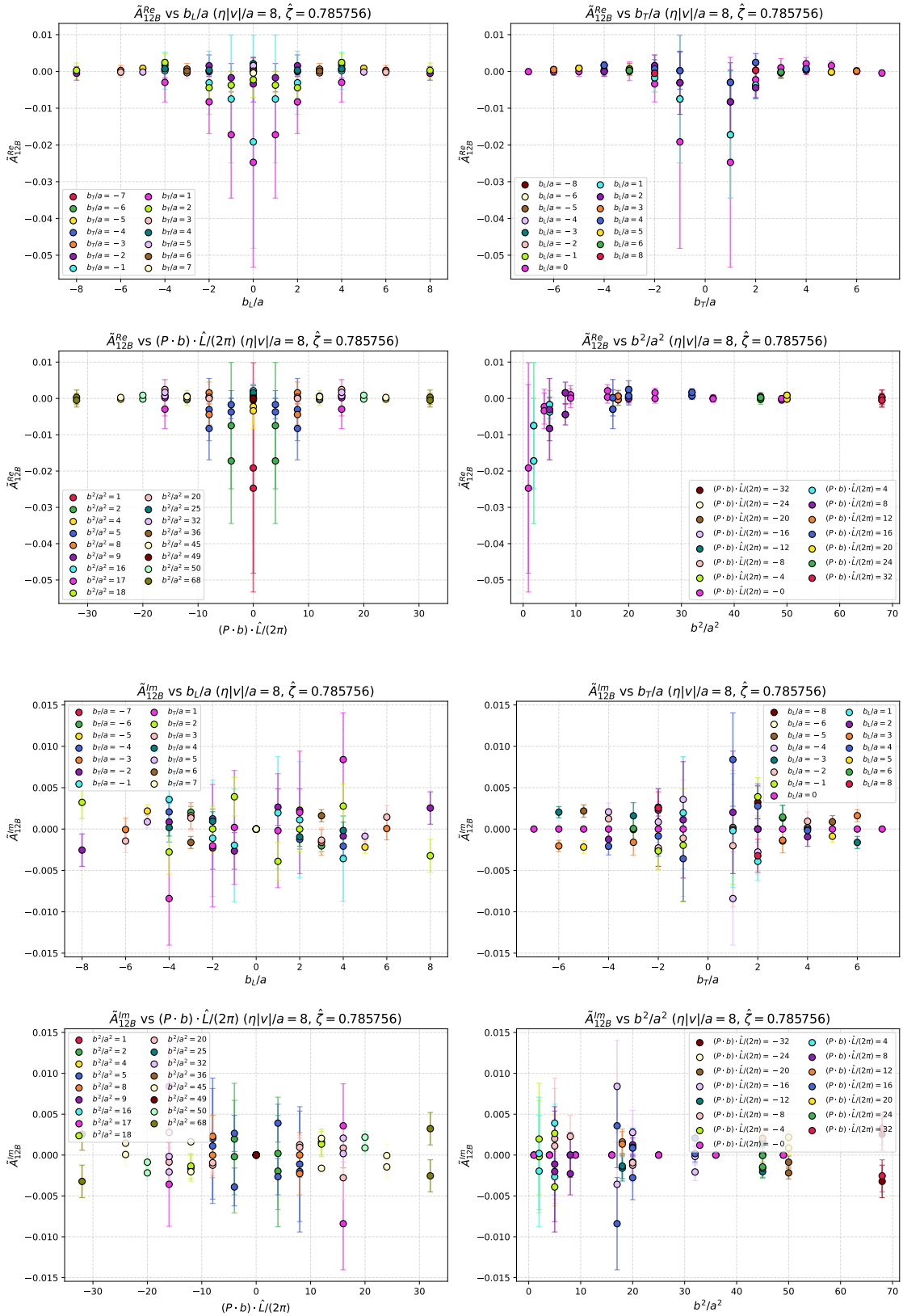



Figure 34: Same as Fig. 22, but for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ ($\hat{\zeta} = 0.785756$).

With the real and imaginary parts of $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ verified across all four simulated Collins-Soper momenta, these path-averaged, interpolated amplitude sets are the data fitted in Section 6; the fitted models then enter the Fourier-transform integral of Eq. (5.6), from which the $x$-dependent Sivers shift is extracted.

## 6 FITTING METHODOLOGIES AND STABILITY ANALYSIS

All amplitudes fitted in this chapter are for the isovector $u-d$ combination. As discussed in the previous section (Sec. 5), the lattice data show varying statistical precision among the invariant amplitudes. The real part of the unpolarized amplitude, $\widetilde{A}^{\mathrm{Re}}_{2B}$, has small statistical uncertainties across the simulated position space, whereas the imaginary unpolarized amplitude $\widetilde{A}^{\mathrm{Im}}_{2B}$ and the real Sivers amplitude $\widetilde{A}^{\mathrm{Re}}_{12B}$ have larger gauge noise. We already discarded the imaginary Sivers amplitude $\widetilde{A}^{\mathrm{Im}}_{12B}$ because it is consistent with zero. We first establish a stable global fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ as the baseline for the remaining numerical analysis, as it corresponds to the leading-twist unpolarized TMD and has the highest data quality.

We parameterize the amplitudes in the Cartesian space $(b_L, b_T)$ rather than the Lorentz-invariant variable space $(P\cdot b, b^2)$. Fitting in $(b_L, b_T)$ provides a larger, more uniformly distributed set of discrete data points and resolves kinematic degeneracies. Because the squared separation distance $b^2 = b_L^2 + b_T^2$ is non-negative, multiple distinct combinations of longitudinal and transverse separations (such as a large $b_L$ with a small $b_T$, or a small $b_L$ with a large $b_T$) map to the same invariant $b^2$. Fitting in the $(P\cdot b, b^2)$ basis would therefore compress these distinct orthogonal spatial dependencies into a single coordinate. Fitting over the $(b_L, b_T)$ grid allows the model to separate the longitudinal decay profile from the transverse decay profile.

Section 6.1 shows that a simple Gaussian model fails for $\widetilde{A}^{\mathrm{Re}}_{2B}$ and makes the Fourier integral over $P\cdot b$ diverge. Section 6.2 removes the data with $|\boldsymbol{b}| < 3a$, which contain discretization artifacts. Section 6.3 explains why the fits must include the staple extent $\eta$ and reach the asymptotic limit $\eta \to \infty$. Section 6.4 uses symbolic regression (`PySR`) with an asymptotic penalty as a guide to the functional form, which points to a power-law tail $(1+b^2)^{-j}$ multiplied by $e^{-f\eta}$. Section 6.5 tests exponential decay in $|\boldsymbol{b}|$ and in $P\cdot b$ as an alternative. Section 6.6 compares Gaussian and power-law fits over $6 \le \eta \le 10$ and fits individual $\eta$ slices to find the $\eta$-dependence of the parameters $c$ and $d$, and Section 6.7 selects the model for $c(\eta)$ and gives the models for the three amplitudes.

Section 6.8 fits the three amplitudes simultaneously with the full Jackknife covariance matrix, cancels the shared $e^{-f\eta}$ factor, and evaluates the $x$-dependent Sivers shift at each of the four momenta. Section 6.9 applies separable longitudinal and transverse models to all three amplitudes.

### 6.1 Failure of Simple Gaussian Model

For the real parts of the amplitudes ($\widetilde{A}^{\mathrm{Re}}_{iB}$), Hermitian conjugation requires the function to be even in the longitudinal separation $b_L$ (Eq. (4.28)). As a starting point, we model the real amplitude with a standard 2D Gaussian centered at the origin

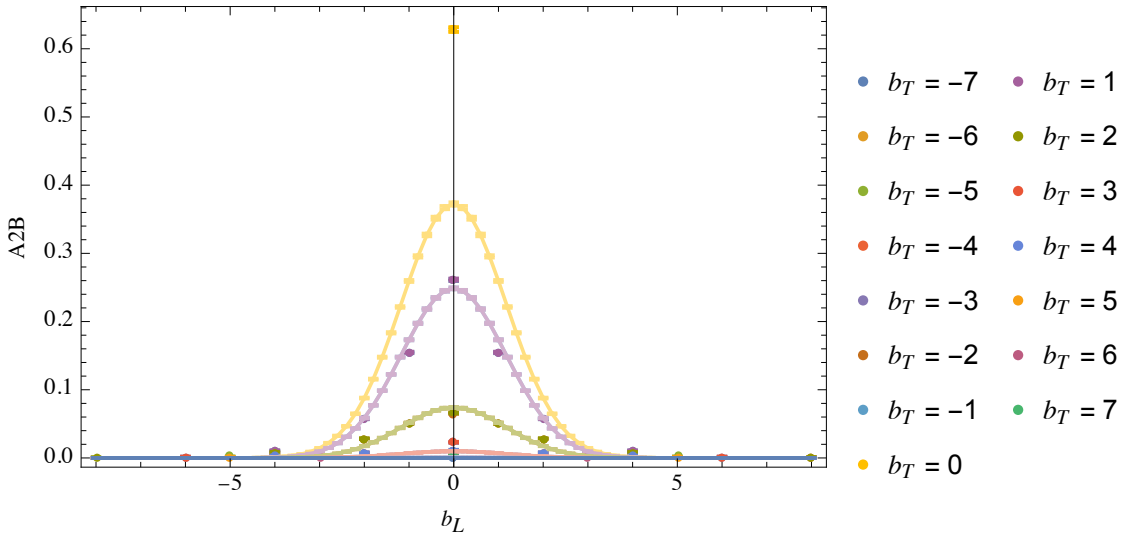
$$\widetilde{A}^{\mathrm{Re}}_{iB} = \gamma \exp\left[-\alpha b_L^2 - \beta b_T^2\right] \tag{6.1}$$

To transform this into the invariant variable space, we substitute $b_T^2 = b^2 - b_L^2$ and apply the Lorentz-invariant projection $b_L = (P \cdot b)/(-P_L)$ derived in Section 5.8. Squaring this substitution gives

$$\widetilde{A}^{\mathrm{Re}}_{iB} = \gamma \exp\left[-\frac{(\alpha - \beta)}{P_L^2}(P \cdot b)^2 - \beta b^2\right] \tag{6.2}$$

For the Fourier integral $\int d(P\cdot b) e^{ix(P\cdot b)} \widetilde{A}_{iB}$ to converge, the argument of the exponential must remain negative as $(P \cdot b) \to \pm\infty$. This requires $(\alpha - \beta) > 0$. If $\alpha < \beta$, the amplitude grows exponentially at large longitudinal separations and the integral diverges.

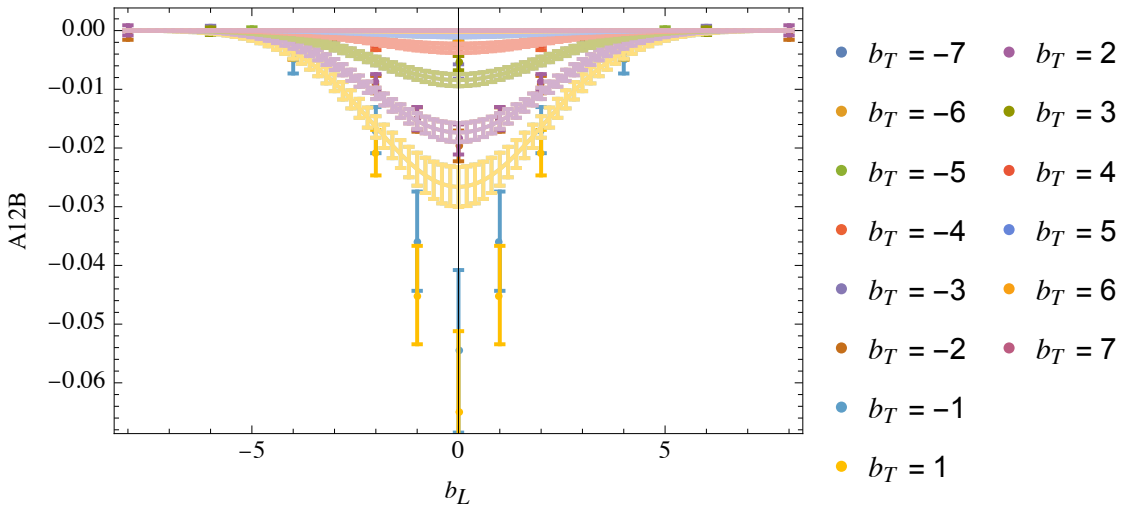
However, fitting this simple Gaussian model to our lattice data yields $\alpha < \beta$ for the unpolarized real amplitude, $\widetilde{A}^{\mathrm{Re}}_{2B}$, as shown for $\widetilde{A}^{\mathrm{Re}}_{2B}$ and $\widetilde{A}^{\mathrm{Re}}_{12B}$ in Figures 35-36.




```
Re A2B => P_L = -1 ,eta = 8
 Fitting into  A e^(-α b_L^2-β b_T^2)  -> (A, α, β) = {⟨0.3724(26)⟩_2902 J, ⟨0.3612(26)⟩_2902 J, ⟨0.4039(29)⟩_2902 J}
 chisquareDoF = 235.509

In (b.P,b^2) spcae:  Re A2B => P_L = -1 ,eta = 8
 Fitting into  A_1 e^(-α_1 (b.P)^2) e^(-β_1 b^2)  -> (A_1, α_1, β_1) = {⟨0.3724(26)⟩_2902 J, ⟨-0.0427(27)⟩_2902 J, ⟨0.4039(29)⟩_2902 J}
```

Figure 35: Failure of Gaussian model fit for the $\widetilde{A}_{2B}^{\mathrm{Re}}$ amplitude, with $\alpha - \beta = -0.0427(27)$ and $\chi^2/\mathrm{dof} \approx 235$.




```
Re A12B => P_L = -1 ,eta = 8
 Fitting into  A e^(-α b_L^2-β b_T^2)  -> (A, α, β) = {⟨-0.0307(45)⟩_2902 J, ⟨0.121(21)⟩_2902 J, ⟨0.144(22)⟩_2902 J}
 chisquareDoF = 1.43445

In (b.P,b^2) spcae:  Re A12B => P_L = -1 ,eta = 8
 Fitting into  A_1 e^(-α_1 (b.P)^2) e^(-β_1 b^2)  -> (A_1, α_1, β_1) = {⟨-0.0307(45)⟩_2902 J, ⟨-0.022(25)⟩_2902 J, ⟨0.144(22)⟩_2902 J}
```

Figure 36: Failure of Gaussian model fit for the $\widetilde{A}_{12B}^{\mathrm{Re}}$ amplitude, with $\alpha - \beta = -0.022(25)$ and $\chi^2/\mathrm{dof} \approx 1.4$.

This means the amplitude decays faster transversely than longitudinally, so the expression in $(P{\cdot}b, b^2)$ space is not integrable. Figures 35–36 also show poor fit quality. A 2D Gaussian decay profile is too

rigid to describe the non-local operator.

### 6.2 Discretization Effects and Short-Distance Data Exclusion

The Gaussian fit of Section 6.1 includes quark separations $\boldsymbol{b}$ at short distance ($|\boldsymbol{b}| \sim 1a$ or $2a$) as well as in the medium range ($|\boldsymbol{b}| \geq 3a$). The short-distance regime is governed by different physics, namely perturbative effects, including incipient short-distance divergences that require additional renormalization, while we aim to probe the medium-range non-perturbative TMD physics. A fit that includes the short-distance data would have to account for this crossover between regimes even without discretization artifacts. Before exploring further models, we therefore account for the limitations of discretized spacetime and keep only the data that probe non-perturbative TMD physics.

As discussed in Chapter 3, Lattice QCD defines the theory on a discrete grid with a finite lattice spacing $a$. Observables evaluated at length scales comparable to $a$ contain discretization artifacts. At short quark-separation distances ($|\boldsymbol{b}| \sim 1a$ or $2a$), the continuous $O(4)$ rotational symmetry of Euclidean QCD breaks down to the discrete hypercubic group of the lattice. The step-wise Wilson lines connecting these closely spaced fields also introduce $\mathcal{O}(a)$ and $\mathcal{O}(a^2)$ finite-spacing errors [77, 3]. Fitting continuum models to these short-distance points would parameterize ultraviolet lattice artifacts rather than the nucleon's non-perturbative structure.

To separate our parameterizations from these short-distance effects, we apply a spatial cut to our dataset before fitting. We exclude data points where the magnitude of the quark spatial separation is less than three lattice units:

$$|\boldsymbol{b}| = \sqrt{b_L^2 + b_T^2} \geq 3a \ . \tag{6.3}$$

With the dataset restricted to $|\boldsymbol{b}| \geq 3a$, the non-local operator spans enough lattice sites for the step-wise gauge link to approach a continuous trajectory. The fits and subsequent Fourier transforms then depend on long-range dynamics rather than the ultraviolet regulator.

The lattice data themselves show the difference between the two regions (Figure 37). At fixed $b^2$, the short-distance unpolarized amplitude rises with $|P \cdot b|$ by about 14% for $b^2/a^2 = 5$, while the

long-distance amplitude stays flat within uncertainties for $b^2/a^2 = 20$ and 45.

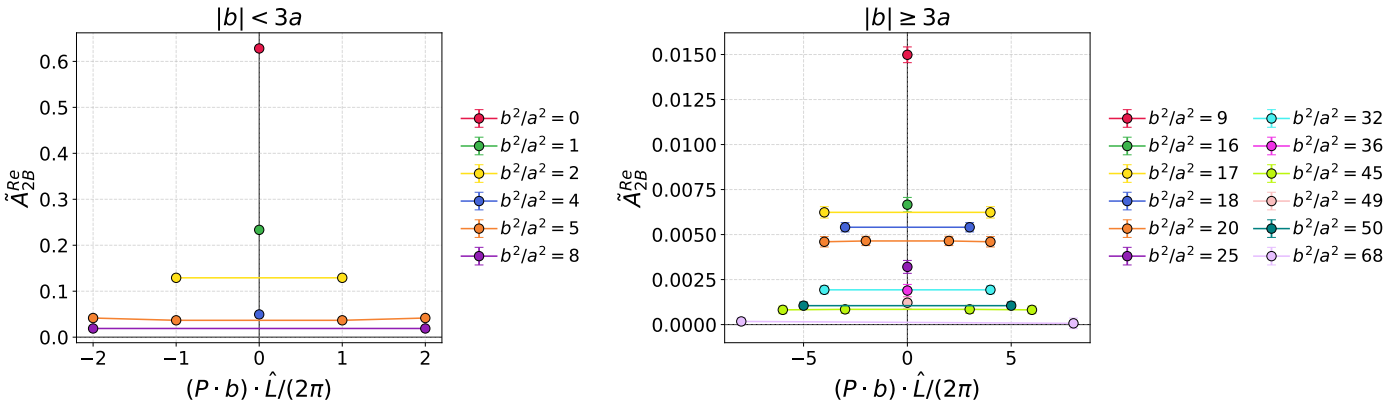



Figure 37: The $\widetilde{A}^{\mathrm{Re}}_{2B}$ amplitude at fixed momentum ($P_L = -1$) and fixed staple extent $\eta|\boldsymbol{v}|/a = 9$, for short-distance ($|\boldsymbol{b}| < 3a$, left) and long-distance ($|\boldsymbol{b}| \geq 3a$, right) data. Points with the same $b^2$ are connected for illustration purposes.

The fits we do are global fits over the full $(b_L, b_T, \eta)$ grid, equivalently over $(P \cdot b, b^2, \eta)$. They do not model the amplitude at any single value of $b^2$, so what matters is the behavior of the data set as a whole rather than the trend of each individual $b^2$ group. Isolated groups at $|\boldsymbol{b}| \geq 3a$ can follow the short-distance trend within their uncertainties, as $b^2/a^2 = 20$ does at $\eta|\boldsymbol{v}|/a = 6$, 7 and 8. However, the fit is constrained by all data at $|\boldsymbol{b}| \geq 3a$ together. Therefore, all later stability analyses, global fits, and extracted phenomenological parameters in this chapter use only the $|\boldsymbol{b}| \geq 3a$ dataset.

### 6.3 Global fit in $(b_L, b_T, \eta)$ (equivalently in $(P \cdot b, b^2, \eta)$)

As discussed in Section 4.3, we require the staple extent $\eta$ to reach infinity, $\eta|v| \to \pm\infty$, as a crucial condition, thereby representing the generalized Sivers shifts (Eq. (4.48)) for SIDIS and DY. The first Mellin-moment ($x$-integrated) calculation of generalized Sivers shifts in Ref. [2] shows that the Sivers shift ratio reaches an asymptotic plateau in the region $|\eta||v| = 7a \ldots 12a$. This means the Sivers TMD and the unpolarized TMD share a common $\eta$ profile that cancels in the asymptotic limit, giving the Sivers shift ratio a plateau at large $\eta$.

In the current work, we extract the $x$ dependence (we will not integrate over $x$). Since $(P \cdot b)$ (the Fourier conjugate of $x$) and the staple extent $\eta$ are independent for fixed $\hat{\zeta}$ (Eq.(5.9)), we expect the $x$-dependent Sivers TMD and the unpolarized TMD to share a common $\eta$ profile that cancels in the asymptotic limit. This expected $\eta$ profile does not necessarily factorize into $\eta$ and $(b_L, b_T)$ for lattice

amplitudes, since cancellation occurs only in the asymptotic limit. We will explore a Global fit in $(P_L, b_T, \eta)$ (equivalently in $(P \cdot b, b^2, \eta)$) for amplitudes which includes factorized $\eta$ profiles and also coupled to $(P_L, b_T)$ anzertz. Also, note that we are interested in the asymptotic limit; we include the relatively larger (approaching SIDIS and DY limits) $\eta$ range $6 \leq \eta \leq 10$ for all later stability analyses, global fits, and extracted phenomenological parameters from lattice amplitudes.

### 6.4 Physics-Informed Symbolic Regression via PySR

We utilized machine learning (ML) methods, specifically Symbolic Regression (SR), to explore multidimensional functional forms for lattice amplitudes as an aid.

Unlike traditional regression, which optimizes the parameters of a pre-determined function, symbolic regression searches the space of mathematical expressions and their parameters simultaneously. We use `PySR` [93], a framework that uses Genetic Programming to discover interpretable equations from data by balancing accuracy and simplicity.

Section 6.4.1 describes the search configuration, Section 6.4.2 the custom loss function, and Section 6.4.3 the resulting Pareto front.

#### 6.4.1 Expression Search and Configuration

`PySR` [93, 117, 118] uses a genetic algorithm to iteratively evolve mathematical expressions represented as binary trees. The algorithm evaluates a population of expressions against the lattice data using a custom loss function over thousands of generations.

Table 11 lists the `PySR` search configuration: the permitted mathematical operators, the nesting constraints, and the main hyperparameters. The operator library includes a custom Lorentzian-type binary operator, $\mathrm{lor}(x, y) = 1/(1 + x^2 + y^2)^2$, alongside standard arithmetic and exponential functions. Nesting constraints prevent the algorithm from generating physically meaningless double-exponential or double-decay structures.

Table 11: Hyperparameter and operator configuration for the Physics-Informed `PySR` Symbolic Regression search.

| **Configuration Parameter** | **Specification / Value** |
|---|---|
| Binary Operators | $+$, $-$, $\times$, $/$, $\text{pow}(x,y) = x^y$, $\text{lor}(x,y) = \frac{1}{(1+x^2+y^2)^2}$ |
| Unary Operators | $\text{square}(x) = x^2$, $\exp(x)$, $\text{decay}(x) = \exp(-x)$ |
| Operator Constraints | $\text{pow} : (-1, 1)$, $\text{lor} : (10, 10)$, $\text{decay} : 10$ |
| Nesting Constraints | $\exp \to \exp : 0$, $\text{decay} \to \text{decay} : 0$, $\exp \to \text{decay} : 0$ |
| Populations | 30 |
| Evolution Generations | 100 |
| Maximum Tree Complexity | 28 |
| Loss Function | Physics-Informed Weighted $\chi^2/\text{dof} + \epsilon\,\mathcal{P}_{\text{asymptotic}}$ ($\epsilon = 0.1$) |
| Input Features | $n$ (Staple length $\eta$), $b_L$ (Longitudinal separation), $b_T$ (Transverse separation) |
| Target Observable | $\widetilde{A}_{2B}^{\text{Re}}(\eta, b_L, b_T)$ (Lattice momentum $P_1 = 1$, $b \geq 3a$, $\eta \in [6, 10]$) |

### 6.4.2 The Physics-Informed Custom Loss Function

Standard symbolic regression optimizes metrics like Mean Squared Error (MSE). However, a genetic algorithm might discover a polynomial or diverging exponential function that fits the finite $(b_L, b_T)$ lattice grid well, but diverges as $b_L \to \infty$. As shown in Section 6.1, this would cause the subsequent Fourier transform over $P \cdot b$ to diverge.

We implemented a custom loss function within `PySR` to enforce the required asymptotic decay. The reduced chi-squared statistic determines the base accuracy of the model:

$$\mathcal{L}_{\text{base}} = \frac{\chi^2}{\text{dof}} = \frac{1}{N-k} \sum_{i=1}^{N} \left( \frac{\widetilde{A}_{2B}^{\text{Re, lattice}}(i) - \widetilde{A}_{2B}^{\text{Re, model}}(i)}{\sigma_i} \right)^2 \tag{6.4}$$

Here, $N$ is the number of lattice data points, $k$ is the number of degrees of freedom in the generated expression, and $\sigma_i$ is the statistical error of the $i$-th lattice point.

To enforce the asymptotic decay required for Fourier integrability, we introduce a penalty term. We evaluate the proposed model equation at large values of the longitudinal separation, $b_L^{\text{large}}$, while keeping the transverse separations $b_T$ fixed to physical slices. We compute the sum of the absolute model predictions in this asymptotic regime:

$$\mathcal{P}_{\text{asymptotic}} = \sum_{b_T \in \{b_T^{\text{fixed}}\}} \sum_{b_L \in \{b_L^{\text{large}}\}} \left| \widetilde{A}_{2B}^{\text{Re, model}}(b_L, b_T) \right| \tag{6.5}$$

If a generated equation diverges or grows at large $b_L$, the penalty term $\mathcal{P}_{\text{asymptotic}}$ increases, lowering the equation's fitness score. If the equation decays toward zero, the penalty vanishes.

We combine these terms into the total objective function for `PySR`, with the penalty scaled by a small regularization parameter $\epsilon$:

$$\mathcal{L}_{\text{custom}} = \mathcal{L}_{\text{base}} + \epsilon\, \mathcal{P}_{\text{asymptotic}} \tag{6.6}$$

$$\mathcal{L}_{\text{custom}} = \frac{\chi^2}{\text{dof}} + \epsilon \sum_{b_T \in \{b_T^{\text{fixed}}\}} \sum_{b_L \in \{b_L^{\text{large}}\}} \left| \widetilde{A}_{2B}^{\text{Re, model}}(b_L, b_T) \right| \tag{6.7}$$

By evaluating fitness with $\mathcal{L}_{\text{custom}}$, `PySR` restricts the search to mathematical expressions that fit the finite-lattice data and satisfy the asymptotic decay constraints. The Julia implementation of this objective function is shown below. The asymptotic penalty evaluates each candidate expression at five test coordinates in the regime $b_L \gg 1$, spanning staple extents $\eta \in \{6, 8, 10\}$ and longitudinal separations up to $b_L = 80a$. We set the regularization weight to $\epsilon = 0.1$.

```
using Symbolics

function eval_loss(tree, dataset::Dataset{T,L}, options)::L where {T,L}
    prediction, flag = eval_tree_array(tree, dataset.X, options)
    if !flag
        return L(Inf)
    end

    # 1. Base Chi-Squared per degree of freedom
    wmse = sum(((prediction .- dataset.y) .^ 2) .* dataset.weights) / (dataset.n - 3)

    # 2. Physics boundary condition: penalty for growth at large b
    points = [
        [10.0, 50.0, 1.0],
        [10.0, 80.0, 1.0],
        [10.0, 10.0, 1.0],
        [8.0,  20.0, 1.0],
        [6.0,  30.0, 2.0]
    ]

    penalty = 0.0
    for p in points
        Xp = reshape(p, :, 1)
        y_pred, ok = eval_tree_array(tree, Xp, options)
        if ok
            penalty += (y_pred[1])^2
        else
            penalty += 1e6
        end
    end

    return L(wmse + 0.1 * penalty)
end
```

### 6.4.3 Pareto Front of Discovered Models

Symbolic regression outputs a Pareto front of candidate models: a ranked list of equations that trade off mathematical simplicity (measured by expression tree complexity $C$) and predictive accuracy (measured

by the custom loss $\mathcal{L}_{\text{custom}}$). Table 12 lists the Pareto front for the real unpolarized amplitude $\widetilde{A}_{2B}^{\text{Re}}$.

Table 12: Pareto front of analytical candidate models discovered by `PySR` for the real amplitude $\widetilde{A}_{2B}^{\text{Re}}(\eta, b_L, b_T)$, ranked by mathematical complexity and loss. The optimal physics-informed models ($C = 14$ and $C = 16$) are highlighted.

| **Complexity** | **Loss** | **Discovered Analytical Expression $\widetilde{A}_{2B}^{\text{Re}}(\eta, b_L, b_T)$** | **Physical Interpretation** |
|---|---|---|---|
| 1 | 276.43 | 0.00504 | Flat constant baseline |
| 3 | 171.56 | $(0.4988)^{\eta} \approx e^{-0.695\eta}$ | Pure exponential in $\eta$ only |
| 4 | 167.57 | $\exp\left(-\eta^{0.808}\right)$ | Stretched exponential in $\eta$ |
| 7 | 68.93 | $\frac{11.58-\eta}{(1+b_L^2+b_T^2)^2}$ | Linear in $\eta$ × standard Lorentzian |
| 9 | 30.81 | $\frac{10.16-\eta}{(1+b_L^2+b_T^2)^{1.78}}$ | Linear in $\eta$ × generalized Lorentzian |
| 11 | 5.16 | $\frac{2473.5}{\eta^4(1+b_L^2+b_T^2)^{1.44}}$ | Power-law in $\eta$ ($\eta^{-4}$) × Power-law in $b$ |
| 12 | 4.76 | $\left(1-\frac{14.49}{\eta}\right)^2 \frac{1}{(1+b_L^2+b_T^2)^{1.46}}$ | Rational in $\eta$ × Power-law in $b$ |
| **14** | **3.93** | $\mathbf{\frac{39.71}{(1+b_L^2+b_T^2)^{1.39}}\, e^{-0.534\eta}}$ | **Optimal factorized:** $e^{-0.534\eta} \times (1+b^2)^{-1.39}$ |
| **16** | **2.03** | $\mathbf{\frac{245.5}{(1+b_L^2+b_T^2)^{1.26}}\, e^{-0.468\eta} - 0.00127}$ | **Best-fit factorized with offset ($\chi^2/\text{dof} = 2.03$)** |
| 17 | 1.99 | $\frac{1}{\eta}\left(\frac{10.46}{\eta}\right)^{2.34} \frac{1.68}{(1+b_L^2+b_T^2)^{1.21}} - 0.00166$ | High-order power-law $\eta^{-3.34}\times$ Power-law in $b$ |
| 18 | 1.91 | $\frac{0.28}{\eta^2}(17.98-\eta)^2 \frac{1}{(1+b_L^2+b_T^2)^{1.25}} - 0.00137$ | Polynomial in $\eta$ × Power-law in $b$ |
| 19 | 1.76 | $\frac{1}{\eta}\left(\frac{10.38}{\eta}\right)^{2.37}\left(\frac{1.39}{(1+b^2)^{1.13}} - 0.00322\right) - 0.00149$ | Asymptotic refinement |
| 21 | 1.76 | $\frac{1}{\eta}\left(\frac{10.30}{\eta}\right)^{2.43} \frac{1.40}{(1+b^2)^{1.13}} - \frac{0.00319}{\eta} - 0.00139$ | Multi-scale refinement |

At the lowest complexities ($C \leq 4$), `PySR` isolates the overall $\eta$-dependence of the amplitude as an exponential decay $\sim (0.499)^{\eta} \approx e^{-0.695\eta}$. At intermediate complexities ($C = 7$ to 12), the algorithm introduces spatial $b$-dependence through Lorentzian-type power-law denominators $(1+b_L^2+b_T^2)^{-j}$ alongside linear or power-law $\eta$-models. The algorithm does not select a Gaussian spatial profile $e^{-cb^2}$ at any point along the Pareto front. Instead, it converges on the power-law tail structure.

The models at complexities $C = 14$ and $C = 16$ best balance accuracy and structure. The $C = 14$ expression has a factorized form:

$$\widetilde{A}_{2B}^{\text{Re}} = \frac{39.71}{(1+b_L^2+b_T^2)^{1.39}}\, e^{-0.534\eta}\ , \tag{6.8}$$

which is a product of an exponential decay in $\eta$ and a generalized Lorentzian power-law decay in

$b^2 = b_L^2 + b_T^2$. The $C = 16$ model introduces a small additive constant offset of $-0.00127$, reducing the loss to $\mathcal{L}_{\text{custom}} = 2.03$:

$$\widetilde{A}_{2B}^{\text{Re}} = \frac{245.5}{(1 + b_L^2 + b_T^2)^{1.26}}\, e^{-0.468\eta} - 0.00127 \ . \tag{6.9}$$

Beyond $C = 16$, added complexity yields only marginal improvements in loss while reducing interpretability.

The power-law tail $(1 + b^2)^{-j}$ decays more slowly than a Gaussian but, for $j > 1/2$, is absolutely integrable over the real line, so unlike the Gaussian model of Section 6.1 it guarantees convergent Fourier integrals. The factorized structure $e^{-f\eta} \times g(b^2)$ also allows the shared $\eta$-dependent exponential to cancel analytically in the Sivers shift ratio, which we use in Section 6.8.

As shown in Figure 38, the factorized $C = 14$ and $C = 16$ expressions track the lattice data throughout the accessible $(b_L, b_T)$ grid across all staple extents. The pure power-law-$\eta$ models ($C = 11, 17$) show slight systematic tension at the boundary $\eta$ values, where their $\eta^{-n}$ dependence departs from the exponential decay.

The straight-line diagnostic in Figure 39 confirms the preference for a power-law tail over Gaussian fall-off: the lattice data curve upward at large $b^2$ rather than following the Gaussian's expected straight line, consistent with the generalized Lorentzian power-law tail discovered by the genetic algorithm.

The `PySR` search also has a structural limitation. Every model on the front from $C = 7$ to $C = 21$ depends on $b_L$ and $b_T$ only through $b_L^2 + b_T^2 = b^2$, with no separate parameter for $b_L$. Since $b_L = (P \cdot b)/(-P_L)$, these expressions are independent of $P \cdot b$ once mapped to the Lorentz-invariant variables, and they cannot describe the $P \cdot b$ dependence that the Fourier transform to $x$ requires. `PySR` therefore served as a guide for the $b^2$ and $\eta$ dependence, not as the final model. Symbolic regression scales poorly with the number of inputs; a recent `PySR` extraction of the unpolarized TMD from Drell–Yan data likewise needed a factorized ansatz to keep the search tractable [119]. Starting from Gaussian, exponential decay, and power-law forms, we systematically explored functions with $P \cdot b$ dependence in Section 6.6 and tested separable alternatives in Section 6.9.

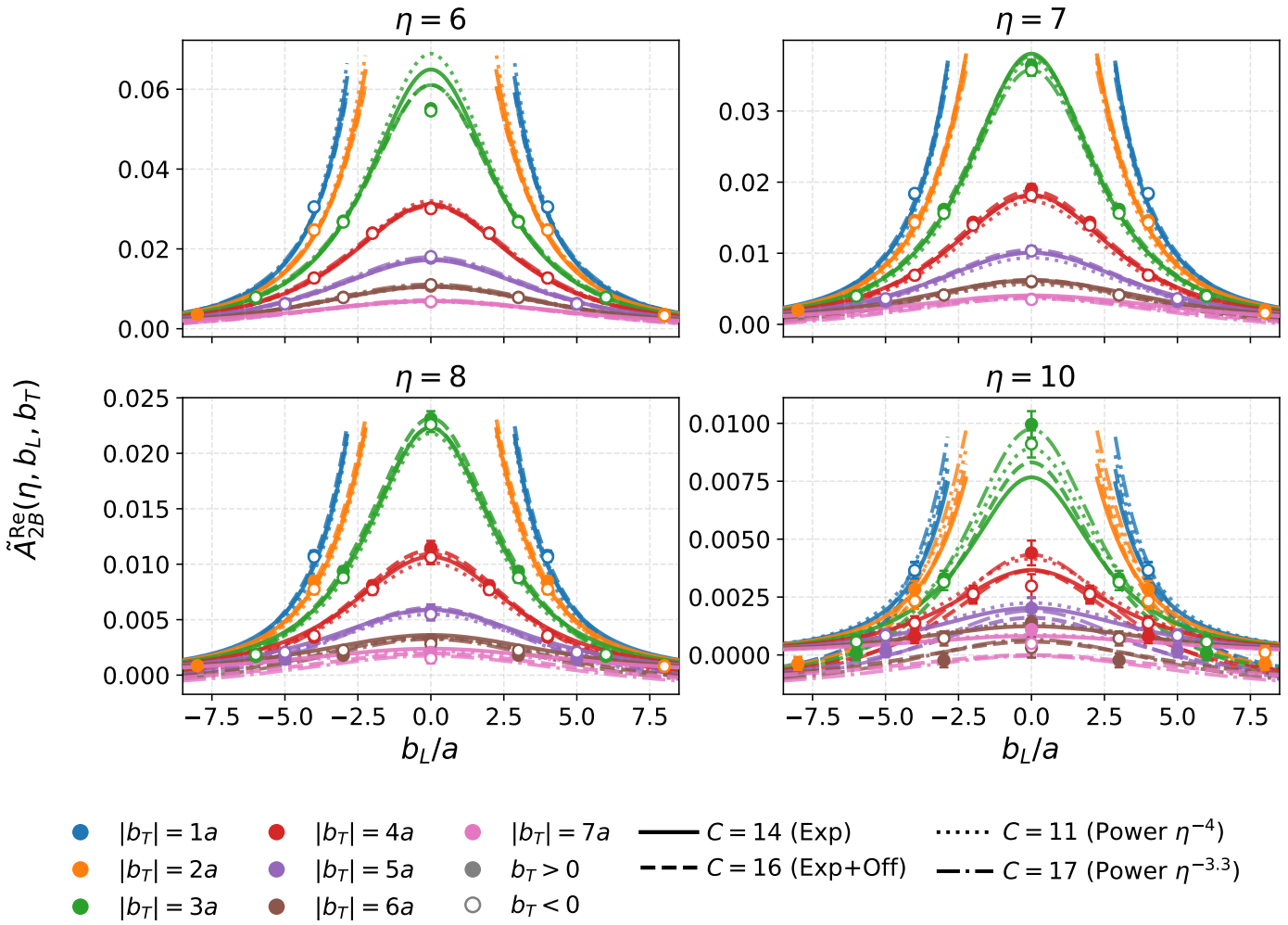



Figure 38: The top PySR analytical models ($C = 11, 14, 16, 17$) compared with the lattice data for $\widetilde{A}_{2B}^{\mathrm{Re}}$ at fixed staple extents $\eta = 6, 7, 8, 10$ (with $P_1 = -1$ and $|\boldsymbol{b}| \geq 3a$). Each panel shows the amplitude as a function of the longitudinal separation $b_L/a$ at fixed transverse separations $b_T$, with the four Pareto-front models overlaid. The model curves are drawn only in the fitted region $|\boldsymbol{b}| \geq 3a$. The two exponential-$\eta$ models ($C = 14$, solid; $C = 16$, dashed) give the most balanced description of the data across all $\eta$ values, while the pure power-law-$\eta$ models ($C = 11$, dotted; $C = 17$, dot-dashed) deviate slightly and systematically at the ends of the $\eta$ range.

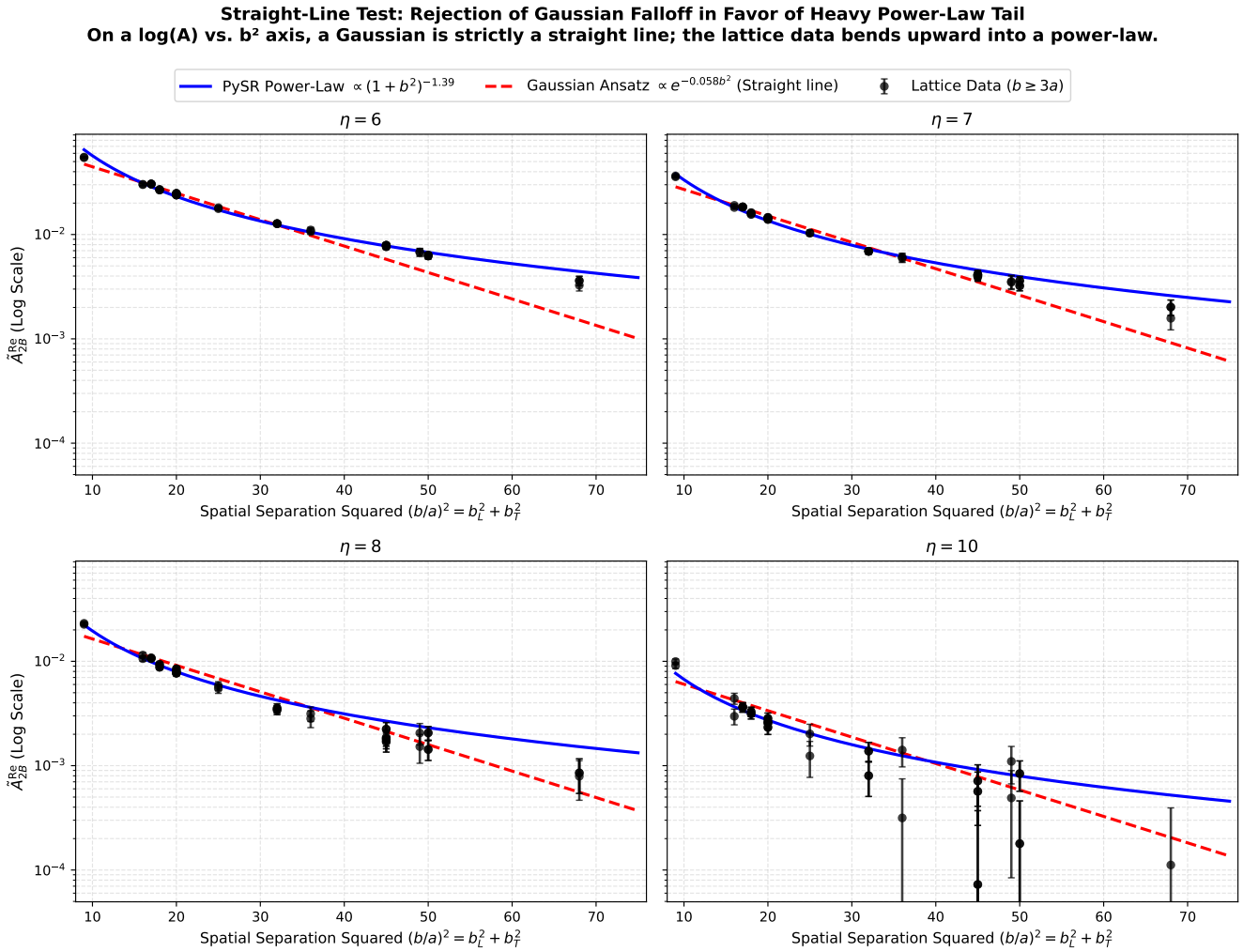



Figure 39: Diagnostic straight-line test for the Gaussian hypothesis. If the amplitude followed a pure Gaussian profile $\widetilde{A}_{2B}^{\mathrm{Re}} \propto e^{-c\,b^2}$, the data would appear as a straight line on a logarithmic ordinate versus $b^2$ (red dashed line, best Gaussian fit with $c = 0.058$). The lattice data (black markers) curve upward away from the Gaussian line at large $b^2$, indicating a heavier-than-Gaussian tail, which the `PySR` power-law form $(1 + b^2)^{-1.39}$ (blue solid line) accurately captures. The curvature is direct, model-independent evidence that the spatial decay of the non-local quark bilinear follows an algebraic power law rather than a Gaussian exponential.

### 6.5 Failure of Exponential Decay Model

An exponential fall-off is a physically motivated alternative to the power law. For Euclidean quark bilinears with a straight Wilson line, the correlator at large separation is a sum of exponential decays $e^{-\Lambda|z|}$ with power-law corrections in $1/|z|$, and the mass gap $\Lambda \approx 0.5$ GeV is set by the binding energy of a heavy-light meson [120]. That derivation does not cover staple-shaped links with $b_T \neq 0$, so we test exponential decay directly on our data. The Pareto front in Table 12, however, contains no exponential decay in $|\boldsymbol{b}|$. The search used $(\eta, b_L, b_T)$ as inputs and had no square-root operator, so $e^{-c|\boldsymbol{b}|}$ has to be built from several nodes ($C \approx 19$ including the $\eta$ factor), while the lor operator of Table 11 gives a power law in $b^2$ at much lower complexity. Fitting $e^{-c|\boldsymbol{b}|}e^{-f\eta}$ directly to the same data gives a loss of about 2.63, and adding a constant offset, as in the $C = 16$ model, lowers it to 2.26. Both are above the loss of the simpler $C = 16$ model (2.03), so neither form would have stayed on the front. We therefore tested exponential decay directly, first in $|\boldsymbol{b}|$ and then in the longitudinal variable $P \cdot b$.

In Figure 40 the data lie close to the straight line of $e^{-c|\boldsymbol{b}|}$ over the accessible range $3 \leq |\boldsymbol{b}|/a \leq 8.2$, so $\widetilde{A}^{\mathrm{Re}}_{2B}$ alone does not favor the power law.

To test an exponential decay along the longitudinal direction, we fit separable models that multiply a longitudinal profile, $e^{-c|P\cdot b|}$ or $e^{-c(P\cdot b)^2}$, by a transverse profile, $e^{-d|\boldsymbol{b}|}$, $e^{-db^2}$ or $(1 + db^2)^{-j}$, with the same factor $e^{-f\eta}$ (Table 13). With the Gaussian transverse profile the fit is poor ($\chi^2/\mathrm{dof} \approx 4.98$). The other four combinations reach $\chi^2/\mathrm{dof} \approx 1.9$ to 2.2, but in each of them the longitudinal coefficient $c$ is consistent with zero and its central value is negative. Without a positive $c$ the model does not decay in $P \cdot b$, and the Fourier transform to $x$ diverges.

Letting the longitudinal coefficient depend on $\eta$, as $c(\eta) = c\,(1 - k_1/\eta + k_2/\eta^2)$ (the $\eta$ dependency is explored more systematically below in Section 6.6, nevertheless we use this simpler $c(\eta)$), changes this (Table 14). The fitted $c$ is now positive, so $c(\eta)$ is positive in the asymptotic limit $\eta \to \infty$ that the Sivers shift requires (Section 6.3), and the best combinations reach $\chi^2/\mathrm{dof} \approx 1.32$. We explore the

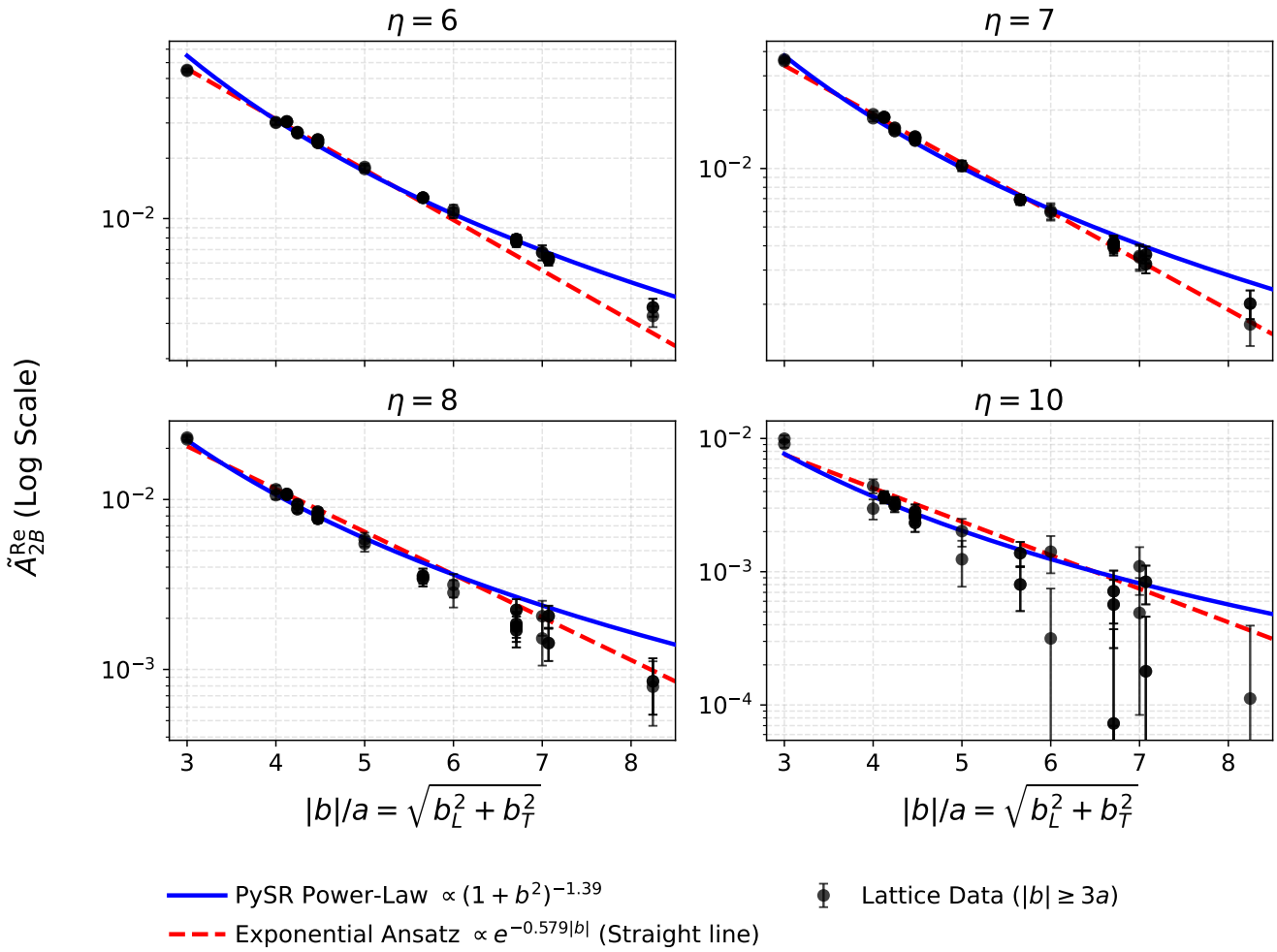



Figure 40: Straight-line test for an exponential decay in $|\boldsymbol{b}| = \sqrt{b^2}$. A profile $\widetilde{A}^{\mathrm{Re}}_{2B} \propto e^{-c|\boldsymbol{b}|}$ is a straight line on a logarithmic ordinate versus $|\boldsymbol{b}|/a$ (red dashed line, best fit $c = 0.579$ with the same $\eta$ factor as the Gaussian baseline of Figure 39). The PySR power law $(1 + b^2)^{-1.39}$ (blue solid line) is shown for comparison. Over $3 \leq |\boldsymbol{b}|/a \leq 8.2$ the exponential describes $\widetilde{A}^{\mathrm{Re}}_{2B}$ at least as well as the power law.

$(b_L, b_T, \eta)$ Fits for Re $\tilde{A}_{2B}$ for $P_L$=−1
with $b \geq 3$ and $6 \geq \eta \geq 10$

| Model Expression | Params List | Fitted Values | $\chi^2$/DoF |
|---|---|---|---|
| $a\, e^{-f\,\eta}\, e^{-c\,\lvert b_L\, P_L\rvert}\, e^{-d\,\sqrt{b_L^2+b_T^2}}$ | {a, c, d, f} | {⟨8.02(56)⟩$_{2902\,J}$, ⟨−0.0021(76)⟩$_{2902\,J}$, ⟨0.579(13)⟩$_{2902\,J}$, ⟨0.5348(84)⟩$_{2902\,J}$} | 2.15678 |
| $a\, e^{-f\,\eta}\, e^{-c\,\lvert b_L\, P_L\rvert}\, e^{-d\,\left(b_L^2+b_T^2\right)}$ | {a, c, d, f} | {⟨2.08(13)⟩$_{2902\,J}$, ⟨0.0199(73)⟩$_{2902\,J}$, ⟨0.0564(15)⟩$_{2902\,J}$, ⟨0.5364(86)⟩$_{2902\,J}$} | 4.98308 |
| $\frac{a\, e^{-f\,\eta}\, e^{-c\,\lvert b_L\, P_L\rvert}}{\left(1+d\,\left(b_L^2+b_T^2\right)\right)^{j2}}$ | {a, c, d, f, j2} | {⟨5.28(60)⟩$_{2902\,J}$, ⟨−0.0089(84)⟩$_{2902\,J}$, ⟨0.094(16)⟩$_{2902\,J}$, ⟨0.5344(84)⟩$_{2902\,J}$, ⟨2.07(13)⟩$_{2902\,J}$} | 1.90543 |
| $a\, e^{-f\,\eta}\, e^{-c\,(b_L\, P_L)^2}\, e^{-d\,\sqrt{b_L^2+b_T^2}}$ | {a, c, d, f} | {⟨8.18(58)⟩$_{2902\,J}$, ⟨−0.0018(17)⟩$_{2902\,J}$, ⟨0.587(14)⟩$_{2902\,J}$, ⟨0.5348(85)⟩$_{2902\,J}$} | 2.12445 |
| $\frac{a\, e^{-f\,\eta}\, e^{-c\,(b_L\, P_L)^2}}{\left(1+d\,\left(b_L^2+b_T^2\right)\right)^{j2}}$ | {a, c, d, f, j2} | {⟨4.99(50)⟩$_{2902\,J}$, ⟨−0.0015(16)⟩$_{2902\,J}$, ⟨0.085(13)⟩$_{2902\,J}$, ⟨0.5343(84)⟩$_{2902\,J}$, ⟨2.14(14)⟩$_{2902\,J}$} | 1.90851 |

Table 13: Separable fits of $\widetilde{A}^{\mathrm{Re}}_{2B}$ at $P_L = -1$ ($|\boldsymbol{b}| \geq 3a$, $6 \leq \eta \leq 10$) with an exponential or Gaussian longitudinal profile in $P \cdot b$ and an exponential, Gaussian or power-law transverse profile. The Gaussian transverse profile gives a poor fit ($\chi^2/\mathrm{dof} \approx 4.98$). In the other models $c$ is consistent with zero within Jackknife uncertainties, so the fits show no longitudinal decay.

choice of $\eta$-dependence for $c$ in Section 6.6.

$(b_L, b_T, \eta)$ Fits for Re $\tilde{A}_{2B}$ for $P_L$=−1
with $b \geq 3$ and $6 \geq \eta \geq 10$

| Model Expression | Params List | Fitted Values | $\chi^2$/DoF |
|---|---|---|---|
| $a\, e^{-f\eta}\, e^{-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\lvert b_L P_L\rvert}\, e^{-d\sqrt{b_L^2+b_T^2}}$ | {a, c, k1, k2, d, f} | {⟨5.28(42)⟩$_{2902\,J}$, ⟨0.51(12)⟩$_{2902\,J}$, ⟨11.60(44)⟩$_{2902\,J}$, ⟨−32.4(27)⟩$_{2902\,J}$, ⟨0.580(13)⟩$_{2902\,J}$, ⟨0.473(11)⟩$_{2902\,J}$} | 1.57561 |
| $a\, e^{-f\eta}\, e^{-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\lvert b_L P_L\rvert}\, e^{-d\left(b_L^2+b_T^2\right)}$ | {a, c, k1, k2, d, f} | {⟨1.36(10)⟩$_{2902\,J}$, ⟨0.53(13)⟩$_{2902\,J}$, ⟨11.09(49)⟩$_{2902\,J}$, ⟨−30.8(28)⟩$_{2902\,J}$, ⟨0.0564(15)⟩$_{2902\,J}$, ⟨0.473(12)⟩$_{2902\,J}$} | 4.38385 |
| $\frac{a\, e^{-f\eta}\, e^{-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\lvert b_L P_L\rvert}}{\left(1+d\left(b_L^2+b_T^2\right)\right)^{j_2}}$ | {a, c, k1, k2, d, f, j2} | {⟨3.53(47)⟩$_{2902\,J}$, ⟨0.49(12)⟩$_{2902\,J}$, ⟨11.79(45)⟩$_{2902\,J}$, ⟨−33.0(26)⟩$_{2902\,J}$, ⟨0.096(16)⟩$_{2902\,J}$, ⟨0.473(11)⟩$_{2902\,J}$, ⟨2.06(12)⟩$_{2902\,J}$} | 1.3183 |
| $a\, e^{-f\eta}\, e^{-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\left(b_L P_L\right)^2}\, e^{-d\sqrt{b_L^2+b_T^2}}$ | {a, c, k1, k2, d, f} | {⟨5.94(48)⟩$_{2902\,J}$, ⟨0.114(30)⟩$_{2902\,J}$, ⟨11.96(44)⟩$_{2902\,J}$, ⟨−34.3(26)⟩$_{2902\,J}$, ⟨0.588(14)⟩$_{2902\,J}$, ⟨0.487(10)⟩$_{2902\,J}$} | 1.54574 |
| $\frac{a\, e^{-f\eta}\, e^{-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\left(b_L P_L\right)^2}}{\left(1+d\left(b_L^2+b_T^2\right)\right)^{j_2}}$ | {a, c, k1, k2, d, f, j2} | {⟨3.68(38)⟩$_{2902\,J}$, ⟨0.110(29)⟩$_{2902\,J}$, ⟨12.02(44)⟩$_{2902\,J}$, ⟨−34.8(26)⟩$_{2902\,J}$, ⟨0.084(13)⟩$_{2902\,J}$, ⟨0.4896(98)⟩$_{2902\,J}$, ⟨2.16(14)⟩$_{2902\,J}$} | 1.35274 |

Table 14: Separable fits as in Table 13, with an $\eta$-dependent longitudinal coefficient $c(\eta) = c\,(1 - k_1/\eta + k_2/\eta^2)$. The fitted $c$ is positive in every model, so $c(\eta)$ is positive as $\eta \to \infty$.

These tests use $\widetilde{A}_{2B}^{\mathrm{Re}}$ alone. In Section 6.9 we apply exponential and Gaussian longitudinal profiles, with exponential and power-law transverse envelopes, to all three amplitudes and discuss there why these separable models, including the ones with exponential decay in $b$, fail for $\widetilde{A}_{2B}^{\mathrm{Im}}$ and $\widetilde{A}_{12B}^{\mathrm{Re}}$.

### 6.6 Systematic Exploration of the Power-Law Model

Guided by the PySR result, we now explore the $\eta$-dependence of the power-law model in more detail. We fit $\widetilde{A}_{2B}^{\mathrm{Re}}$ at momentum $P_L = -1$, with $6 \le \eta \le 10$ and $|\boldsymbol{b}| \ge 3a$. For comparison, we first refit a family of Gaussian ansätze with $\eta$-dependent widths (Table 15); for each model, the figure lists the model expression, the fit parameters, their Jackknife-averaged values, and the resulting $\chi^2$/DoF.

$(b_L, b_T, \eta)$ Fits for Re $\tilde{A}_{2B}$ for $P_L$=-1
with $b \geq 3$ and $6 \geq \eta \geq 10$

| Model Expression | Params List | Fitted Values | $\chi^2$/DoF |
|---|---|---|---|
| $a\, e^{-f\eta}\, e^{-c\,(b_L P_L)^2 - d\,(b_L^2+b_T^2)}$ | {a, c, d, f} | {⟨2.04(13)⟩$_{2902\,J}$, ⟨0.0005 ± 0.0018⟩$_{2902\,J}$, ⟨0.0578(16)⟩$_{2902\,J}$, ⟨0.5367(86)⟩$_{2902\,J}$} | 5.19673 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}\right)(b_L P_L)^2 - d\,(b_L^2+b_T^2)\right)$ | {a, c, kc1, d, f} | {⟨1.48(11)⟩$_{2902\,J}$, ⟨0.0397(89)⟩$_{2902\,J}$, ⟨6.45(29)⟩$_{2902\,J}$, ⟨0.0579(16)⟩$_{2902\,J}$, ⟨0.489(11)⟩$_{2902\,J}$} | 4.64804 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1+\frac{k_1}{\eta}-\frac{k_2}{\sqrt{\eta}}\right)(b_L P_L)^2 - d\,(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, f} | {⟨1.43(11)⟩$_{2902\,J}$, ⟨0.34(11)⟩$_{2902\,J}$, ⟨5.45(38)⟩$_{2902\,J}$, ⟨4.69(16)⟩$_{2902\,J}$, ⟨0.0579(16)⟩$_{2902\,J}$, ⟨0.483(12)⟩$_{2902\,J}$} | 4.58804 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)(b_L P_L)^2 - d\,(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, f} | {⟨1.43(11)⟩$_{2902\,J}$, ⟨0.122(33)⟩$_{2902\,J}$, ⟨11.61(48)⟩$_{2902\,J}$, ⟨-32.8(29)⟩$_{2902\,J}$, ⟨0.0579(16)⟩$_{2902\,J}$, ⟨0.483(12)⟩$_{2902\,J}$} | 4.58644 |
| $a\, e^{-f\eta} \exp\!\left(-c\,(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}\right)(b_L^2+b_T^2)\right)$ | {a, c, d, kd1, f} | {⟨0.742(72)⟩$_{2902\,J}$, ⟨0.0004 ± 0.0018⟩$_{2902\,J}$, ⟨0.1216(90)⟩$_{2902\,J}$, ⟨3.41(18)⟩$_{2902\,J}$, ⟨0.384(17)⟩$_{2902\,J}$} | 4.08215 |
| $a\, e^{-f\eta} \exp\!\left(-c\,(b_L P_L)^2 - d\left(1+\frac{l_1}{\eta}-\frac{l_2}{\sqrt{\eta}}\right)(b_L^2+b_T^2)\right)$ | {a, c, d, kd1, kd2, f} | {⟨0.458(65)⟩$_{2902\,J}$, ⟨0.0002 ± 0.0017⟩$_{2902\,J}$, ⟨0.661(97)⟩$_{2902\,J}$, ⟨4.60(19)⟩$_{2902\,J}$, ⟨4.14(10)⟩$_{2902\,J}$, ⟨0.311(24)⟩$_{2902\,J}$} | 3.67038 |
| $a\, e^{-f\eta} \exp\!\left(-c\,(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}-\frac{l_2}{\eta^2}\right)(b_L^2+b_T^2)\right)$ | {a, c, d, kd1, kd2, f} | {⟨0.459(65)⟩$_{2902\,J}$, ⟨0.0002 ± 0.0017⟩$_{2902\,J}$, ⟨0.287(34)⟩$_{2902\,J}$, ⟨8.91(33)⟩$_{2902\,J}$, ⟨-23.9(13)⟩$_{2902\,J}$, ⟨0.311(24)⟩$_{2902\,J}$} | 3.66774 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, d, kd1, f} | {⟨0.737(70)⟩$_{2902\,J}$, ⟨0.015(10)⟩$_{2902\,J}$, ⟨6.03(69)⟩$_{2902\,J}$, ⟨0.1147(96)⟩$_{2902\,J}$, ⟨3.23(22)⟩$_{2902\,J}$, ⟨0.383(17)⟩$_{2902\,J}$} | 4.04672 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}\right)(b_L P_L)^2 - d\left(1+\frac{l_1}{\eta}-\frac{l_2}{\sqrt{\eta}}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, d, kd1, kd2, f} | {⟨0.462(63)⟩$_{2902\,J}$, ⟨0.015(11)⟩$_{2902\,J}$, ⟨6.06(74)⟩$_{2902\,J}$, ⟨0.647(95)⟩$_{2902\,J}$, ⟨4.71(20)⟩$_{2902\,J}$, ⟨4.18(11)⟩$_{2902\,J}$, ⟨0.313(23)⟩$_{2902\,J}$} | 3.67204 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}-\frac{l_2}{\eta^2}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, d, kd1, kd2, f} | {⟨0.463(63)⟩$_{2902\,J}$, ⟨0.015(11)⟩$_{2902\,J}$, ⟨6.06(74)⟩$_{2902\,J}$, ⟨0.277(34)⟩$_{2902\,J}$, ⟨8.95(33)⟩$_{2902\,J}$, ⟨-24.4(14)⟩$_{2902\,J}$, ⟨0.313(23)⟩$_{2902\,J}$} | 3.66931 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1+\frac{k_1}{\eta}-\frac{k_2}{\sqrt{\eta}}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, f} | {⟨0.568(61)⟩$_{2902\,J}$, ⟨0.66(12)⟩$_{2902\,J}$, ⟨6.79(19)⟩$_{2902\,J}$, ⟨5.222(77)⟩$_{2902\,J}$, ⟨0.130(11)⟩$_{2902\,J}$, ⟨3.61(19)⟩$_{2902\,J}$, ⟨0.344(19)⟩$_{2902\,J}$} | 3.80049 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1+\frac{k_1}{\eta}-\frac{k_2}{\sqrt{\eta}}\right)(b_L P_L)^2 - d\left(1+\frac{l_1}{\eta}-\frac{l_2}{\sqrt{\eta}}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, kd2, f} | {⟨0.464(64)⟩$_{2902\,J}$, ⟨0.06 ± 0.15⟩$_{2902\,J}$, ⟨26.8(87)⟩$_{2902\,J}$, ⟨13.1(35)⟩$_{2902\,J}$, ⟨0.62(13)⟩$_{2902\,J}$, ⟨4.67(32)⟩$_{2902\,J}$, ⟨4.16(17)⟩$_{2902\,J}$, ⟨0.313(24)⟩$_{2902\,J}$} | 3.68333 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1+\frac{k_1}{\eta}-\frac{k_2}{\sqrt{\eta}}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}-\frac{l_2}{\eta^2}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, kd2, f} | {⟨0.464(64)⟩$_{2902\,J}$, ⟨0.05 ± 0.15⟩$_{2902\,J}$, ⟨27.2(89)⟩$_{2902\,J}$, ⟨13.2(35)⟩$_{2902\,J}$, ⟨0.271(43)⟩$_{2902\,J}$, ⟨8.88(52)⟩$_{2902\,J}$, ⟨-24.0(22)⟩$_{2902\,J}$, ⟨0.313(23)⟩$_{2902\,J}$} | 3.68087 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, f} | {⟨0.568(61)⟩$_{2902\,J}$, ⟨0.188(35)⟩$_{2902\,J}$, ⟨13.64(42)⟩$_{2902\,J}$, ⟨-45.8(24)⟩$_{2902\,J}$, ⟨0.130(11)⟩$_{2902\,J}$, ⟨3.61(19)⟩$_{2902\,J}$, ⟨0.344(19)⟩$_{2902\,J}$} | 3.79855 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)(b_L P_L)^2 - d\left(1+\frac{l_1}{\eta}-\frac{l_2}{\sqrt{\eta}}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, kd2, f} | {⟨0.464(64)⟩$_{2902\,J}$, ⟨0.028(45)⟩$_{2902\,J}$, ⟨17.5(51)⟩$_{2902\,J}$, ⟨-71(32)⟩$_{2902\,J}$, ⟨0.62(13)⟩$_{2902\,J}$, ⟨4.65(32)⟩$_{2902\,J}$, ⟨4.15(17)⟩$_{2902\,J}$, ⟨0.313(24)⟩$_{2902\,J}$} | 3.68295 |
| $a\, e^{-f\eta} \exp\!\left(-c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)(b_L P_L)^2 - d\left(1-\frac{l_1}{\eta}-\frac{l_2}{\eta^2}\right)(b_L^2+b_T^2)\right)$ | {a, c, kc1, kc2, d, kd1, kd2, f} | {⟨0.464(64)⟩$_{2902\,J}$, ⟨0.026(45)⟩$_{2902\,J}$, ⟨19.3(61)⟩$_{2902\,J}$, ⟨-83(38)⟩$_{2902\,J}$, ⟨0.271(43)⟩$_{2902\,J}$, ⟨8.88(53)⟩$_{2902\,J}$, ⟨-24.0(22)⟩$_{2902\,J}$, ⟨0.313(23)⟩$_{2902\,J}$} | 3.68043 |

Table 15: 3D jackknife fits of $\widetilde{A}_{2B}^{\mathrm{Re}}(b_L, b_T, \eta)$ to a family of 16 Gaussian ansätze, all built on the common baseline $\widetilde{A}_{2B}^{\mathrm{Re}} = a\, e^{-f\eta} \exp\big[-c\,(b_L P_L)^2 - d\,(b_L^2 + b_T^2)\big]$, in which the coefficients $c$ and $d$ are each independently allowed to be either constant or one of three $\eta$-dependent forms, $c \to c(\eta)$ and/or $d \to d(\eta)$.

The coefficient $c$ of $(P \cdot b)^2$ must be positive and stable for the Fourier transform to $x$ to exist. In Table 15, most of the Gaussian fits give an unstable or negative $c$ and a higher $\chi^2$/DoF, which confirms the conclusion of Section 6.1. By contrast, the power-law ansätze in Table 16 reduce $\chi^2$/DoF

to $\approx 1.5$. We added two more columns for AIC (Akaike information criterion) and Fourier transform of the ansätze. Making $d$ depend on $\eta$ worsens the fit, while making $c$ depend on $\eta$ improves it. This raises a new ambiguity: several $\eta$-dependences give almost the same $\chi^2$/DoF. For each power-law model, the figure lists the model expression, the fit parameters, their Jackknife-averaged values, $\chi^2$/DoF, AIC, and the resulting Fourier transform in $x$.

Gaussian-type fit for Re $\tilde{A}_{2B}$ for $P_L$=-1
$b \geq 3$ are SELECTED for fit!
$6 \geq \eta \geq 10$ are SELECTED for fit!

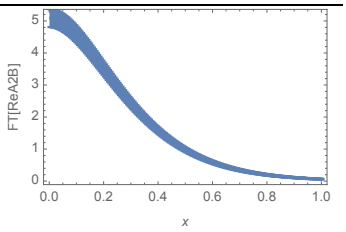
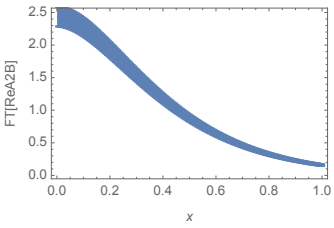
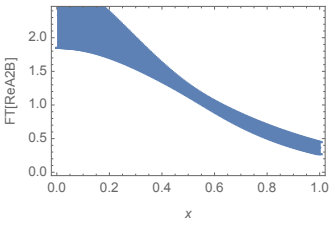
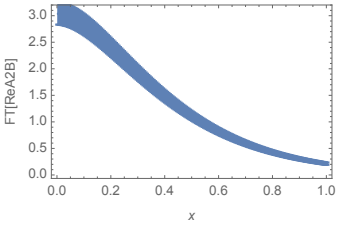
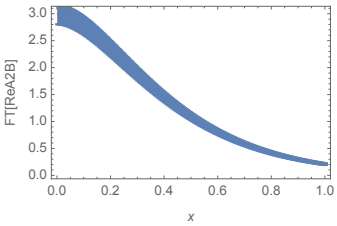
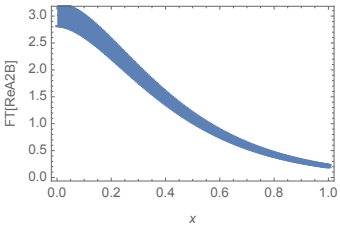
| Model Expression | Params List | Fitted Values | $\chi^2$/DoF | AIC | FT Plot |
|---|---|---|---|---|---|
| $a\, e^{-f\eta}\left(1+(c+d)\, b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f | ⟨2.19(16)⟩$_{2902\,J}$ ⟨0.0218(44)⟩$_{2902\,J}$<br>⟨0.0218(44)⟩$_{2902\,J}$ ⟨0.5382(88)⟩$_{2902\,J}$<br>⟨2.62(33)⟩$_{2902\,J}$ | 24.778 | 5461.17 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k1}{\eta}\right)\right) b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f<br>k1 j | ⟨2.85(30)⟩$_{2902\,J}$ ⟨0.078(14)⟩$_{2902\,J}$<br>⟨0.078(14)⟩$_{2902\,J}$ ⟨0.4625(89)⟩$_{2902\,J}$<br>⟨6.60(15)⟩$_{2902\,J}$ ⟨2.16(15)⟩$_{2902\,J}$ | 2.113 | 474.746 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k2}{\eta^2}-\frac{k1}{\eta}\right)\right) b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2 | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.146(51)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨12.27(50)⟩$_{2902\,J}$ ⟨-36.2(28)⟩$_{2902\,J}$<br>⟨2.06(14)⟩$_{2902\,J}$ | 1.56788 | 355.799 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k3}{\eta^3}-\frac{k2}{\eta^2}-\frac{k1}{\eta}\right)\right) b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>k3 j | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.093(18)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨6.7(52)⟩$_{2902\,J}$ ⟨36(63)⟩$_{2902\,J}$<br>⟨-220(200)⟩$_{2902\,J}$ ⟨2.06(14)⟩$_{2902\,J}$ | 1.57467 | 357.703 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k1\,\mathrm{Log}[\eta]}{\eta}\right)\right) b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f<br>k1 j | ⟨3.90(43)⟩$_{2902\,J}$ ⟨0.092(17)⟩$_{2902\,J}$<br>⟨0.092(17)⟩$_{2902\,J}$ ⟨0.4929(86)⟩$_{2902\,J}$<br>⟨3.616(83)⟩$_{2902\,J}$ ⟨2.07(14)⟩$_{2902\,J}$ | 1.66147 | 375.863 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-e^{-n\eta}\, k2-\frac{k1}{\eta}\right)\right) b_L^2+d\, b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>n j | ⟨3.98(58)⟩$_{2902\,J}$ ⟨0.093(18)⟩$_{2902\,J}$<br>⟨0.093(18)⟩$_{2902\,J}$ ⟨0.495(12)⟩$_{2902\,J}$<br>⟨7.77(98)⟩$_{2902\,J}$ ⟨82(76)⟩$_{2902\,J}$<br>⟨0.64(45)⟩$_{2902\,J}$ ⟨2.06(14)⟩$_{2902\,J}$ | 1.57582 | 357.952 |  |

Table 16: 3D Jackknife fits of $\widetilde{A}_{2B}^{\mathrm{Re}}(b_L, b_T, \eta)$ to a family of power-law ansätze, all built on the common baseline $\widetilde{A}_{2B}^{\mathrm{Re}} = a\, e^{-f\eta}\,(1+(c+d)b_L^2+db_T^2)^{-j}$, in which the coefficient $c$ is either constant or one of several $\eta$-dependent forms, $c \to c(\eta)$.

The fit parameter $c$ determines how wide the amplitudes are in Fourier-transformed space $x$; we explored more $\eta$-dependent forms, $c \to c(\eta)$, which yield wider Fourier transforms in $x$ space.

Gaussian-type fit for Re $\tilde{A}_{2B}$ for $P_L$=-1
$b \geq 3$ are SELECTED for fit!
$6 \geq \eta \geq 10$ are SELECTED for fit!

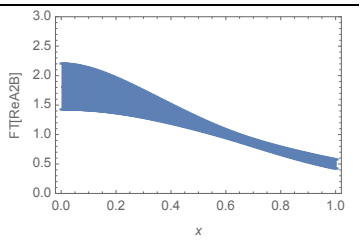
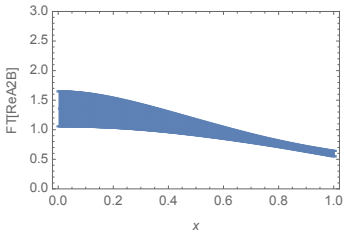
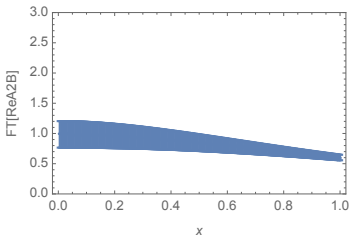
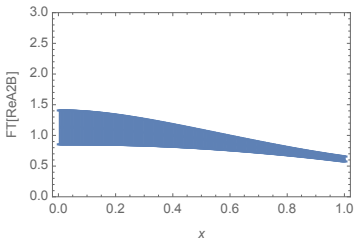
| Model Expression | Params List | Fitted Values | $\chi^2$/DoF | AIC | FT Plot ($\eta \to \infty$) |
|---|---|---|---|---|---|
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k1}{\eta}-\frac{k2\,\mathrm{Log}[\eta]}{\eta}\right)\right)b_L^2+d\,b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>j | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.240(89)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨-4.50(52)⟩$_{2902\,J}$ ⟨5.94(29)⟩$_{2902\,J}$<br>⟨2.06(14)⟩$_{2902\,J}$ | 1.56865 | 355.966 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k1}{\eta}+\frac{k2}{\sqrt{\eta}}\right)\right)b_L^2+d\,b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>j | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.43(17)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨-5.81(33)⟩$_{2902\,J}$ ⟨-4.86(14)⟩$_{2902\,J}$<br>⟨2.06(14)⟩$_{2902\,J}$ | 1.56911 | 356.065 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1-\frac{k1}{\eta}+\frac{k2\,\mathrm{Log}[\eta]}{\sqrt{\eta}}\right)\right)b_L^2+d\,b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>j | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.81(32)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨0.409(88)⟩$_{2902\,J}$ ⟨-1.284(19)⟩$_{2902\,J}$<br>⟨2.06(14)⟩$_{2902\,J}$ | 1.56961 | 356.175 |  |
| $a\, e^{-f\eta}\left(1+\left(d+c\left(1+\frac{k1\sqrt{\eta}}{1+k2\,\eta}\right)\right)b_L^2+d\,b_T^2\right)^{-j}$ | a c<br>d f<br>k1 k2<br>j | ⟨4.01(58)⟩$_{2902\,J}$ ⟨0.59(27)⟩$_{2902\,J}$<br>⟨0.094(18)⟩$_{2902\,J}$ ⟨0.496(12)⟩$_{2902\,J}$<br>⟨-0.855(43)⟩$_{2902\,J}$ ⟨0.179(17)⟩$_{2902\,J}$<br>⟨2.06(14)⟩$_{2902\,J}$ | 1.57002 | 356.264 |  |

Table 17: 3D jackknife fits of $\widetilde{A}_{2B}^{\mathrm{Re}}(b_L, b_T, \eta)$ to a family of power-law ansätze, all built on the common baseline $\widetilde{A}_{2B}^{\mathrm{Re}} = a\, e^{-f\eta}\,(1+(c+d)b_L^2+db_T^2)^{-j}$, in which the coefficient $c$ takes further $\eta$-dependent forms, $c \to c(\eta)$, that yield wider Fourier transforms in $x$ space.

All viable asymptotic functions in $\eta$ have a common power-law profile with a power $j \approx 2$ and an amplitude scale $a \approx 4$, damped by the suppression factor $e^{-0.5\eta}$. To isolate the $\eta$-dependence of the width parameters $c$ and $d$, we fix these structural parameters ($j = 2$, $a = 4$) and fit individual $\eta$ slices sequentially (Figure 41).

**Fit for Re $\tilde{A}_{2B}$ at Fixed $\eta$ ($6 \le \eta \le 10$) & $P_L = -1$**

*Spatial Cut:* $|\vec{b}| \ge 3a$ | *Model:* $(4\,e^{-0.5\,\eta})/(1 + (c + d)b_L^2 + d b_T^2)^2$

| $\eta$ | Parameters {c, d} | $\chi^2$/DoF |
|---|---|---|
| 6 | {-0.0052(18), 0.0964(13)} | 0.925 |
| 7 | {-0.0021(27), 0.0967(17)} | 0.634 |
| 8 | {0.0011(44), 0.1001(25)} | 1.846 |
| 9 | {0.0091(66), 0.0987(36)} | 2.353 |
| 10 | {0.0121(105), 0.1016(57)} | 2.384 |

Figure 41: Sequential Lorentzian fits for $\widetilde{A}_{2B}^{\mathrm{Re}}$ at individual $\eta$ slices, holding the global power $j = 2$ and amplitude $a = 4$ constant.

From these fits, we extract the parameters $c$ and $d$. The fit parameter $c$ is the coefficient of $b_L^2 = (P \cdot b)^2/P_L^2$, while $d$ is a coefficient of the invariant squared separation $b^2$. We plot the extracted values of $c$ and $d$ against our candidate asymptotic models to analyze their $\eta$-dependence:

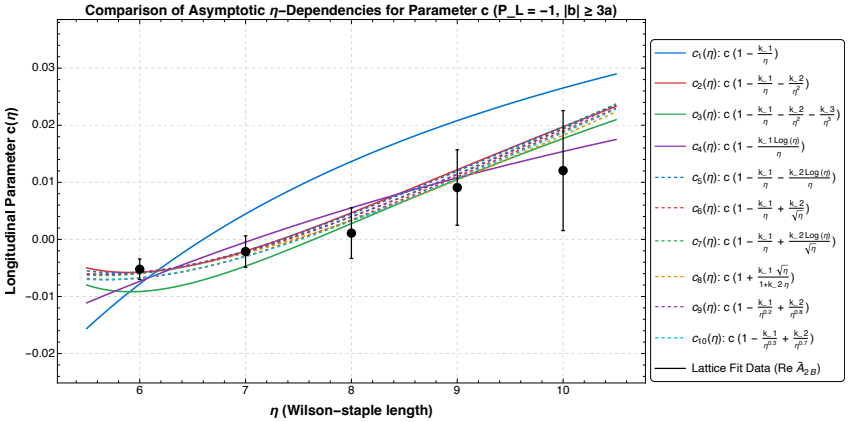



Figure 42: The extracted parameter $c(\eta)$ overlaid with the candidate asymptotic models $c_1(\eta), \ldots, c_{10}(\eta)$. The extracted values approach an asymptote at large $\eta$.

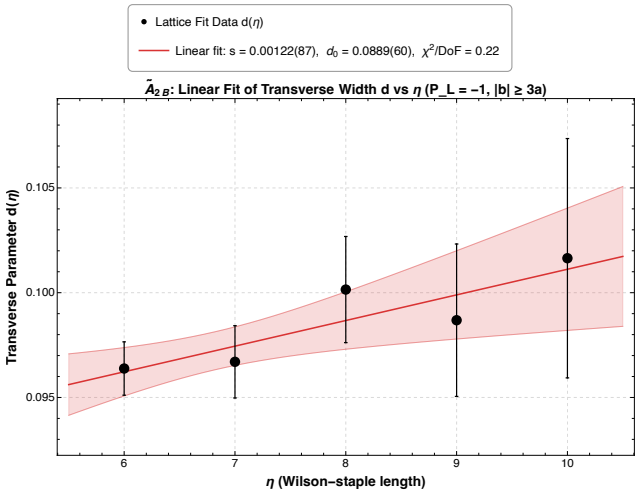



Figure 43: The extracted parameter $d(\eta)$ with a linear fit, slope $s = 0.00122(87)$ ($\chi^2/\mathrm{DoF} = 0.22$), showing no visible dependence on the staple extent $\eta$.

For the unpolarized real amplitude $\widetilde{A}^{\mathrm{Re}}_{2B}$, several of the asymptotic models in Figure 42 describe the extracted $c(\eta)$ equally well. We select $c_2(\eta) = c\left(1 - \frac{k_1}{\eta} - \frac{k_2}{\eta^2}\right)$, for the reasons given in Section 6.7. The transverse parameter $d$ is different. A linear fit $d(\eta) = d_0 + s\,\eta$ gives a slope $s = 0.00122(87)$, within $1.4\sigma$ of zero and small next to $d_0 = 0.0889(60)$, with $\chi^2/\mathrm{DoF} = 0.22$ (Figure 43). We also tried an asymptotic model for $d(\eta)$, but that fit did not converge. To avoid overfitting, we keep $d$ constant in $\eta$.

For the real Sivers amplitude $\widetilde{A}^{\mathrm{Re}}_{12B}$, we repeated the sequential Lorentzian fits at fixed $\eta$ (Table 18). Linear fits in $\eta$ give slopes of $-0.0112(131)$ for $c$ and $0.0052(48)$ for $d$, both consistent with zero within about $1\sigma$ (Figure 44). We therefore model $c$ and $d$ as constants for $\widetilde{A}^{\mathrm{Re}}_{12B}$.

**Fit for Re $\tilde{A}_{12\,B}$ at Fixed $\eta$ ($6 \le \eta \le 10$) & $P_L = -1$**

*Spatial Cut:* $|\vec{b}| \ge 3a$ | *Model:* $(-0.658\, e^{-0.4702\,\eta})/(1 + (c + d)b_L^2 + d b_T^2)^{2.0054}$

| $\eta$ | Parameters {c, d} | $\chi^2$/DoF |
|---|---|---|
| 6 | {0.0236(169), 0.0730(56)} | 0.604 |
| 7 | {0.0388(288), 0.0728(77)} | 0.860 |
| 8 | {0.0174(424), 0.0799(127)} | 0.831 |
| 9 | {−0.0105(575), 0.0925(214)} | 0.730 |
| 10 | {−0.0723(871), 0.1230(479)} | 0.468 |

Table 18: Sequential Lorentzian fits for $\widetilde{A}^{\mathrm{Re}}_{12B}$ at individual $\eta$ slices showing consistent $\chi^2$/dof and stable parameters.

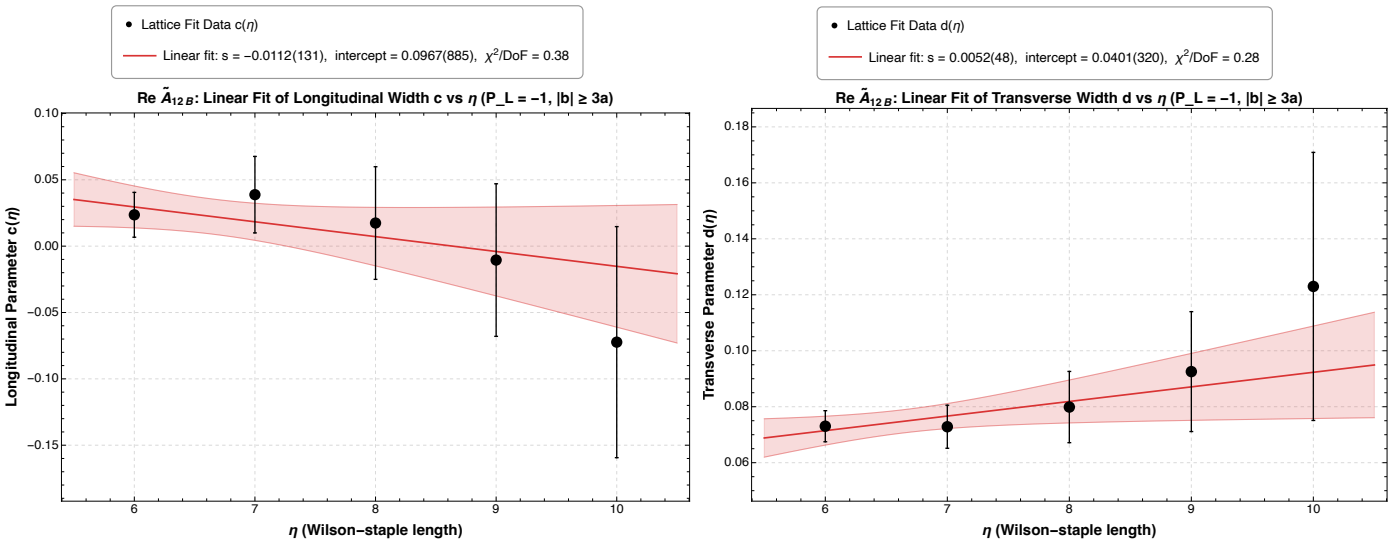



Figure 44: Lattice extracted parameters $c$ (left) and $d$ (right) for $\widetilde{A}^{\mathrm{Re}}_{12B}$ as a function of $\eta$. Both parameters show no visible dependence on $\eta$. Linear fits give slopes $-0.0112(131)$ for $c$ and $0.0052(48)$ for $d$, both consistent with zero.

For the imaginary unpolarized amplitude $\widetilde{A}^{\mathrm{Im}}_{2B}$, the lattice data are noisier (Table 19). The slope of $d$ is $-0.0316(219)$, within $1.5\sigma$ of zero, while the slope of $c$, $0.0765(350)$, differs from zero by about $2\sigma$ (Figure 45). Both linear fits already give $\chi^2/\mathrm{DoF} < 1$, and with these error bars an extra $\eta$-dependence would be poorly constrained, so we again keep $c$ and $d$ constant to avoid overfitting.

**Fit for Im $\tilde{A}_{2B}$ at Fixed $\eta$ ($6 \le \eta \le 10$) & $P_L = -1$**

*Spatial Cut:* $|\vec{b}| \ge 3a$ | *Model:* $(0.267\, b_L\, e^{-0.4702\,\eta})/(1 + (c + d)b_L^2 + d b_T^2)^{1.108}$

| $\eta$ | Parameters {c, d} | $\chi^2$/DoF |
|---|---|---|
| 6 | {0.0129(407), 0.3277(358)} | 2.033 |
| 7 | {0.0521(585), 0.3059(470)} | 1.938 |
| 8 | {0.0671(1104), 0.3346(840)} | 1.252 |
| 9 | {0.2724(1536), 0.2536(987)} | 1.040 |
| 10 | {0.6084(2810), 0.1409(1084)} | 0.645 |

Table 19: Sequential Lorentzian fits for $\widetilde{A}^{\mathrm{Im}}_{2B}$ at individual $\eta$ slices showing consistent $\chi^2$/dof.

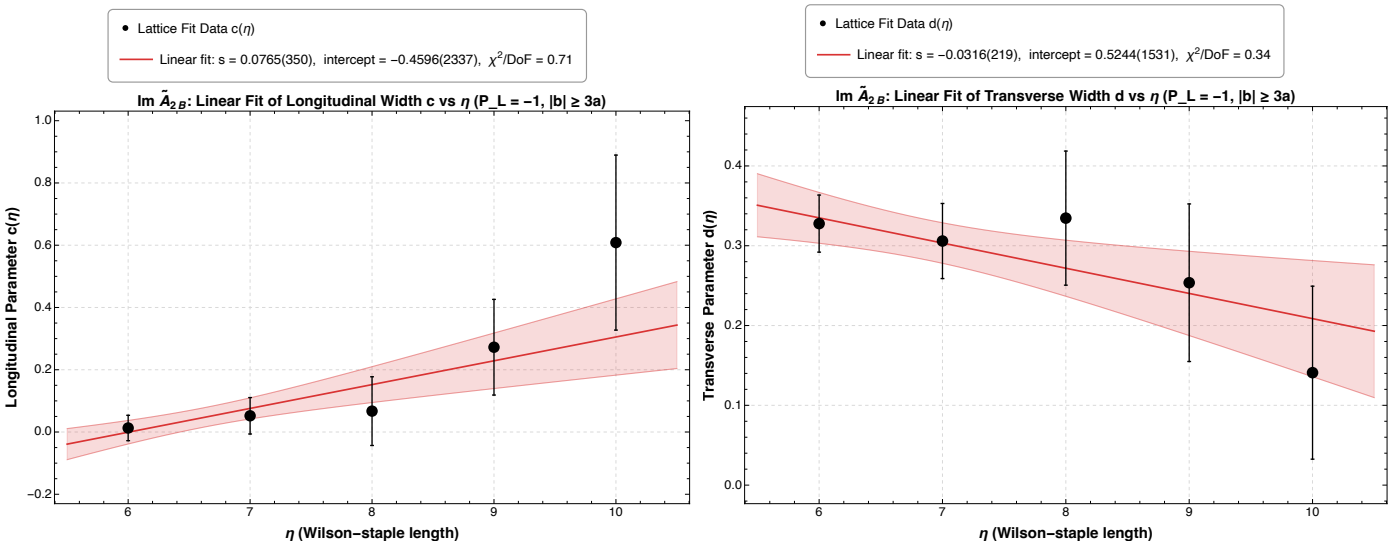



Figure 45: Lattice extracted parameters $c$ (left) and $d$ (right) for $\widetilde{A}_{2B}^{\mathrm{Im}}$ as a function of $\eta$. Linear fits give slopes 0.0765(350) for $c$ and $-0.0316(219)$ for $d$. The data are noisier than for the real amplitudes, and both parameters are kept constant.

### 6.7 Candidate Selection

The $\chi^2$/dof minimizations and the sequential fits at individual $\eta$ slices determine the structural parameterizations for the three surviving invariant amplitudes ($\widetilde{A}_{2B}^{\mathrm{Re}}$, $\widetilde{A}_{2B}^{\mathrm{Im}}$, and $\widetilde{A}_{12B}^{\mathrm{Re}}$; $\widetilde{A}_{12B}^{\mathrm{Im}}$ remains excluded, as noted above).

In selecting the model for $c(\eta)$ for the $\widetilde{A}_{2B}^{\mathrm{Re}}$ amplitude, goodness of fit must be paired with physical boundary conditions on the light-cone momentum fraction $x \in [-1, 1]$:

- **Quark versus Anti-quark contribution for unpolarized TMD**: In light-cone field theory, positive momentum fractions ($x > 0$) correspond to quark distributions $q(x) = f_1(x)$, while negative momentum fractions ($x < 0$) govern anti-quark distributions through $\bar{q}(x) = -f_1(-x)$ [2]. In the physical limit, at larger momentum, the correct candidate model should give a near-zero contribution for anti-quarks, i.e., for unpolarized TMDs' momentum fraction $x \lesssim -0.2$.

$$\left[\tilde{f}_1^{(0)}(x \lesssim -0.2) = 2\widetilde{A}_{2B}^{[x]}(x \lesssim -0.2)\right] \longrightarrow 0 \tag{6.10}$$

  where, $\widetilde{A}_{2B}^{[x]}(x) = \int_{-\infty}^{+\infty} d(b \cdot P) e^{ix(b\cdot P)} \left[\widetilde{A}_{2B}^{\mathrm{Re}}(\cdots) + i\widetilde{A}_{2B}^{\mathrm{Im}}(\cdots)\right]$. This crucial condition involves both $\widetilde{A}_{2B}^{\mathrm{Re}}$ and $\widetilde{A}_{2B}^{\mathrm{Im}}$.

- **Mellin moment ($x$-integrated) calculations**: The previous work on lattice TMDs [2, 84, 85]

calculated the first Mellin moment in $x$ ($x$-integrated TMD: $f^{[1]}(\boldsymbol{k}_{\rm T}^2;\ldots) \equiv \int_{-1}^{1} dx\, f(x,\boldsymbol{k}_{\rm T}^2;\ldots)$) of the Sivers shift, which is accessible by evaluating amplitudes at $(b\cdot P)=0$:

$$\begin{aligned}\langle \boldsymbol{k}_y\rangle_{TU}(\boldsymbol{b}_{\rm T}^2;\hat{\zeta},\eta v\cdot P) \;&\equiv\; m_N \frac{\tilde{f}_{1T}^{\perp[1](1)}(\boldsymbol{b}_{\rm T}^2;\hat{\zeta},\ldots,\eta v\cdot P)}{\tilde{f}_{1}^{[1](0)}(\boldsymbol{b}_{\rm T}^2;\hat{\zeta},\ldots,\eta v\cdot P)} = -m_N \frac{\widetilde{A}_{12B}(-\boldsymbol{b}_{\rm T}^2,0,0,-1/(m_N\hat{\zeta})^2,\eta v\cdot P)}{\widetilde{A}_{2B}(-\boldsymbol{b}_{\rm T}^2,0,0,-1/(m_N\hat{\zeta})^2,\eta v\cdot P)} \\ &\xrightarrow{\boldsymbol{b}_{\rm T}^2=0} \frac{\int dx \int d^2\boldsymbol{k}_{\rm T}\, \boldsymbol{k}_y\; \Phi^{[\gamma^+]}(x,\boldsymbol{k}_{\rm T},P,S;\ldots)}{\int dx \int d^2\boldsymbol{k}_{\rm T} \quad \Phi^{[\gamma^+]}(x,\boldsymbol{k}_{\rm T},P,S;\ldots)}\Bigg|_{\boldsymbol{S}_{\rm T}=(1,0)}\end{aligned} \tag{6.11}$$

The correct candidate model must reproduce the previous results. This means we calculate

$$\widetilde{A}_{2B}^{[x]}(x) = \int_{-\infty}^{+\infty} d(b\cdot P) e^{ix(b\cdot P)} \left[\widetilde{A}_{2B}^{\rm Re}(\cdots) + i\widetilde{A}_{2B}^{\rm Im}(\cdots)\right] \tag{6.12}$$

$$\widetilde{A}_{12B}^{[x]}(x) = \int_{-\infty}^{+\infty} d(b\cdot P) e^{ix(b\cdot P)} \left[\widetilde{A}_{12B}^{\rm Re}(\cdots)\right] \tag{6.13}$$

from candidate model for $\widetilde{A}_{2B}^{\rm Re}$, $\widetilde{A}_{2B}^{\rm Im}$, and $\widetilde{A}_{12B}^{\rm Re}$, then we calculate

$$\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\rm Re} - \widetilde{A}_{2B}^{[x]\rm Im}) \tag{6.14}$$

$$\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\rm Re} \tag{6.15}$$

to calculate $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$. This condition involves all $\widetilde{A}_{2B}^{\rm Re}$, $\widetilde{A}_{2B}^{\rm Im}$ and $\widetilde{A}_{12B}^{\rm Re}$ amplitudes.

The model $c_2(\eta)$ (of Figure 42) for real unpolarized amplitude $\widetilde{A}_{2B}^{\rm Re} = \left[\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}\right]$ combined with the imaginary unpolarized amplitude model $\widetilde{A}_{2B}^{\rm Im} = \left[\frac{a''b_Le^{-f\eta}}{\left[1+(c''+d'')b_L^2+d''b_T^2\right]^{j''}}\right]$ suppresses anti-quark contributions in above mentioned regime and has the lowest AIC (and $\chi^2$/DoF). The row 1 and column 3 of Figures 52, 60, 68, 76 plot the $\tilde{f}_1^{(0)}(x) = 2\widetilde{A}_{2B}^{[x]}(x)$, which shows near-zero contribution for anti-quark ($x \lesssim -0.2$) for higher momentum.

The selected model for $\widetilde{A}_{2B}^{\rm Re}$, $\widetilde{A}_{2B}^{\rm Im}$ and along with the model for $\widetilde{A}_{12B}^{\rm Re} = \left[\frac{-a'e^{-f\eta}}{\left[1+(c'+d')b_L^2+d'b_T^2\right]^{j'}}\right]$, produce Mellin moment ($x$-integrated) calculations of Sivers shift TMD $\langle \boldsymbol{k}_y\rangle_{TU} = m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ which is consistent previously published results. Tables 21, 23, 25, 27, calculate the $\int dx\, \widetilde{A}_{2B}^{[x]\rm Re}$, $\int dx\, (\widetilde{A}_{2B}^{[x]\rm Im})$, $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\rm Re} - \widetilde{A}_{2B}^{[x]\rm Im})$, $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\rm Re}$, and finally the Mellin moment of Sivers shift TMD $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ for integration limit of $[-\infty,\infty]$, $[-1,1]$, $[-1,0]$, and $[0,1]$ which are consistent with previously published results.

In summary the final candidate analytical models in Cartesian $(b_L, b_T)$ space for the three amplitudes are:

$$\widetilde{A}^{\mathrm{Re}}_{2B} = \left[\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}\right] \tag{6.16}$$

$$\widetilde{A}^{\mathrm{Re}}_{12B} = \left[\frac{-a'e^{-f\eta}}{[1+(c'+d')b_L^2+d'b_T^2]^{j'}}\right] \tag{6.17}$$

$$\widetilde{A}^{\mathrm{Im}}_{2B} = \left[\frac{a''b_Le^{-f\eta}}{[1+(c''+d'')b_L^2+d''b_T^2]^{j''}}\right] \tag{6.18}$$

### 6.8 Simultaneous Fitting Strategy

The structural models in Eq. (6.16)-(6.18) show that all three surviving amplitudes share an $\eta$-dependence damped by a common exponential decay factor, $e^{-f\eta}$. Because the Sivers shift is a ratio, this shared factor cancels in the $x$-dependent result.

We evaluate the Sivers shift of Eq. (5.6). With $\widetilde{A}^{\mathrm{Im}}_{12B}$ dropped (Section 5.9) and only the even-parity combinations surviving the symmetric integration, it reduces to

$$\langle \boldsymbol{k}_y\rangle_{TU} = -m_N\frac{\int d(b\cdot P)\cos\left(x(b\cdot P)\right)\widetilde{A}^{\mathrm{Re}}_{12B}(\cdots)}{\int d(b\cdot P)\left[\cos\left(x(b\cdot P)\right)\widetilde{A}^{\mathrm{Re}}_{2B}(\cdots)-\sin\left(x(b\cdot P)\right)\widetilde{A}^{\mathrm{Im}}_{2B}(\cdots)\right]}\ . \tag{6.19}$$

Substituting the analytical models from Eq. (6.16)-(6.18), the shared $e^{-f\eta}$ term is independent of the integration variable $b\cdot P$ and cancels between the numerator and denominator integrals. Applying the mapping of Section 5.8 ($b_L = P\cdot b/(-P_L)$, $b_T^2 = b^2 - b_L^2$), we arrive at the final integrands in the invariant frame:

$$\begin{aligned}
\langle \boldsymbol{k}_y\rangle_{TU} &= m_N\frac{\int d(b\cdot P)\cos\left(x(b\cdot P)\right)\frac{a'}{D^{\mathrm{Re}}_{12B}}}{\int d(b\cdot P)\left[\cos\left(x(b\cdot P)\right)\frac{a}{D^{\mathrm{Re}}_{2B}}-\sin\left(x(b\cdot P)\right)\frac{a''(P\cdot b)/(-P_L)}{D^{\mathrm{Im}}_{2B}}\right]}\ ,\\
D^{\mathrm{Re}}_{12B} &= \left[1+(c')\frac{(P\cdot b)^2}{P_L^2}+d'b^2\right]^{j'}\ ,\\
D^{\mathrm{Re}}_{2B} &= \left[1+\left(c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)\frac{(P\cdot b)^2}{P_L^2}+db^2\right]^{j}\ ,\\
D^{\mathrm{Im}}_{2B} &= \left[1+(c'')\frac{(P\cdot b)^2}{P_L^2}+d''b^2\right]^{j''}\ .
\end{aligned} \tag{6.20}$$

We enforce this shared $\eta$-dependence with a simultaneous global fit in which the parameter $f$ is the same in all three amplitude models ($\widetilde{A}^{\mathrm{Re}}_{2B}$, $\widetilde{A}^{\mathrm{Im}}_{2B}$, and $\widetilde{A}^{\mathrm{Re}}_{12B}$). This keeps the three models theoretically consistent and stabilizes the parameter extraction.

Section 6.8.1 describes the correlated fit used at each momentum.

### 6.8.1 Simultaneous Fit with Full Covariance Matrix

Because we extract the amplitudes from the same gauge field configurations and evaluate them across nearby spatial coordinates $(b_L, b_T)$, their statistical fluctuations are strongly correlated. An uncorrelated $\chi^2$ minimization weights these data points incorrectly, which underestimates the uncertainties and biases the central values.

We therefore perform a correlated $\chi^2$ minimization with the full covariance matrix, which accounts for the cross-correlations in the lattice data, both across different quark spatial separations and across the real and imaginary amplitude components.

We estimate these correlations using Jackknife resampling. Let $y_i^{(k)}$ represent the lattice evaluation of the $i$-th data point (spanning all spatial coordinates across $\widetilde{A}_{2B}^{\mathrm{Re}}$, $\widetilde{A}_{2B}^{\mathrm{Im}}$, and $\widetilde{A}_{12B}^{\mathrm{Re}}$) in the $k$-th Jackknife bin. The full Jackknife covariance matrix $C_{ij}$ is defined as [121]:

$$C_{ij} = \frac{N-1}{N} \sum_{k=1}^{N} \left( y_i^{(k)} - \bar{y}_i \right) \left( y_j^{(k)} - \bar{y}_j \right) \ , \tag{6.21}$$

where $N$ is the total number of Jackknife samples and $\bar{y}_i = \frac{1}{N} \sum_k y_i^{(k)}$ is the sample mean.

The global simultaneous fit minimizes the correlated $\chi^2$ objective function with respect to the model parameters $\mathbf{p}$:

$$\chi^2(\mathbf{p}) = \sum_{i,j} \left( \bar{y}_i - F_i(\mathbf{p}) \right) C_{ij}^{-1} \left( \bar{y}_j - F_j(\mathbf{p}) \right) \ , \tag{6.22}$$

where $F_i(\mathbf{p})$ represents the analytical models evaluated at the corresponding kinematics.

We use the `lsqfit` [122] non-linear least squares fitting package and the `gvar` library for Gaussian error propagation. We construct the full Jackknife covariance matrix that captures the cross-correlations between $\widetilde{A}_{2B}^{\mathrm{Re}}$, $\widetilde{A}_{2B}^{\mathrm{Im}}$, and $\widetilde{A}_{12B}^{\mathrm{Re}}$. The `gvar` framework binds the combined lattice data array to the inverse covariance matrix.

The following subsections give the results of these correlated simultaneous fits at each of the discrete lattice momenta.

### 6.8.2 Simultaneous Fit Results for $\hat{\zeta} = 0.090772$ ($\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$)

The extracted parameters from the simultaneous fit for the lowest simulated nucleon momentum, $P_1 = -1$ ($\hat{\zeta} = 0.090772$), are in Table 20.

Table 20: Simultaneous Fit with common $f$: for $P_L = -1$, $b_{min} \geq 3$, $\eta \in [6, 10]$. Global $\chi^2$/DoF = 1.85(10)

| Amplitude | Model Expression | Params List | Fitted Values |
|---|---|---|---|
| $\widetilde{A}_{2B}^{\mathrm{Re}}$ | $\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, k_1, k_2, j, f\}$ | $[3.863(11), 0.336(40), 0.1630(74), 12.288(23), -36.1965(45), 1.627(25), 0.4702(46)]$ |
| $\widetilde{A}_{2B}^{\mathrm{Im}}$ | $\frac{ab_Le^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.267(13), 0.2253(69), 0.2205(66), 1.108(21), 0.4702(46)]$ |
| $\widetilde{A}_{12B}^{\mathrm{Re}}$ | $\frac{-ae^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.658(22), 0.0773(46), 0.0747(43), 2.0054(90), 0.4702(46)]$ |

Figures 46–48 show the 3D fit plots on the raw lattice data in both $\{b_L, b_T, \eta\}$-space and $\{(b\cdot P), b^2, \eta\}$-space. Because we exclude the short-distance data ($|\boldsymbol{b}| < 3a$), the fit describes the long-range dynamics, and ultraviolet lattice artifacts do not influence the results. Figures 49–51 show the fit plots on the raw lattice data in both $\{b_L, b_T, \eta = 8\}$-space and $\{(b \cdot P), b^2, \eta = 8\}$-space at a specific $\eta$ slice ($\eta = 8$).

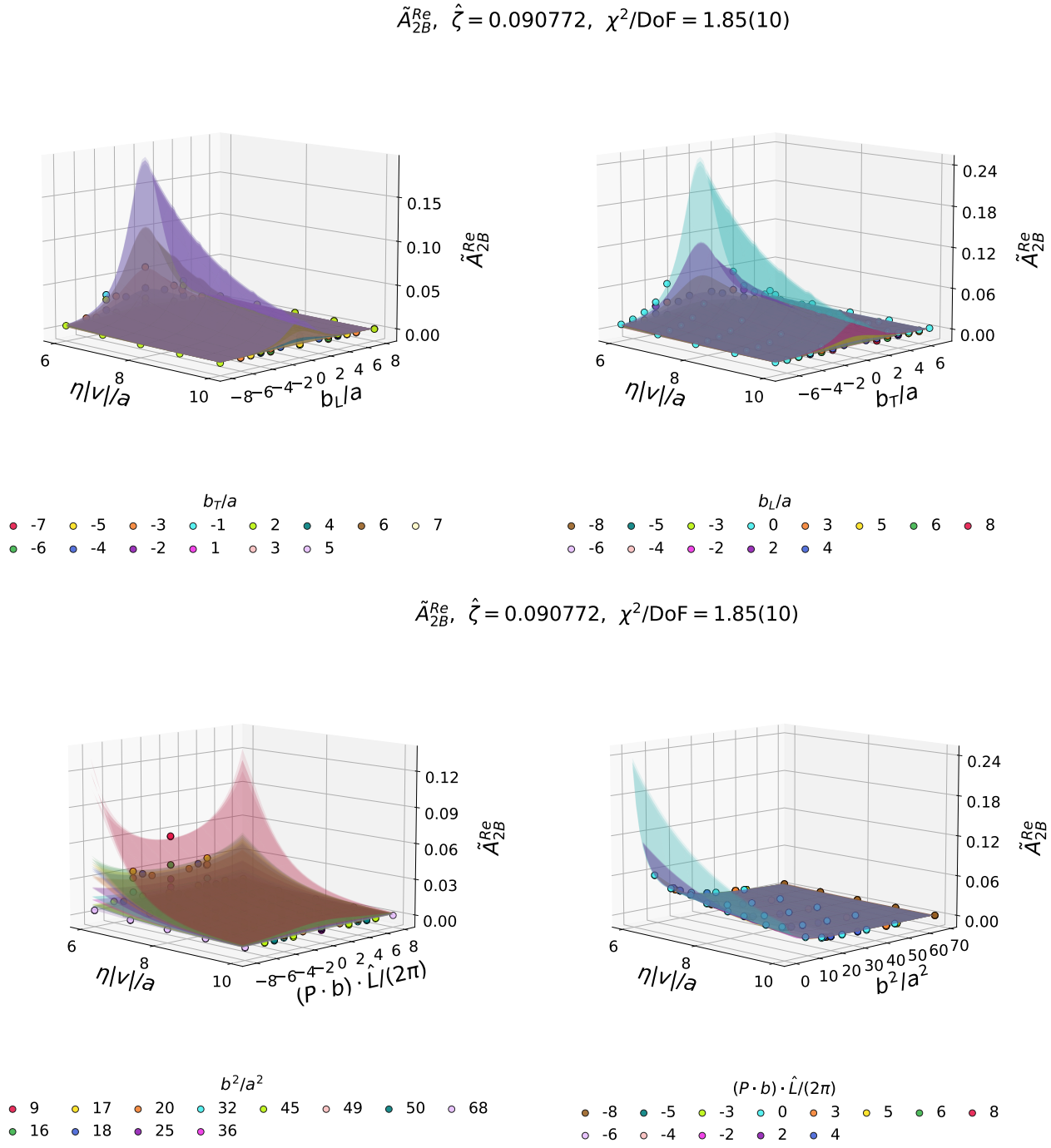



Figure 46: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

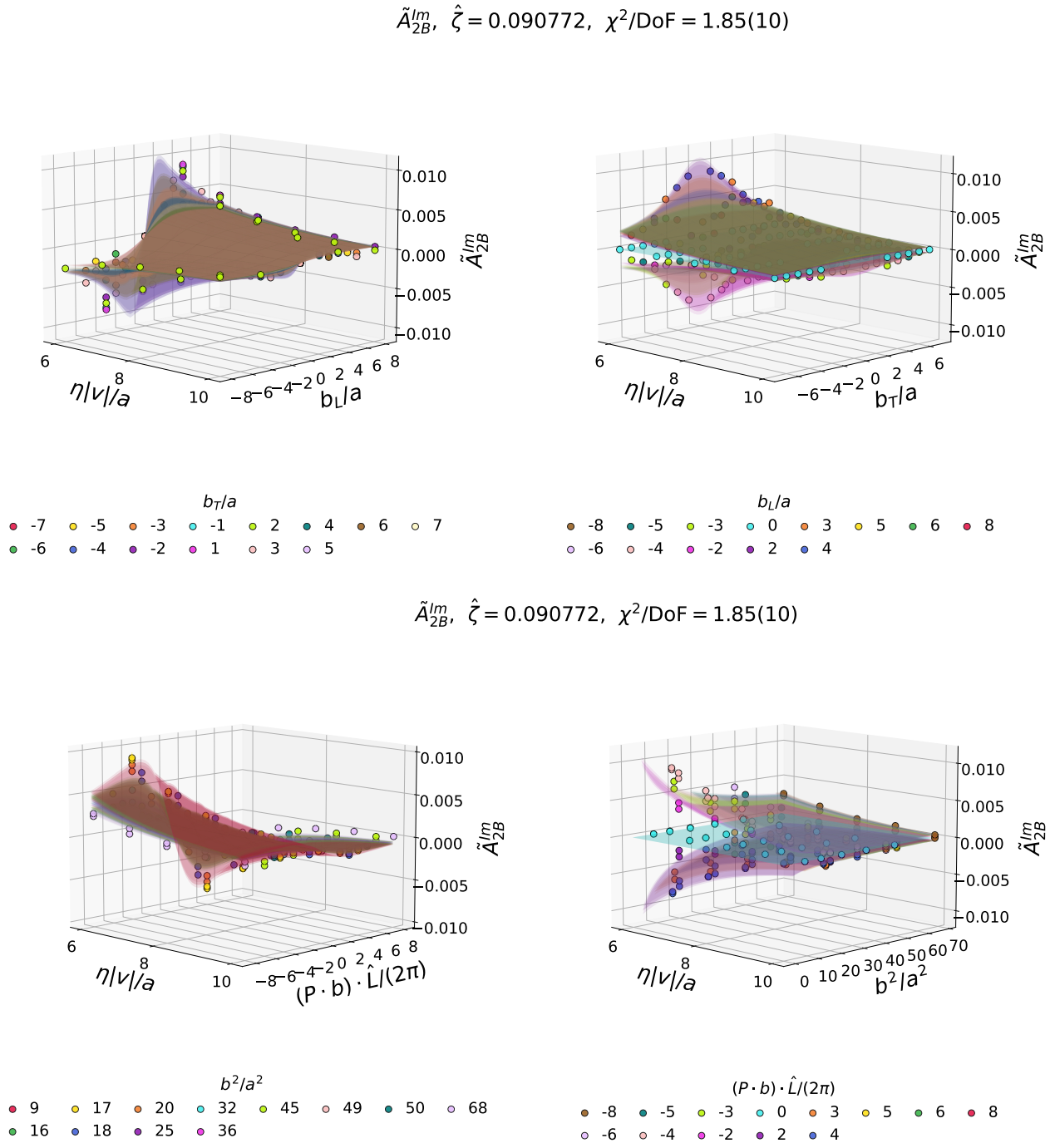



Figure 47: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-1,0,0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

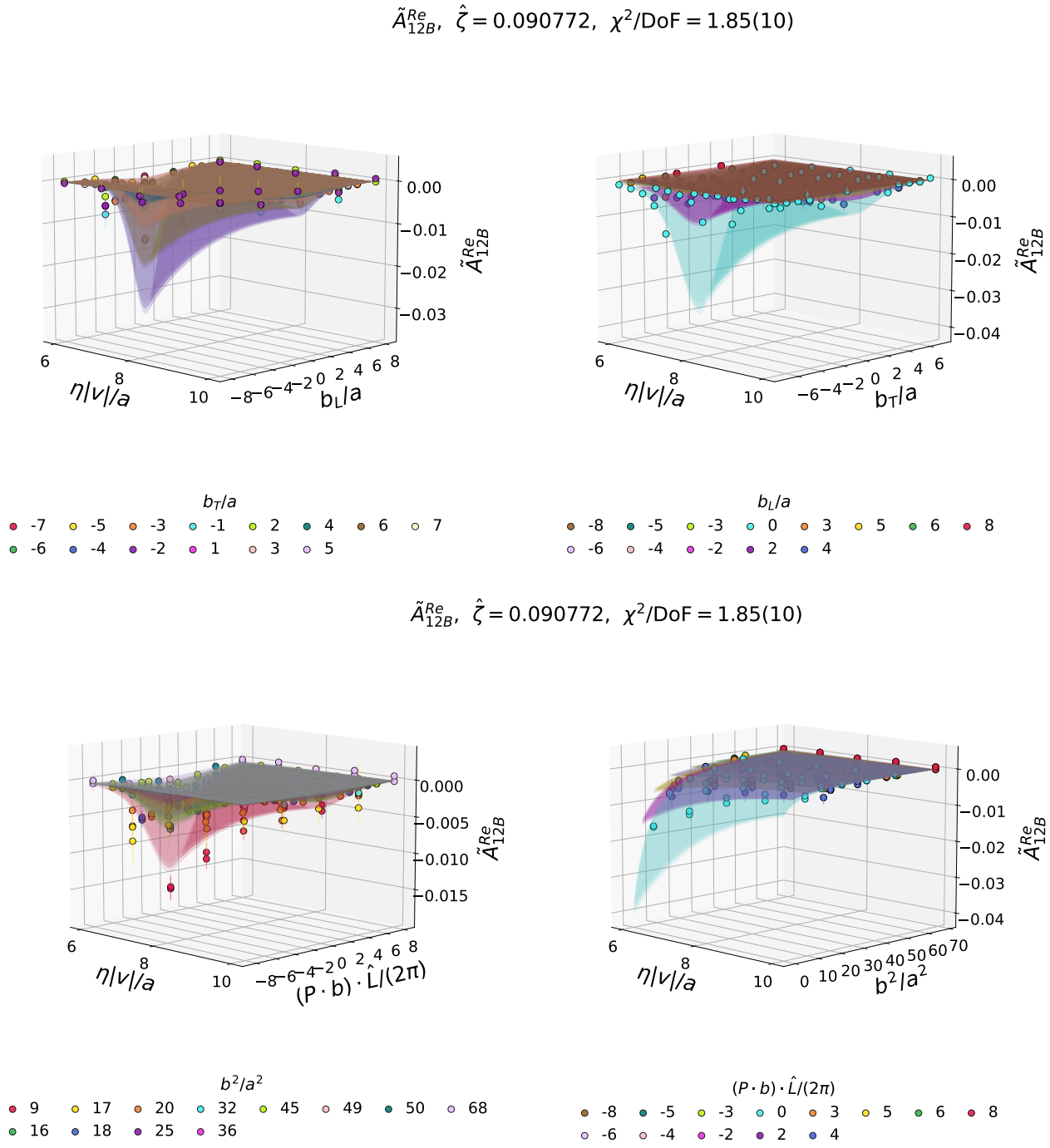



Figure 48: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

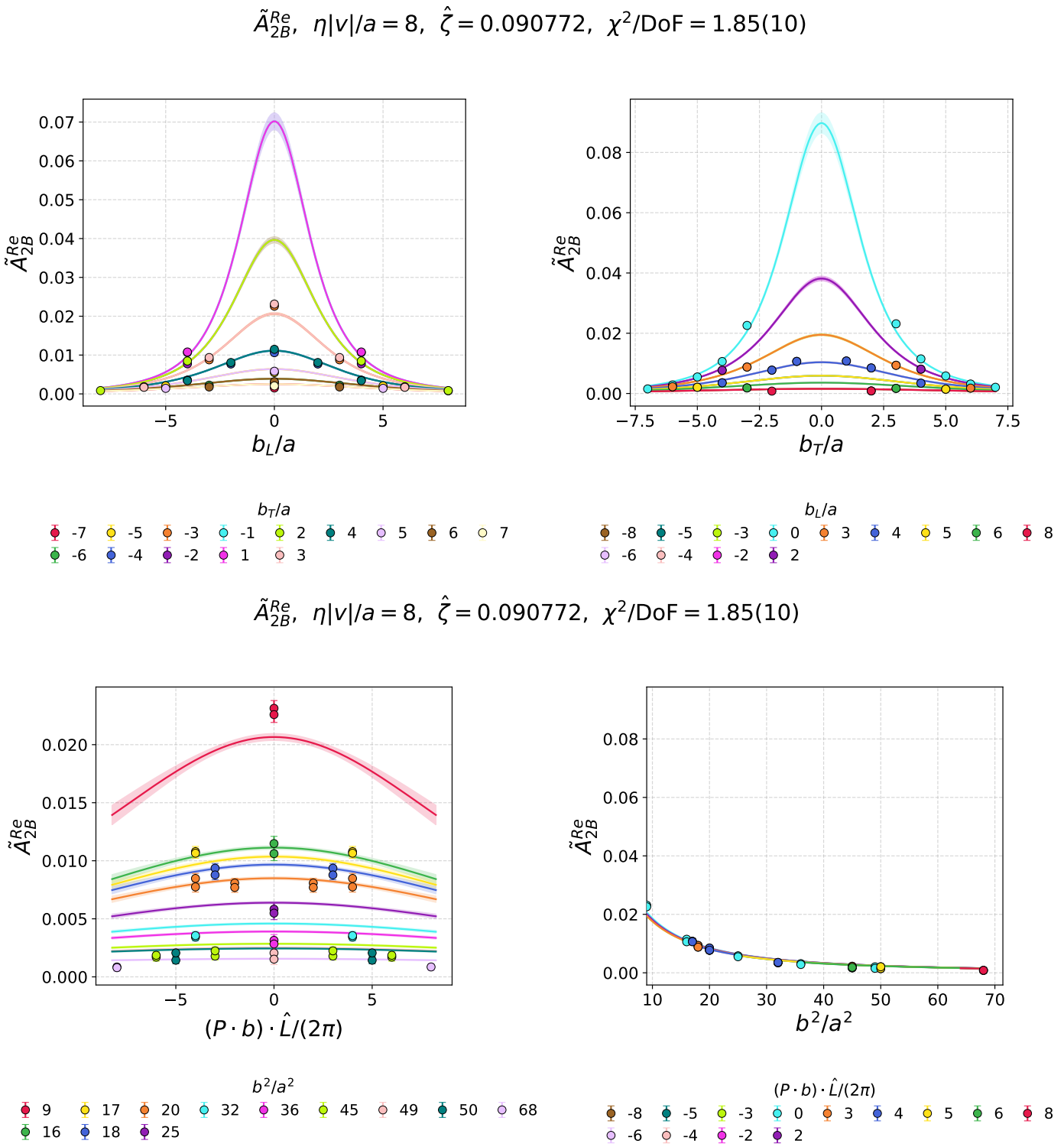
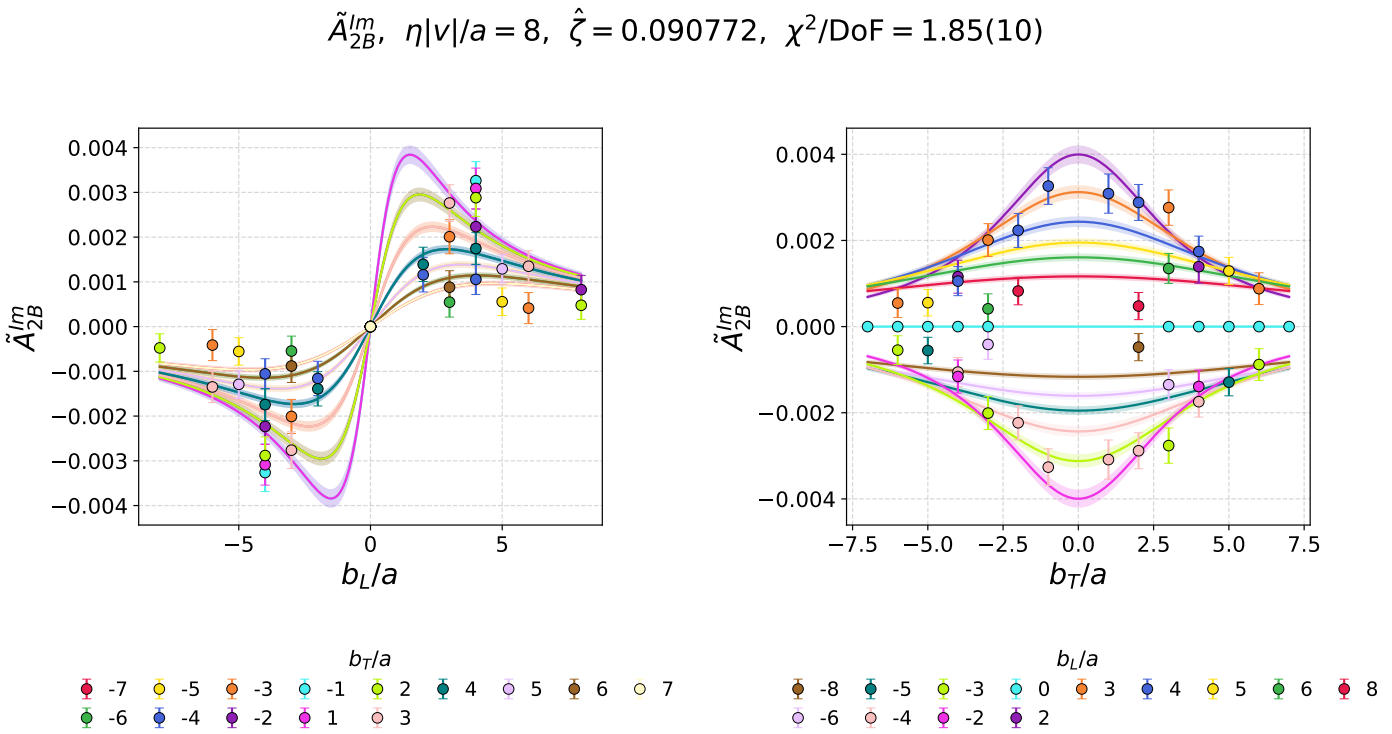



Figure 49: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.



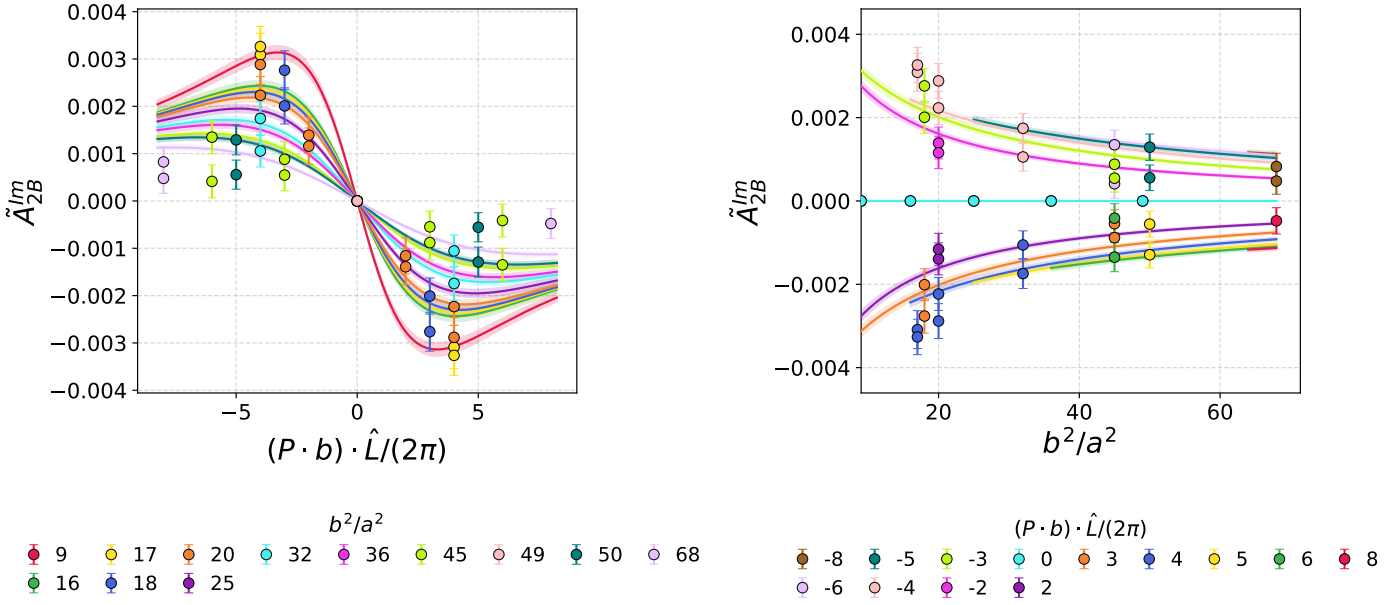




Figure 50: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

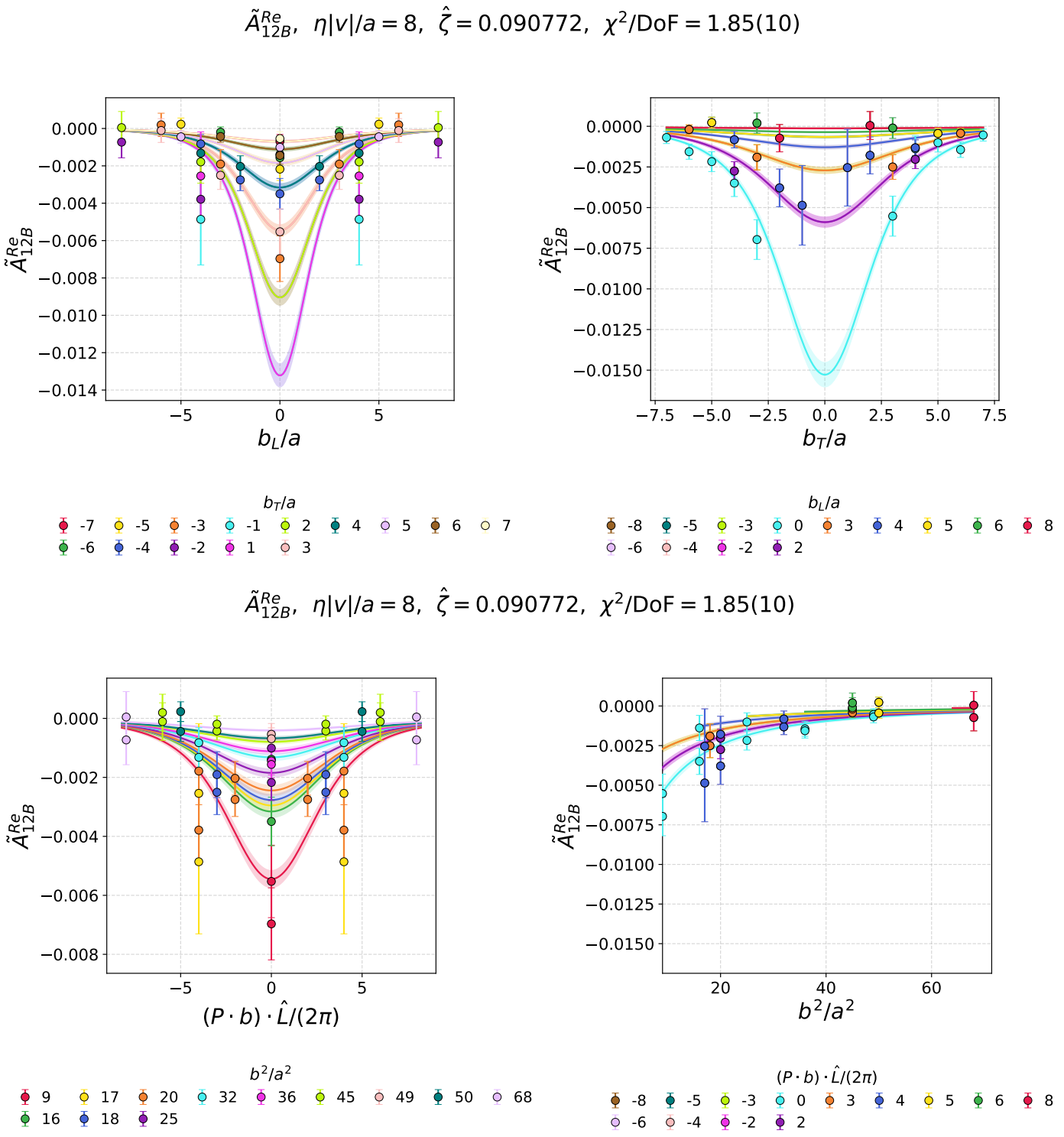



Figure 51: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-1, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

From the continuous parametrization of the amplitudes, we evaluate the Fourier integrals over the invariant scalar $P \cdot b$,

$$\widetilde{A}_{2B}^{[x]\mathrm{Re}}(x, b^2; \ldots) = \int_{-\infty}^{+\infty} \frac{d(b\cdot P)}{2\pi} e^{ix(b\cdot P)}\, \widetilde{A}_{2B}^{\mathrm{Re}}((b\cdot P), b^2; \ldots)/\widetilde{\mathcal{S}}(b^2; \ldots) \tag{6.23}$$

$$i\widetilde{A}_{2B}^{[x]\mathrm{Im}}(x, b^2; \ldots) = \int_{-\infty}^{+\infty} \frac{d(b\cdot P)}{2\pi} e^{ix(b\cdot P)}\, \widetilde{A}_{2B}^{\mathrm{Im}}((b\cdot P), b^2; \ldots)/\widetilde{\mathcal{S}}(b^2; \ldots) \tag{6.24}$$

$$\widetilde{A}_{12B}^{[x]\mathrm{Re}}(x, b^2; \ldots) = \int_{-\infty}^{+\infty} \frac{d(b\cdot P)}{2\pi} e^{ix(b\cdot P)}\, \widetilde{A}_{12B}^{\mathrm{Re}}((b\cdot P), b^2; \ldots)/\widetilde{\mathcal{S}}(b^2; \ldots) \tag{6.25}$$

Note that, since $\widetilde{A}_{12B}^{\mathrm{Im}}$ is odd in $(b \cdot P)$, its Fourier transform gives an imaginary solution (as discussed in Eq. (5.8)). We can calculate the Fourier-transformed TMDs using Eqs. (4.43)-(4.44):

$$\tilde{f}_1^{(0)} = 2\left[\widetilde{A}_{2B}^{[x]\mathrm{Re}} + i(i\widetilde{A}_{2B}^{[x]\mathrm{Im}})\right] = 2\left[\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}}\right], \tag{6.26}$$

$$\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}. \tag{6.27}$$

As we detailed in Section 6.7, Tables 21, calculate the $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$, and finally the Mellin moment of Sivers shift TMD $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ for integration limit of $[-\infty, \infty]$, $[-1, 1]$, $[-1, 0]$, and $[0, 1]$; the results are consistent with previous studies of $x$-moment lattice calculations [2, 85].

Table 21: Numerical Integration of Fourier Transforms: $P_L = -1, |b_T| = 3a \approx 0.34$ fm

| Integral | $[-\infty, \infty]$ | $[-1, 1]$ | $[-1, 0]$ | $[0, 1]$ |
|---|---|---|---|---|
| $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ | 0.889(24) | 0.780(29) | 0.390(14) | 0.390(14) |
| $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | -1(108)e-17 | 2(282)e-17 | 0.1223(58) | -0.1223(58) |
| $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 1.778(47) | 1.560(58) | 0.535(25) | 1.024(36) |
| $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ | 0.469(23) | 0.454(23) | 0.227(12) | 0.227(12) |
| $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ | 0.284(14) | 0.313(18) | 0.456(29) | 0.238(13) |

Figs. 52 and 53 show the $x$-dependence obtained from the Fourier transforms of the fitted amplitudes: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$, $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, the Sivers shift $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$ ([momentum fraction] $\times$ [Sivers shift]). Figure 52 is evaluated at fixed $b^2 = 9a^2$, and Figure 53 shows the same quantities as functions of $x$ and $b^2$. The bands are the Jackknife uncertainties propagated from the fit parameters.

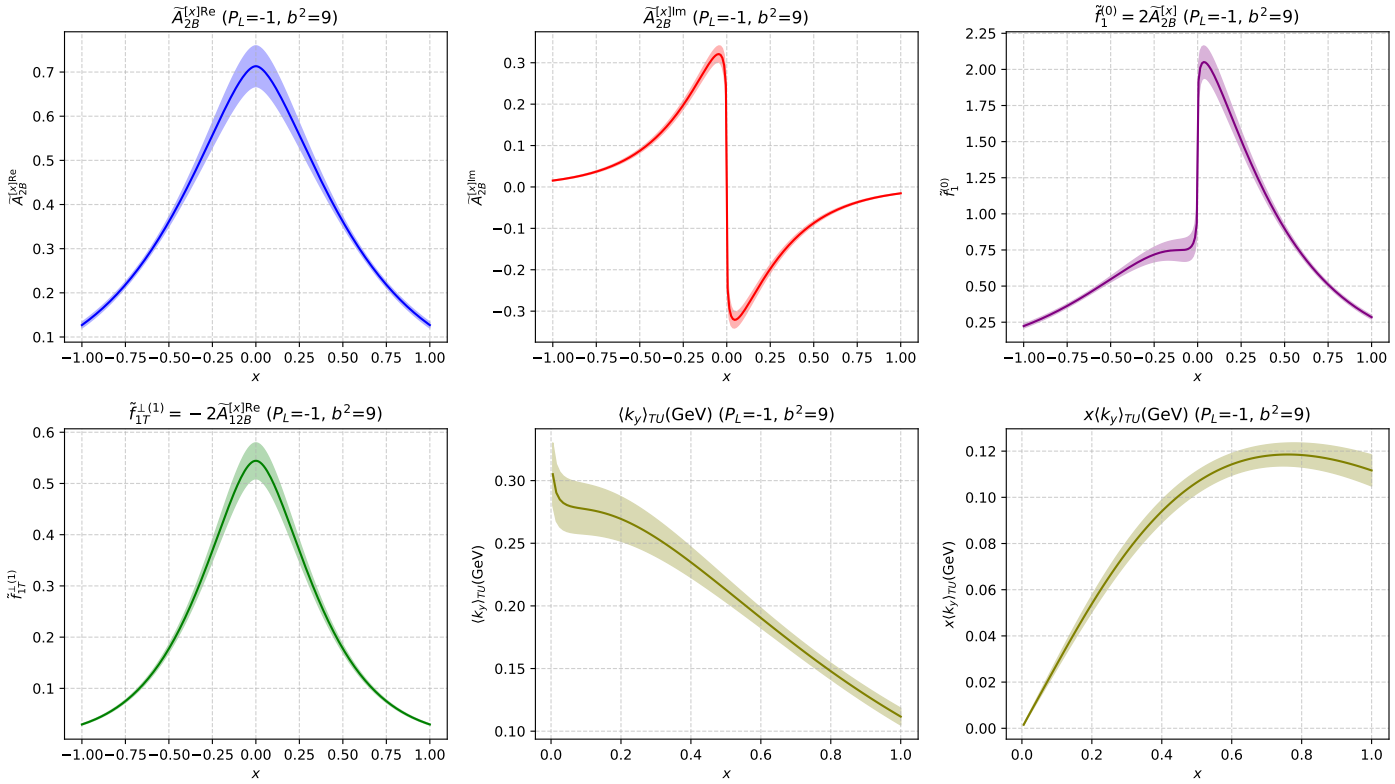



Figure 52: $x$-dependence at $b^2 = 9a^2$ for $P_L = -1$. Top row: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, and $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$. Bottom row: $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$. Bands show the Jackknife uncertainty.

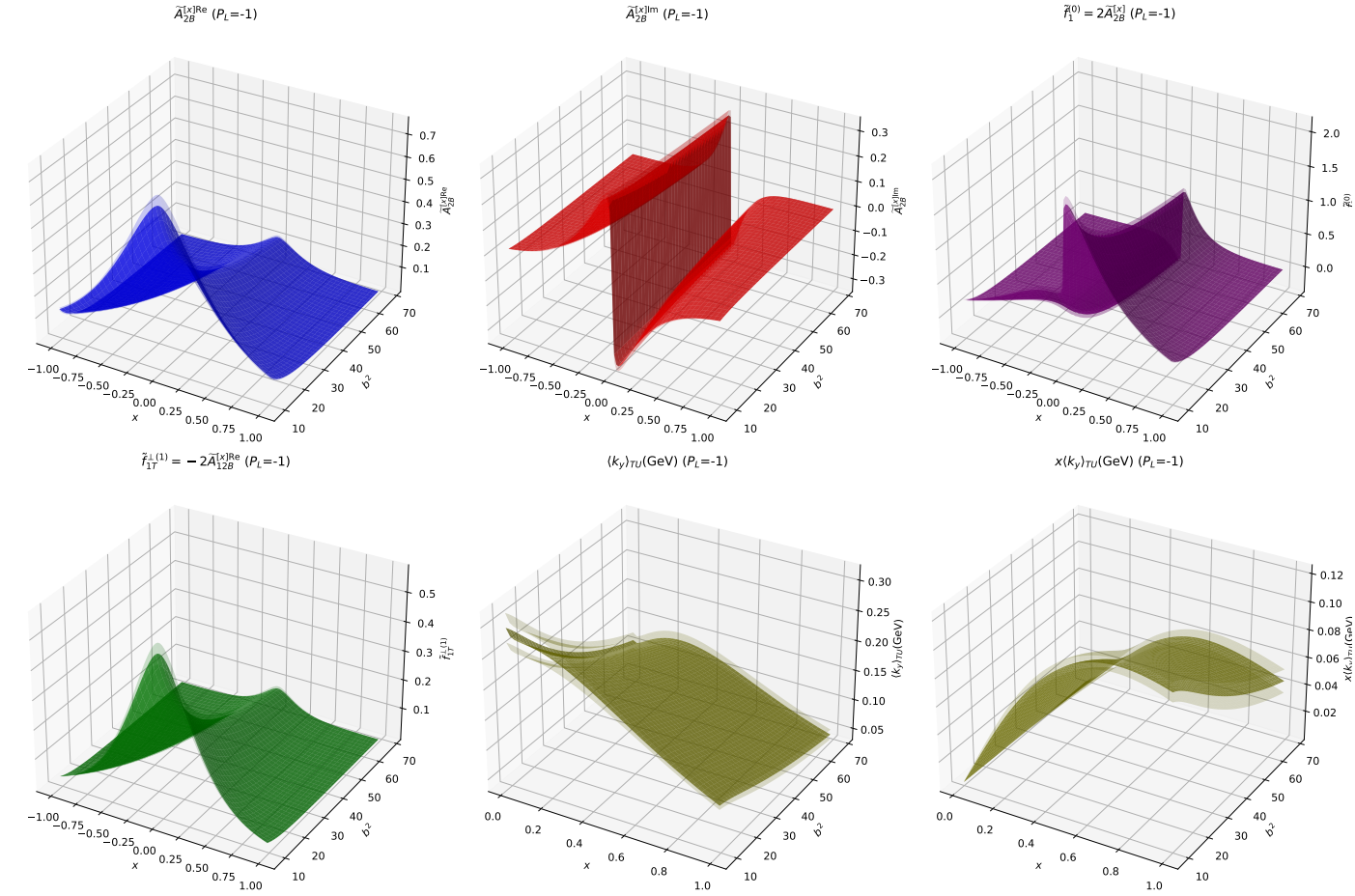



Figure 53: The quantities of Figure 52 as functions of $x$ and $b^2/a^2 \in [9, 68]$ for $P_L = -1$.

### 6.8.3 Simultaneous Fit Results for $\hat{\zeta} = 0.294367$ ($\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$)

The extracted parameters from the simultaneous fit for the nucleon momentum, $P_1 = -2$ ($\hat{\zeta} = 0.294367$), are in Table 22.

Table 22: Simultaneous Fit with common $f$: for $P_L = -2$, $b_{min} \geq 3$, $\eta \in [6, 10]$. Global $\chi^2$/DoF = 1.197(85)

| Amplitude | Model Expression | Params List | Fitted Values |
|---|---|---|---|
| $\widetilde{A}^{\mathrm{Re}}_{2B}$ | $\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, k_1, k_2, j, f\}$ | $[3.9833(80), 0.166(35), 0.1020(46), 12.305(14), -36.1919(27), 2.061(20), 0.5357(66)]$ |
| $\widetilde{A}^{\mathrm{Im}}_{2B}$ | $\frac{ab_Le^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.368(12), 0.1346(69), 0.1305(67), 1.141(20), 0.5357(66)]$ |
| $\widetilde{A}^{\mathrm{Re}}_{12B}$ | $\frac{-ae^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.632(19), 0.0769(54), 0.0741(52), 2.0141(69), 0.5357(66)]$ |

Figures 54–59 show the fit on lattice data, and Figures 60 and 61 show the resulting $x$-dependence.

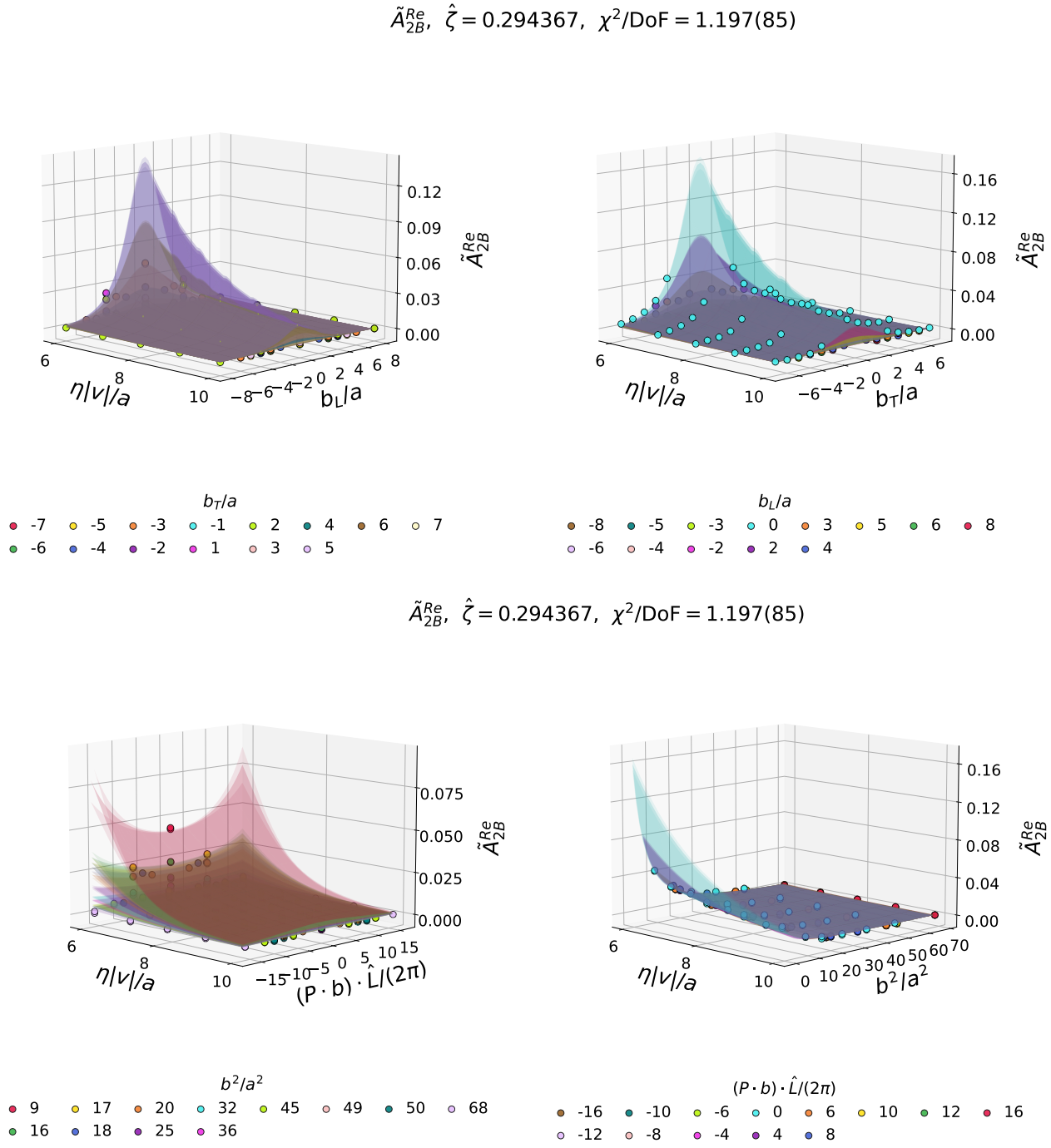



Figure 54: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-2,0,0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

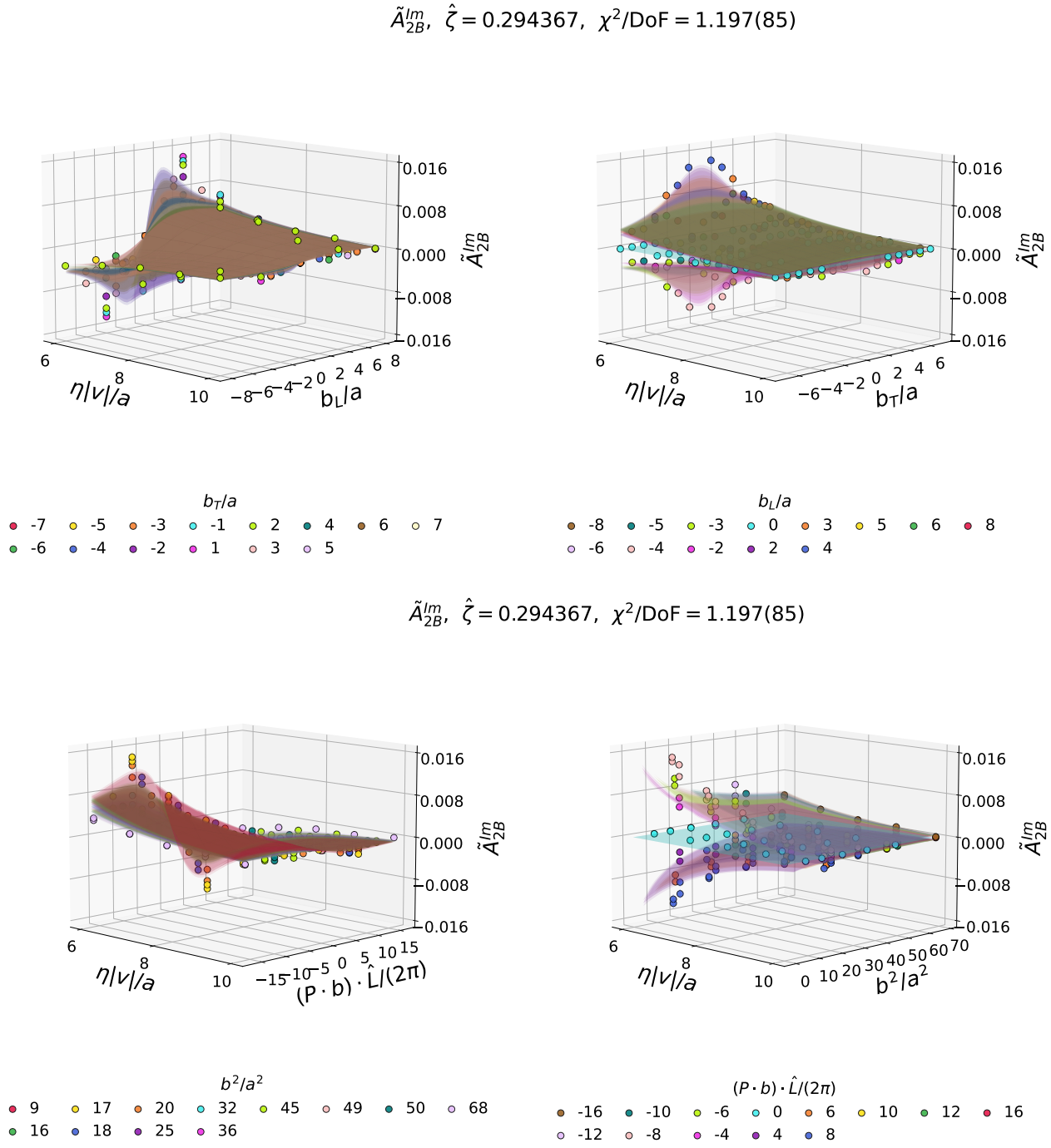



Figure 55: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

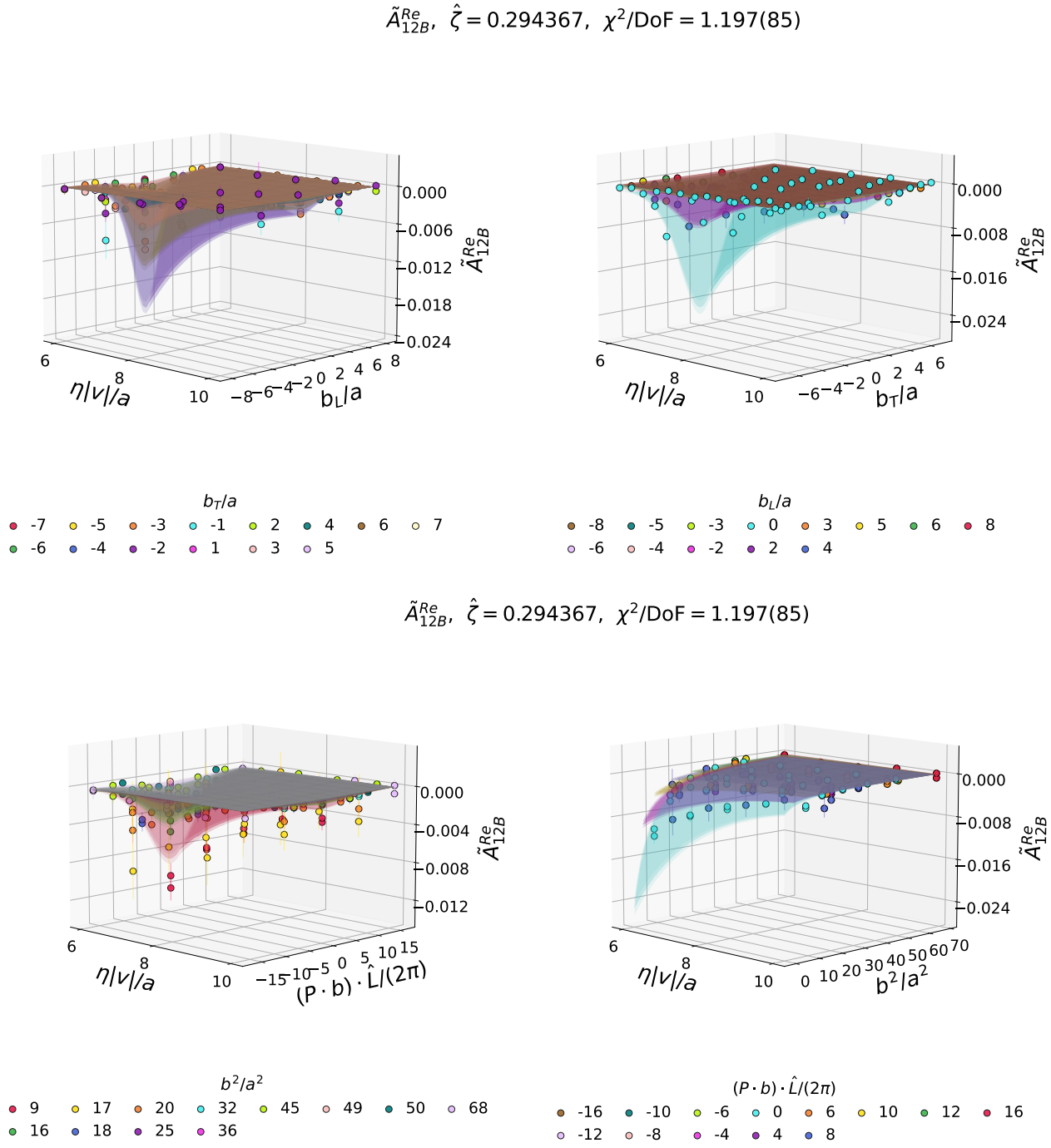



Figure 56: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

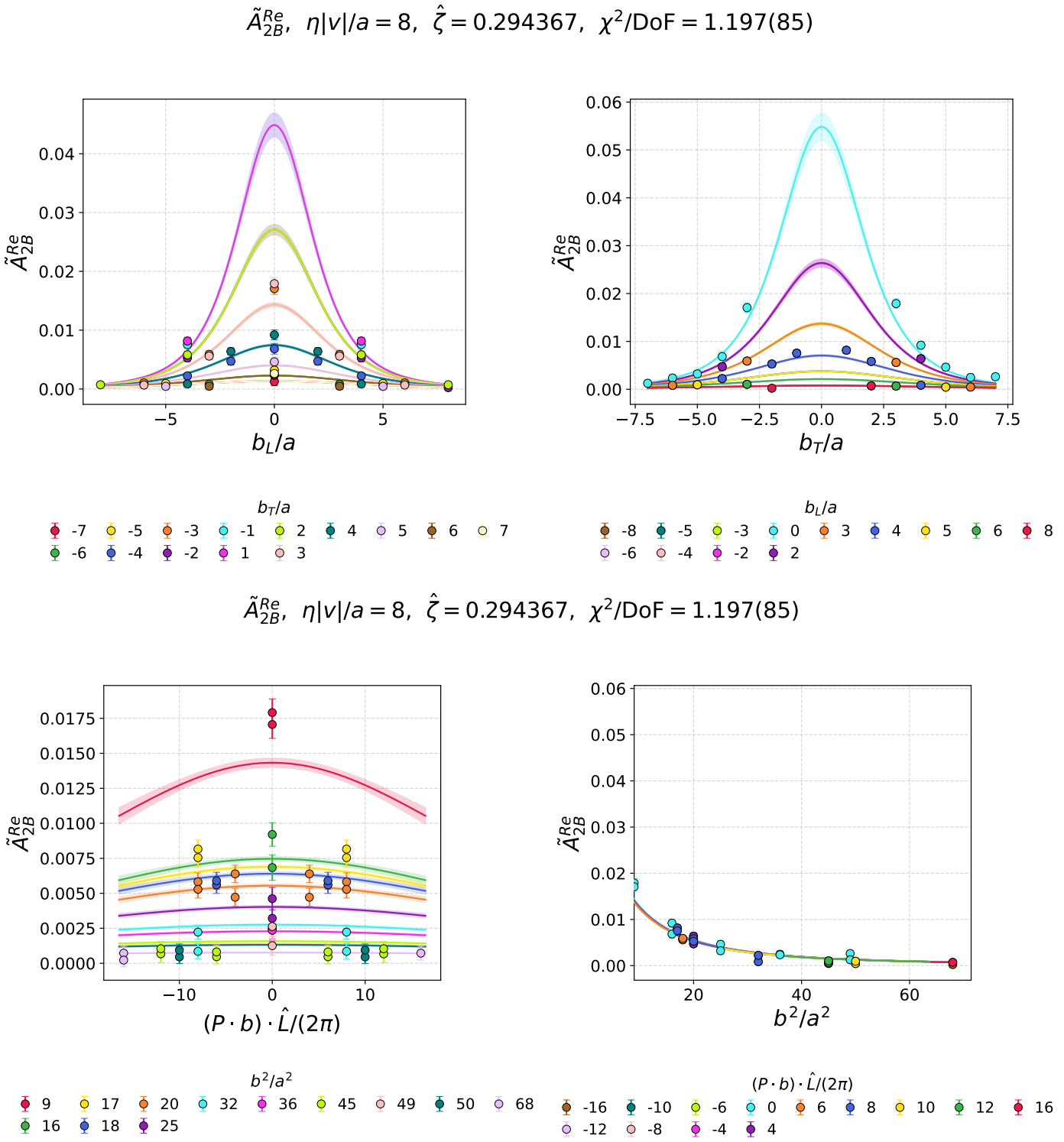



Figure 57: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

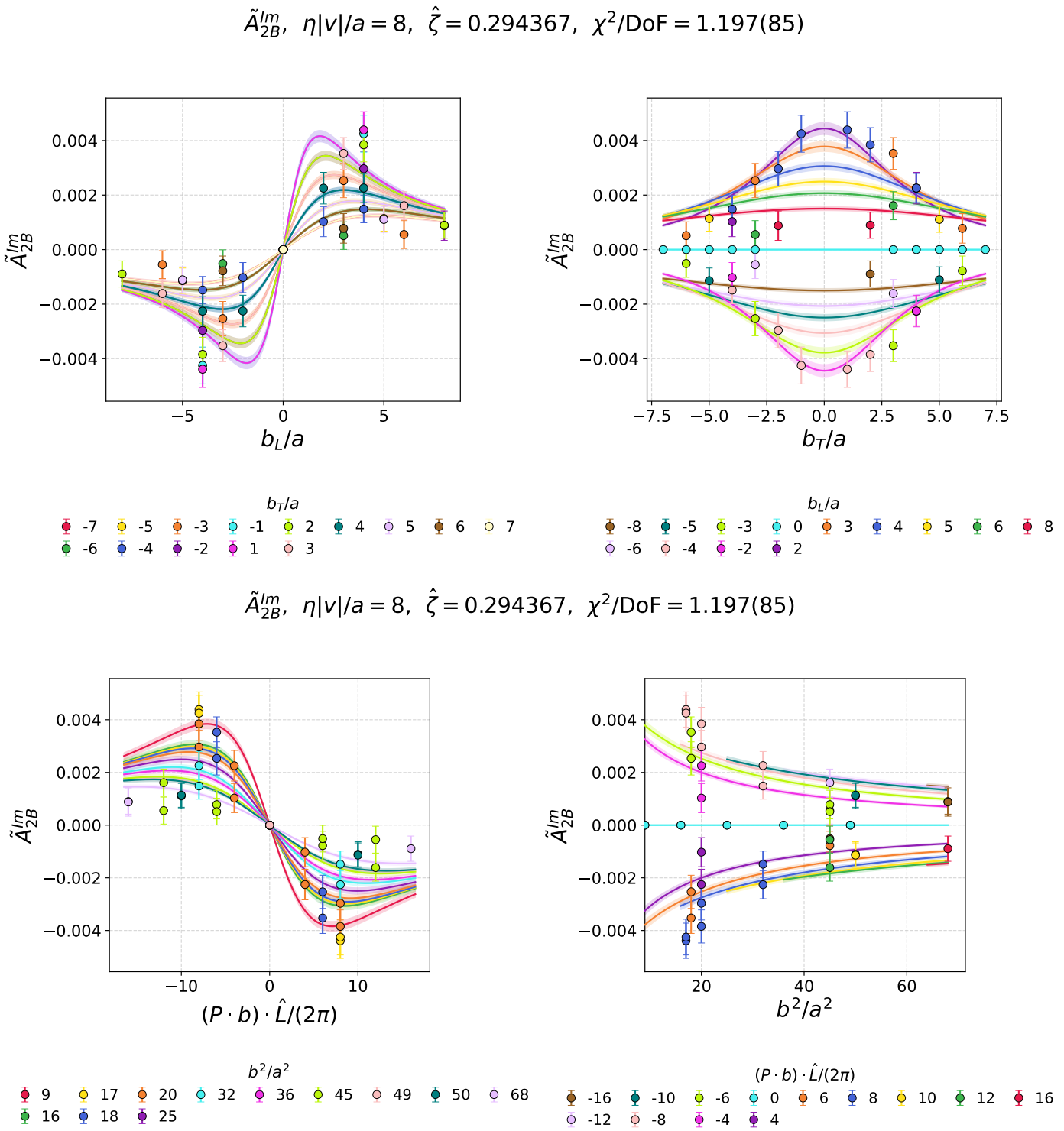



Figure 58: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

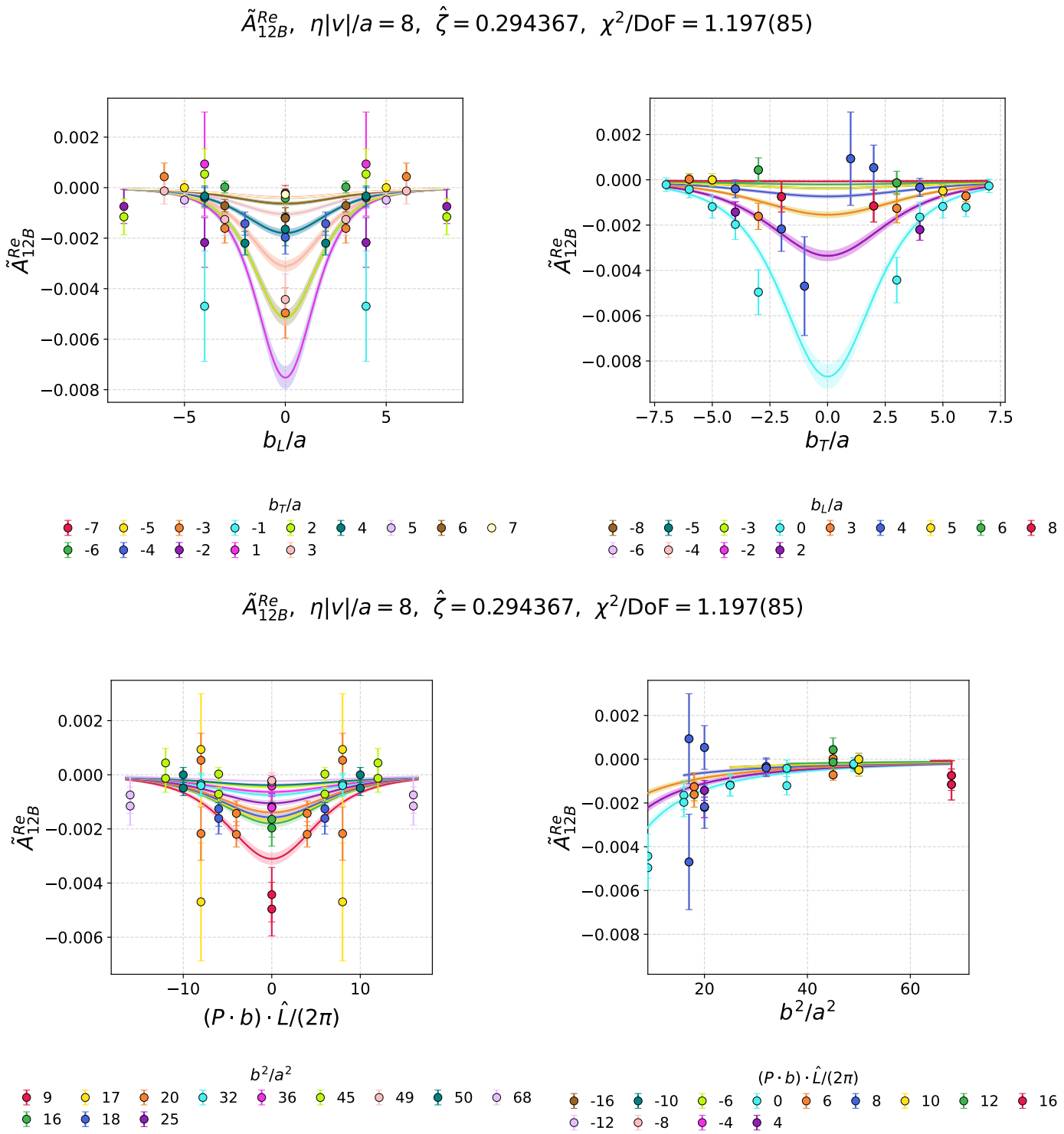



Figure 59: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-2, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

From the continuous parametrization of the amplitudes, we evaluate the Fourier integrals over the invariant scalar $P \cdot b$. As we detailed in Section 6.7, Tables 23, calculate the $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$, and finally the Mellin moment of Sivers shift TMD $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ for integration limit of $[-\infty, \infty]$, $[-1, 1]$, $[-1, 0]$, and $[0, 1]$; the results are consistent with previous studies of $x$-moment lattice calculations [2, 85].

Table 23: Numerical Integration of Fourier Transforms: $P_L = -2, |b_T| = 3a \approx 0.34$ fm

| Integral | $[-\infty, \infty]$ | $[-1, 1]$ | $[-1, 0]$ | $[0, 1]$ |
|---|---|---|---|---|
| $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ | 1.041(41) | 1.035(43) | 0.518(22) | 0.518(22) |
| $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | -6(176)e-17 | 7(444)e-17 | 0.257(13) | -0.257(13) |
| $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 2.081(82) | 2.070(86) | 0.520(34) | 1.550(63) |
| $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ | 0.451(27) | 0.451(27) | 0.225(14) | 0.225(14) |
| $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ | 0.233(14) | 0.234(14) | 0.466(36) | 0.1565(95) |

Figs. 60 and 61 show the $x$-dependence obtained from the Fourier transforms of the fitted amplitudes: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$, $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, the Sivers shift $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$ ([momentum fraction] $\times$ [Sivers shift]). Figure 60 is evaluated at fixed $b^2 = 9a^2$, and Figure 61 shows the same quantities as functions of $x$ and $b^2$. The bands are the Jackknife uncertainties propagated from the fit parameters.

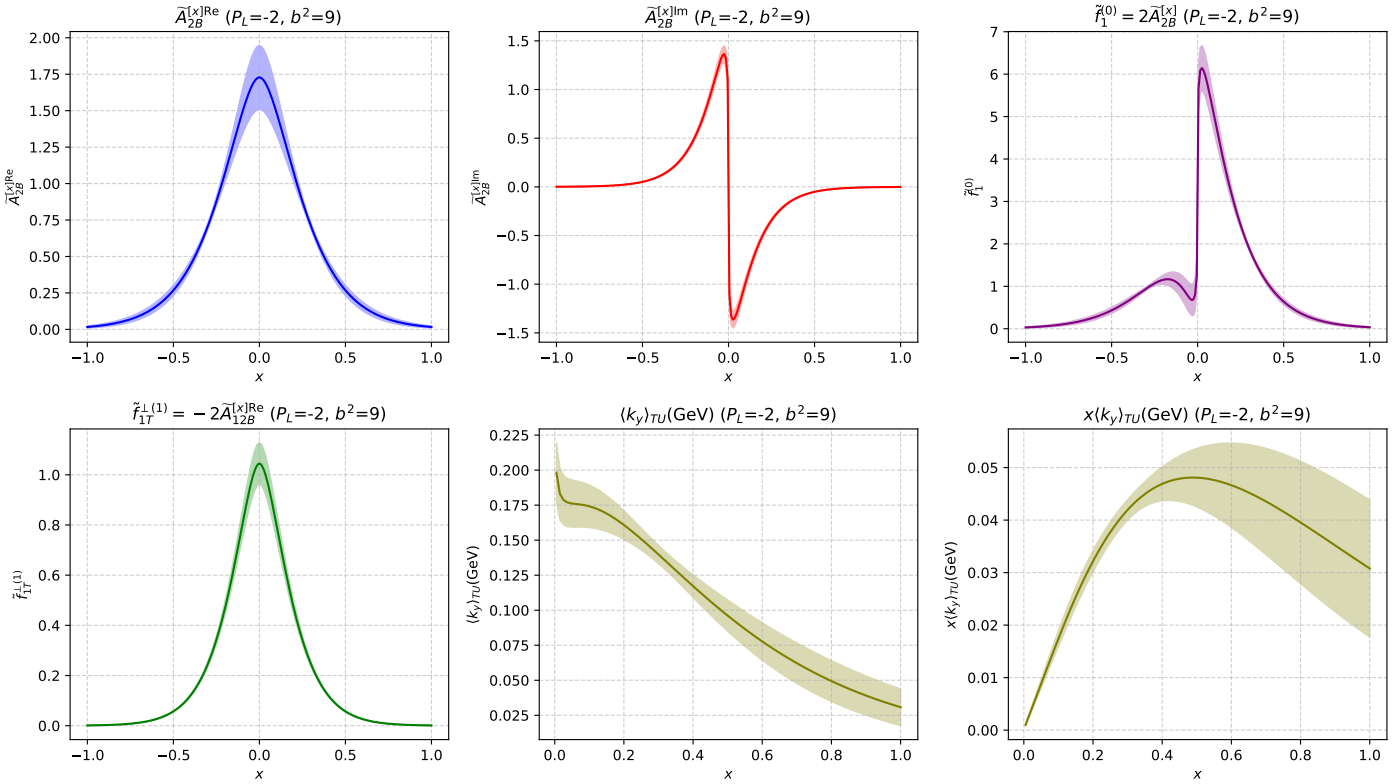



Figure 60: $x$-dependence at $b^2 = 9a^2$ for $P_L = -2$. Top row: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $i\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, and $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$. Bottom row: $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$. Bands show the Jackknife uncertainty.

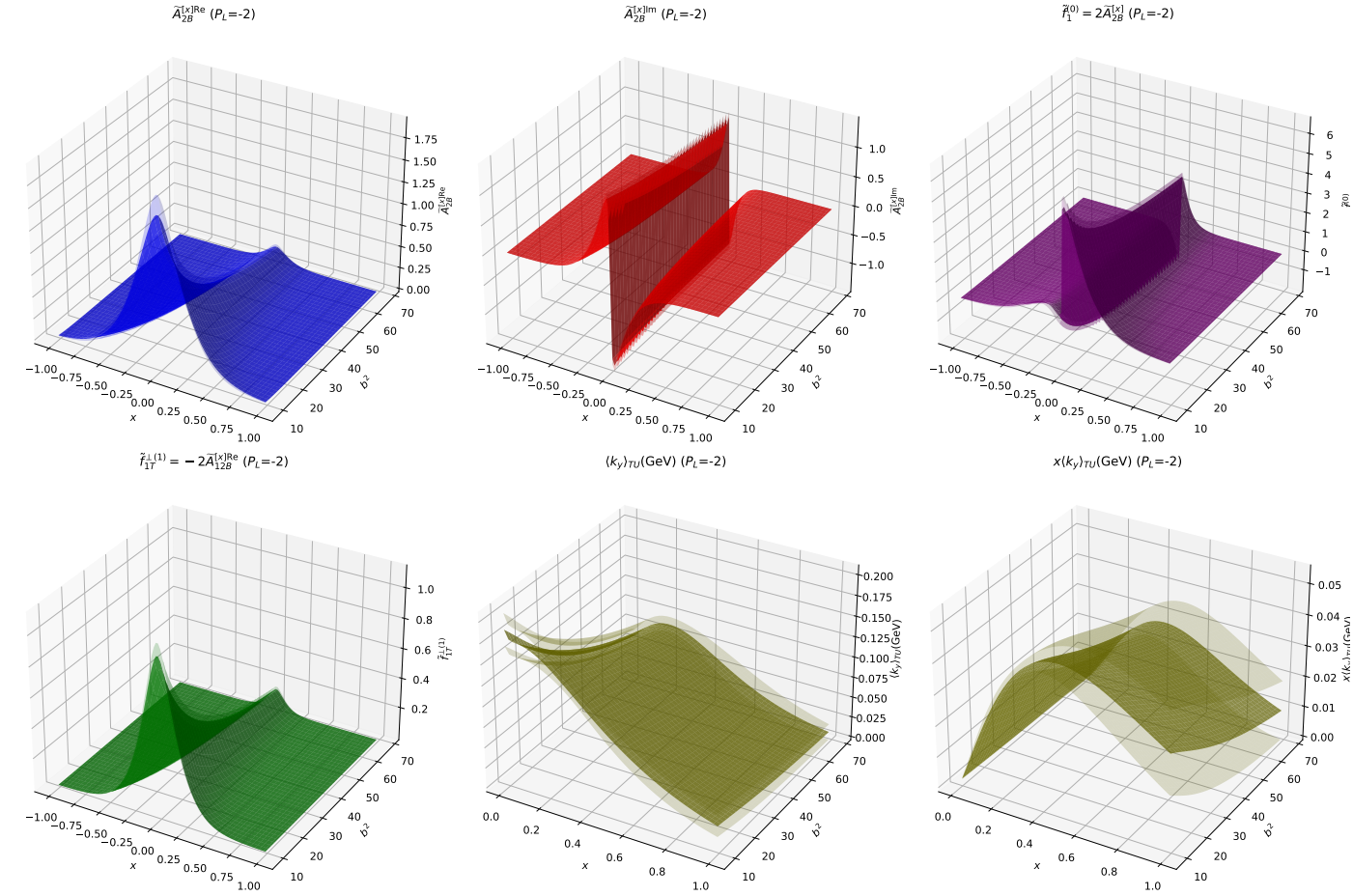



Figure 61: The quantities of Figure 60 as functions of $x$ and $b^2/a^2 \in [9, 68]$ for $P_L = -2$.

### 6.8.4 Simultaneous Fit Results for $\hat{\zeta} = 0.539684$ ($\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$)

The extracted parameters from the simultaneous fit for the nucleon momentum, $P_1 = -3$ ($\hat{\zeta} = 0.539684$), are in Table 24.

Table 24: Simultaneous Fit with common $f$: for $P_L = -3$, $b_{min} \geq 3$, $\eta \in [6, 10]$. Global $\chi^2$/DoF = 0.712(66)

| Amplitude | Model Expression | Params List | Fitted Values |
|---|---|---|---|
| $\widetilde{A}_{2B}^{\mathrm{Re}}$ | $\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, k_1, k_2, j, f\}$ | $[4.0061(31), 0.45(12), 0.136(11), 12.284(11), -36.1973(19), 2.085(10), 0.540(15)]$ |
| $\widetilde{A}_{2B}^{\mathrm{Im}}$ | $\frac{ab_Le^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.328(11), 0.163(15), 0.147(13), 1.057(15), 0.540(15)]$ |
| $\widetilde{A}_{12B}^{\mathrm{Re}}$ | $\frac{-ae^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.569(13), 0.100(17), 0.094(16), 1.9969(49), 0.540(15)]$ |

Figures 62–67 show the fit on lattice data, and Figures 68 and 69 show the resulting $x$-dependence.

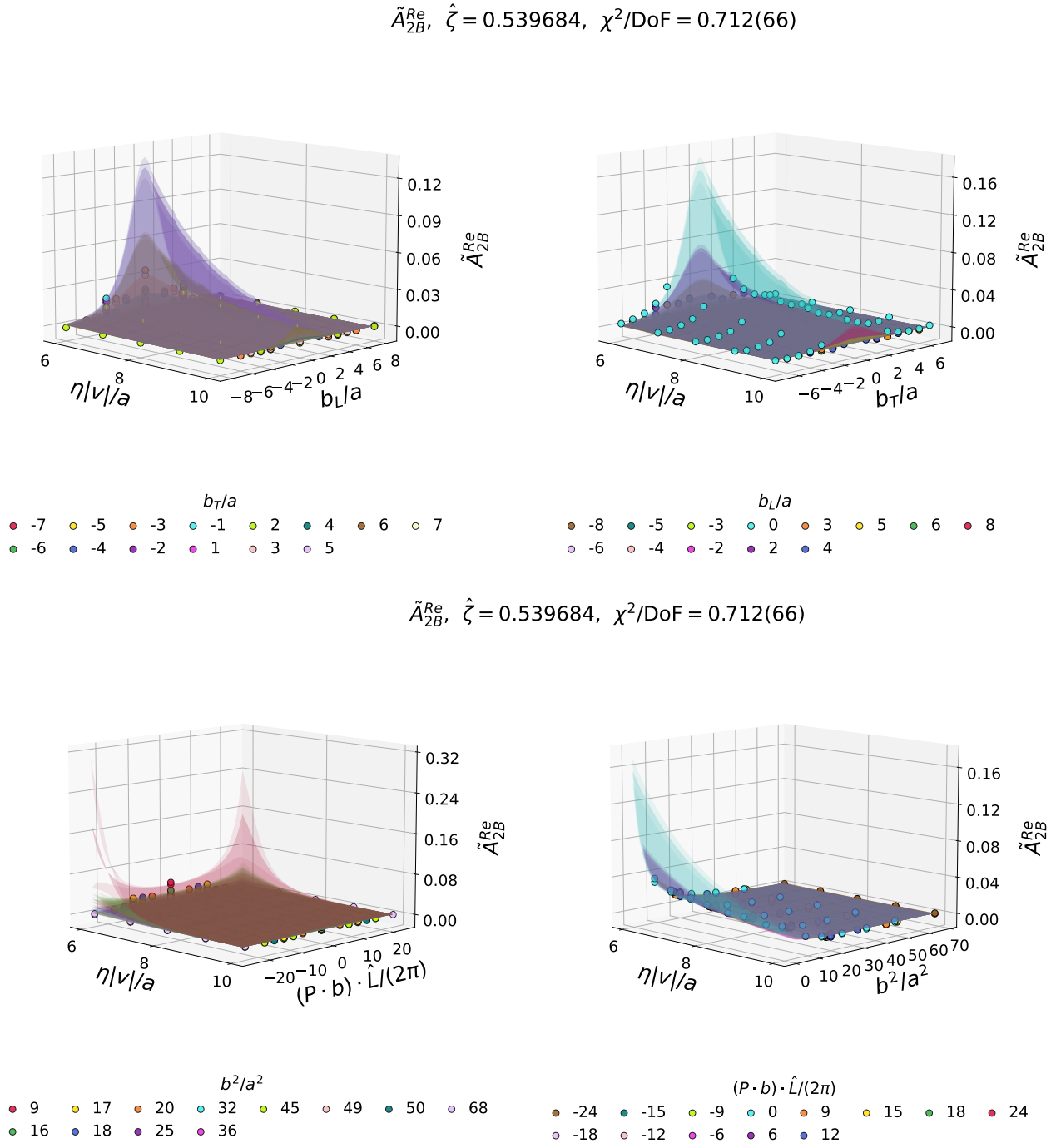



Figure 62: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

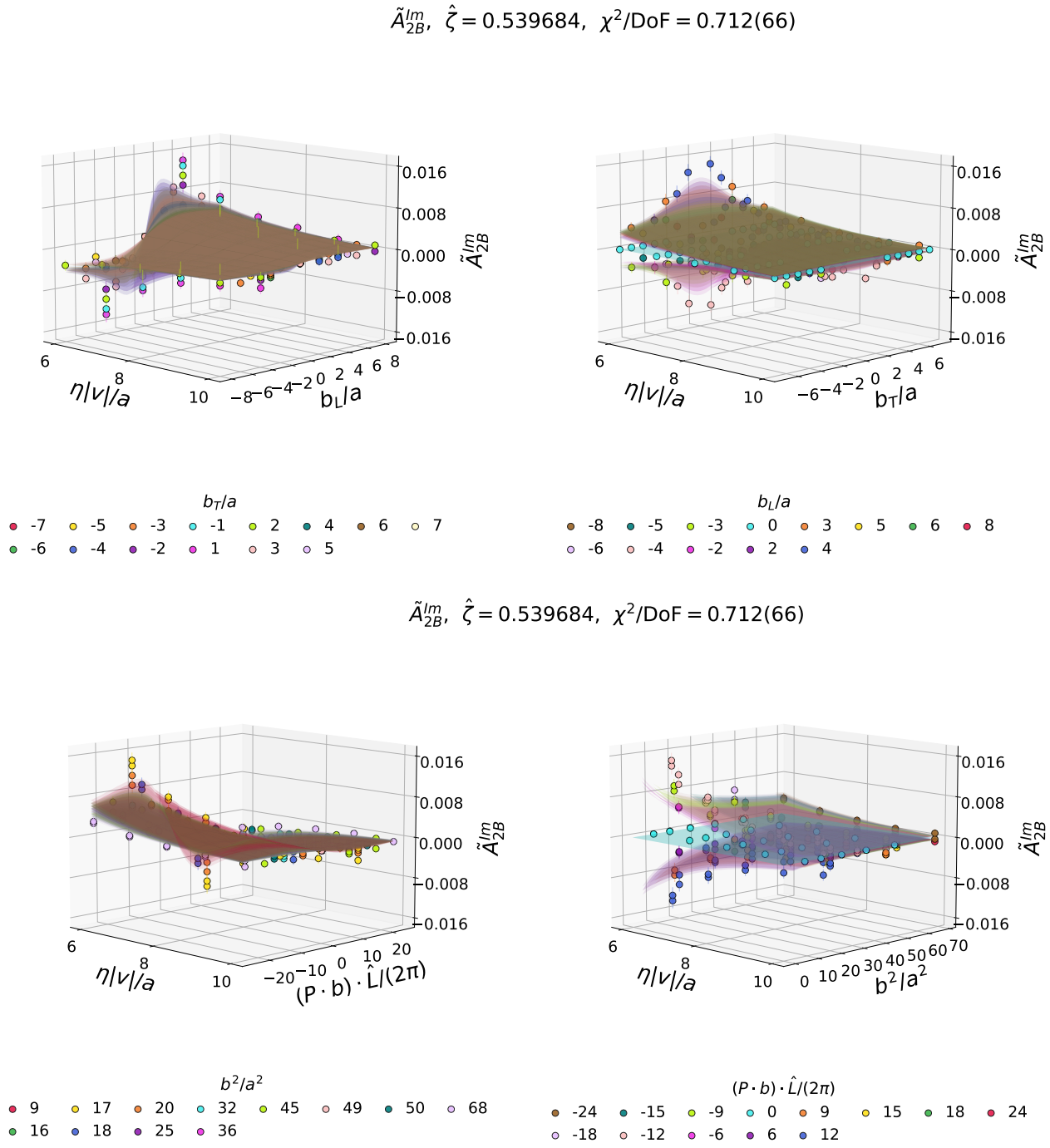



Figure 63: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

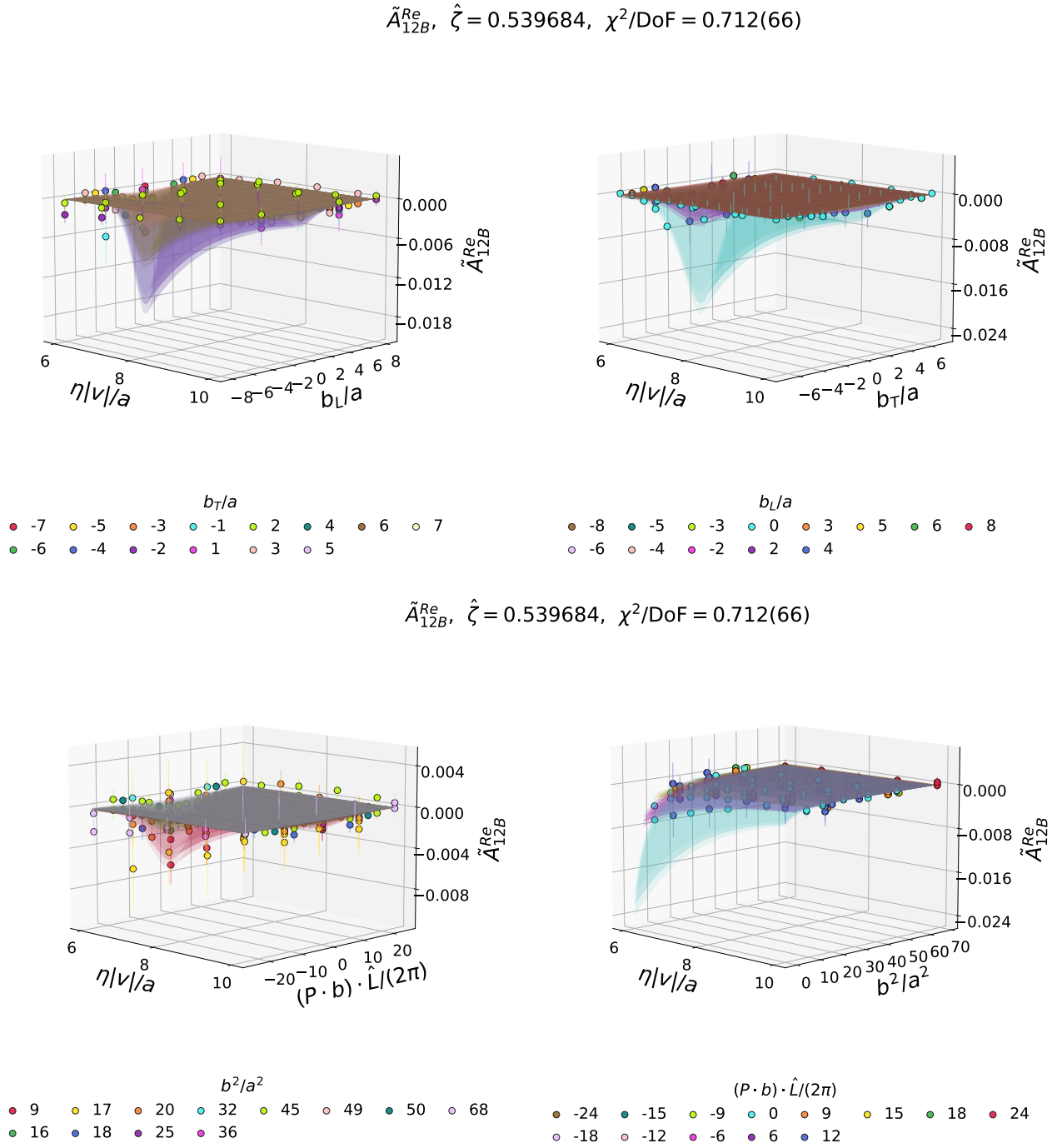



Figure 64: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

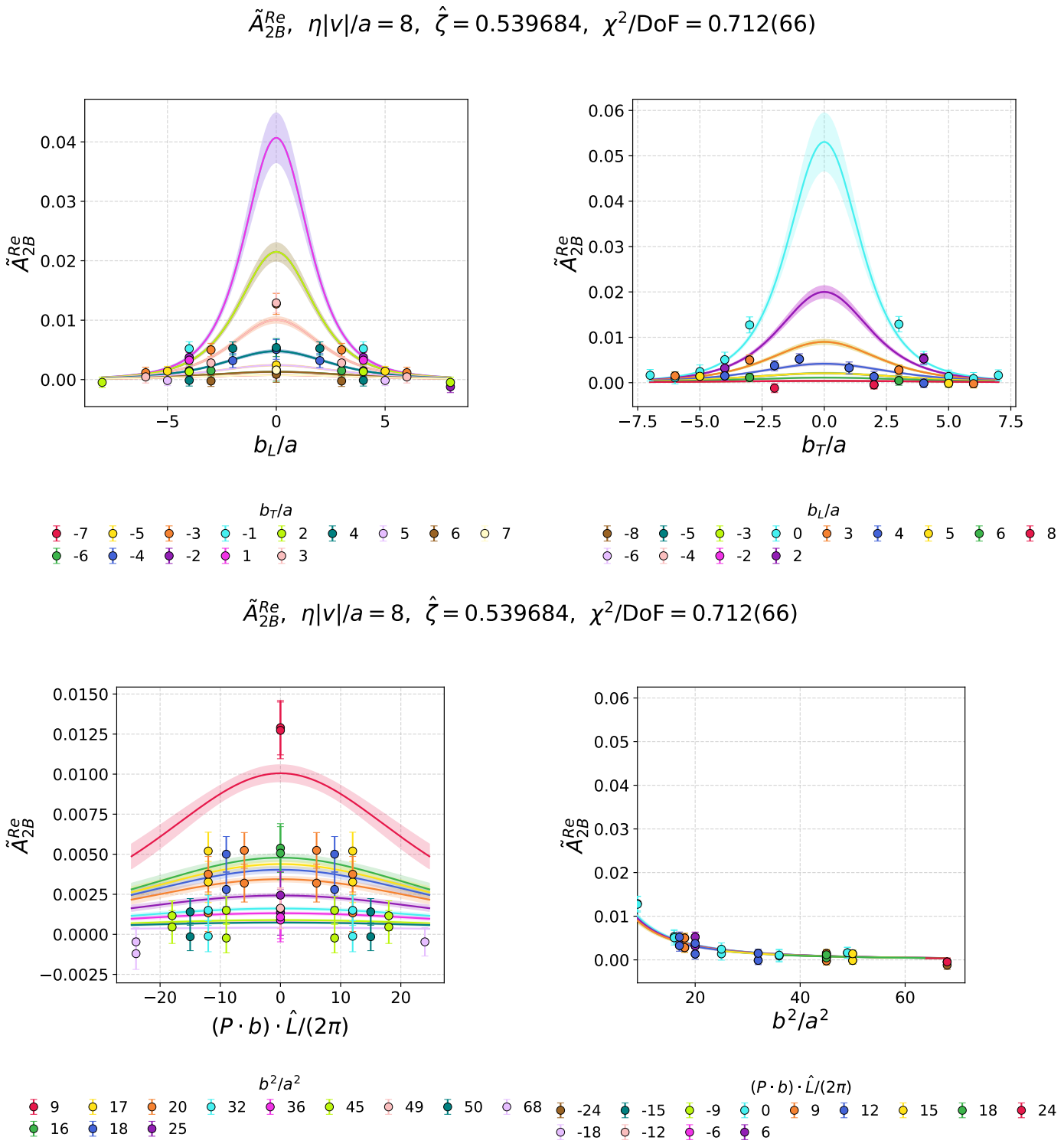



Figure 65: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

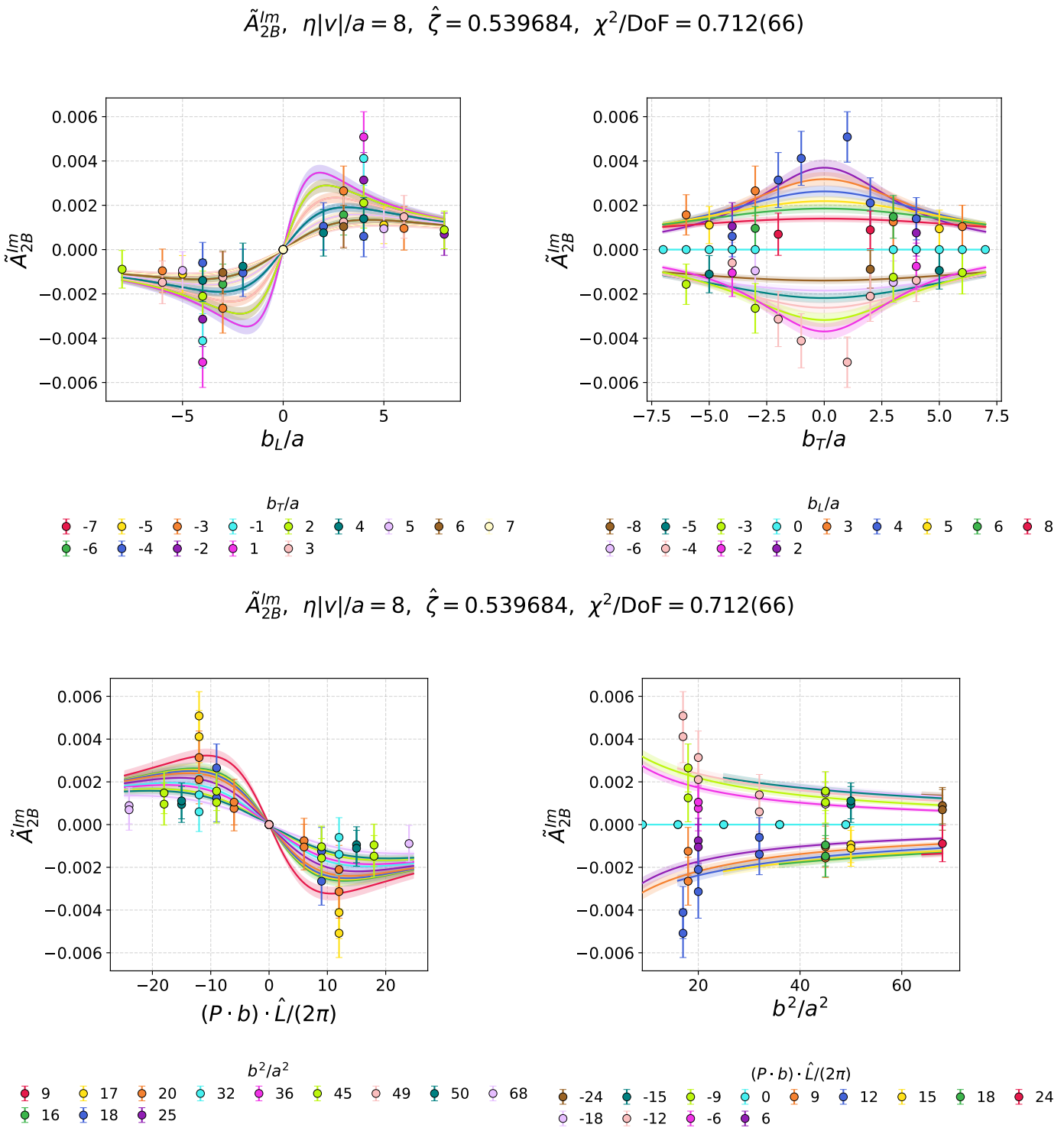



Figure 66: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

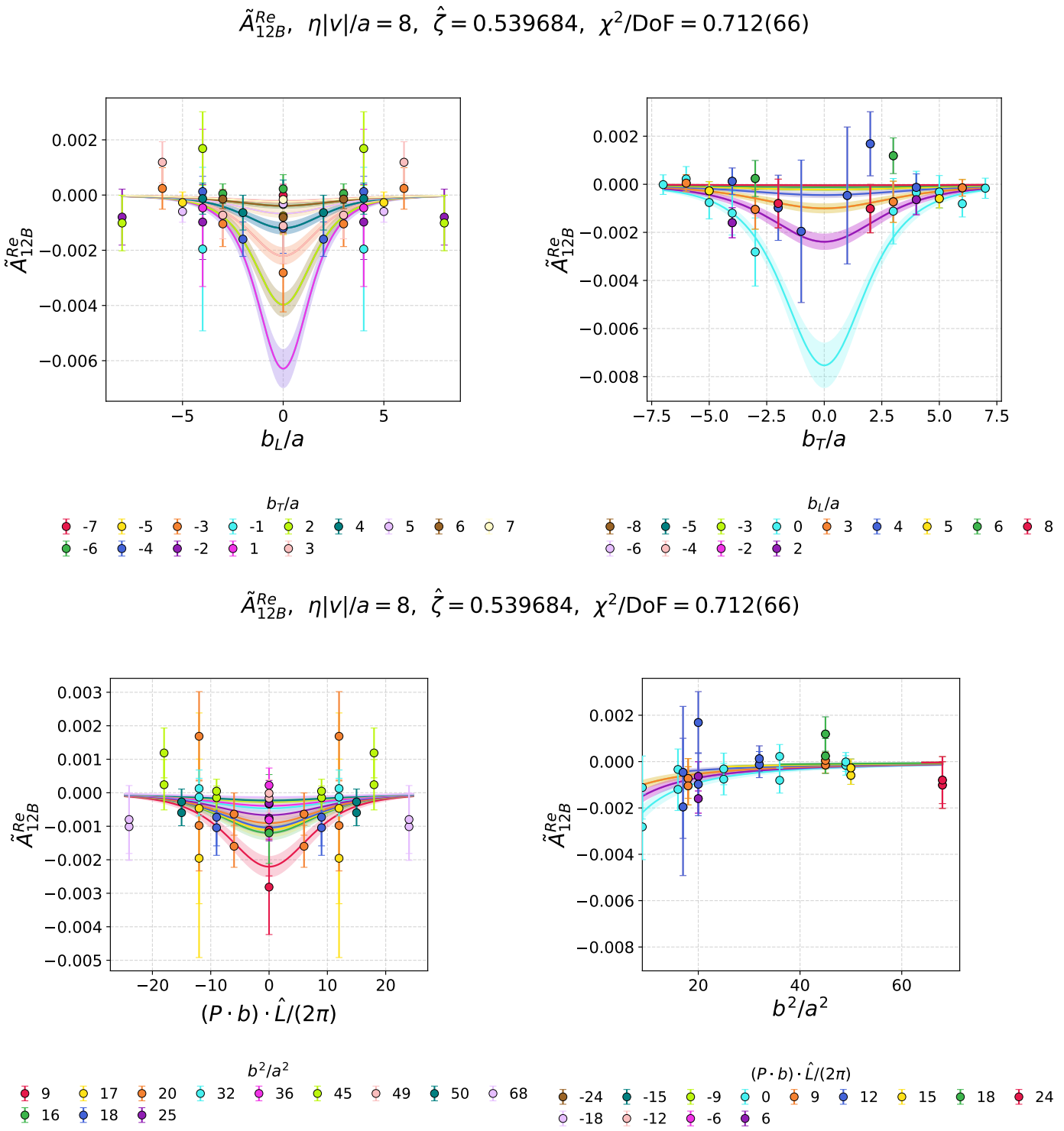



Figure 67: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-3, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

From the continuous parametrization of the amplitudes, we evaluate the Fourier integrals over the invariant scalar $P \cdot b$. As we detailed in Section 6.7, Tables 25, calculate the $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$, and finally the Mellin moment of Sivers shift TMD $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ for integration limit of $[-\infty, \infty]$, $[-1, 1]$, $[-1, 0]$, and $[0, 1]$; the results are consistent with previous studies of $x$-moment lattice calculations [2, 85].

Table 25: Numerical Integration of Fourier Transforms: $P_L = -3, |b_T| = 3a \approx 0.34$ fm

| Integral | $[-\infty, \infty]$ | $[-1, 1]$ | $[-1, 0]$ | $[0, 1]$ |
|---|---|---|---|---|
| $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ | 0.759(69) | 0.754(70) | 0.377(35) | 0.377(35) |
| $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | -1(320)e-17 | 2(33)e-16 | 0.234(23) | -0.235(23) |
| $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 1.52(14) | 1.51(14) | 0.286(55) | 1.22(11) |
| $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ | 0.333(51) | 0.333(51) | 0.167(25) | 0.167(25) |
| $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ | 0.236(32) | 0.238(32) | 0.63(13) | 0.147(20) |

Figs. 68 and 69 show the $x$-dependence obtained from the Fourier transforms of the fitted amplitudes: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$, $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, the Sivers shift $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$ ([momentum fraction] $\times$ [Sivers shift]). Figure 68 is evaluated at fixed $b^2 = 9a^2$, and Figure 69 shows the same quantities as functions of $x$ and $b^2$. The bands are the Jackknife uncertainties propagated from the fit parameters.

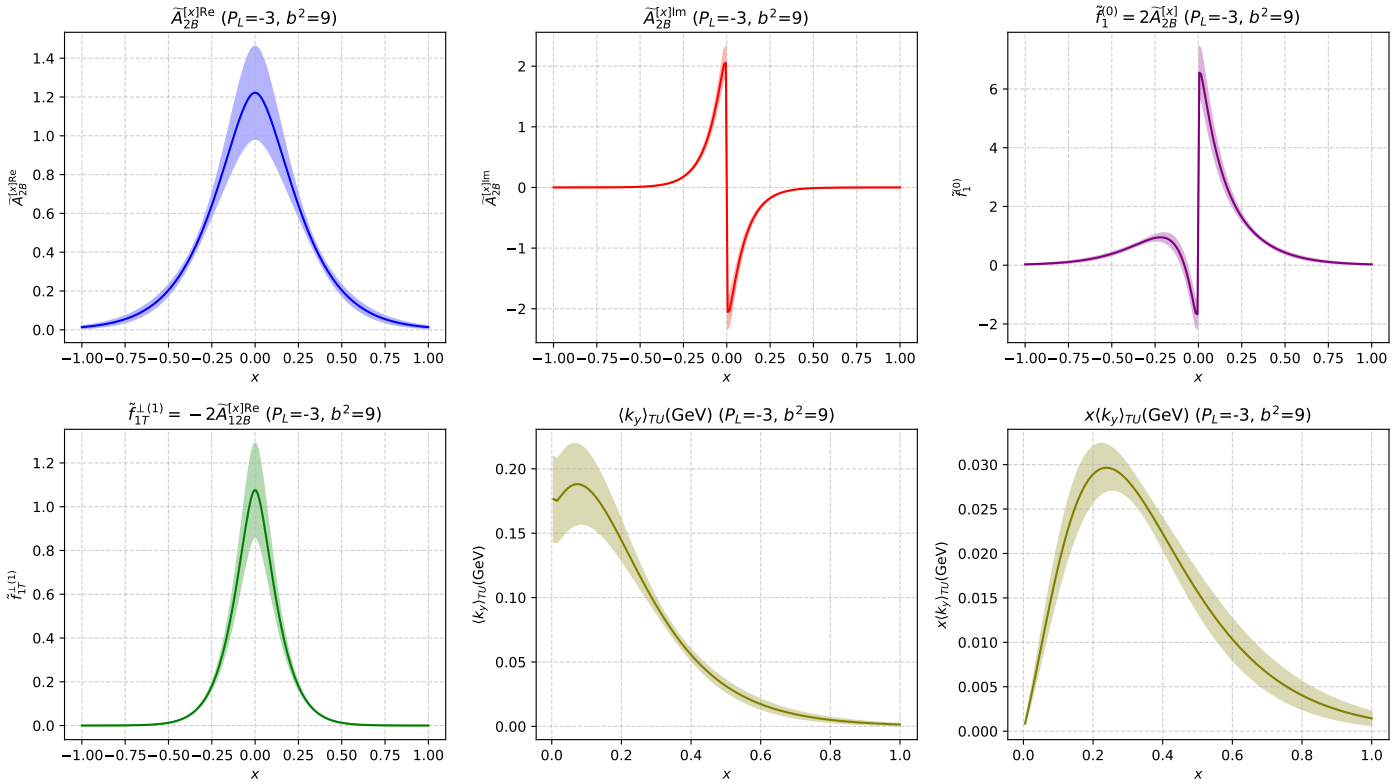



Figure 68: $x$-dependence at $b^2 = 9a^2$ for $P_L = -3$. Top row: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $i\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, and $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$. Bottom row: $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$. Bands show the Jackknife uncertainty.

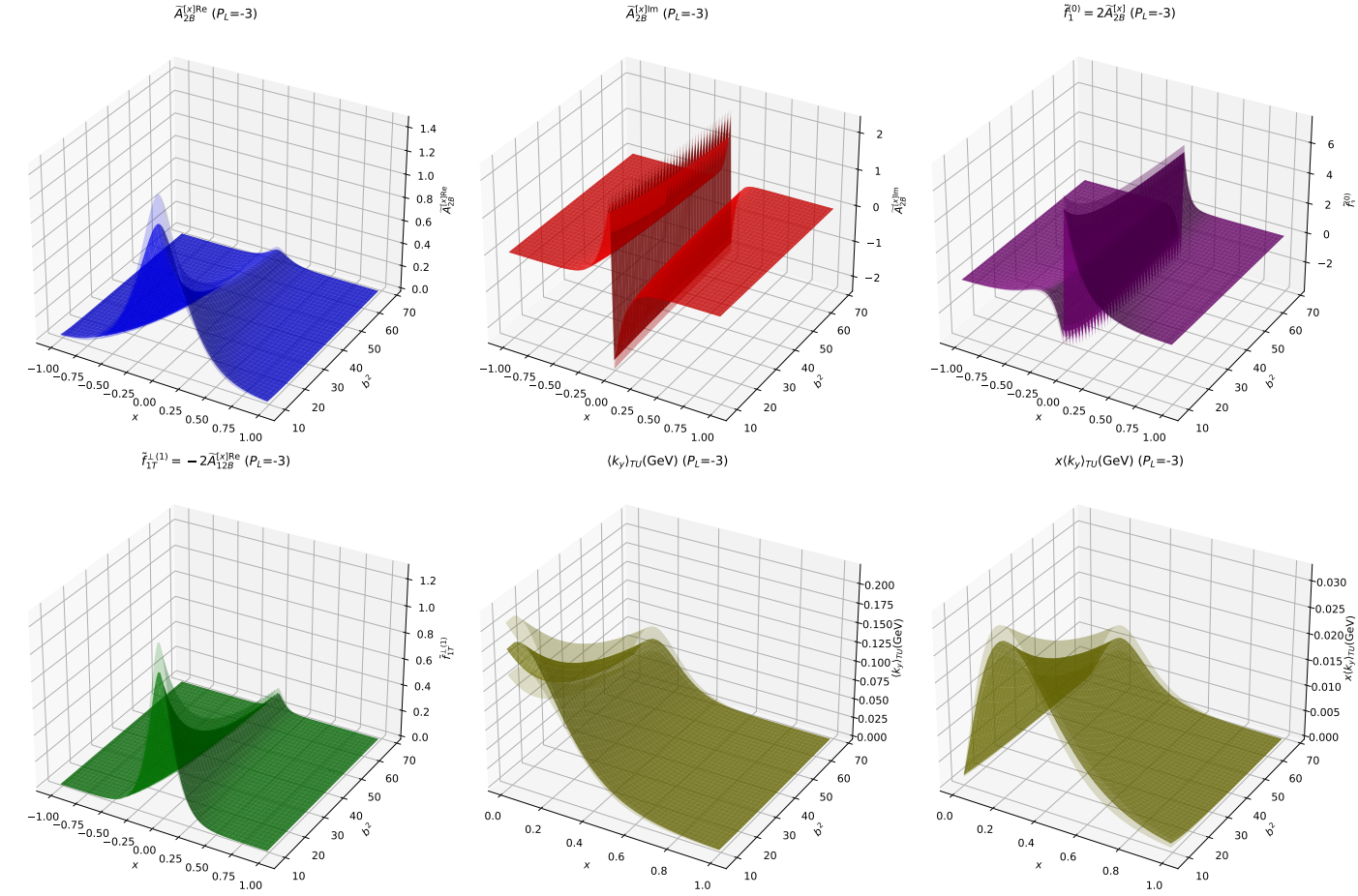



Figure 69: The quantities of Figure 68 as functions of $x$ and $b^2/a^2 \in [9, 68]$ for $P_L = -3$.

### 6.8.5 Simultaneous Fit Results for $\hat{\zeta} = 0.785756$ ($\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$)

The extracted parameters from the simultaneous fit for the nucleon momentum, $P_1 = -4$ ($\hat{\zeta} = 0.785756$), are in Table 26.

Table 26: Simultaneous Fit with common $f$: for $P_L = -4$, $b_{min} \geq 3$, $\eta \in [6, 10]$. Global $\chi^2$/DoF = 0.434(51)

| Amplitude | Model Expression | Params List | Fitted Values |
|---|---|---|---|
| $\widetilde{A}^{\mathrm{Re}}_{2B}$ | $\frac{ae^{-f\eta}}{\left[1+\left(d+c\left(1-\frac{k_1}{\eta}-\frac{k_2}{\eta^2}\right)\right)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, k_1, k_2, j, f\}$ | $[4.0086(15), 0.27(21), 0.122(13), 12.2751(58), -36.19925(89), 2.0654(56), 0.577(19)]$ |
| $\widetilde{A}^{\mathrm{Im}}_{2B}$ | $\frac{ab_Le^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.269(13), 0.264(47), 0.196(14), 1.0084(79), 0.577(19)]$ |
| $\widetilde{A}^{\mathrm{Re}}_{12B}$ | $\frac{-ae^{-f\eta}}{\left[1+(c+d)b_L^2+db_T^2\right]^j}$ | $\{a, c, d, j, f\}$ | $[0.5824(55), 0.116(19), 0.102(18), 2.0059(34), 0.577(19)]$ |

Figures 70–75 show the fit on lattice data, and Figures 76 and 77 show the resulting $x$-dependence.

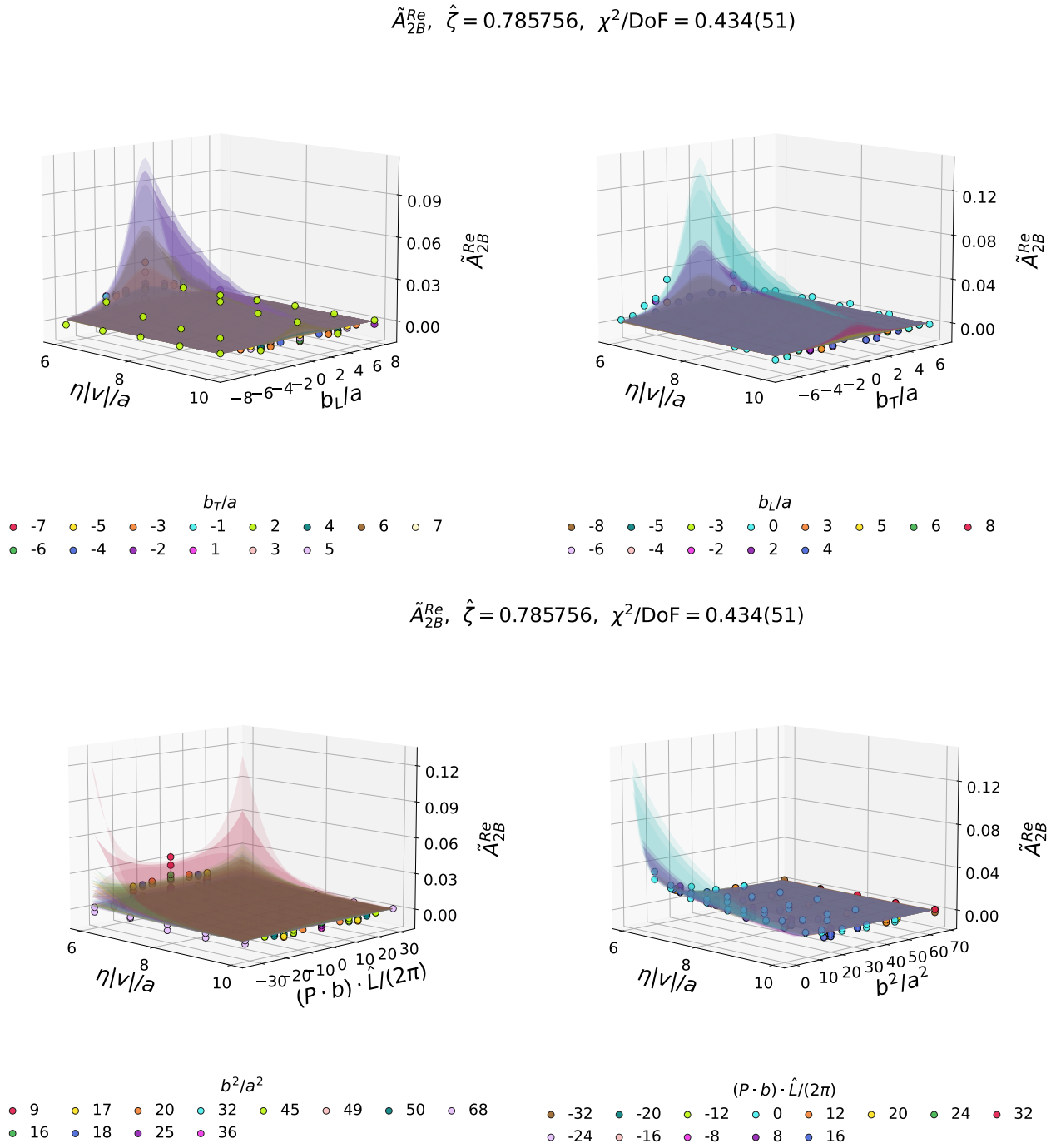



Figure 70: Fit for $\widetilde{A}_{2B}^{\mathrm{Re}}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

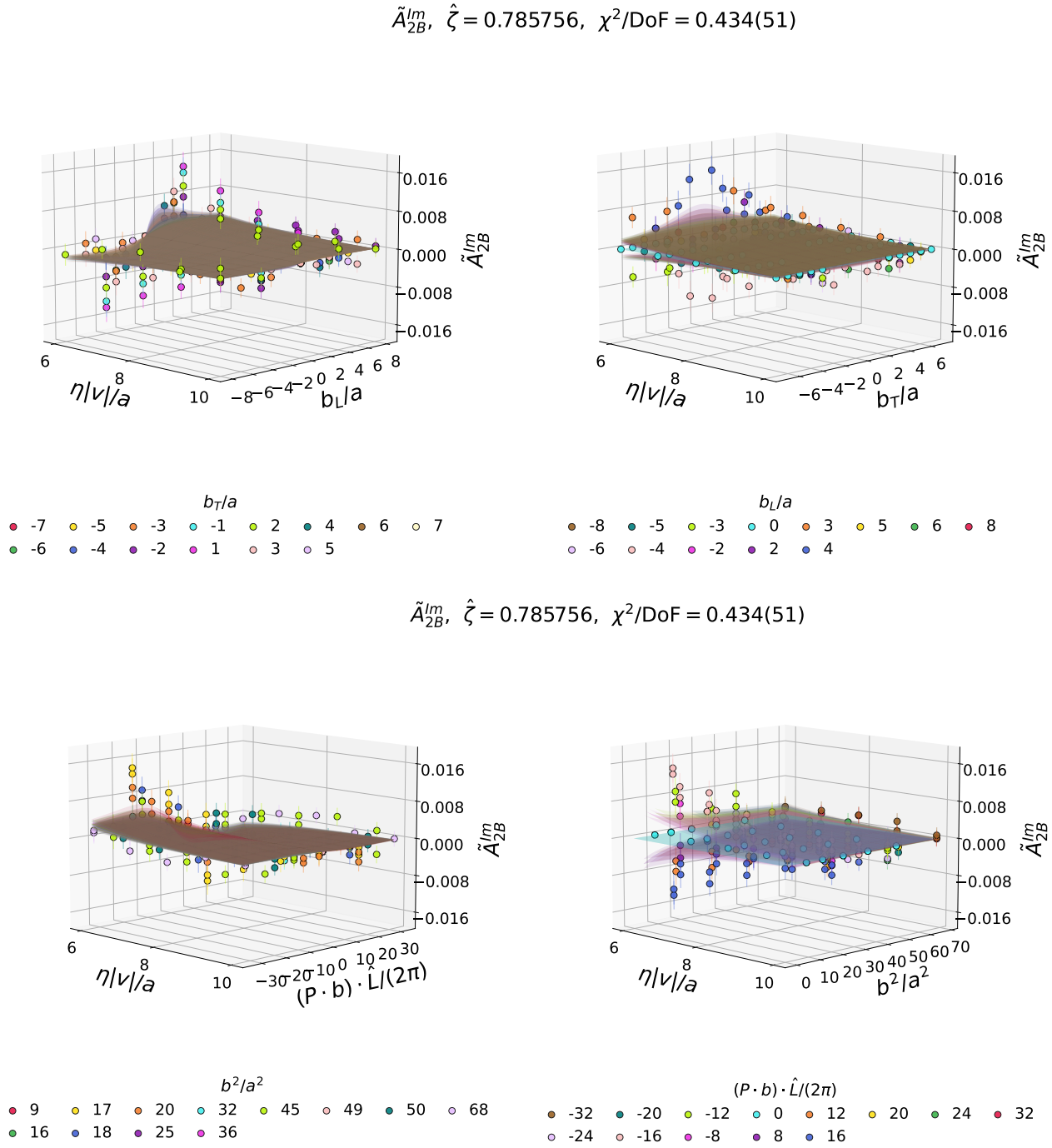



Figure 71: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ amplitude for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

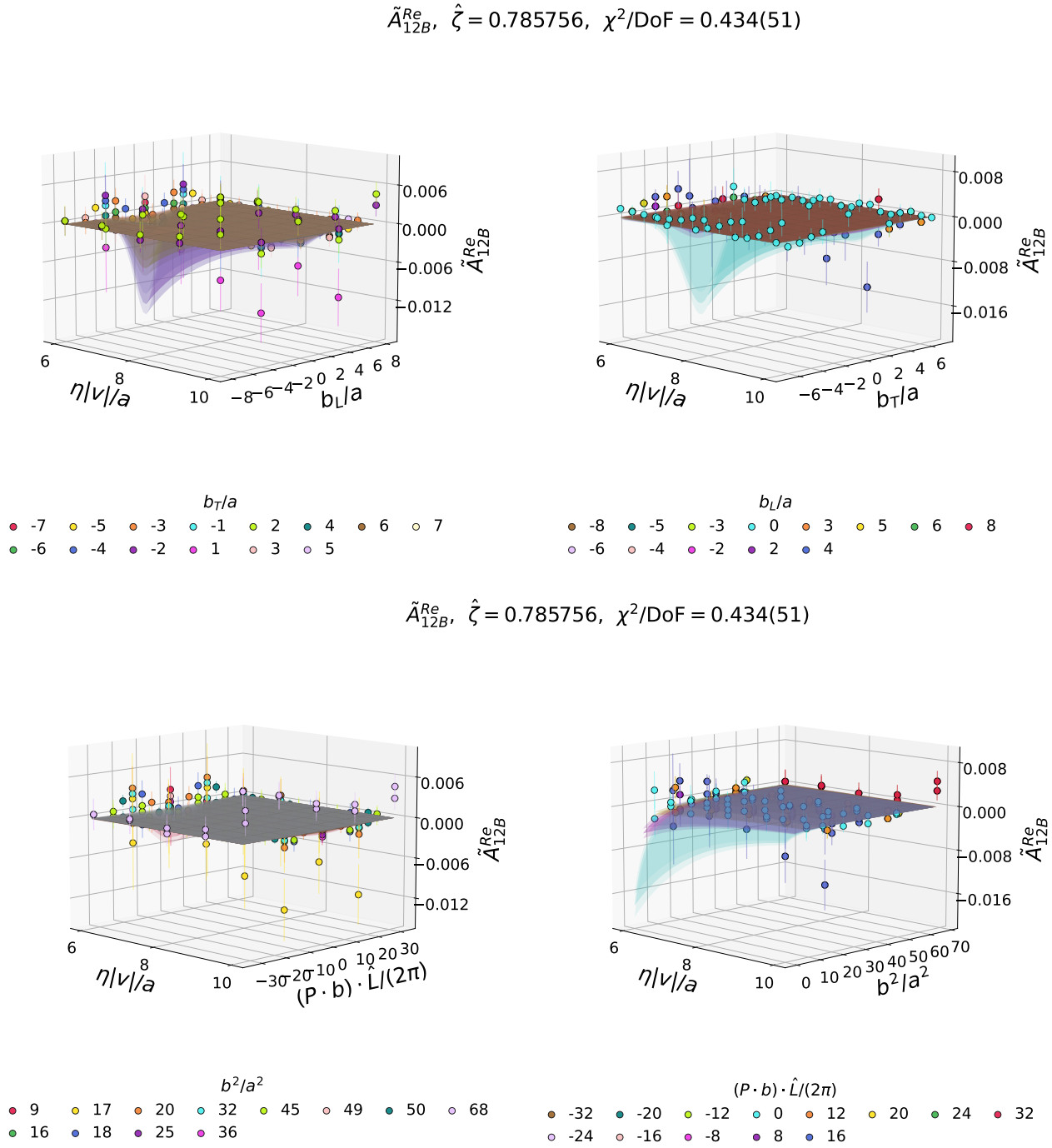



Figure 72: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ amplitude for $\boldsymbol{P}\cdot a\hat{L}/(2\pi) = \{-4,0,0\}$. The top row displays the $(b_L/a, b_T/a, \eta|\boldsymbol{v}|/a)$ space, while the bottom row displays the $((\boldsymbol{P}\cdot\boldsymbol{b})\cdot\hat{L}/(2\pi), b^2/a^2, \eta|\boldsymbol{v}|/a)$ space with the global simultaneous fit overlaid.

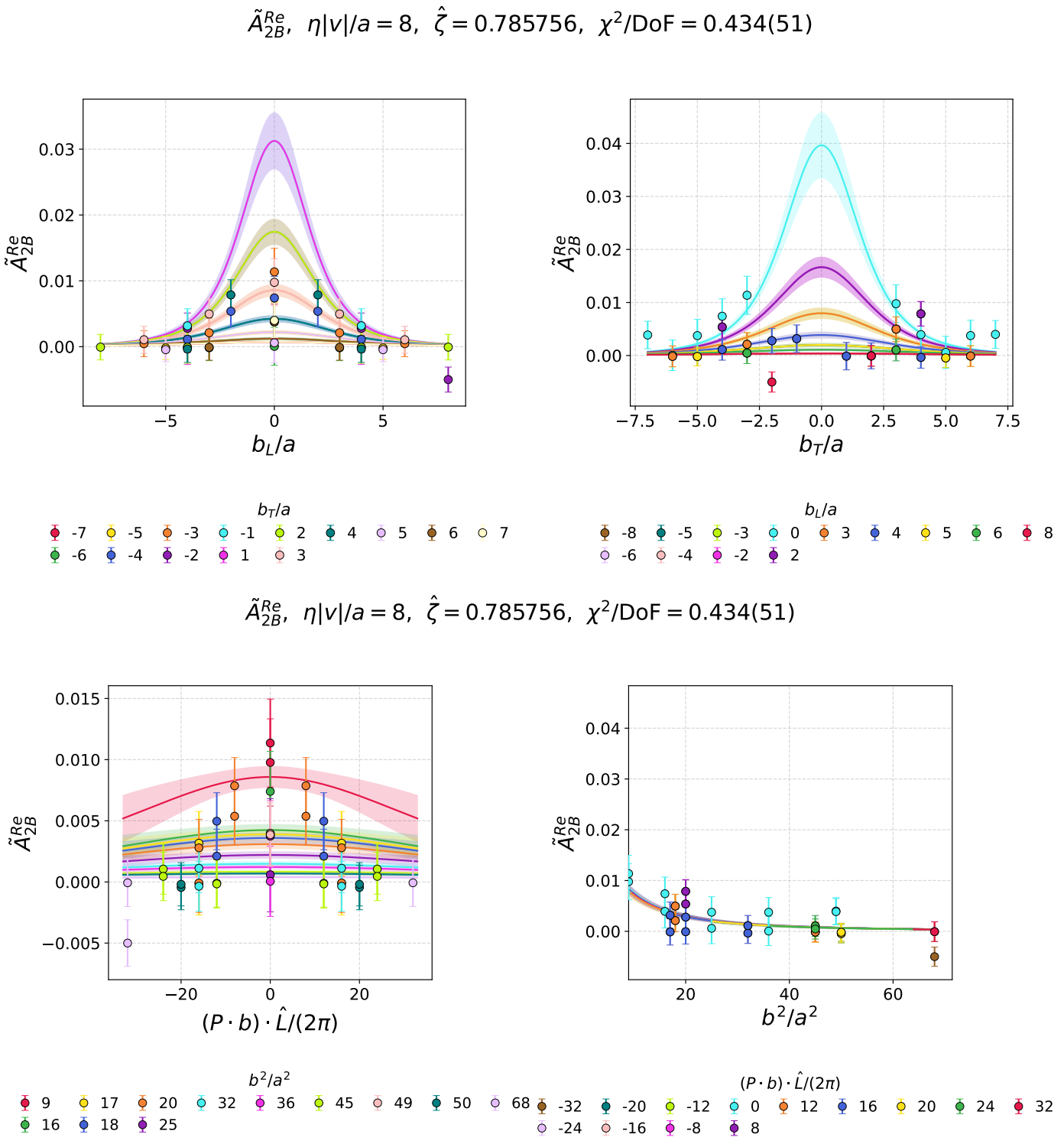



Figure 73: Fit for $\widetilde{A}^{\mathrm{Re}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

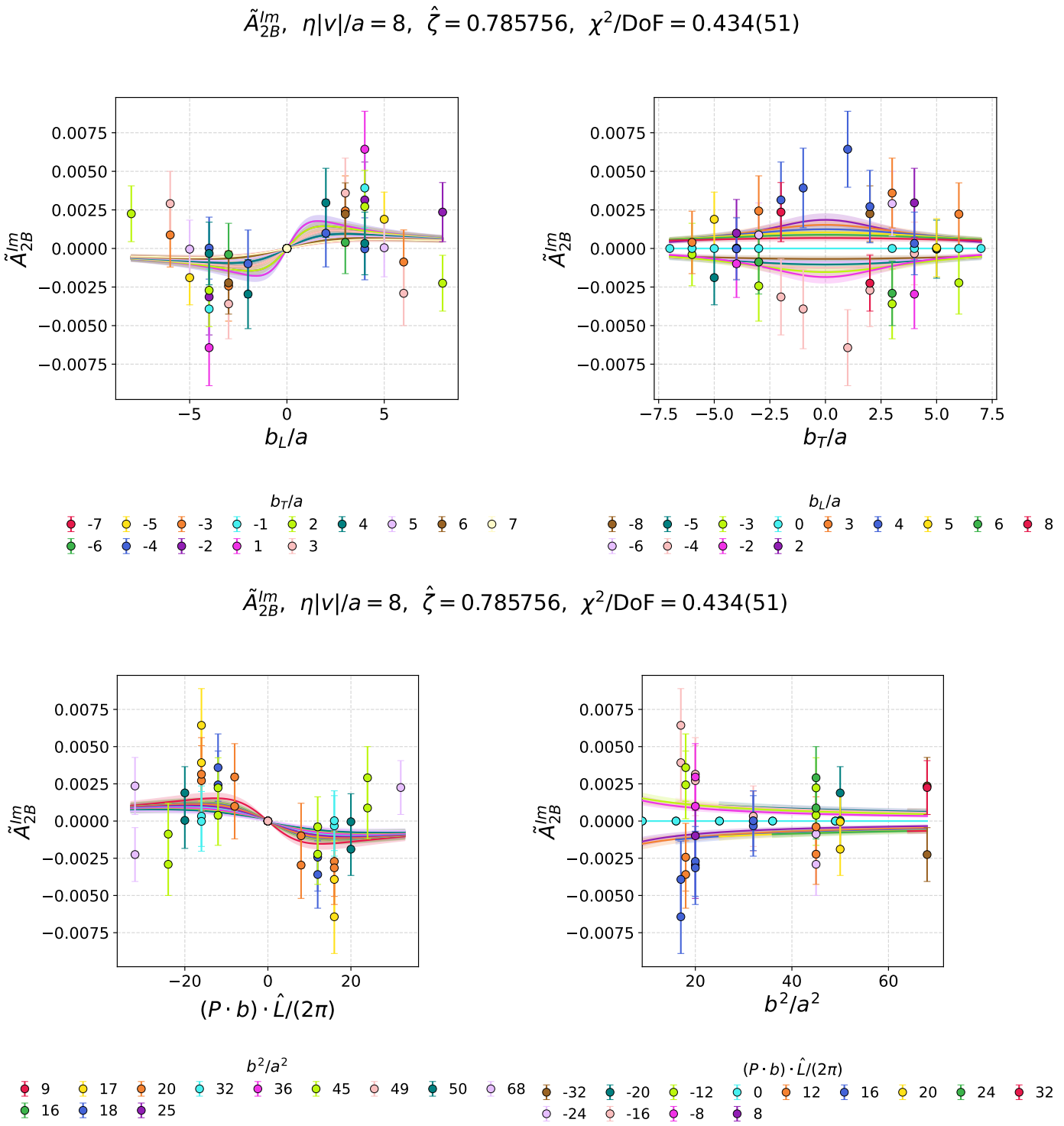



Figure 74: Fit for $\widetilde{A}^{\mathrm{Im}}_{2B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

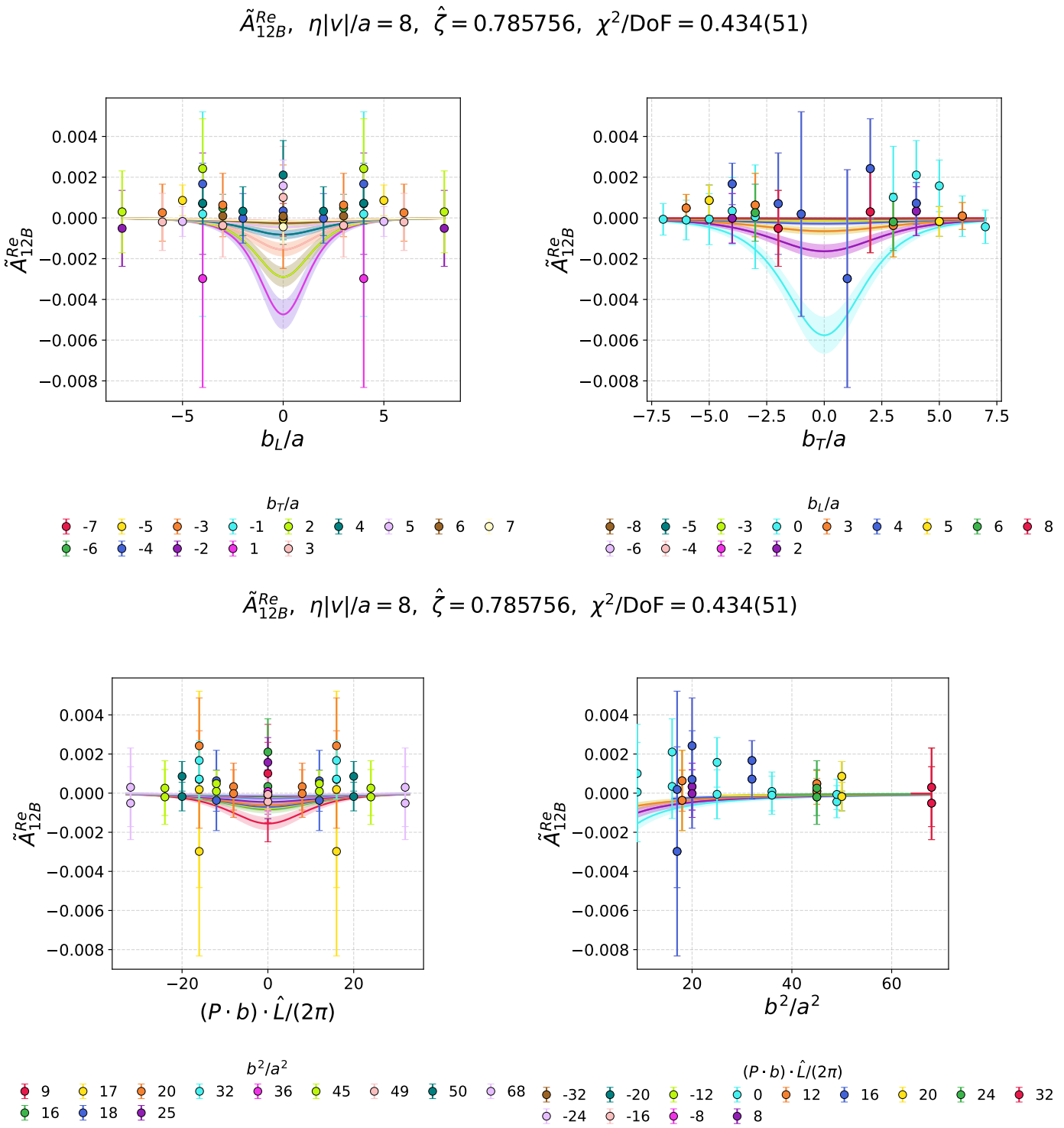



Figure 75: Fit for $\widetilde{A}^{\mathrm{Re}}_{12B}$ for $\boldsymbol{P} \cdot a\hat{L}/(2\pi) = \{-4, 0, 0\}$ at a fixed staple extent $\eta|\boldsymbol{v}|/a = 8$. The top row displays the $(b_L/a, b_T/a)$ space, while the bottom row displays the $((\boldsymbol{P} \cdot \boldsymbol{b}) \cdot \hat{L}/(2\pi), b^2/a^2)$ space with the fit.

From the continuous parametrization of the amplitudes, we evaluate the Fourier integrals over the invariant scalar $P \cdot b$. As we detailed in Section 6.7, Tables 27, calculate the $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$, $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$, and finally the Mellin moment of Sivers shift TMD $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ for integration limit of $[-\infty, \infty]$, $[-1, 1]$, $[-1, 0]$, and $[0, 1]$; the results are consistent with previous studies of $x$-moment lattice calculations [2, 85].

Table 27: Numerical Integration of Fourier Transforms: $P_L = -4, |b_T| = 3a \approx 0.34$ fm

| Integral | $[-\infty, \infty]$ | $[-1, 1]$ | $[-1, 0]$ | $[0, 1]$ |
|---|---|---|---|---|
| $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ | 0.87(10) | 0.87(10) | 0.434(52) | 0.434(52) |
| $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 3(207)e-17 | 2(21)e-16 | 0.153(26) | -0.154(26) |
| $\int dx\, \tilde{f}_1^{(0)} = 2\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 1.74(21) | 1.74(21) | 0.56(12) | 1.18(12) |
| $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ | 0.315(55) | 0.315(55) | 0.157(28) | 0.157(28) |
| $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ | 0.195(37) | 0.195(37) | 0.301(73) | 0.144(27) |

Figs. 76 and 77 show the $x$-dependence obtained from the Fourier transforms of the fitted amplitudes: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$, $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, the Sivers shift $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$ ([momentum fraction] $\times$ [Sivers shift]). Figure 76 is evaluated at fixed $b^2 = 9a^2$, and Figure 77 shows the same quantities as functions of $x$ and $b^2$. The bands are the Jackknife uncertainties propagated from the fit parameters.

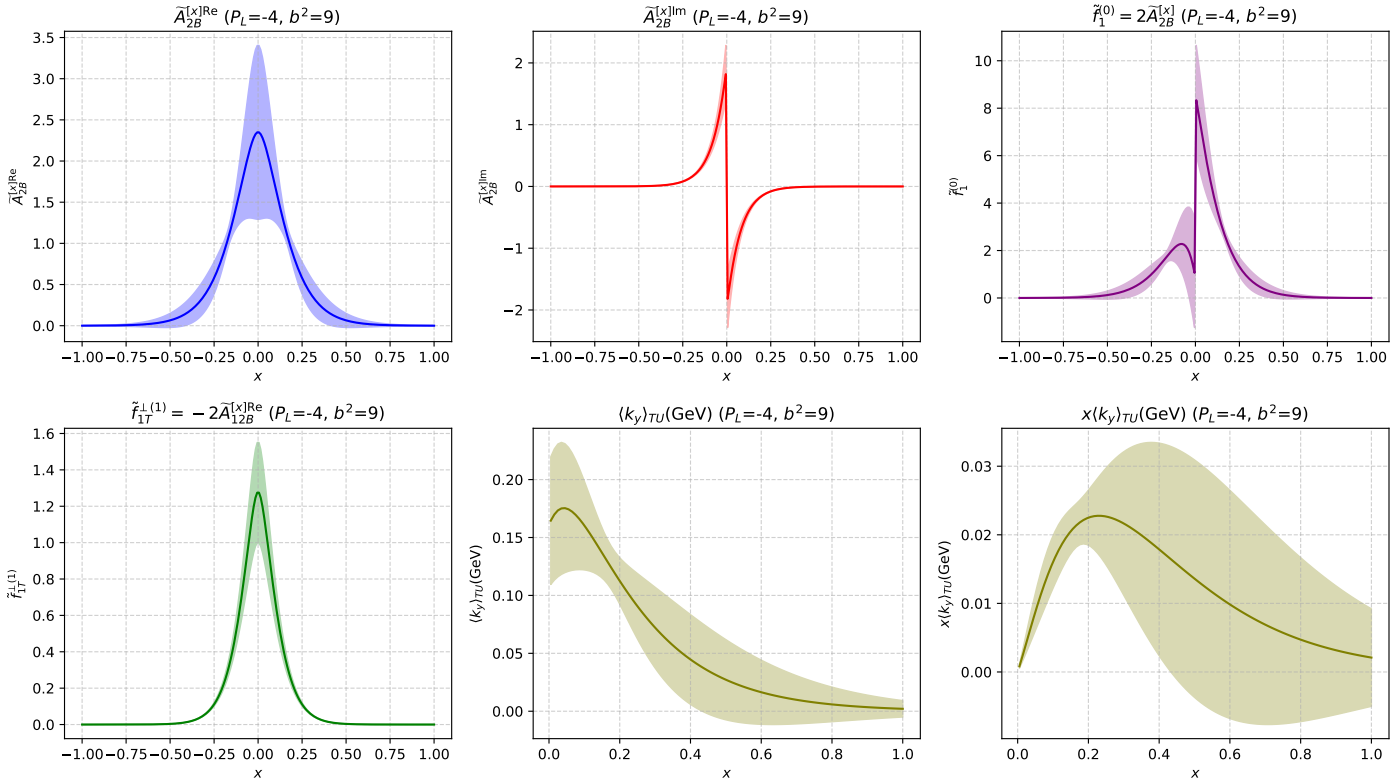



Figure 76: $x$-dependence at $b^2 = 9a^2$ for $P_L = -2$. Top row: $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$, $i\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, and $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$. Bottom row: $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, $\langle k_y \rangle_{TU}$, and $x\langle k_y \rangle_{TU}$. Bands show the Jackknife uncertainty.

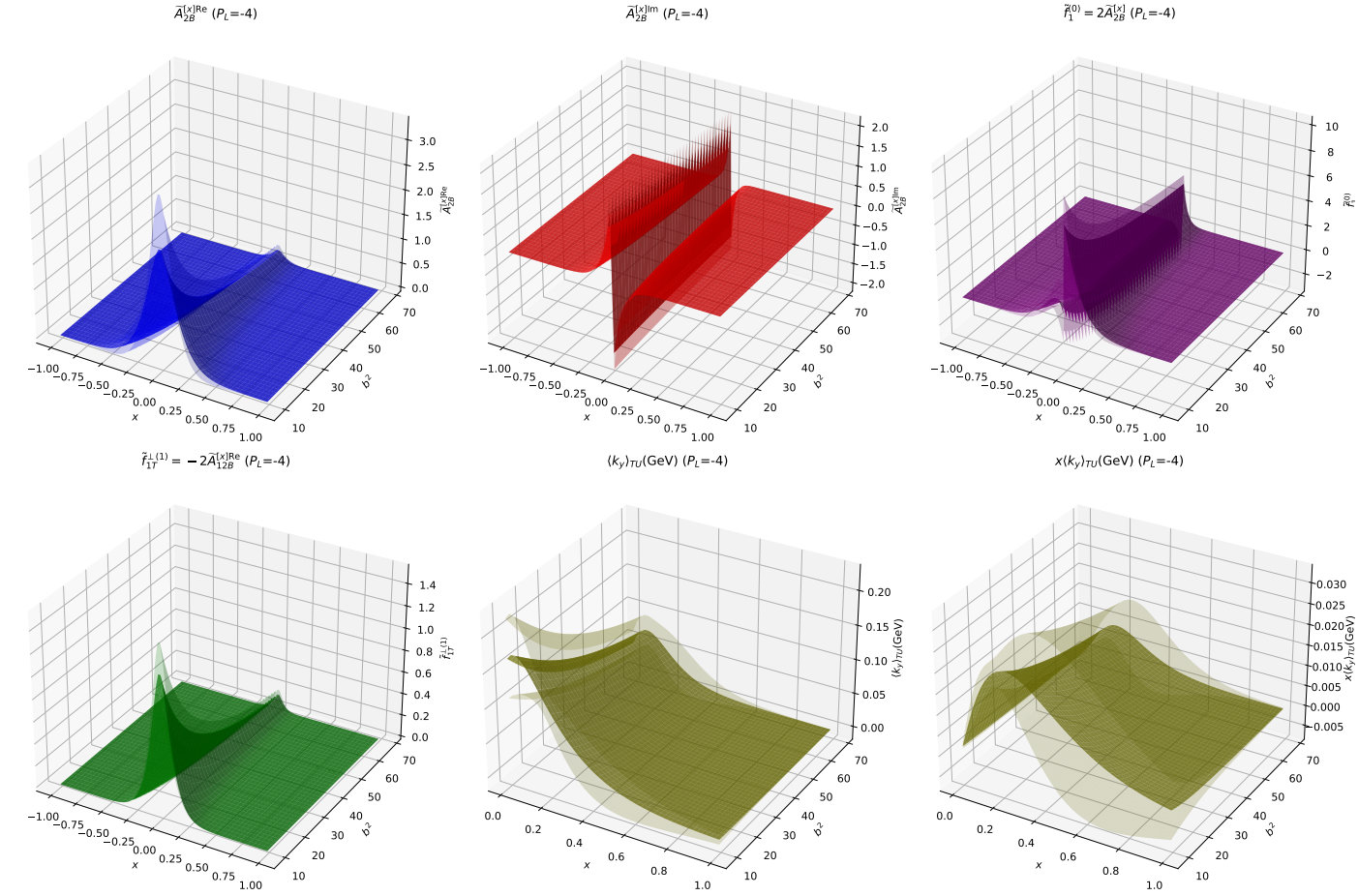



Figure 77: The quantities of Figure 76 as functions of $x$ and $b^2/a^2 \in [9, 68]$ for $P_L = -2$.

### 6.9 Separable Longitudinal and Transverse Models

We now apply separable longitudinal and transverse models to all three amplitudes and compare them with the selected power-law fit. Section 6.5 tested such profiles on $\widetilde{A}_{2B}^{\mathrm{Re}}$ alone; here we use the four lowest-$\chi^2$ combinations.

The four ansätze combine either an exponential ($e^{-c|P\cdot b|}$) or Gaussian ($e^{-c(P\cdot b)^2}$) longitudinal profile with either an exponential ($e^{-d|b|}$) or power-law ($(1+db^2)^{-j}$) transverse envelope:

1. `ExpPb_Expb`:

$$\widetilde{A}_{2B}^{\mathrm{Re}} = a_{\mathrm{re}}\, e^{-c_{\mathrm{re}}|P\cdot b| - d_{\mathrm{re}}|b|}\, e^{-f\eta}\ , \tag{6.28}$$

$$\widetilde{A}_{2B}^{\mathrm{Im}} = a_{\mathrm{im}}\,(P\cdot b)\, e^{-c_{\mathrm{im}}|P\cdot b| - d_{\mathrm{im}}|b|}\, e^{-f\eta}\ , \tag{6.29}$$

$$\widetilde{A}_{12B}^{\mathrm{Re}} = -a_{\mathrm{reA12B}}\, e^{-c_{\mathrm{reA12B}}|P\cdot b| - d_{\mathrm{reA12B}}|b|}\, e^{-f\eta}\ . \tag{6.30}$$

2. `ExpPb_PowerLawb`:

$$\widetilde{A}_{2B}^{\mathrm{Re}} = \frac{a_{\mathrm{re}}\, e^{-c_{\mathrm{re}}|P\cdot b| - f\eta}}{(1+d_{\mathrm{re}}b^2)^{j_{\mathrm{re}}}}\ , \tag{6.31}$$

$$\widetilde{A}_{2B}^{\mathrm{Im}} = \frac{a_{\mathrm{im}}\,(P\cdot b)\, e^{-c_{\mathrm{im}}|P\cdot b| - f\eta}}{(1+d_{\mathrm{im}}b^2)^{j_{\mathrm{im}}}}\ , \tag{6.32}$$

$$\widetilde{A}_{12B}^{\mathrm{Re}} = \frac{-a_{\mathrm{reA12B}}\, e^{-c_{\mathrm{reA12B}}|P\cdot b| - f\eta}}{(1+d_{\mathrm{reA12B}}b^2)^{j_{\mathrm{reA12B}}}}\ . \tag{6.33}$$

3. `GaussianPb_Expb`:

$$\widetilde{A}_{2B}^{\mathrm{Re}} = a_{\mathrm{re}}\, e^{-c_{\mathrm{re}}(P\cdot b)^2 - d_{\mathrm{re}}|b|}\, e^{-f\eta}\ , \tag{6.34}$$

$$\widetilde{A}_{2B}^{\mathrm{Im}} = a_{\mathrm{im}}\,(P\cdot b)\, e^{-c_{\mathrm{im}}(P\cdot b)^2 - d_{\mathrm{im}}|b|}\, e^{-f\eta}\ , \tag{6.35}$$

$$\widetilde{A}_{12B}^{\mathrm{Re}} = -a_{\mathrm{reA12B}}\, e^{-c_{\mathrm{reA12B}}(P\cdot b)^2 - d_{\mathrm{reA12B}}|b|}\, e^{-f\eta}\ . \tag{6.36}$$

4. `GaussianPb_PowerLawb`:

$$\widetilde{A}_{2B}^{\mathrm{Re}} = \frac{a_{\mathrm{re}}\, e^{-c_{\mathrm{re}}(P\cdot b)^2 - f\eta}}{(1+d_{\mathrm{re}}b^2)^{j_{\mathrm{re}}}}\ , \tag{6.37}$$

$$\widetilde{A}_{2B}^{\mathrm{Im}} = \frac{a_{\mathrm{im}}\,(P\cdot b)\, e^{-c_{\mathrm{im}}(P\cdot b)^2 - f\eta}}{(1+d_{\mathrm{im}}b^2)^{j_{\mathrm{im}}}}\ , \tag{6.38}$$

$$\widetilde{A}_{12B}^{\mathrm{Re}} = \frac{-a_{\mathrm{reA12B}}\, e^{-c_{\mathrm{reA12B}}(P\cdot b)^2 - f\eta}}{(1+d_{\mathrm{reA12B}}b^2)^{j_{\mathrm{reA12B}}}}\ . \tag{6.39}$$

Each model has a closed-form Fourier transform, so the resulting $x$-space distributions are easy to inspect. We performed global simultaneous Jackknife fits on these four ansatze using the same kinematic cuts ($|\boldsymbol{b}| \geq 3a$, $\eta \in [6, 10]$) and shared area-law factor $e^{-f\eta}$. The fit quality and extracted parameters are summarized in Table 28.

Table 28: Comparison of the four separable longitudinal-transverse models across nucleon momenta $P_L$, with extracted parameters shown at $P_L = -1$. While models with power-law transverse envelopes describe $\widetilde{A}^{\mathrm{Re}}_{2B}$ passably, all four separable ansätze fail on $\widetilde{A}^{\mathrm{Im}}_{2B}$ and $\widetilde{A}^{\mathrm{Re}}_{12B}$.

| **Model Ansatz** | **$\chi^2$/dof Across Momenta** | | | | **Key Fitted Parameters at $P_L = -1$** | | | **Primary Failure Mode** |
|---|---|---|---|---|---|---|---|---|
| | $P_L = -1$ | $P_L = -2$ | $P_L = -3$ | $P_L = -4$ | $c_{\mathrm{im}}$ | $c_{\mathrm{reA12B}}$ | $a_{\mathrm{im}}$ | |
| `ExpPb_Expb` | 1.406(92) | 0.956(75) | 0.622(61) | 0.455(53) | 0.0010(262) | 0.045(23) | −0.167(16) | $c_{\mathrm{im}} \to 0$; unphysical sign on $a_{\mathrm{im}}$. |
| `ExpPb_PowerLawb` | 1.286(89) | 0.938(75) | 0.623(61) | 0.447(53) | 0.0045(248) | 0.028(24) | −0.167(31) | Fits $\widetilde{A}^{\mathrm{Re}}_{2B}$ well, but $c_{\mathrm{im}}$ and $c_{\mathrm{reA12B}}$ vanish within error; flips sign of $a_{\mathrm{im}}$. |
| `GaussianPb_Expb` | 2.389(121) | 2.005(111) | 0.945(76) | 0.453(52) | 0.568(171) | 0.0038(69) | +0.050(121) | Poor $\chi^2$/dof; $c_{\mathrm{reA12B}}$ pinned near zero; Gaussian over-damping at large $x$. |
| `GaussianPb_PowerLawb` | 2.316(120) | 1.981(111) | 0.948(76) | 0.448(52) | 0.602(48) | 0.0016(61) | +0.168(92) | Poor global fit at low momenta; $c_{\mathrm{reA12B}} \approx 0$; unphysical truncation of valence tails. |
| **Power-law in $((P \cdot b), b^2)$** | 1.85(10) | 1.197(85) | 0.712(66) | 0.434(51) | **0.2253(69)** | **0.0773(46)** | **+0.267(13)** | **All parameters stable, positive, and well-constrained across all amplitudes.** |

The numerical results show several ways in which the separable models fail. In `ExpPb_PowerLawb`, the real unpolarized amplitude fits with $\chi^2/\mathrm{dof} \approx 1.286$. Because $\widetilde{A}^{\mathrm{Re}}_{2B}$ dominates the statistical weight, the fit quality alone can make the model look adequate. The same parameterization fails for the imaginary amplitude $\widetilde{A}^{\mathrm{Im}}_{2B}$, however: the longitudinal decay width $c_{\mathrm{im}} = 0.0045 \pm 0.0248$ is zero within errors. The model then loses its longitudinal decay profile and gives an essentially flat distribution in $P \cdot b$. The overall amplitude also flips to an unphysical negative sign ($a_{\mathrm{im}} = -0.167$).

For the real Sivers amplitude $\widetilde{A}^{\mathrm{Re}}_{12B}$, the longitudinal parameter $c_{\mathrm{reA12B}}$ is similarly suppressed across all separable models ($0.028 \pm 0.024$ for `ExpPb_PowerLawb` and $0.0016 \pm 0.0061$ for `GaussianPb_PowerLawb`). The separable models cannot capture the spin-dependent correlator's distinct longitudinal curvature without forcing the decay parameter to its lower boundary. Separating out an exponential $|P \cdot b|$ dependence introduces a discontinuous first derivative (a cusp) at $P \cdot b = 0$. For an odd-parity

amplitude like $\widetilde{A}_{2B}^{\mathrm{Im}}$, which must vanish linearly through the origin ($\widetilde{A}_{2B}^{\mathrm{Im}} \propto b_L$), the non-analytic factor $e^{-c|P\cdot b|}$ conflicts with the smooth behavior of the Euclidean correlation function. The fit responds by driving $c_{\mathrm{im}} \to 0$ to minimize the cusp penalty.

In the Lorentz-invariant variable space $(P\cdot b, b^2)$, the longitudinal separation $b_L = (P\cdot b)/(-P_L)$ and transverse separation $b_T = \sqrt{b^2 - b_L^2}$ are geometrically coupled. Separable models artificially decouple $P\cdot b$ and $b^2$. In contrast, coupling $(P\cdot b)^2$ and $b^2$ within a single denominator,

$$D = \left[1 + c(\eta)\frac{(P\cdot b)^2}{P_L^2} + d\, b^2\right]^j \ , \tag{6.40}$$

preserves smooth analyticity at $P\cdot b = 0$, satisfies power-law integrability, and yields stable, positive parameters across all three amplitudes, matching the values already obtained from the global fit above (e.g., $c_{\mathrm{im}} = 0.2253(69)$, $c_{\mathrm{reA12B}} = 0.0773(46)$, $a_{\mathrm{im}} = +0.267(13)$). This confirms that the coupled power-law structure adopted for the global fit is preferred over the four separable forms tested above.

In the next section, we extrapolate the lattice calculation modeled by the adopted power-law ansatz to $\hat{\zeta} \to \infty$.

# 7 EXTRAPOLATIONS TO THE PHYSICAL LIMIT

All quantities in this chapter are for the isovector $u-d$ combination. Combining the results obtained at each momentum in Section 6.8, we extrapolate them to $\hat{\zeta} \to \infty$, which for the Sivers shift is the physical limit. Section 7.1 shows that the amplitudes, the TMD combinations, and the Sivers shift change systematically with $1/\hat{\zeta}$ across the four simulated momenta. Section 7.2 extrapolates the fit parameters to $1/\hat{\zeta} \to 0$ using $P_L = -2, -3, -4$, reconstructs the amplitudes and the Sivers shift in this limit, and lists the numerical Fourier-transform integrals over $x$ at $|b_T| = 3a$. Section 7.3 tests factorization, first in $(P \cdot b)$ and $b^2$ on the lattice data at finite momentum, and then in $x$ and $|b_T|$ for the extrapolated Sivers shift and amplitudes.

## 7.1 Motivation for the $\hat{\zeta} \to \infty$ Extrapolation

Chapter 6 established stable global simultaneous fits to the invariant amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ at each of our four simulated nucleon momenta, corresponding to the finite Collins-Soper parameters $\hat{\zeta} = 0.090772, 0.294367, 0.539684, 0.785756$ for $P_L = -1, -2, -3, -4$ respectively (Table 2). As discussed in Section 5.2, the fully quantitative predictions for, or a comparison with, phenomenological and experimental TMD studies employing QCD factorization would require lattice data for much larger Collins-Soper parameters, $\hat{\zeta} \gg 1$. With $\hat{\zeta}^2 = -(v \cdot P)^2/(v^2 m_N^2)$, the limit $\hat{\zeta} \to \infty$ corresponds to the limit of a light-like staple direction $v$, or an infinite rapidity $y_v \to -\infty$ in Eq. (4.12).

Figures 78–79 show why we need this extrapolation. At fixed transverse quark separation $|b_T| = 3a \approx 0.34$ fm (the TMD-frame separation, $|b_T|^2 = b^2 = b_L^2 + b_T^2$ in the lattice components of Section 2.10), we evaluate the reconstructed amplitudes $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$ and $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$, the corresponding unpolarized and Sivers TMD combinations $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$ and $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, and the resulting Sivers shift $\langle k_y \rangle_{TU}$ and $x\langle k_y \rangle_{TU}$, as continuous functions of $x$ at each of our four simulated momenta. Plotted against $1/\hat{\zeta}$, all six quantities vary systematically with the momentum.

For the Sivers shift defined in Eq. (4.48), where the soft factors in the TMD definitions [31, 32, 23] cancel out, large values of $\hat{\zeta}$ can also be accessed through large nucleon momenta. Clearly, the limit

of an infinite Collins-Soper parameter is in practice not accessible on the lattice, so that we must rely on results for a limited range of $\hat{\zeta}$, as in Figures 78–79. From the perturbative prediction for the $\zeta$ dependence, cf., e.g., Ref. [32], we would expect that the ratios of TMDs should become independent of $\hat{\zeta}$ as $\hat{\zeta} \to \infty$.

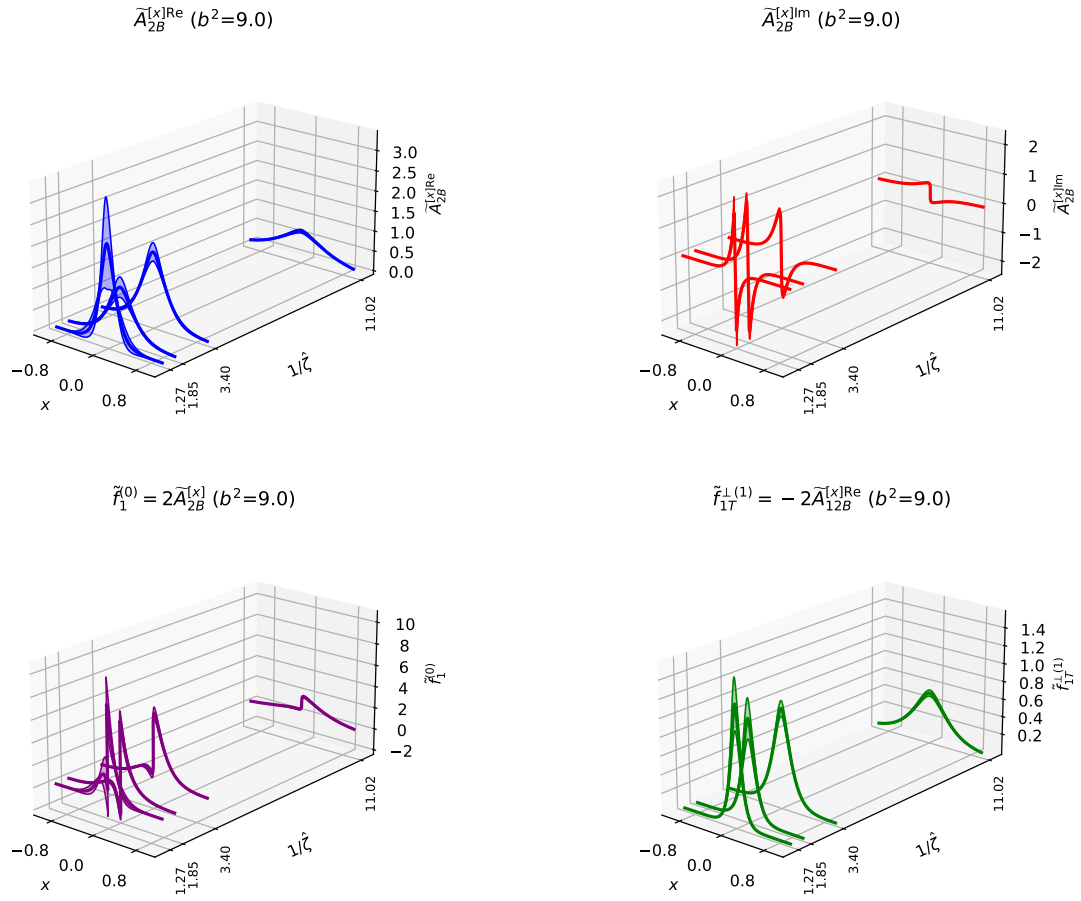



Figure 78: Discrete $\hat{\zeta}$-slices of the reconstructed amplitudes $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$ and $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$ and the unpolarized and Sivers TMD combinations $\tilde{f}_1^{(0)} = 2\widetilde{A}_{2B}^{[x]}$ and $\tilde{f}_{1T}^{\perp(1)} = -2\widetilde{A}_{12B}^{[x]\mathrm{Re}}$, plotted against both $x$ and $1/\hat{\zeta}$ at fixed transverse separation $|b_T| = 3a \approx 0.34$ fm. Each ribbon corresponds to one of the four simulated momenta $P_L = -1, -2, -3, -4$ (equivalently $1/\hat{\zeta} = 11.02,\ 3.40,\ 1.85,\ 1.27$). The systematic drift of every panel with $1/\hat{\zeta}$ motivates the extrapolation to the light-cone limit $\hat{\zeta} \to \infty$ performed below.

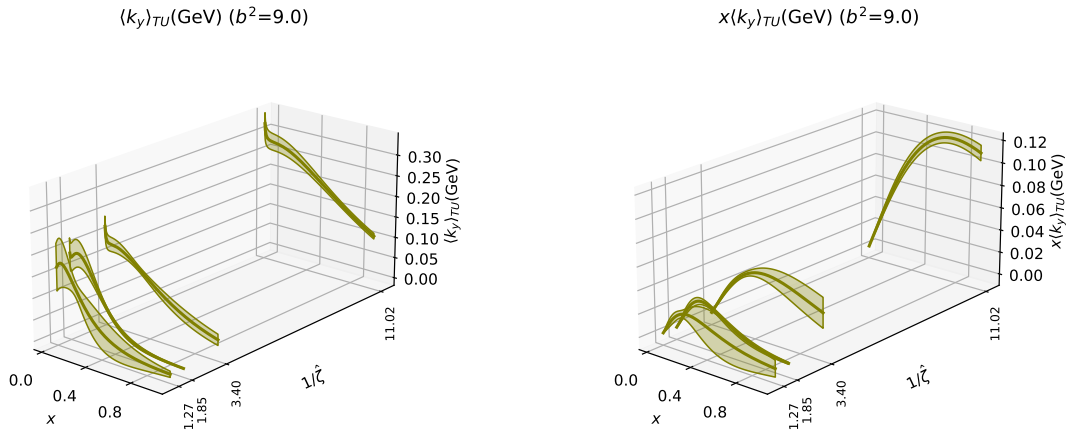



Figure 79: Sivers shift $\langle k_y \rangle_{TU}$ and $x\langle k_y \rangle_{TU}$ for the same four momenta and the same $|b_T|$ as Fig. 78, restricted to the quark region $x \geq 0$.

### 7.2 Extrapolation $\hat{\zeta} \to \infty$

The lowest simulated momentum, $P_L = -1$ ($\hat{\zeta} = 0.090772$), lies at $1/\hat{\zeta} = 11.02$, far from the other three momenta ($1/\hat{\zeta} = 3.40, 1.85, 1.27$) and from the asymptotic region $1/\hat{\zeta} \to 0$ that the extrapolation targets. A single point that far out would dominate the two-parameter fits introduced below and bias the extrapolated intercept rather than constrain it. We therefore exclude $P_L = -1$ and extrapolate using only the three higher momenta $P_L = -2, -3, -4$.

Rather than extrapolate the amplitudes point by point in $x$, we extrapolate each of the twelve parameters of the global simultaneous-fit models directly: the four parameters $\{a, c, d, j\}$ appearing in each of the power-law models for $\widetilde{A}^{\mathrm{Re}}_{2B}$, $\widetilde{A}^{\mathrm{Im}}_{2B}$, and $\widetilde{A}^{\mathrm{Re}}_{12B}$ (Eq. (6.16)-(6.18)). In these models $c$ multiplies $b_L^2 = (P \cdot b)^2 / P_L^2$ and $a_{\mathrm{im}}$ multiplies $b_L = (P \cdot b)/(-P_L)$, so the parameters $c_{\mathrm{re}}$, $c_{\mathrm{im}}$, and $c_{\mathrm{reA12B}}$ are first rescaled by $1/P_L^2$ and $a_{\mathrm{im}}$ by $1/P_L$. After rescaling, they multiply powers of the invariant $P \cdot b$ directly (up to an overall sign for $a_{\mathrm{im}}$), so all three momenta can be fit on a common footing. For the resulting $\hat{\zeta}$-dependence of each rescaled parameter, we tested two functional forms (as explored in Ref. [2]),

$$p(\hat{\zeta}) = c + \frac{d}{\hat{\zeta}} \qquad \text{and} \qquad p(\hat{\zeta}) = c + \frac{d}{\hat{\zeta}^2}, \tag{7.1}$$

using correlated Jackknife fits to the three data points. The quadratic ansatz consistently gave a lower $\chi^2/\mathrm{dof}$ than the linear form across the twelve parameters, and we adopt it for every extrapolation shown below. The extrapolated value of each parameter at $\hat{\zeta} \to \infty$ is its fitted intercept $c$ at $1/\hat{\zeta}^2 = 0$.

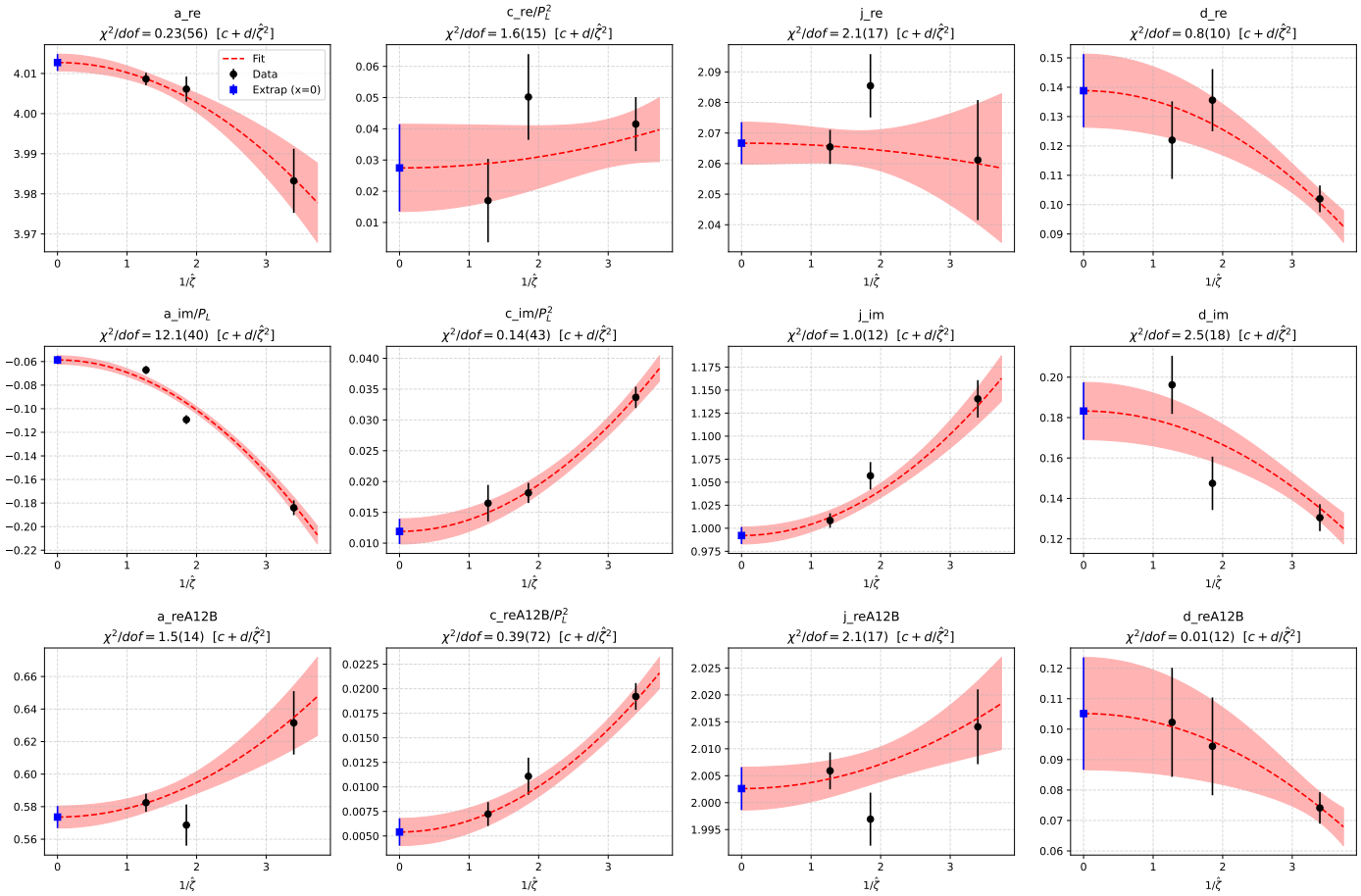



Figure 80: Extrapolation of the twelve simultaneous-fit parameters to $\hat{\zeta} \to \infty$, using only the three higher momenta $P_L = -2, -3, -4$ (black points, plotted against $1/\hat{\zeta}$). The dashed red curve and shaded band show the fitted ansatz $c + d/\hat{\zeta}^2$ of Eq. (7.1), with each panel's Jackknife $\chi^2$/dof noted in its title; the blue square at $1/\hat{\zeta} = 0$ marks the extrapolated value. The parameters $c_{\rm re}$, $c_{\rm im}$, and $c_{\rm reA12B}$ are shown rescaled by $1/P_L^2$, and $a_{\rm im}$ by $1/P_L$, as described in the text.

Reconstructing the amplitudes from these twelve extrapolated parameters gives the unpolarized and Sivers TMD combinations directly at $\hat{\zeta} \to \infty$, shown in Fig. 81 as continuous functions of $x$ at fixed $|b_T| = 3a \approx 0.34$ fm.

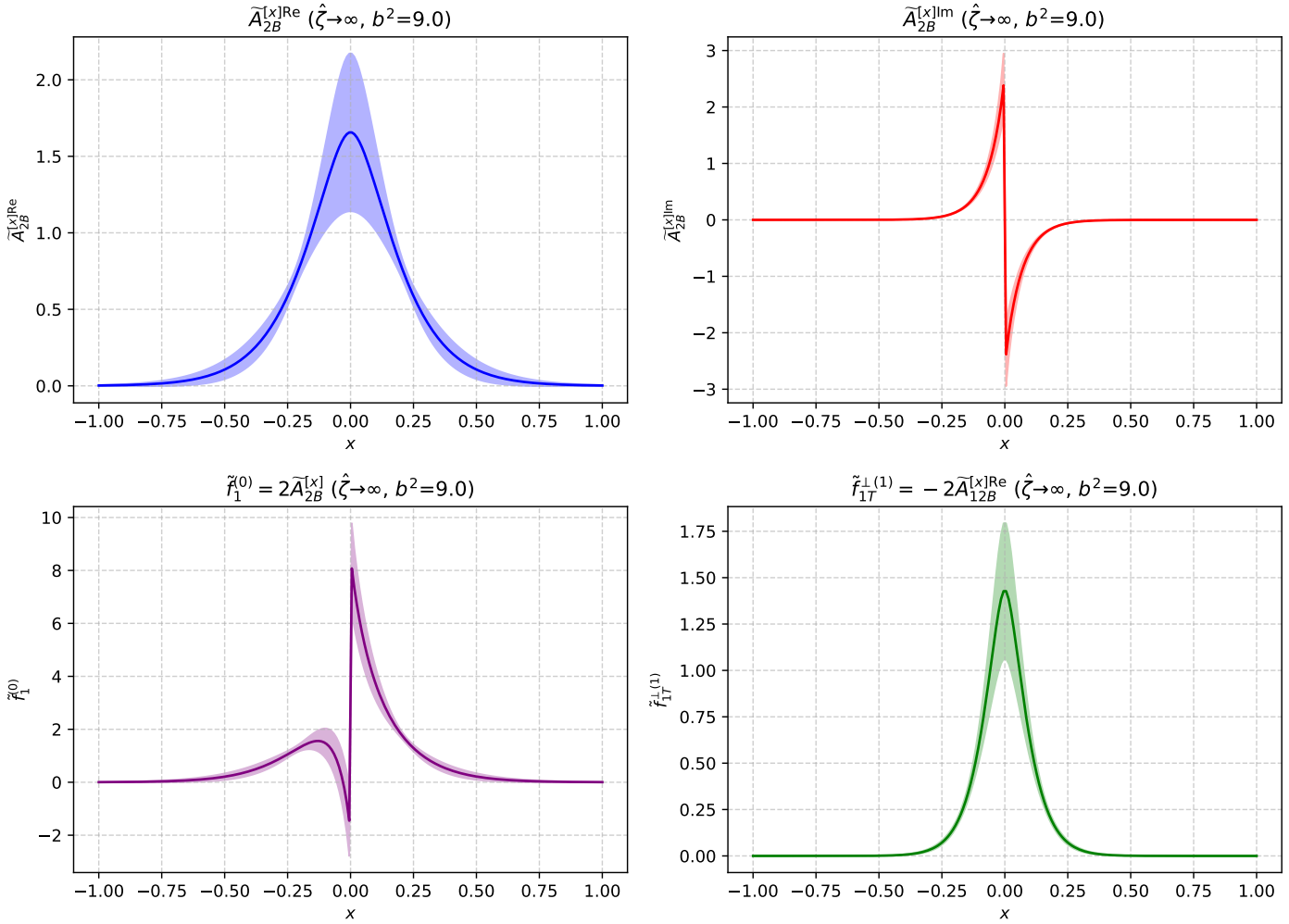



Figure 81: Unpolarized and Sivers TMD amplitudes reconstructed from the extrapolated ($\hat{\zeta} \to \infty$) fit parameters of Fig. 80, evaluated as continuous functions of $x$ at fixed transverse separation $|b_T| = 3a \approx 0.34$ fm: $\widetilde{A}^{[x]\mathrm{Re}}_{2B}$, $\widetilde{A}^{[x]\mathrm{Im}}_{2B}$, the full unpolarized combination $\tilde{f}^{(0)}_1 = 2\widetilde{A}^{[x]}_{2B}$, and the Sivers combination $\tilde{f}^{\perp(1)}_{1T} = -2\widetilde{A}^{[x]\mathrm{Re}}_{12B}$. Shaded bands are the Jackknife uncertainty propagated from the twelve extrapolated parameters.

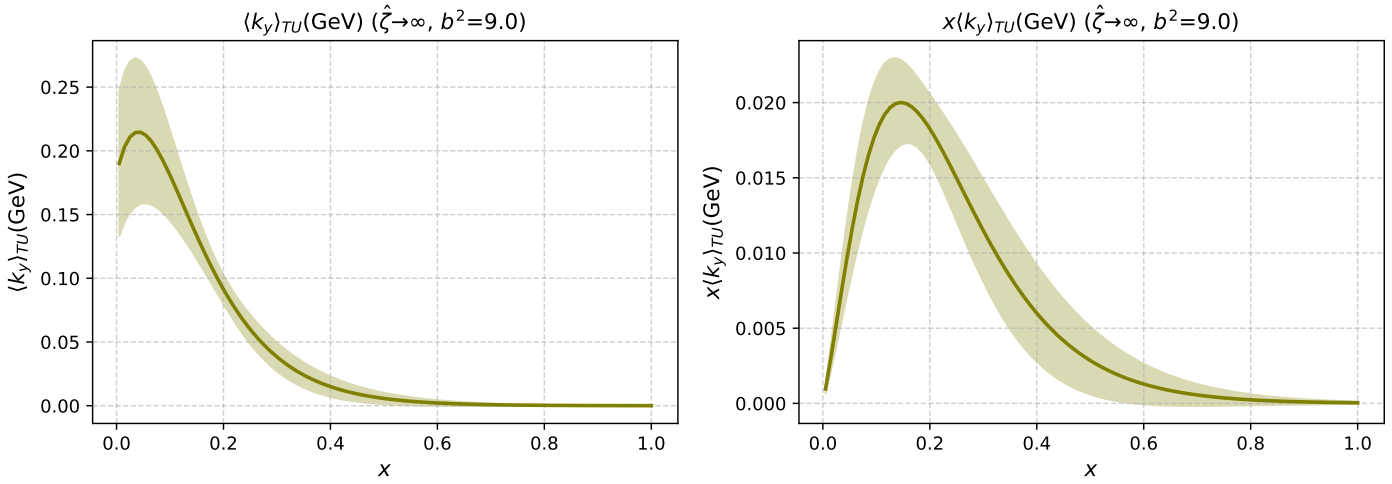



Figure 82: Isovector ($u-d$) Sivers shift $\langle k_y\rangle_{TU}$ and $x\langle k_y\rangle_{TU}$ built from the extrapolated amplitudes of Fig. 81, restricted to the quark region $x \geq 0$. Shaded bands are the Jackknife uncertainty.

Table 29 lists the corresponding numerical Fourier-transform integrals at $|b_T| = 3a \approx 0.34$ fm, evaluated over the full range $x \in \{-\infty, \infty\}$ as well as the restricted ranges $\{-1, 1\}$, $\{-1, 0\}$, and $\{0, 1\}$ used to isolate, respectively, the quark-plus-antiquark, antiquark, and quark contributions to the $x$-integrals $\int dx\, \tilde{f}_1^{(0)}$ and $\int dx\, \tilde{f}_{1T}^{\perp(1)}$.

Table 29: Numerical integration of the Fourier transforms at $|b_T| = 3a \approx 0.34$ fm, evaluated using the extrapolated ($\hat{\zeta} \to \infty$) amplitudes of Fig. 81, over the full range and three restricted $x$-ranges isolating the quark, antiquark, and combined contributions. The first four rows are unrenormalized; the multiplicative renormalization factor cancels only in the fifth row, where the ratio is formed.

| Integral | $[-\infty, \infty]$ | $[-1, 1]$ | $[-1, 0]$ | $[0, 1]$ |
|---|---|---|---|---|
| $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ | 0.751(79) | 0.751(80) | 0.375(40) | 0.375(40) |
| $\int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | $6(96) \times 10^{-17}$ | $3(24) \times 10^{-16}$ | 0.165(27) | −0.169(29) |
| $\int dx\, \tilde{f}_1^{(0)} = 2 \int dx\, (\widetilde{A}_{2B}^{[x]\mathrm{Re}} - \widetilde{A}_{2B}^{[x]\mathrm{Im}})$ | 1.50(16) | 1.50(16) | 0.421(81) | 1.09(11) |
| $\int dx\, \tilde{f}_{1T}^{\perp(1)} = -2 \int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ | 0.302(53) | 0.302(53) | 0.151(27) | 0.151(27) |
| $m_N \frac{\int dx\, \tilde{f}_{1T}^{\perp(1)}}{\int dx\, \tilde{f}_1^{(0)}}$ | 0.217(38) | 0.217(38) | 0.387(89) | 0.150(26) |

Restricting the integration to $x \in \{-1, 0\}$ or $\{0, 1\}$ isolates the antiquark and quark contributions respectively; because $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$ is an odd function of $x$ while $\widetilde{A}_{2B}^{[x]\mathrm{Re}}$ and $\widetilde{A}_{12B}^{[x]\mathrm{Re}}$ are even, the antiquark and quark ranges give equal $\int dx\, \widetilde{A}_{2B}^{[x]\mathrm{Re}}$ and $\int dx\, \widetilde{A}_{12B}^{[x]\mathrm{Re}}$ but opposite-sign $\int dx\, (i\widetilde{A}_{2B}^{[x]\mathrm{Im}})$, so the combined $\{-1, 1\}$ integral of the imaginary term is consistent with zero within uncertainty. The integrals over the full range $\{-\infty, \infty\}$ agree with those over the physically motivated range $\{-1, 1\}$ well within uncertainties. The extrapolated amplitudes therefore decay fast enough in $x$ that truncating at the light-cone-momentum-fraction cutoff is negligible.

Figure 83 extends this reconstruction to the full dependence on both $x$ and transverse separation, while Fig. 85 compares the extrapolated result with the four simulated momenta, of which $P_L = -2, -3, -4$ enter the extrapolation.

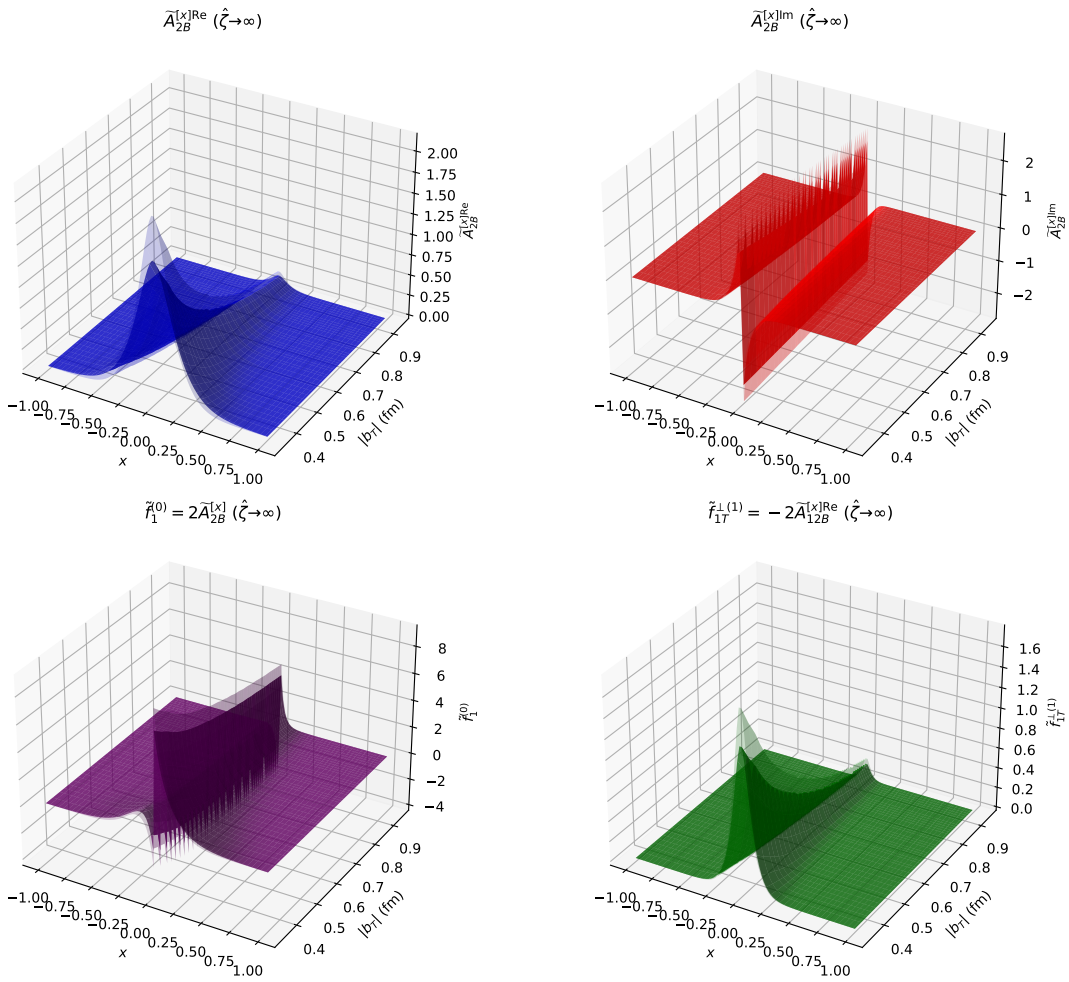



Figure 83: The four extrapolated ($\hat{\zeta}\to\infty$) amplitude quantities of Fig. 81, shown as surfaces in $x$ and $|b_T|$ (from $3a \approx 0.34$ fm to $\sqrt{68}\,a \approx 0.94$ fm). The joint $x$-$|b_T|$ dependence shows how the reconstructed amplitudes change with transverse quark separation at $\hat{\zeta}\to\infty$.

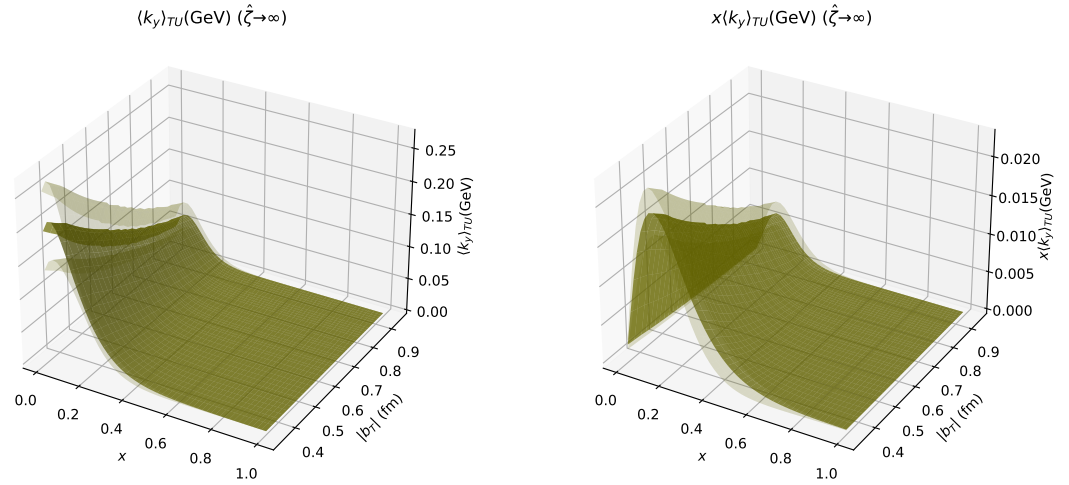



Figure 84: The extrapolated isovector ($u-d$) Sivers shift $\langle k_y\rangle_{TU}$ and $x\langle k_y\rangle_{TU}$ of Fig. 82, shown as surfaces in $x \geq 0$ and $|b_T|$ over the same range.

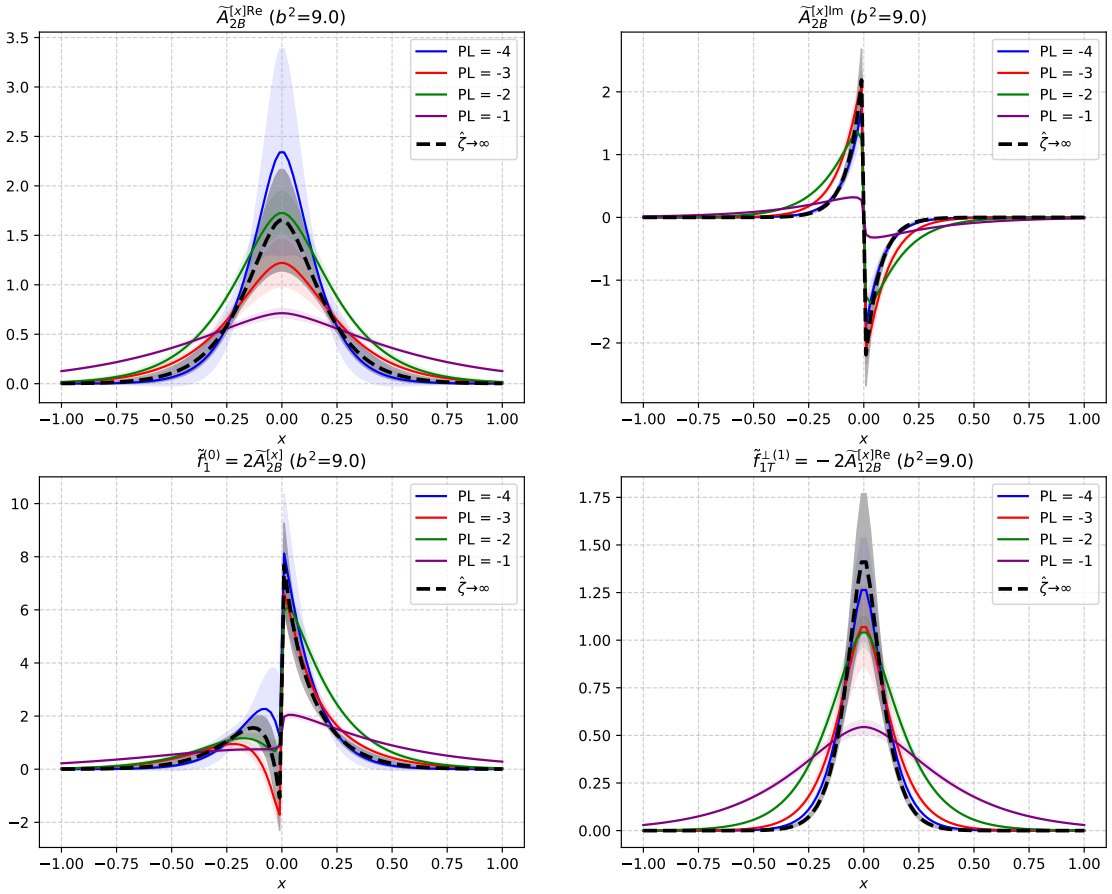



Figure 85: Comparison, at fixed $|b_T| = 3a \approx 0.34$ fm, between the four amplitude quantities evaluated at each of the four simulated momenta ($P_L = -1, -2, -3, -4$, colored curves, each using its own unextrapolated fit parameters) and the extrapolated $\hat{\zeta} \to \infty$ result (black dashed curve), which uses only $P_L = -2, -3, -4$. The extrapolated curve differs from all three finite-momentum curves used in the extrapolation, including the one at the largest simulated momentum; the fit-parameter extrapolation of Fig. 80 captures momentum dependence beyond the $P_L = -4$ result.

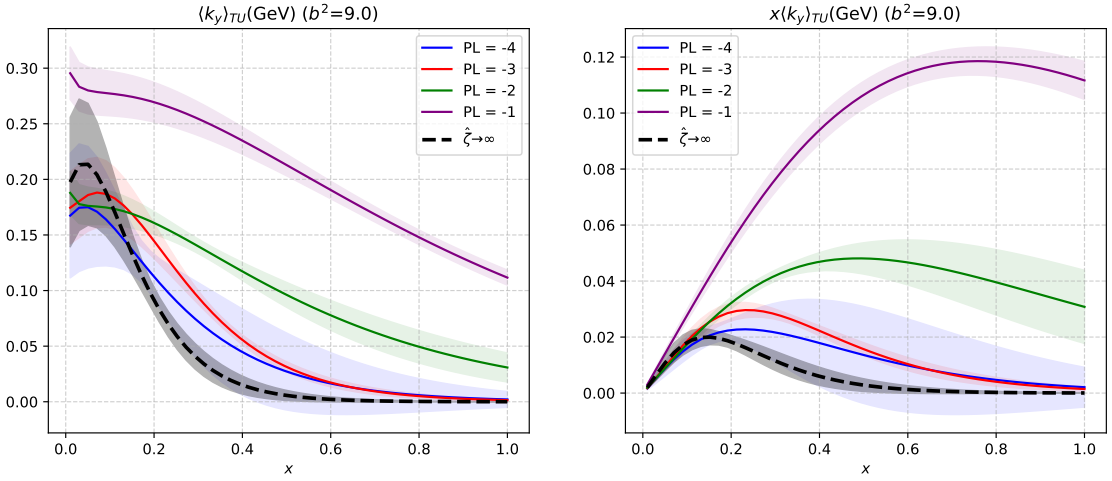



Figure 86: The same comparison as Fig. 85 for the isovector $(u - d)$ Sivers shift $\langle k_y\rangle_{TU}$ and $x\langle k_y\rangle_{TU}$ $(x \geq 0)$.

## 7.3 Testing Factorization in $x$ and $|b_T|$ at $\hat{\zeta} \to \infty$

We test factorization at two levels. First, we check whether the lattice amplitudes factorize in $(P \cdot b)$ and $b^2$ at finite momentum. We then ask whether the extrapolated distributions factorize in the momentum fraction $x$ and the transverse separation $|b_T|$.

The selected amplitude model and the other ansätze of Chapter 6 are not products of separate functions of $(P \cdot b)$ and $b^2$. The Gaussian model of Section 6.1, which is such a product, fails; the `PySR` search of Section 6.4 found no such product; and the separable models of Sections 6.6 and 6.9 fail as well. Section 7.3.1 checks the question directly on the lattice data at finite momentum.

### 7.3.1 Ratio Test in $(P \cdot b, b^2)$ on the Lattice Data

We first check directly on the lattice data whether an amplitude can be written as a product of longitudinal and transverse profiles, $\widetilde{A}(P \cdot b, b^2) = F(P \cdot b)\, G(b^2)$. If it can, any ratio taken at a common $b^2$ between two values of $P \cdot b$ is a constant in $b^2$, and any ratio taken at a common $P \cdot b$ between two values of $b^2$ does not depend on $P \cdot b$. We form

$$T(P \cdot b, b^2) = \frac{\widetilde{A}(P \cdot b, b^2)}{\widetilde{A}(P \cdot b = 4, b^2)} \stackrel{?}{=} \frac{F(P \cdot b)}{F(4)} \,, \tag{7.2}$$

$$B(P \cdot b, b^2) = \frac{\widetilde{A}(P \cdot b, b^2)}{\widetilde{A}(P \cdot b, b^2 = 18)} \stackrel{?}{=} \frac{G(b^2)}{G(18)} \,, \tag{7.3}$$

where the second equality in each line holds only under factorization. $T$ should be flat in $b^2$ and $B$ should be flat in $P \cdot b$. Both ratios are computed on each Jackknife sample, and the error comes from the Jackknife spread of the ratio.

The limitation of this test lies in how the data are sampled. At fixed $\eta$ and $P_L$, the $(b_L, b_T)$ points computed on the lattice map onto an irregular set of $(P \cdot b, b^2)$ pairs, listed in Table 30 for $\eta|v|/a = 8$ and $P_L = -1$. There are 34 pairs, but the rows at $\pm P \cdot b$ are symmetric copies of each other (identical for the real amplitudes and opposite in sign for $\widetilde{A}^{\mathrm{Im}}_{2B}$), so only $P \cdot b \geq 0$ carries independent information. At each $(P \cdot b, b^2)$, the two points with $\pm b_T$ are independent measurements, and we average them sample by sample. After the cut $|\boldsymbol{b}| \geq 3a$, that is $b^2 \geq 9$, only three values of $P \cdot b$ have more than one $b^2$:

$P \cdot b = 0$ with $b^2 = 9, 16, 25, 36, 49$; $P \cdot b = 3$ with $b^2 = 18, 45$; and $P \cdot b = 4$ with $b^2 = 17, 20, 32$. No two of these rows share two $b^2$ values, so the ratios in Eqs. (7.2) and (7.3) cannot be built from data alone.

| $P \cdot b$ | $b^2/a^2$ |
|---|---|
| 0 | 0, 1, 4, **9, 16, 25, 36, 49** |
| $\pm1$ | 2, 5 |
| $\pm2$ | 5, 8, 20 |
| $\pm3$ | **18, 45** |
| $\pm4$ | **17, 20, 32** |
| $\pm5$ | 50 |
| $\pm6$ | 45 |
| $\pm8$ | 68 |

Table 30: All 34 $(P \cdot b, b^2)$ pairs available at $\eta|v|/a = 8$ and $P_L = -1$, where $P \cdot b = -P_L\, b_L$ and $b^2 = b_L^2 + b_T^2$. Grey entries fail the cut $|\boldsymbol{b}| \geq 3a$. Bold entries are the data used in the factorization test of Figure 87.

To address this, we place the three rows on a common grid $b^2 = 18, 20, 25, 32$ that spans all three rows. On the $P \cdot b = 0$ and $P \cdot b = 4$ rows, we use quadratic interpolation in $b^2$: the value at a grid point comes from the parabola through the three data points on the same row that lie closest to it. The $P \cdot b = 3$ row has only two data points, $b^2 = 18$ and 45, so there we use linear interpolation between them. There is no extrapolation. As a check, Figure 87 also shows the ratios obtained with linear interpolation on all three rows, between the nearest data points above and below each grid value. Table 31 lists which grid values are data and which are interpolated: 4 of the 12 are data, and the other 8 are interpolated. For $\widetilde{A}_{2B}^{\mathrm{Im}}$, which is odd in $P \cdot b$ and vanishes at $P \cdot b = 0$, only the $P \cdot b = 3$ and 4 rows are used.

| $P \cdot b$ | $b^2 = 18$ | $b^2 = 20$ | $b^2 = 25$ | $b^2 = 32$ |
|---|---|---|---|---|
| 0 | lin. $[16, 25]$<br>quad. $(9, 16, 25)$ | lin. $[16, 25]$<br>quad. $(9, 16, 25)$ | **data** | lin. $[25, 36]$<br>quad. $(16, 25, 36)$ |
| 3 | **data** | lin. $[18, 45]$ | lin. $[18, 45]$ | lin. $[18, 45]$ |
| 4 | lin. $[17, 20]$<br>quad. $(17, 20, 32)$ | **data** | lin. $[20, 32]$<br>quad. $(17, 20, 32)$ | **data** |

Table 31: Origin of each grid value used in Figure 87. "Quad. $(b_1^2, b_2^2, b_3^2)$" means the parabola in $b^2$ through the data at those three points on the same $P \cdot b$ row; "lin. $[b_a^2, b_b^2]$" means linear interpolation in $b^2$ between the data at $b_a^2$ and $b_b^2$. Where a cell has two entries, the upper one gives the linear points (circles) and the lower one the quadratic points (triangles) of Figure 87. The pattern is the same for all three amplitudes.

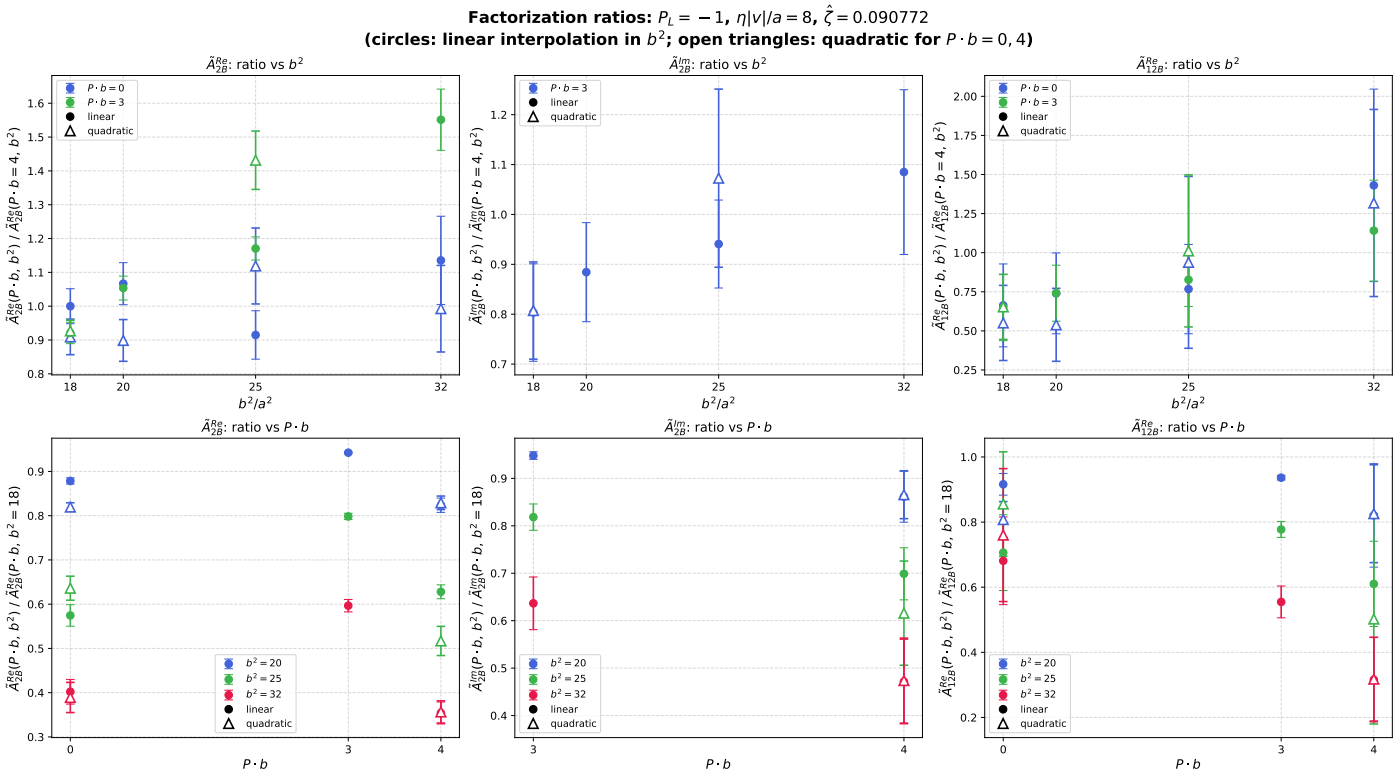



Figure 87: Factorization ratio test at $\eta|v|/a = 8$, $P_L = -1$ ($\hat{\zeta} = 0.090772$) for $\widetilde{A}^{\mathrm{Re}}_{2B}$, $\widetilde{A}^{\mathrm{Im}}_{2B}$ and $\widetilde{A}^{\mathrm{Re}}_{12B}$. Top row: $T(P \cdot b, b^2)$ of Eq. (7.2), the ratio to the $P \cdot b = 4$ row, plotted against $b^2$. Bottom row: $B(P \cdot b, b^2)$ of Eq. (7.3), the ratio to $b^2 = 18$, plotted against $P \cdot b$. Under factorization, every set of points should be flat. Filled circles use linear interpolation in $b^2$ on all three rows. Open triangles use quadratic interpolation on the $P \cdot b = 0$ and 4 rows and linear interpolation on the $P \cdot b = 3$ row (Table 31). A triangle is omitted where both schemes give the same value, which happens for every point that involves only the $P \cdot b = 3$ row and data points. Error bars are Jackknife errors and do not include any uncertainty from the interpolation.

From Figure 87, in certain region $P \cdot b = 0$, the ratio $T(P \cdot b, b^2) = \frac{\widetilde{A}(P \cdot b, b^2)}{\widetilde{A}(P \cdot b = 4, b^2)}$ is flat but for $P \cdot b = 3$, the ratio $T$ is increasing monotonically. The same behavior is captured in both linear and quadratic interpolation. The other ratio $B(P \cdot b, b^2) = \frac{\widetilde{A}(P \cdot b, b^2)}{\widetilde{A}(P \cdot b, b^2 = 18)}$ is not flat, and there exists a cut in continuity at $P \cdot b = 3$ which could also arise from systematic uncertainty from interpolation and limitation of data.

The ratio test is thus only a qualitative check. Where the interpolated values rest on closely spaced data, the $P \cdot b = 0$ and 4 rows are compatible with factorization, but the sampling is too sparse to settle the question. The separable fits of Section 6.9 lead to the same conclusion: no separable form describes all three amplitudes.

### 7.3.2 Factorization in $x$ and $|b_T|$ at $\hat{\zeta} \to \infty$

Figures 83 and 84 show the joint dependence of the extrapolated amplitudes and Sivers shift on both $x$ and $|b_T|$. We ask whether this joint dependence is separable, that is, whether the extrapolated Sivers shift factorizes as

$$\langle k_y \rangle_{TU}(x, |b_T|) \stackrel{?}{=} F(x)\, G(|b_T|) \ , \tag{7.4}$$

with all of the $x$-shape carried by $F(x)$ and all of the $|b_T|$-shape by $G(|b_T|)$. A recent symbolic-regression extraction of the non-perturbative unpolarized TMD from Drell–Yan data finds a mild but genuine correlation between $x$ and $b_T$ rather than strict separability [119], so the question is worth testing directly on the lattice. Such a factorization is not automatic in our fit ansatz for $\widetilde{A}^{\mathrm{Re}}_{2B}$, $\widetilde{A}^{\mathrm{Im}}_{2B}$, and $\widetilde{A}^{\mathrm{Re}}_{12B}$ (Eq. (6.16)-(6.18)). After Fourier transformation, each amplitude enters only through the combination $d_X(b^2) = 1 + d_X\, b^2$ inside a Bessel function that also carries the $x$-dependence,

$$\widetilde{A}^{[x]}_X(x, b^2) \ \propto \ N_X(b^2)\ |x|^{-1/2+j_X}\ K_{\nu_X}\Big(|x|\sqrt{d_X(b^2)/c_X}\Big) \ , \tag{7.5}$$

so $|b_T|$ rescales the argument of the same Bessel function that carries the $x$-dependence, rather than multiplying it by an overall $x$-independent prefactor. The Sivers shift is a ratio of three such amplitudes ($\widetilde{A}^{[x]\mathrm{Re}}_{2B}$, $\widetilde{A}^{[x]\mathrm{Im}}_{2B}$, $\widetilde{A}^{[x]\mathrm{Re}}_{12B}$), each with its own independently fitted $(j_X, d_X)$; an exact factorization would require these parameters to coincide across the three amplitudes so that the $x$- and $|b_T|$-dependence separates. Whether the extrapolated fit comes close enough to that coincidence is a numerical question, which we answer below.

Section 7.3.3 defines the relative distance from a factorized form, and Sections 7.3.4 and 7.3.5 apply it to the Sivers shift and to the individual amplitudes.

### 7.3.3 Quantifying the Distance from a Factorized Form

Given a function of two variables $f(x, |b_T|)$, we quantify how far it is from a factorized form $g(x)h(|b_T|)$ by finding the closest such pair in the least-squares sense and reporting the residual. Explicitly, we minimize

$$J = \int dx\, d|b_T|\, \big(f(x, |b_T|) - g(x)h(|b_T|)\big)^2 \tag{7.6}$$

over all functions $g$ and $h$. Setting the functional derivatives with respect to $g(x)$ and $h(|b_T|)$ to zero gives

$$0 = \frac{\delta J}{\delta g(x)} = \int d|b_T|\, f(x,|b_T|)\, h(|b_T|) - g(x) \int d|b_T|\, h^2(|b_T|)\ , \tag{7.7}$$

$$0 = \frac{\delta J}{\delta h(|b_T|)} = \int dx\, f(x,|b_T|)\, g(x) - h(|b_T|) \int dx\, g^2(x)\ , \tag{7.8}$$

which can be solved for $h$ in terms of $g$,

$$h(|b_T|) = \frac{\int dx\, f(x,|b_T|)\, g(x)}{\int dx\, g^2(x)}\ , \tag{7.9}$$

and reinserted to eliminate $h$, giving a closed, self-consistent equation for $g$ alone:

$$g(x) = \left[\int d|b_T| \left[\frac{\int dx'\, f(x',|b_T|)\, g(x')}{\int dx'\, g^2(x')}\right]^2\right]^{-1} \int d|b_T|\, f(x,|b_T|)\, \frac{\int dx'\, f(x',|b_T|)\, g(x')}{\int dx'\, g^2(x')}\ . \tag{7.10}$$

We solve Eq. (7.10) iteratively. Starting from an initial guess $g = 1$, we evaluate the right-hand side to obtain a refined $g$, reinsert it, and repeat until $g$ (and the resulting relative distance below) stops changing within a fixed tolerance.

We then compare the residual $\sqrt{J}$ of the optimal separable pair $g(x)h(|b_T|)$ with the overall size of $f$, $N = \int dx\, d|b_T|\, f^2(x,|b_T|)$, which gives the relative distance from a factorized form:

$$\text{relative distance} = \sqrt{J/N}\ . \tag{7.11}$$

A relative distance of zero indicates exact factorization; the closer to zero, the better $F(x)G(|b_T|)$ approximates the true joint $(x,|b_T|)$ dependence. In practice, we replace the continuum integrals with sums over a discrete $(x,|b_T|)$ grid, apply the iteration of Eq. (7.10) independently to each Jackknife sample of the extrapolated amplitudes, and reduce the resulting distribution of relative distances to a Jackknife mean and error. We repeat the full test in four $x$-windows, $[0,1]$, $[0.1,0.9]$, $[0.2,0.8]$, and $[0.3,0.8]$, to check that the conclusion is not an artifact of the amplitudes' behavior near $x = 0$.

### 7.3.4 Factorization Test for the Sivers Shift

The extrapolated fit parameters already suggest, before any distance test, that exact factorization is unlikely. As noted in Section 7.3.2, factorization requires the Bessel-function order $j_X$ and width

parameter $d_X$ to coincide across the three input amplitudes; the extrapolated values are

$$j_{\rm re} = 2.0667(69), \quad j_{\rm im} = 0.9921(92), \quad j_{\rm reA12B} = 2.0026(40) \ , \tag{7.12}$$

$$d_{\rm re} = 0.139(12), \quad d_{\rm im} = 0.183(14), \quad d_{\rm reA12B} = 0.105(18) \ . \tag{7.13}$$

The two $j$ parameters entering the real amplitudes, $j_{\rm re}$ and $j_{\rm reA12B}$, agree with each other, but $j_{\rm im} \approx 1$ differs clearly from both. The three $d$ parameters lie closer together but are still resolved from one another well outside their Jackknife uncertainties. This mismatch in the parameters, especially in $j$, gives a separate reason to expect a nonzero residual from exact factorization.

Table 32 lists the resulting relative distance of the extrapolated Sivers shift $\langle k_y \rangle_{TU}(x, |b_T|)$ from its best separable approximation, over $|b_T| \in [0.3421, 0.9403]$ fm (60 points, fixed across all four $x$-windows).

Table 32: Relative distance of the extrapolated isovector $(u - d)$ Sivers shift $\langle k_y \rangle_{TU}(x, |b_T|)$ from its optimal factorized approximation $F(x)G(|b_T|)$, Eq. (7.11), over $|b_T| \in [0.3421, 0.9403]$ fm for four choices of $x$-window.

| $x$-window | Relative distance |
|---|---|
| $[0.0, 1.0]$ | 0.0568(78) |
| $[0.1, 0.9]$ | 0.0714(95) |
| $[0.2, 0.8]$ | 0.0758(75) |
| $[0.3, 0.8]$ | 0.0620(88) |

In every window the relative distance is at the level of 5.7-7.6% and nonzero well outside its Jackknife uncertainty. The Sivers shift is therefore close to, but not exactly, factorized. The residual stays stable as the $x$-window narrows away from $x = 0$, so behavior at the endpoint does not drive it. Figure 88 shows the optimal separable factors $g(x)$ and $h(|b_T|)$ for all four windows and compares the $g(x)h(|b_T|)$ reconstruction with the actual data at the two extreme $|b_T|$ values in the fit range.

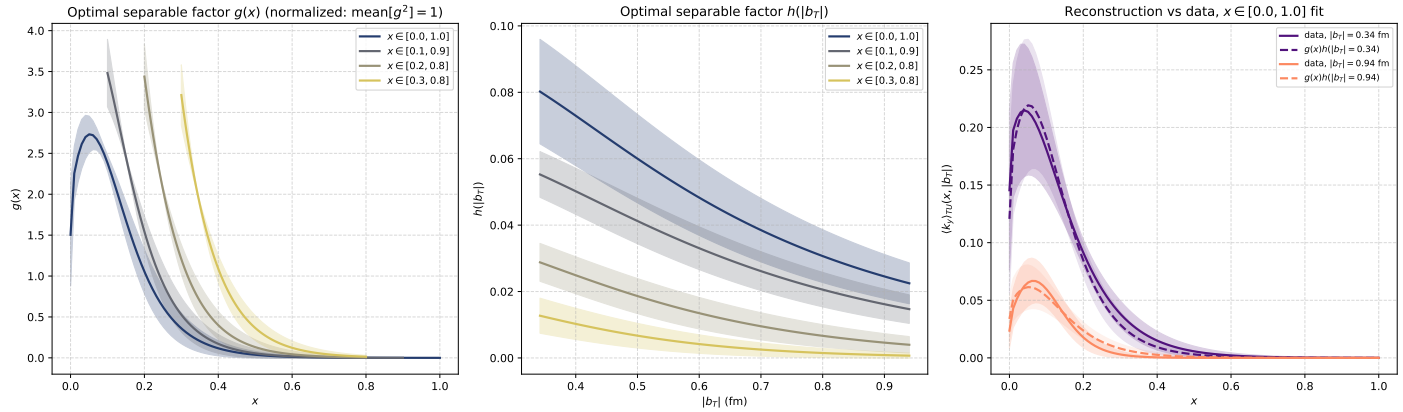



Figure 88: Left and center: the optimal separable factors $g(x)$ and $h(|b_T|)$ (Eqs. (7.9)-(7.10), normalized so mean$[g^2] = 1$) for the extrapolated isovector $(u - d)$ Sivers shift, shown for all four $x$-windows of Table 32. Right: the reconstructed $g(x)h(|b_T|)$ (dashed) compared to the actual extrapolated Sivers shift (solid) at the smallest and largest fitted $|b_T|$, using the $x \in [0, 1]$ fit; the visible gap between the solid and dashed curves is the residual quantified in Table 32.

Ratios of the Sivers shift at fixed $x$ or fixed $|b_T|$ show the same non-factorization more directly. If $\langle k_y\rangle_{TU}(x, |b_T|) = F(x)G(|b_T|)$ exactly, then the ratios

$$R_b(x) \equiv \frac{\langle k_y\rangle_{TU}(x, |b_T|)}{\langle k_y\rangle_{TU}(x, |b_T|_{\text{ref}})} = \frac{G(|b_T|)}{G(|b_T|_{\text{ref}})} \ , \tag{7.14}$$

$$R_x(|b_T|) \equiv \frac{\langle k_y\rangle_{TU}(x, |b_T|)}{\langle k_y\rangle_{TU}(x_{\text{ref}}, |b_T|)} = \frac{F(x)}{F(x_{\text{ref}})} \tag{7.15}$$

must be flat in $x$ for every $|b_T|$ and reference $|b_T|_{\text{ref}}$, and flat in $|b_T|$ for every $x$ and $x_{\text{ref}}$, respectively. Requiring this for every pair of references is both necessary and sufficient for factorization, so the ratio test is a complete test rather than a heuristic proxy. Unlike the lattice-data test of Section 7.3.1, which is limited by the sparse $(P \cdot b, b^2)$ sampling, these ratios use the continuous reconstructed functions, so they can be formed for any pair of references. Figures 89 and 90 show that neither ratio is flat.

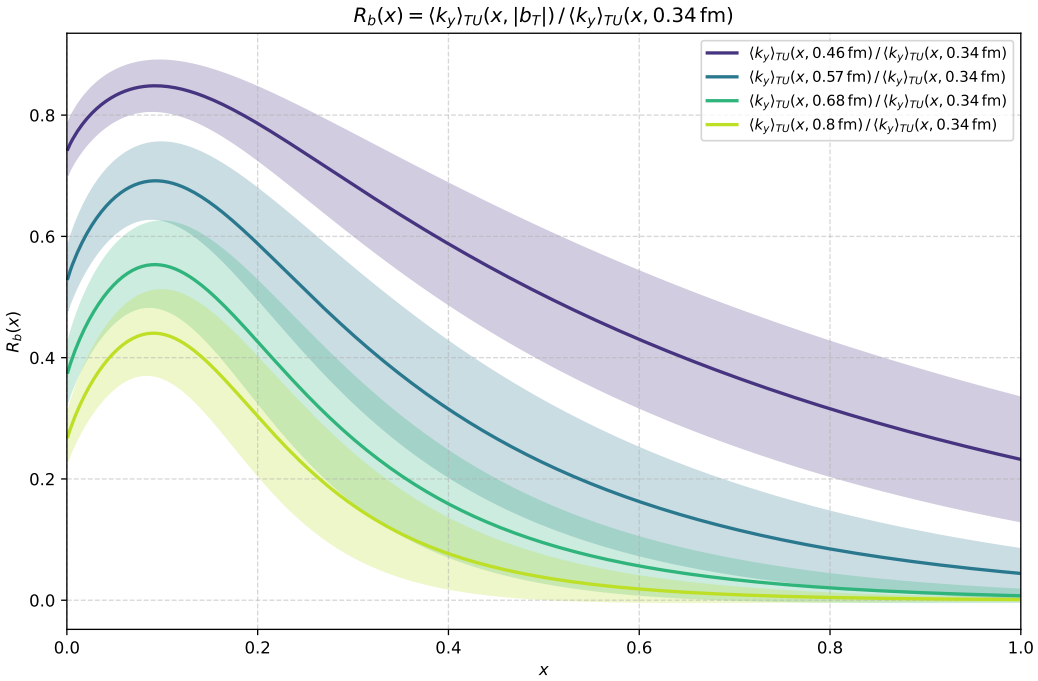



Figure 89: Isovector $(u-d)$: $R_b(x) = \langle k_y \rangle_{TU}(x, |b_T|) / \langle k_y \rangle_{TU}(x, 0.34\text{ fm})$ for four values of $|b_T|$, as a function of $x$. Under exact factorization each curve would be flat. Instead, every curve rises from its $x \to 0$ value to a peak near $x \approx 0.1$ before decaying, so the $|b_T|$-dependence of the Sivers shift itself depends on $x$.

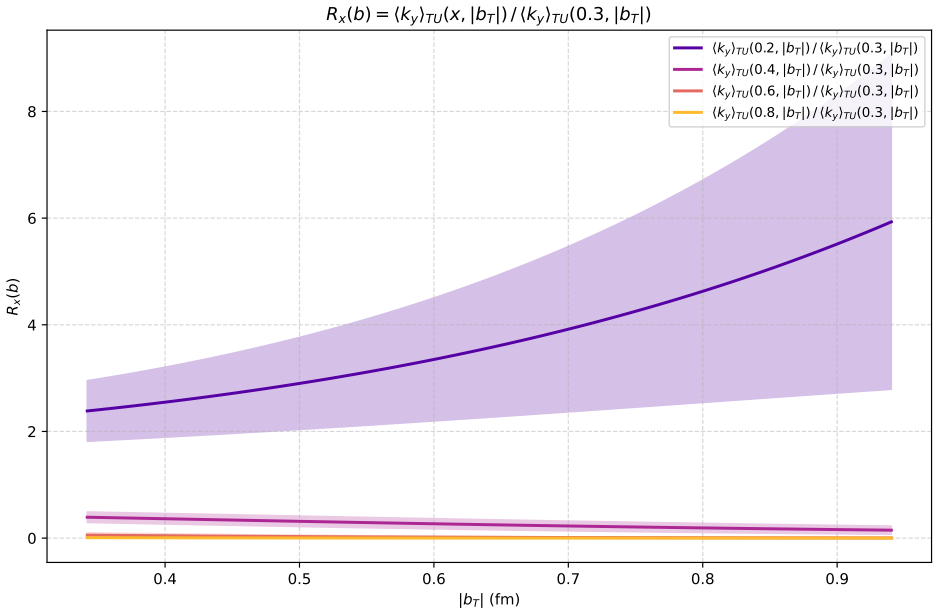



Figure 90: Isovector $(u-d)$: $R_x(|b_T|) = \langle k_y \rangle_{TU}(x, |b_T|) / \langle k_y \rangle_{TU}(0.3, |b_T|)$ for four values of $x$, as a function of $|b_T|$. The $x = 0.2$ curve changes by more than a factor of two across the fitted $|b_T|$ range instead of staying flat, so the $x$-dependence of the Sivers shift itself depends on $|b_T|$, consistent with Fig. 89.

Finally, to check that the $\hat{\zeta} \to \infty$ extrapolation itself does not introduce this non-factorization, we repeat the $R_b(x)$ and $R_x(|b_T|)$ ratio tests on the raw, unextrapolated fit parameters at each finite momentum. Figures 91 and 92 show that the same qualitative pattern, non-flat ratios in both $x$ and

$|b_T|$, is already present at every simulated momentum. The departure from flatness grows with $|P_L|$, most visibly in the $x = 0.2$ curve of Fig. 92.

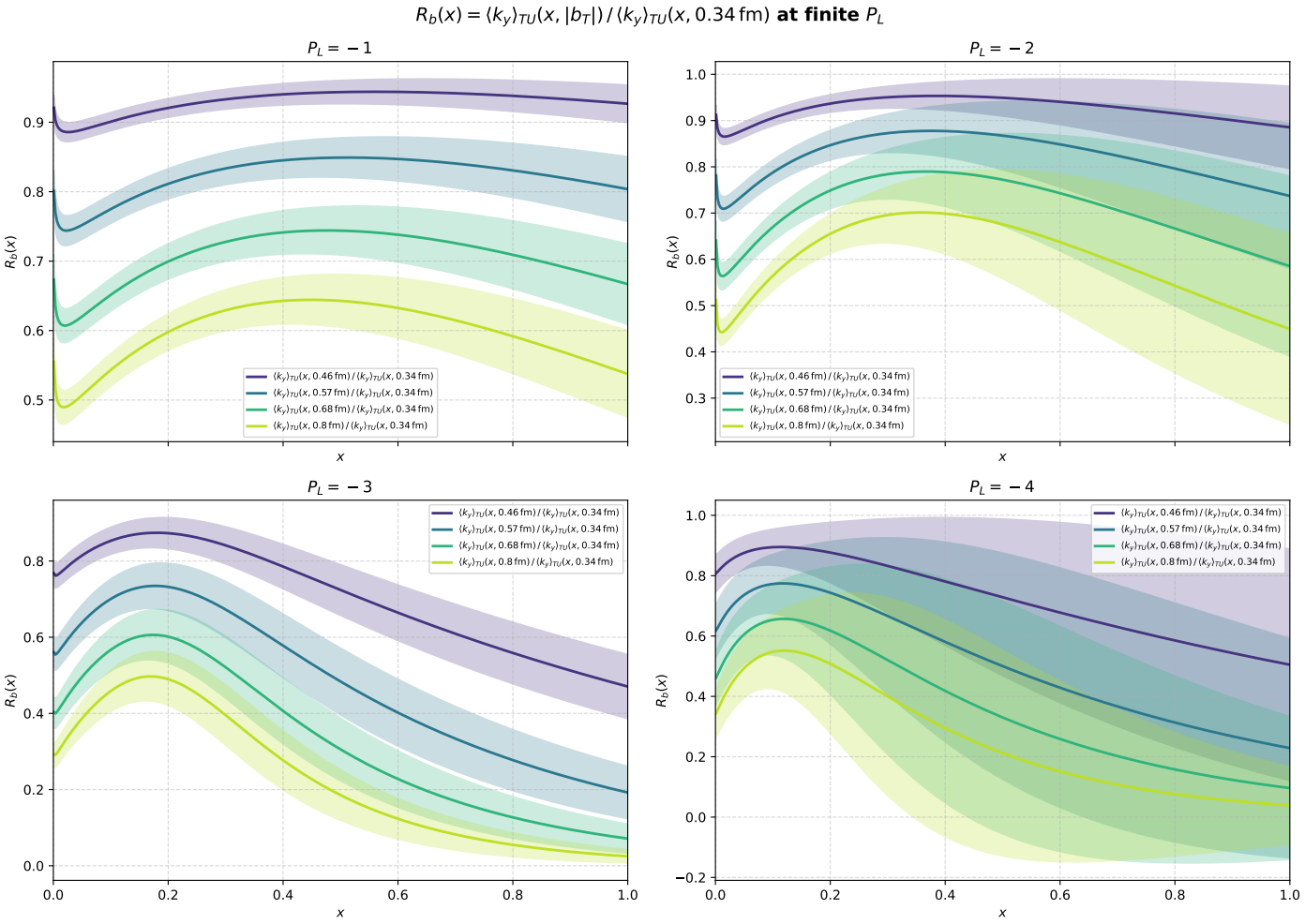



Figure 91: The same ratio test as Fig. 89, $R_b(x)$, repeated at each finite momentum individually using its own raw (unextrapolated) fit parameters. None of the four panels shows a flat curve, so the non-factorization is a feature of the fitted amplitudes and not an artifact of the $\hat{\zeta} \to \infty$ extrapolation.

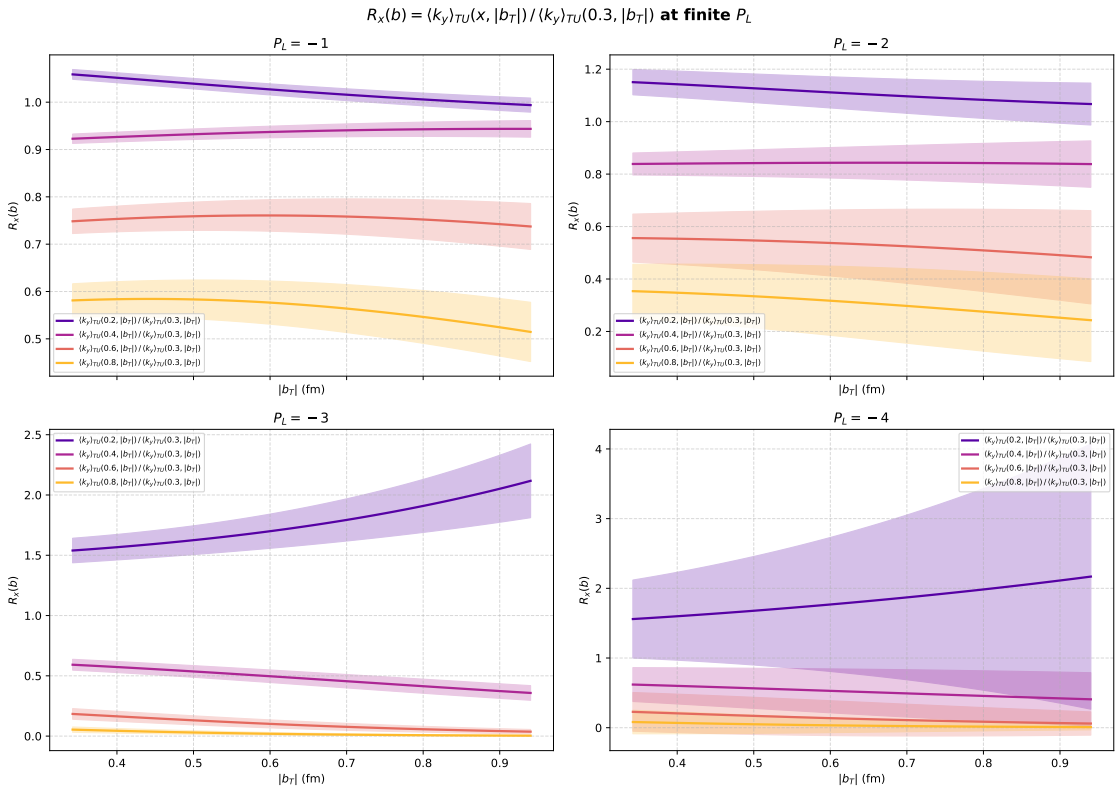



Figure 92: The same ratio test as Fig. 90, $R_x(|b_T|)$, repeated at each finite momentum individually. The departure from flatness grows with $|P_L|$, most visibly in the $x = 0.2$ curve, which is consistent with the parameter mismatch identified in the extrapolated $j_X$ and $d_X$ values above.

### 7.3.5 Factorization Test at the Amplitude Level

The Sivers-shift test above probes factorization of the full ratio $\langle k_y \rangle_{TU} = -m_N \widetilde{A}^{[x]\mathrm{Re}}_{12B} / (\widetilde{A}^{[x]\mathrm{Re}}_{2B} - i\widetilde{A}^{[x]\mathrm{Im}}_{2B})$, in which some of the non-factorized behavior of the individual amplitudes could in principle cancel. We therefore repeat the same distance test of Section 7.3.3 on each extrapolated amplitude separately: $\widetilde{A}^{[x]\mathrm{Re}}_{12B}$, $\widetilde{A}^{[x]\mathrm{Re}}_{2B}$, $i\widetilde{A}^{[x]\mathrm{Im}}_{2B}$, and the combination $\widetilde{A}^{[x]\mathrm{Re}}_{2B} - i\widetilde{A}^{[x]\mathrm{Im}}_{2B}$ (proportional to $\tilde{f}_1^{(0)}$) that appears in the Sivers-shift denominator. Table 33 lists the resulting relative distances, using the same $|b_T| \in [0.3421, 0.9403]$ fm grid and four $x$-windows as the Sivers-shift test.

Table 33: Relative distance of each extrapolated amplitude (and their combination $\widetilde{A}^{[x]\mathrm{Re}}_{2B} - i\widetilde{A}^{[x]\mathrm{Im}}_{2B}$) from its optimal factorized approximation, over the same $|b_T| \in [0.3421, 0.9403]$ fm grid and $x$-windows as Table 32.

| Amplitude | $x \in [0.0, 1.0]$ | $x \in [0.1, 0.9]$ | $x \in [0.2, 0.8]$ | $x \in [0.3, 0.8]$ |
|---|---|---|---|---|
| $\widetilde{A}^{[x]\mathrm{Re}}_{12B}$ | 0.0804(18) | 0.0573(26) | 0.0397(24) | 0.0303(21) |
| $\widetilde{A}^{[x]\mathrm{Re}}_{2B}$ | 0.08138(65) | 0.0718(36) | 0.0581(55) | 0.0478(56) |
| $i\widetilde{A}^{[x]\mathrm{Im}}_{2B}$ | 0.107(18) | 0.0868(30) | 0.0578(28) | 0.0425(23) |
| $\widetilde{A}^{[x]}_{2B}$ | 0.1943(63) | 0.0926(59) | 0.0594(40) | 0.0470(47) |

Every amplitude has a nonzero relative distance from factorized form in every window. For the widest window, $x \in [0, 1]$, the amplitude-level residuals (8-19%) exceed the corresponding Sivers-ratio residual (5.7%, Table 32), so the ratio structure of the Sivers shift partially, but not completely, cancels the non-factorized behavior of the individual amplitudes. All four amplitude-level residuals shrink as the $x$-window narrows away from $x = 0$. This is consistent with the Bessel-function singularity structure of each amplitude near the origin, which differs between amplitudes because of the parameter mismatch identified in Section 7.3.4.

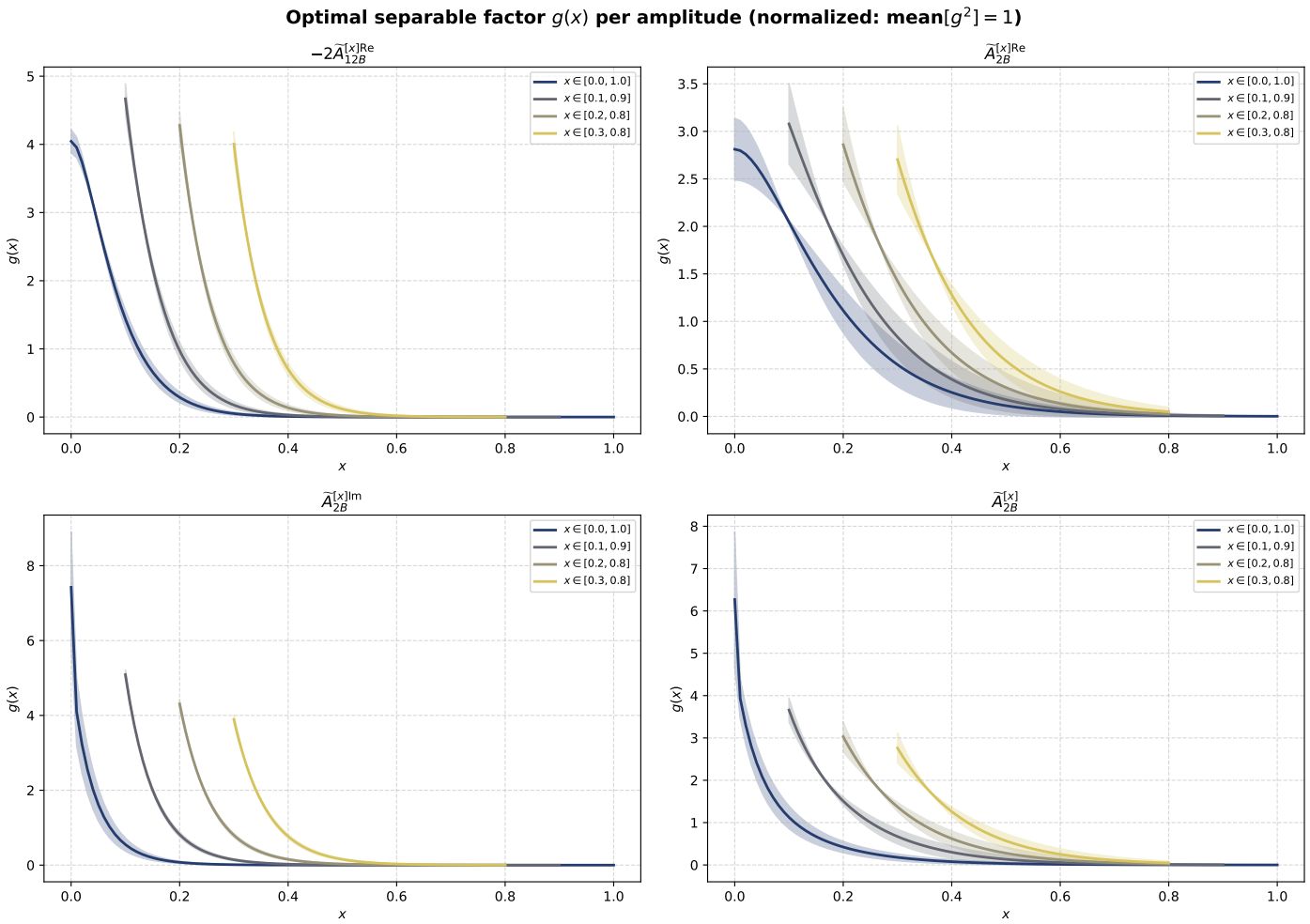



Figure 93: Optimal separable factor $g(x)$ for each of the four amplitude combinations of Table 33, for all four $x$-windows (normalized so mean$[g^2] = 1$). Each amplitude's $g(x)$ is dominated by a sharp rise as $x \to 0$, which steepens as the window narrows and reflects the Bessel-function behavior at small $x$.

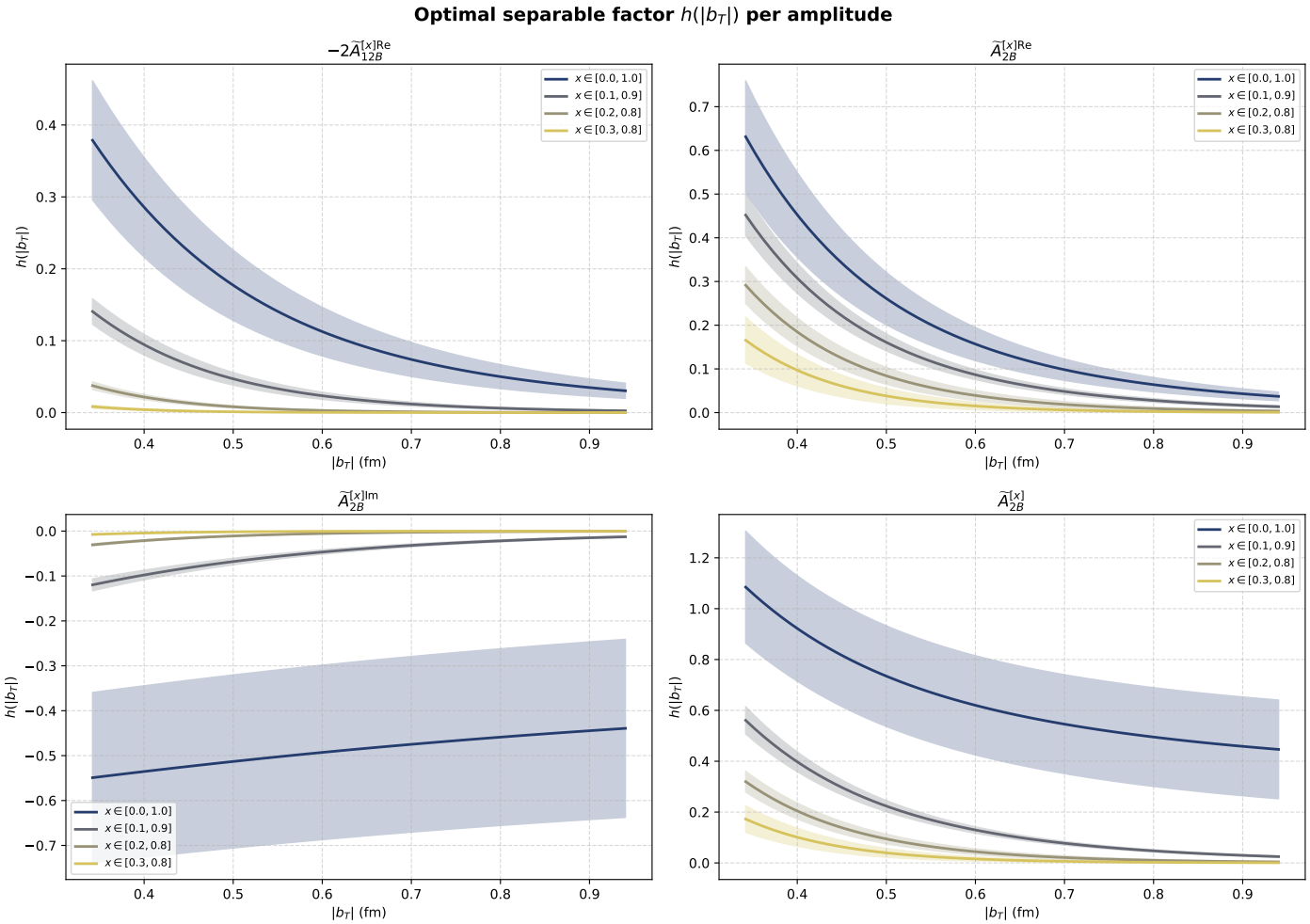



Figure 94: Optimal separable factor $h(|b_T|)$ for each of the four amplitude combinations of Table 33, for all four $x$-windows. For all four amplitudes, $h(|b_T|)$ falls smoothly with increasing $|b_T|$, but its magnitude and slope depend on the $x$-window used to define the fit, which itself signals imperfect factorization.

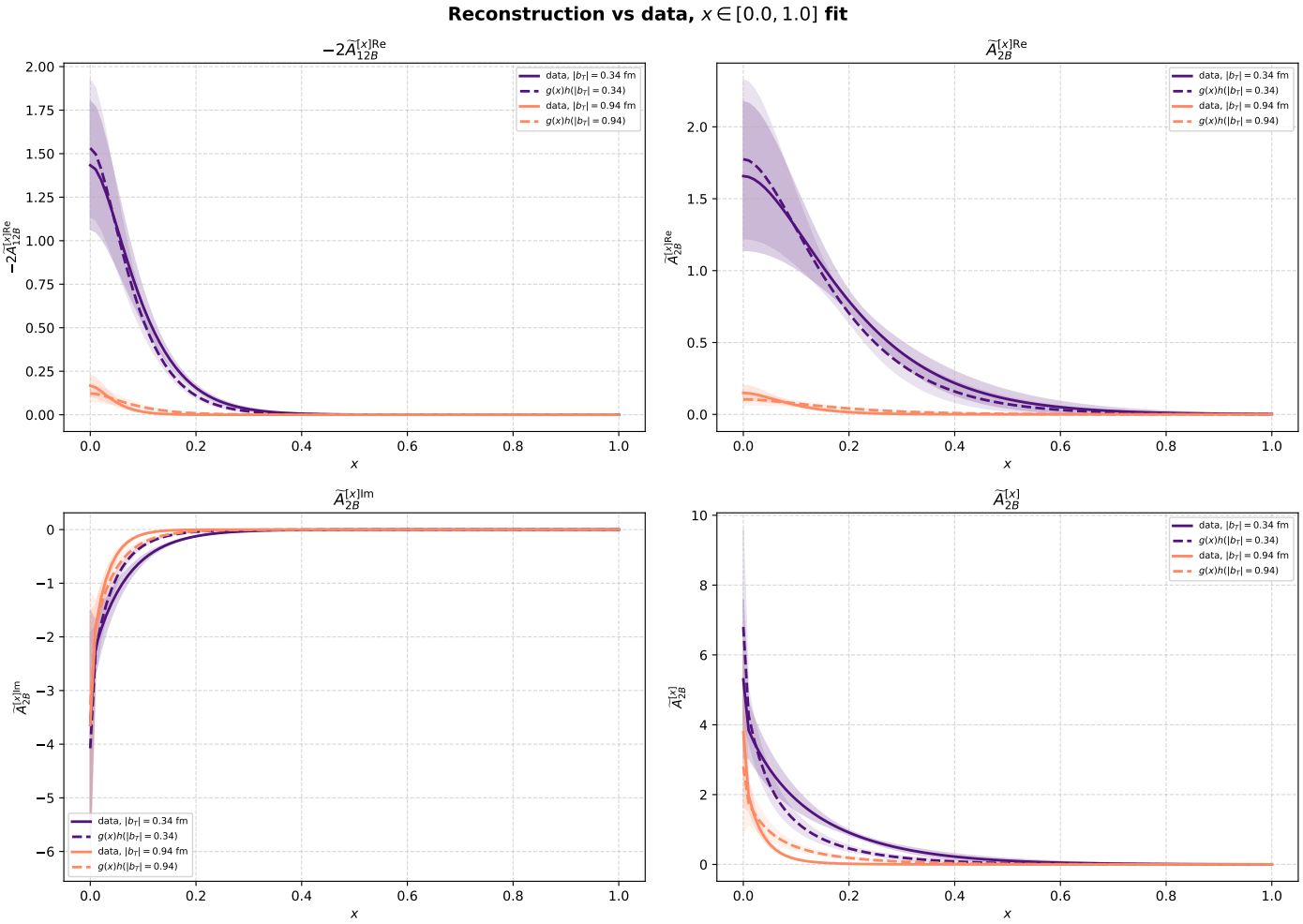



Figure 95: Reconstructed $g(x)h(|b_T|)$ (dashed) against the actual extrapolated amplitude data (solid) at the smallest and largest fitted $|b_T|$, using the $x \in [0, 1]$ fit, for each of the four amplitude combinations of Table 33. The largest reconstruction gap appears near $x \to 0$ for every amplitude, consistent with the $x \in [0, 1]$ column of Table 33 having the largest relative distance in every row.

The Sivers-shift and amplitude-level tests agree. Neither the individual amplitudes nor the Sivers-shift ratio built from them is exactly factorized in $x$ and $|b_T|$, but the deviation from a factorized form is a modest, well-resolved residual and not a qualitative failure. The Sivers shift itself is factorized to within roughly 5.7-7.6% in the relative $L_2$ sense. This is smaller than the 8-19% residuals of the amplitudes that enter it, which points to a partial cancellation of non-factorized behavior in the ratio. The residual traces back directly to the extrapolated fit parameters: the Bessel-function order $j_{\mathrm{im}} \approx 1$ governing $\widetilde{A}_{2B}^{[x]\mathrm{Im}}$ differs from $j_{\mathrm{re}}, j_{\mathrm{reA12B}} \approx 2$ governing the two real amplitudes, and the three width parameters $d_X$ are likewise resolved from one another well outside their Jackknife uncertainties. We conclude that an exactly factorized ansatz $F(x)G(|b_T|)$ would be a good, but not exact, approximation to the extracted Sivers shift.

This conclusion agrees qualitatively with the recent extraction of the non-perturbative unpolarized TMD from Drell–Yan data [119], in which an $x$–$b_T$ cross term survives a sparsity prior that pushes it toward zero. The two results are complementary rather than directly comparable. That extraction constrains the TMD for $b_T \lesssim 2.5$ GeV$^{-1}$ ($\approx 0.49$ fm), while our range $|b_T| = 0.34$–$0.94$ fm extends well beyond it, and our individual amplitudes are unrenormalized, so only the Sivers-shift ratio is a physical quantity. The comparison is therefore qualitative.

# 8 CONCLUSION AND OUTLOOK

This thesis set out to compute the dependence of the isovector $(u-d)$ generalized Sivers shift on the longitudinal momentum fraction $x$ directly from lattice QCD. We summarize the four central chapters, state the main results, and describe how the same method extends to other TMD observables.

## 8.1 Summary

Chapter 4 defined the quark TMD correlator with a staple-shaped gauge link and decomposed its Fourier-transformed, $b$-dependent form into Lorentz-invariant amplitudes. The Fourier-transformed TMDs $\tilde{f}_1$ and $\tilde{f}_{1T}^{\perp(1)}$ are given by the amplitudes $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$, and the $x$-dependent generalized Sivers shift, Eq. (4.48), is the ratio of their Fourier transforms over $b{\cdot}P$. In this ratio, the soft factor and the multiplicative, $\Gamma$-independent renormalization cancel.

Chapter 5 described the lattice calculation on a $32^3 \times 96$ ensemble with $a = 0.11403$ fm at four nucleon momenta, $P_L = -1, \ldots, -4$, which correspond to $\hat{\zeta} = 0.090772$ to $0.785756$. The invariants are not independent: Eq. (4.32) ties $v{\cdot}b$ to $b{\cdot}P$. We chose a lattice geometry that satisfies this constraint over a continuous range of $b{\cdot}P$ (Section 5.3). From the lattice correlators we extracted the real and imaginary parts of $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ and checked their parity in $b_L$. The imaginary Sivers amplitude $\widetilde{A}_{12B}^{\mathrm{Im}}$ is consistent with zero and was dropped.

Chapter 6 built a global fit of the lattice amplitudes by exploring several functional forms, with symbolic regression (`PySR`) as a guide. After we removed data with $|\boldsymbol{b}| < 3a$, `PySR` pointed to a power-law tail in $b^2$. A Gaussian model fails because its Fourier transform over $P \cdot b$ diverges. An exponential decay in $|\boldsymbol{b}|$ or in $P \cdot b$ describes $\widetilde{A}_{2B}^{\mathrm{Re}}$ alone but fails once it is applied to all three amplitudes. We fixed the $\eta$-dependence of the power-law parameters, selected a model for each amplitude, and fit the three amplitudes simultaneously with the full Jackknife covariance matrix to obtain the $x$-dependent Sivers shift at each momentum. Separable longitudinal and transverse models failed for $\widetilde{A}_{2B}^{\mathrm{Im}}$ and $\widetilde{A}_{12B}^{\mathrm{Re}}$.

Chapter 7 extrapolated the fit parameters to $\hat{\zeta} \to \infty$ with the form $c + d/\hat{\zeta}^2$, using $P_L = -2, -3, -4$, and reconstructed the amplitudes and the isovector Sivers shift in this limit (see Section 8.2). It then

tested factorization, first qualitatively on the lattice data in $(P \cdot b)$ and $b^2$, and then for the extrapolated results in $x$ and $|b_T|$. The Sivers shift is factorized to within about 5.7-7.6% in the relative $L_2$ sense, smaller than the 8-19% residuals of the amplitudes that enter it. A factorized form $F(x)G(|b_T|)$ is therefore a good, but not exact, approximation to the extracted Sivers shift.

### 8.2 Main results

- It is feasible to obtain the $x$-dependence of TMD ratios from lattice QCD. This is the main result of this thesis: we extracted the isovector $(u-d)$ **Sivers shift** $\langle k_y \rangle_{TU}$ as a function of $x$ at four nucleon momenta and extrapolated it to the physical limit $\hat{\zeta} \to \infty$ (Figures 82 and 84). At $|b_T| = 3a \approx 0.34$ fm, $\langle k_y \rangle_{TU}$ peaks at about 0.21 GeV at small $x$ and falls to near zero by $x \approx 0.5$, while $x\langle k_y \rangle_{TU}$ peaks at about 0.02 GeV near $x \approx 0.15$. The overall sign reverses between the SIDIS/DY limits while the magnitude is unchanged, so the magnitude and $x$-shape are the result. Figure 84 shows how both change with $|b_T|$ between 0.34 and 0.94 fm.

- The range of accessible $b{\cdot}P$ is limited:

$$\frac{v{\cdot}b}{v{\cdot}P} = b{\cdot}P \frac{R(\hat{\zeta}^2)}{m_N^2} \tag{8.1}$$

  In spite of this constraint, we improved the analysis. Orthogonal transverse vectors $\boldsymbol{v}_\mathrm{T} \perp \boldsymbol{b}_\mathrm{T}$ satisfy the constraint over a continuous range of $b{\cdot}P$, interpolation between lattice paths gives access to the off-axis staple directions, and the four momenta allow the extrapolation to $\hat{\zeta} \to \infty$.

### 8.3 Outlook

Extracting the $x$-dependence of the Sivers shift required the $x$-dependence of both the unpolarized TMD $f_1$ and the Sivers TMD $f_{1T}^{\perp}$. The same method applies to the other leading-twist TMDs of Eqs. (2.41)-(2.42), which come from the Dirac structures $\gamma^+\gamma_5$ and $i\sigma^{i+}\gamma_5$ instead of $\gamma^+$. Their $x$-integrated generalized shifts were defined and computed in Refs. [2, 82]. Following Eq. (4.48), their

$x$-dependent versions require non-zero $b{\cdot}P$ and a Fourier transform over it:

$$\begin{aligned}
\langle \boldsymbol{k}_y \rangle_{UT}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \eta v{\cdot}P) \;&\equiv\; m_N \frac{\tilde{h}_1^{\perp(1)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{(0)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)} \\
&=\; m_N \frac{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{4B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)}{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{2B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)} \\
&\xrightarrow{\boldsymbol{b}_{\rm T}^2=0}\; \frac{\sum_{\Lambda=\pm1} \int d^2\boldsymbol{k}_{\rm T}\, \boldsymbol{k}_y\; \Phi^{[\gamma^+ + s^j i\sigma^{j+}\gamma^5]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}{\sum_{\Lambda=\pm1} \int d^2\boldsymbol{k}_{\rm T} \quad \Phi^{[\gamma^+ + s^j i\sigma^{j+}\gamma^5]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}\bigg|_{\boldsymbol{s}_{\rm T}=(1,0)}\,, \qquad (8.2)
\end{aligned}$$

$$\begin{aligned}
\langle \boldsymbol{k}_x \rangle_{TL}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \eta v{\cdot}P) \;&\equiv\; m_N \frac{\tilde{g}_{1T}^{(1)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{(0)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)} \\
&=\; -m_N \frac{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{7B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)}{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{2B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)} \\
&\xrightarrow{\boldsymbol{b}_{\rm T}^2=0}\; \frac{\int d^2\boldsymbol{k}_{\rm T}\, \boldsymbol{k}_x\; \Phi^{[\gamma^+ + \lambda\gamma^+\gamma^5]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}{\int d^2\boldsymbol{k}_{\rm T} \quad \Phi^{[\gamma^+ + \lambda\gamma^+\gamma^5]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}\bigg|_{\boldsymbol{S}_{\rm T}=(1,0),\ \lambda=1}\,, \qquad (8.3)
\end{aligned}$$

$$\begin{aligned}
\frac{\tilde{h}_1^{(0)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)}{\tilde{f}_1^{(0)}(\boldsymbol{b}_{\rm T}^2, x; \hat{\zeta}, \ldots, \eta v{\cdot}P)} \;&=\; -\frac{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{9Bm}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)}{\int d(b{\cdot}P)\, e^{ix(b\cdot P)}\, \widetilde{A}_{2B}\big(-\boldsymbol{b}_{\rm T}^2, b{\cdot}P, (b{\cdot}P)R(\hat{\zeta}^2)/m_N^2, -1/(m_N\hat{\zeta})^2, \eta v{\cdot}P\big)} \\
&\xrightarrow{\boldsymbol{b}_{\rm T}^2=0}\; \frac{\int d^2\boldsymbol{k}_{\rm T}\; \Phi^{[s^j i\sigma^{j+}\gamma^5]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}{\int d^2\boldsymbol{k}_{\rm T}\; \Phi^{[\gamma^+]}(x, \boldsymbol{k}_{\rm T}, P, S; \ldots)}\bigg|_{\boldsymbol{S}_{\rm T}=(1,0),\ \boldsymbol{s}_{\rm T}=(1,0)}\,. \qquad (8.4)
\end{aligned}$$

Here $\widetilde{A}_{4B}$, $\widetilde{A}_{7B}$ and $\widetilde{A}_{9Bm}$ are the corresponding combinations of invariant amplitudes in the decompositions of the correlator for the Dirac structures $i\sigma^{\mu\nu}\gamma^5$ and $\gamma^\mu\gamma^5$, defined in Refs. [2, 82], just as $\widetilde{A}_{2B}$ and $\widetilde{A}_{12B}$ are for $\gamma^\mu$. Unlike the $x$-integrated shifts, which need only $b{\cdot}P = 0$, each of these ratios needs the amplitudes over a range of $b{\cdot}P$, with the kinematic constraint of Eq. (4.32) handled as in this thesis.

- The generalized Boer-Mulders shift $\langle \boldsymbol{k}_y \rangle^{\rm BM} = \langle \boldsymbol{k}_y \rangle_{UT}$, Eq. (8.2), describes transversely polarized quarks in an unpolarized proton. Like the Sivers shift, it is T-odd, so we expect opposite signs in the SIDIS and DY limits $\eta v{\cdot}P \to \pm\infty$. In the limit $\boldsymbol{b}_{\rm T}^2 = 0$, it measures, at fixed $x$, the dipole moment of the quark distribution orthogonal to the quark polarization. The sum over proton helicities $\sum_{\Lambda=\pm1}$ represents the unpolarized target.

- The generalized shift $\langle \boldsymbol{k}_x \rangle_{TL}$, Eq. (8.3), belongs to the "worm-gear" function $g_{1T}$. It describes a dipole deformation of the transverse momentum distribution caused by the correlation between the quark helicity and the transverse proton spin. It is T-even, so the SIDIS and DY limits should agree. This shift has been computed on the lattice with straight gauge links [105, 81, 82], and

its $x$-dependence would show how much the staple-shaped links in the limits $\eta v{\cdot}P \to \pm\infty$ change that result.

- The ratio $\tilde{h}_1^{(0)}/\tilde{f}_1^{(0)}$, Eq. (8.4), is an $x$-dependent "generalized tensor charge". It is also T-even. Because it involves no $\boldsymbol{k}$-weighting and relates directly to the transversity and unpolarized distributions, it is a clean observable. $\tilde{h}_1$ has been studied on the lattice with straight gauge links [105, 81, 82], which makes this ratio a good test of the change from straight to staple-shaped gauge links as a function of $x$.

## A QLUA: Gamma $\Gamma(n)$

The binary notation for 16 basis matrices:

$$\Gamma(n) = \gamma_1^{n_0}\gamma_2^{n_1}\gamma_3^{n_2}\gamma_4^{n_3} \tag{A.1}$$

where, the binary representation of $[n]_{10}$ is $[n_3n_2n_1n_0]_2$

$$[n]_{10} = n_3 2^3 + n_2 2^2 + n_1 2^1 + n_0 2^0 \tag{A.2}$$

The below table contains all 16 gamma matrices with binary notations.

| $[n]_{10}$ | $[n_3n_2n_1n_0]_2$ | $\Gamma(n)$ |
|---|---|---|
| 0 | 0000 | $\mathbb{1}$ |
| 1 | 0001 | $\gamma_1$ |
| 2 | 0010 | $\gamma_2$ |
| 3 | 0011 | $\gamma_1\gamma_2$ |
| 4 | 0100 | $\gamma_3$ |
| 5 | 0101 | $\gamma_1\gamma_3$ |
| 6 | 0110 | $\gamma_2\gamma_3$ |
| 7 | 0111 | $\gamma_1\gamma_2\gamma_3{=}\gamma_5\gamma_4$ |
| 8 | 1000 | $\gamma_4$ |
| 9 | 1001 | $\gamma_1\gamma_4$ |
| 10 | 1010 | $\gamma_2\gamma_4$ |
| 11 | 1011 | $\gamma_1\gamma_2\gamma_4 = -\gamma_5\gamma_3$ |
| 12 | 1100 | $\gamma_3\gamma_4$ |
| 13 | 1101 | $\gamma_1\gamma_3\gamma_4 = \gamma_5\gamma_2$ |
| 14 | 1110 | $\gamma_2\gamma_3\gamma_4 = -\gamma_5\gamma_1$ |
| 15 | 1111 | $\gamma_1\gamma_2\gamma_3\gamma_4 = \gamma_5$ |

## References


[1] A. Accardi *et al.*, Eur. Phys. J. A **52**, 268 (2016), arXiv:1212.1701 [nucl-ex] .

[2] B. U. Musch, P. Hägler, M. Engelhardt, J. W. Negele, and A. Schäfer, Phys. Rev. D **85**, 094510 (2012), arXiv:1111.4249 [hep-lat] .

[3] H.-W. Lin and H. B. Meyer, eds., *Lattice QCD for Nuclear Physics*, Lect. Notes Phys., Vol. 889 (Springer, 2015).

[4] R. Hofstadter, Rev. Mod. Phys. **28**, 214 (1956).

[5] E. D. Bloom *et al.*, Phys. Rev. Lett. **23**, 930 (1969).

[6] M. Breidenbach, J. I. Friedman, H. W. Kendall, E. D. Bloom, D. H. Coward, H. C. DeStaebler, J. Drees, L. W. Mo, and R. E. Taylor, Phys. Rev. Lett. **23**, 935 (1969).

[7] J. D. Bjorken, Phys. Rev. **179**, 1547 (1969).

[8] R. P. Feynman, Phys. Rev. Lett. **23**, 1415 (1969).

[9] J. D. Bjorken and E. A. Paschos, Phys. Rev. **185**, 1975 (1969).

[10] C. G. Callan, Jr. and D. J. Gross, Phys. Rev. Lett. **22**, 156 (1969).

[11] C.-N. Yang and R. L. Mills, Phys. Rev. **96**, 191 (1954).

[12] H. Fritzsch, M. Gell-Mann, and H. Leutwyler, Phys. Lett. B **47**, 365 (1973).

[13] D. J. Gross and F. Wilczek, Phys. Rev. Lett. **30**, 1343 (1973).

[14] H. D. Politzer, Phys. Rev. Lett. **30**, 1346 (1973).

[15] S. D. Drell and T.-M. Yan, Phys. Rev. Lett. **25**, 316 (1970), [Erratum: Phys.Rev.Lett. 25, 902 (1970)].

[16] D. Müller, D. Robaschik, B. Geyer, F. M. Dittes, and J. Hořejši, Fortsch. Phys. **42**, 101 (1994), arXiv:hep-ph/9812448 .

[17] A. V. Radyushkin, Phys. Rev. D **56**, 5524 (1997), arXiv:hep-ph/9704207 .

[18] X.-D. Ji, Phys. Rev. Lett. **78**, 610 (1997), arXiv:hep-ph/9603249 .

[19] M. Diehl, Phys. Rept. **388**, 41 (2003), arXiv:hep-ph/0307382 .

[20] M. Burkardt, Phys. Rev. D **62**, 071503 (2000), [Erratum: Phys.Rev.D 66, 119903 (2002)], arXiv:hep-ph/0005108 .

[21] J. C. Collins and D. E. Soper, Nucl. Phys. B **194**, 445 (1982).

[22] P. J. Mulders and R. D. Tangerman, Nucl. Phys. B **461**, 197 (1996), [Erratum: Nucl.Phys.B 484, 538–540 (1997)], arXiv:hep-ph/9510301 .

[23] J. Collins, *Foundations of perturbative QCD*, Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology, Vol. 32 (Cambridge University Press, 2011).

[24] R. N. Cahn, Phys. Lett. B **78**, 269 (1978).

[25] J. C. Collins and D. E. Soper, Nucl. Phys. B **193**, 381 (1981), [Erratum: Nucl.Phys.B 213, 545 (1983)].

[26] J. C. Collins, D. E. Soper, and G. F. Sterman, Nucl. Phys. B **250**, 199 (1985).

[27] J. P. Ralston and D. E. Soper, Nucl. Phys. B **152**, 109 (1979).

[28] R. D. Tangerman and P. J. Mulders, Phys. Rev. D **51**, 3357 (1995), arXiv:hep-ph/9403227 .

[29] A. Kotzinian, Nucl. Phys. B **441**, 234 (1995), arXiv:hep-ph/9412283 .

[30] D. Boer and P. J. Mulders, Phys. Rev. D **57**, 5780 (1998), arXiv:hep-ph/9711485 .

[31] X.-D. Ji, J.-P. Ma, and F. Yuan, Phys. Rev. D **71**, 034005 (2005), arXiv:hep-ph/0404183 .

[32] S. M. Aybat and T. C. Rogers, Phys. Rev. D **83**, 114042 (2011), arXiv:1101.5057 [hep-ph] .

[33] A. V. Belitsky, X.-d. Ji, and F. Yuan, Phys. Rev. D **69**, 074014 (2004), arXiv:hep-ph/0307383 .

[34] S. Meissner, A. Metz, and M. Schlegel, JHEP **08**, 056 (2009), arXiv:0906.5323 [hep-ph] .

[35] C. Lorcé and B. Pasquini, Phys. Rev. D **84**, 014015 (2011), arXiv:1106.0139 [hep-ph] .

[36] D. Boer *et al.*, "Gluons and the quark sea at high energies: Distributions, polarization, tomography," (2011), arXiv:1108.1713 [nucl-th] .

[37] R. Abdul Khalek *et al.*, Nucl. Phys. A **1026**, 122447 (2022), arXiv:2103.05419 [physics.ins-det] .

[38] J. Ashman *et al.* (European Muon), Phys. Lett. B **206**, 364 (1988).

[39] R. L. Jaffe and A. Manohar, Nucl. Phys. B **337**, 509 (1990).

[40] E. Leader and C. Lorcé, Phys. Rept. **541**, 163 (2014), arXiv:1309.4235 [hep-ph] .

[41] G. Bunce *et al.*, Phys. Rev. Lett. **36**, 1113 (1976).

[42] D. L. Adams *et al.* (FNAL-E704), Phys. Lett. B **264**, 462 (1991).

[43] G. L. Kane, J. Pumplin, and W. Repko, Phys. Rev. Lett. **41**, 1689 (1978).

[44] D. W. Sivers, Phys. Rev. D **41**, 83 (1990).

[45] A. Bacchetta, U. D'Alesio, M. Diehl, and C. A. Miller, Phys. Rev. D **70**, 117504 (2004), arXiv:hep-ph/0410050 .

[46] J. C. Collins, Nucl. Phys. B **396**, 161 (1993), arXiv:hep-ph/9208213 .

[47] S. J. Brodsky, D. S. Hwang, and I. Schmidt, Phys. Lett. B **530**, 99 (2002), arXiv:hep-ph/0201296 .

[48] J. C. Collins, Phys. Lett. B **536**, 43 (2002), arXiv:hep-ph/0204004 .

[49] S. J. Brodsky, D. S. Hwang, and I. Schmidt, Nucl. Phys. B **642**, 344 (2002), arXiv:hep-ph/0206259 .

[50] A. V. Belitsky, X. Ji, and F. Yuan, Nucl. Phys. B **656**, 165 (2003), arXiv:hep-ph/0208038 .

[51] D. Boer, P. J. Mulders, and F. Pijlman, Nucl. Phys. B **667**, 201 (2003), arXiv:hep-ph/0303034 .

[52] Z.-B. Kang and J.-W. Qiu, Phys. Rev. Lett. **103**, 172001 (2009), arXiv:0903.3629 [hep-ph] .

[53] M. Burkardt, Phys. Rev. D **66**, 114005 (2002), arXiv:hep-ph/0209179 .

[54] M. Burkardt, Nucl. Phys. A **735**, 185 (2004), arXiv:hep-ph/0302144 .

[55] M. Burkardt, Phys. Rev. D **69**, 091501 (2004), arXiv:hep-ph/0402014 .

[56] J.-w. Qiu and G. F. Sterman, Phys. Rev. Lett. **67**, 2264 (1991).

[57] X. Ji, J.-W. Qiu, W. Vogelsang, and F. Yuan, Phys. Rev. Lett. **97**, 082002 (2006), arXiv:hep-ph/0602239 .

[58] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, and M. Schlegel, JHEP **02**, 093 (2007), arXiv:hep-ph/0611265 .

[59] A. Airapetian *et al.* (HERMES), Phys. Rev. Lett. **103**, 152002 (2009), arXiv:0906.3918 [hep-ex] .

[60] A. Airapetian *et al.* (HERMES), JHEP **12**, 010 (2020), arXiv:2007.07755 [hep-ex] .

[61] M. Alekseev *et al.* (COMPASS), Phys. Lett. B **673**, 127 (2009), arXiv:0802.2160 [hep-ex] .

[62] C. Adolph *et al.* (COMPASS), Phys. Lett. B **744**, 250 (2015), arXiv:1408.4405 [hep-ex] .

[63] X. Qian *et al.* (Jefferson Lab Hall A), Phys. Rev. Lett. **107**, 072003 (2011), arXiv:1106.0363 [nucl-ex] .

[64] L. Adamczyk *et al.* (STAR), Phys. Rev. Lett. **116**, 132301 (2016), arXiv:1511.06003 [nucl-ex] .

[65] M. Aghasyan *et al.* (COMPASS), Phys. Rev. Lett. **119**, 112002 (2017), arXiv:1704.00488 [hep-ex] .

[66] C. Adolph *et al.* (COMPASS), Phys. Lett. B **770**, 138 (2017), arXiv:1609.07374 [hep-ex] .

[67] M. Anselmino, M. Boglione, U. D'Alesio, A. Kotzinian, F. Murgia, and A. Prokudin, Phys. Rev. D **72**, 094007 (2005), [Erratum: Phys.Rev.D 72, 099903 (2005)], arXiv:hep-ph/0507181 .

[68] M. Anselmino, M. Boglione, U. D'Alesio, A. Kotzinian, S. Melis, F. Murgia, A. Prokudin, and C. Turk, Eur. Phys. J. A **39**, 89 (2009), arXiv:0805.2677 [hep-ph] .

[69] M. G. Echevarria, A. Idilbi, Z.-B. Kang, and I. Vitev, Phys. Rev. D **89**, 074013 (2014), arXiv:1401.5078 [hep-ph] .

[70] A. Bacchetta, F. Delcarro, C. Pisano, and M. Radici, Phys. Lett. B **827**, 136961 (2022), arXiv:2004.14278 [hep-ph] .

[71] M. Bury, A. Prokudin, and A. Vladimirov, Phys. Rev. Lett. **126**, 112002 (2021), arXiv:2012.05135 [hep-ph] .

[72] M. Bury, A. Prokudin, and A. Vladimirov, JHEP **05**, 151 (2021), arXiv:2103.03270 [hep-ph] .

[73] J. Cammarota, L. Gamberg, Z.-B. Kang, J. A. Miller, D. Pitonyak, A. Prokudin, T. C. Rogers, and N. Sato (Jefferson Lab Angular Momentum), Phys. Rev. D **102**, 054002 (2020), arXiv:2002.08384 [hep-ph] .

[74] R. Boussarie *et al.*, "TMD Handbook," (2023), arXiv:2304.03302 [hep-ph] .

[75] J. Dudek *et al.*, Eur. Phys. J. A **48**, 187 (2012), arXiv:1208.1244 [hep-ex] .

[76] K. G. Wilson, Phys. Rev. D **10**, 2445 (1974).

[77] C. Gattringer and C. B. Lang, *Quantum chromodynamics on the lattice*, Lect. Notes Phys., Vol. 788 (Springer, 2010).

[78] M. Göckeler, R. Horsley, E.-M. Ilgenfritz, H. Perlt, P. E. L. Rakow, G. Schierholz, and A. Schiller, Phys. Rev. D **54**, 5705 (1996), arXiv:hep-lat/9602029 .

[79] H.-W. Lin *et al.*, Prog. Part. Nucl. Phys. **100**, 107 (2018), arXiv:1711.07916 [hep-ph] .

[80] M. Constantinou *et al.*, Prog. Part. Nucl. Phys. **121**, 103908 (2021), arXiv:2006.08636 [hep-ph] .

[81] P. Hägler, B. U. Musch, J. W. Negele, and A. Schäfer, EPL **88**, 61001 (2009), arXiv:0908.1283 [hep-lat] .

[82] B. U. Musch, P. Hägler, J. W. Negele, and A. Schäfer, Phys. Rev. D **83**, 094507 (2011), arXiv:1011.1213 [hep-lat] .

[83] M. Engelhardt, P. Hägler, B. Musch, J. Negele, and A. Schäfer, Phys. Rev. D **93**, 054501 (2016), arXiv:1506.07826 [hep-lat] .

[84] B. Yoon, T. Bhattacharya, M. Engelhardt, J. Green, R. Gupta, P. Hägler, B. Musch, J. Negele, A. Pochinsky, and S. Syritsyn, in *33rd International Symposium on Lattice Field Theory*, PoS, Vol. LATTICE2015 (SISSA, 2016) p. 116, arXiv:1601.05717 [hep-lat] .

[85] B. Yoon, M. Engelhardt, R. Gupta, T. Bhattacharya, J. R. Green, B. U. Musch, J. W. Negele, A. V. Pochinsky, A. Schäfer, and S. N. Syritsyn, Phys. Rev. D **96**, 094508 (2017), arXiv:1706.03406 [hep-lat] .

[86] X. Ji, Phys. Rev. Lett. **110**, 262002 (2013), arXiv:1305.1539 [hep-ph] .

[87] X. Ji, P. Sun, X. Xiong, and F. Yuan, Phys. Rev. D **91**, 074009 (2015), arXiv:1405.7640 [hep-ph] .

[88] X. Ji, Y.-S. Liu, Y. Liu, J.-H. Zhang, and Y. Zhao, Rev. Mod. Phys. **93**, 035005 (2021), arXiv:2004.03543 [hep-ph] .

[89] M. A. Ebert, I. W. Stewart, and Y. Zhao, Phys. Rev. D **99**, 034505 (2019), arXiv:1811.00026 [hep-ph] .

[90] P. Shanahan, M. Wagman, and Y. Zhao, Phys. Rev. D **104**, 114502 (2021), arXiv:2107.11930 [hep-lat] .

[91] M. Schlemmer, A. Vladimirov, C. Zimmermann, M. Engelhardt, and A. Schäfer, JHEP **08**, 004 (2021), arXiv:2103.16991 [hep-lat] .

[92] M.-H. Chu *et al.* (Lattice Parton (LPC)), Phys. Rev. D **106**, 034509 (2022), arXiv:2204.00200 [hep-lat] .

[93] M. Cranmer, "Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl," (2023), arXiv:2305.01582 [astro-ph.IM] .

[94] M. E. Peskin and D. V. Schroeder, *An Introduction to quantum field theory* (Addison-Wesley, Reading, USA, 1995).

[95] S. Navas *et al.* (Particle Data Group), Phys. Rev. D **110**, 030001 (2024).

[96] M. Creutz, Phys. Rev. D **21**, 2308 (1980).

[97] S. J. Brodsky, H.-C. Pauli, and S. S. Pinsky, Phys. Rept. **301**, 299 (1998), arXiv:hep-ph/9705477 .

[98] V. N. Gribov and L. N. Lipatov, Sov. J. Nucl. Phys. **15**, 438 (1972).

[99] G. Altarelli and G. Parisi, Nucl. Phys. B **126**, 298 (1977).

[100] Y. L. Dokshitzer, Sov. Phys. JETP **46**, 641 (1977).

[101] K. Goeke, A. Metz, and M. Schlegel, Phys. Lett. B **618**, 90 (2005), arXiv:hep-ph/0504130 .

[102] J. C. Collins, Acta Phys. Polon. B **34**, 3103 (2003), arXiv:hep-ph/0304122 .

[103] A. Bacchetta, M. Boglione, A. Henneman, and P. J. Mulders, Phys. Rev. Lett. **85**, 712 (2000), arXiv:hep-ph/9912490 .

[104] J. C. Collins, T. C. Rogers, and A. M. Stasto, Phys. Rev. D **77**, 085009 (2008), arXiv:0708.2833 [hep-ph] .

[105] B. U. Musch, *Transverse momentum distributions inside the nucleon from lattice QCD*, Ph.D. thesis, TU Munich (2009), arXiv:0907.2381 [hep-lat] .

[106] S. M. Aybat, J. C. Collins, J.-W. Qiu, and T. C. Rogers, Phys. Rev. D **85**, 034043 (2012), arXiv:1110.6428 [hep-ph] .

[107] A. Bacchetta, D. Boer, M. Diehl, and P. J. Mulders, JHEP **08**, 023 (2008), arXiv:0803.0227 [hep-ph] .

[108] D. Boer, L. Gamberg, B. Musch, and A. Prokudin, JHEP **10**, 021 (2011), arXiv:1107.5294 [hep-ph] .

[109] X. Ji, Y. Liu, and Y.-S. Liu, Nucl. Phys. B **955**, 115054 (2020), arXiv:1910.11415 [hep-ph] .

[110] M. A. Ebert, S. T. Schindler, I. W. Stewart, and Y. Zhao, JHEP **04**, 178 (2022), arXiv:2201.08401 [hep-ph] .

[111] A. V. Radyushkin, Phys. Rev. D **96**, 034025 (2017), arXiv:1705.01488 [hep-ph] .

[112] H. B. Nielsen and M. Ninomiya, Nucl. Phys. B **185**, 20 (1981), [Erratum: Nucl.Phys.B 195, 541 (1982)].

[113] B. Sheikholeslami and R. Wohlert, Nucl. Phys. B **259**, 572 (1985).

[114] S. Duane, A. D. Kennedy, B. J. Pendleton, and D. Roweth, Phys. Lett. B **195**, 216 (1987).

[115] A. Idilbi, X.-D. Ji, J.-P. Ma, and F. Yuan, Phys. Rev. D **70**, 074021 (2004), arXiv:hep-ph/0406302 .

[116] A. Accardi, A. Bacchetta, W. Melnitchouk, and M. Schlegel, JHEP **11**, 093 (2009), arXiv:0907.2942 [hep-ph] .

[117] J. R. Koza, *Genetic Programming: On the Programming of Computers by Means of Natural Selection* (MIT Press, Cambridge, MA, USA, 1992).

[118] M. Schmidt and H. Lipson, Science **324**, 81 (2009).

[119] C. Granger, A. Bacchetta, V. Bertone, C. Bissolotti, M. Cerutti, M. Radici, S. Rodini, L. Rossi, and C. Fanelli, "Symbolic Extraction of Non-Perturbative Transverse-Momentum-Dependent Distributions from Drell-Yan Data," (2026), arXiv:2607.24710 [hep-ph] .

[120] X. Ji, Y. Liu, and Y. Su, "Asymptotic Long-Distance Expansion of Euclidean Correlators in Lattice Parton Applications," (2026), arXiv:2601.12189 [hep-lat] .

[121] B. Yoon, Y.-C. Jang, C. Jung, and W. Lee, "Covariance fitting of highly correlated data in lattice QCD," (2011), arXiv:1101.2248 [hep-lat] .

[122] G. P. Lepage, B. Clark, C. T. H. Davies, K. Hornbostel, P. B. Mackenzie, C. Morningstar, and H. Trottier, Nucl. Phys. B Proc. Suppl. **106**, 12 (2002), arXiv:hep-lat/0110175 .